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NANOTECHNOLOGY AND NANOSCIENCE FOR SUSTAINABILITY

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PREFACE

The word “**nanoscience**” refers to the study, manipulation, and engineering of matter, particles, and structures on the nanometer scale. Important properties of materials, such as the electrical, optical, thermal, and mechanical properties, are determined by the way molecules and atoms assemble on the nanoscale into larger structures. Moreover, in nanometer-sized structures, these properties are often different than on a macroscale because quantum mechanical effects become important. **Nanotechnology** is the application of nanoscience leading to the use of new nanomaterials and nanosize components in useful products. Nanotechnology will eventually provide us with the ability to design custom-made materials and products with new enhanced properties, new nanoelectronics components, new types of “smart” medicines and sensors, and even interfaces between electronics and biological systems. These scientific disciplines are situated at the interface between physics, chemistry, materials science, microelectronics, biochemistry, and biotechnology.

Therefore, it is inevitable that for sustainable development in 21st century, nanoscience has to play a pivotal role. It is also a rapidly developing area of modern science to cater to the needs of the 21st century. It is widely believed that a clean environment is the need of the hour, and this has driven the nanoscientists to develop newer, cheaper, and more effective nanomaterials for environmental remediation, leading to numerous research publications in the last few years. Keeping this in mind, we have compiled this book to give the scientific community a comprehensive idea about the recent progress in this field.

This book contains both review articles along with some interesting research work involving the designing of new nanomaterials for various applications, highlighting the future scope of this field. We hope this book will be of particular interest to nanoscientists and will provide new ideas for future research in the field of nanoscience.

We would like to offer sincere thanks to the scientific reviewers of this project, **Dr. Ashim Jyoti Thakur, Professor, Department of Chemical Science, Tezpur University**, and **Dr. Palash Moni Saikia, Principal, Darrang College**, for providing their insightful comments and suggestions for improving the manuscripts.

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Nanoscience and Nanotechnology : An Emerging and Promising Area

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Abstract :

Nanoscience and Nanotechnology are rapidly growing fields and have paved the way for new opportunities in many areas of research and application which involve structures, devices, and systems with novel features and functionalities due to the arrangement of their atoms on a nanoscale. Areas in which nanotechnology is making a difference include energy, environmental protection, resource management, and health care through the development of smart materials and connected devices. Nanotechnology and materials are also expected to contribute to the realization of a sustainable society by ensuring water purification, reducing CO₂ emissions, and promoting material circulation with recycling approaches. Within a few years, Nanotechnology and Nanoscience have become of fundamental importance as far as applications are concerned.

Keywords : Catalysis, Nanoscience, Nanoparticles, Nanotechnology, Quantum dots, Sustainability

Introduction :

Nanoscience and nanotechnology have garnered a lot of interest over time due to the astounding advancements followed by innovative and intriguing applications of nanomaterials. Both have emerged as transformative potential technologies providing sustainable solutions and alternatives to conventional counterparts [1]. Nanoscience is an emerging area of science that has the potential to reshape the world around us. Interdisciplinary in nature, it is the convergence of physics, materials science, and biology, which deal with the manipulation of particles and materials at extremely small scales, i.e. at atomic and molecular scales [2]. On the other hand, nanotechnology is the design, production, and application of structures, devices, and systems at the nanoscale [3]. It is one of the most promising technologies of the 21st century that comprises the synthesis, engineering, and utilization of materials whose size is above molecular dimensions and below macroscopic ones and ranges from 1 to 100 nm [4] at least in one dimension ((height, width, or length). This is the convention of International System of Units (SI) for nanomaterials. Literally, 'nano' (Greek origin) means

dwarf and, by extension, very small.

Nanotechnology also encompasses several disciplines including biology, chemical, mechanical, and electronics engineering to comprehend, manipulate, and build devices with remarkable functionalities and qualities at the atomic, molecular, or supramolecular levels [5]. Nanotechnology can also be defined as the creation of any application on the nanometre scale. Since, a nanometer is one-billionth of a meter, and hence, nanotechnology is all about reliably manufacturing technology at such tiny scales [6].

Literatures documented that the concept of nanotechnology was introduced in 1959 by the legendary American physicist and Nobel laureate Richard Feynman. He rightly visioned at that time that there's plenty of room at the bottom. Subsequently, the term "nanotechnology" was formally defined by Japanese scientist Norio Taniguchi in 1974 [7]. Today, it is a rapidly expanding field with the potential to revolutionise different sectors and increase our understanding of matter's fundamental features. From the industry point of view, Nanotechnology has gained popularity as it offers better-built and smarter products with exciting features. For instance, in the medicine and health care sector, the application of nanotechnology is amazingly enormous as it has been used in diagnosis and to combat various diseases, including cardiovascular diseases and cancer [8]. Objects at such a smaller scale display different, interesting and unique size dependant physical, chemical, or biological properties such as electrical conductance, chemical reactivity, magnetism, optical effects and physical strength from their bulk counterpart at micro and macro scale. At the nanoscale whether natural or engineered gravity becomes less important. On the other hand, properties like surface tension and van der Waals forces play very important roles. Because of the changes in the behaviour of materials at the nanoscale, it has become possible to develop some extraordinary products which could not have been possible to develop from macro-scale particles.

Nanoparticles :

Nanoparticles are the important building components of nanotechnology and a basic term used in nanotechnology. Nanoparticles are objects ranging in size from 1 to 100nm at least in one dimension and are the most fundamental component in the fabrication of a nanostructure. Typically, they are of carbon, metal, metal oxides, organic matter etc. origin. Other examples include CdS, ZnS, silicon, and germanium etc.

It is interesting to note that compared to their bulk counterparts, nanoparticles have higher surface charge, absorption capacity, reactivity, surface area, sensitivity, stability, and strength [9]. In recent years manufacturing of nanoscale materials for instance, metallic nanoparticles have gained enormous attention due to their unique features [10]. Metallic nanoparticles have different size dependant physical and chemical properties from bulk metals i.e. lower melting points, higher specific surface areas, specific optical properties, mechanical strengths, and magnetization properties that appear attractive to various industries [11].

Nanocomposites :

Nanocomposites are a class of nanomaterials, where one or more nanostructured materials (organic or inorganic) are incorporated to improve the particular property of a material. One of the striking features of nanocomposites is that they exhibit synergistic properties that are not present in the individual components. Nanocomposites are applied in various branches of science and different industries, such as target drug delivery, imaging, agriculture, sensors, batteries, artificial implants, and aircraft. Some examples of nanocomposite components include nanofibers, nanorods, nanoparticles, and carbon nanotubes [12].

Quantum dots :

Semiconductor nanoparticles also known as quantum dots were theorized in the 1970s and first produced in the early 1980s. Quantum effects come into play when semiconductor particles are made small enough which limits the energies at which electrons and holes can exist in the particles [13]. As energy is related to wavelength this means that the optical properties of the particle can be finely tuned depending on its size. Thus, particles can be made to emit or absorb specific wavelengths of light, merely by controlling their size. Quantum dots have found applications in composites, solar cells, and fluorescent biological labels which use both small particle size and tuneable energy levels [14].

Nanomaterials :

Nanomaterials (NM) is defined as material with any internal or external structures on the nanoscale dimension. They are synthesized using nanotechnology principles focusing on the creation, examination, and understanding of materials at the nanoscale. Some examples of nanomaterials are quantum dots, nanocomposite, carbon nanotubes, graphene, nanowire, and fullerene. Nanomaterials find distinctive features as they provide a large surface area-to-volume ratio. Nanomaterials can be added to pre-existing products that improve the composite objects by lending some of their unique characteristics [15]. Nanomaterials have typical optical, electrical, and magnetic properties at the nanoscale, and can be applied in the field of electronics and medicine, amongst others. Apart from other uses nanomaterials also find applications in catalysis, water treatment, energy storage, medicine, and agriculture. The beneficial effect of nanomaterials can be found in cosmetics, sports goods, tires, electronic devices, and other everyday items. Notably, the usage of nanomaterials dates back to 4500 years when natural asbestos nanofibres were utilized for ceramic matrices. The oldest and richest culture globally, the Egyptians, realized the importance of nanomaterials 4000 years ago [16,17].

Nanomaterials represent a significant domain within nanotechnology,

Classification of nanomaterials :

Nanomaterials are the fundamental elements of nanotechnology. Interestingly, there is not a single classification of nanomaterials. Nanomaterials can be divided into five categories depending on their size, place of origin, structural configuration, pore size, and potential toxicity [1].

Classification of nanomaterials based on their origin :

Considering their origin, nanomaterials are divided into two groups - natural and artificial nanomaterials.

Natural nanomaterials :

Naturally occurring viruses, protein molecules, minerals like clay, natural colloids like milk and blood, fog, gelatine, mineralized natural materials like shells, corals, and bones, insect wings and opals, spider silk, lotus leaves, gecko feet, volcanic ash, and ocean spray are the examples of natural nanomaterials [1].

Artificial nanomaterials :

Carbon nanotubes and semiconductor nanoparticles like quantum dots (QDs) are examples of artificial nanomaterials that are made consciously using precise mechanical and manufacturing procedures [1].

Classification of Nanomaterials based on Structural Configuration :

Nanomaterials are categorized into their respective categories based on their morphology, i.e. their structure, size, and shape:

Organic nanomaterials :

Organic nanomaterials result when organic compounds are converted into nanoscale. Examples include micelles, dendrimers, liposomes, nanogels, polymeric nanoparticles and layered biopolymers. Certain organic nanoparticles such as micelles and liposomes are non-toxic and biodegradable and can be broken down naturally. Organic nanoparticles can perform as an excellent option as a carrier of pharmaceuticals and are frequently used in the process of transporting target medications to their intended locations [1].

Inorganic nanomaterials :

Inorganic nanoparticles do not contain carbon and have the advantages of being hydrophilic, non-toxic, and biocompatible with living systems. The stability of inorganic nanoparticles is superior to that of organic analogs. Magnetic nanoparticles (MNPs) including maghemite ($\gamma\text{-Fe}_2\text{O}_3$) or magnetite (Fe_3O_4) are the examples of most notable inorganic nanomaterials. Nanoparticles of elements such as Cd, Al, Cu, Ag, Pb, Co, Zn, Au, Fe, and compounds such as FeO, Ag_2O , ZnO, Bi_2O_3 , CuO, CaO, and Al_2O_3 are well-known examples of inorganic nanoparticles. In 2015, Thomas and co-workers found that Ag_2O nanoparticles had antibacterial capabilities against *E. coli*. Similarly, ZnO nanoparticles demonstrated strong bactericidal efficacy against gram-negative and gram-positive bacteria and spores. Noble metal nanoparticles have been used in diverse biomedical applications, including those for cancer treatment, to improve the effectiveness of radiotherapy, fight germs and fungi, perform thermal ablation, deliver drugs, and transport genes [1].

Carbon-based nanomaterials :

Carbon-based nanomaterials are formed entirely of carbon. Carbon nanotubes, graphene, carbon nanofibers, fullerenes, and carbon black are common examples of carbon-based nanomaterials. Owing to the presence of a variety of allotropes of carbon, carbon-based materials have been debated as valuable treasures in recent times. The well-known allotropic phases of carbon are amorphous diamond, graphite, and recently discovered allotropes like opportune graphene quantum dots (GQDs), fullerene, graphene oxide (GO), and carbon nanotubes (CNTs). Amorphous carbon is one of the most common forms of carbon. Single-walled carbon nanotubes (SWCNTs) and multi-walled carbon nanotubes (MWCNTs) are the two categories that can be used to categorize carbon nanotubes [1]. As a sustainable ecofriendly material, carbon nanomaterial is expected to replace many other materials in different fields in the coming days.

Classification of nanomaterials based on their dimensions :

Nanomaterials undergo four types of classification based on the dimensionality (D) of their features, resulting in distinctions of Zero-dimensional (0D), One-dimensional (1D), 2D, and Three-dimensional (3D) nanomaterials. [1, 18].

Zero-dimensional nanomaterials :

These nanomaterials have either all three dimensions (x, y, and z) within the nanoscale range or are not dimensional outside the nanometric range ($>10\text{nm}$). Within the domain of 0D nanostructures, one encounters nanoparticles, nanospheres, quantum dots, isolated molecules, and atoms—point structures showcasing nanoscale dimensions in all directions [1, 19]. They are found to exhibit various shapes and forms and are amorphous or crystalline, single crystalline or polycrystalline, metallic, or ceramic in nature [1].

One-dimensional nanomaterials :

In this class, nanomaterials have two of their three dimensions (x, y) in the nanoscale range, but the other dimension of the nanostructure is outside the non-metric range ($>10\text{nm}$). Nanofibers, nanotubes, nanohorns, nanorods, thin films, and nanowires are examples of needle-shaped 1D nanomaterials. They can be amorphous or crystalline (single or polycrystalline), chemically pure or impure, stand alone materials, or embedded within another medium, such as metallic, ceramic, or polymeric 1D nanoparticles [1].

Two-dimensional nanomaterials :

Two-dimensional (2D) nanomaterials have two dimensions outside the nanometer range, but one dimension is at the nanoscale (between 1 and 100nm). 2D nanomaterials can be amorphous or crystalline of various chemical compositions, deposited on a substrate, or integrated into a surrounding matrix material (metallic or polymeric). Coatings and thin-film multilayers, nanosheets or nanowalls, free particles, tubes, fibers, ultrafine-grained over layers, wires, nanoplates, nanobelts, and nanodiscs, and platelets are examples of 2D nanomaterials [1, 15, 16, 18, 19].

Three-dimensional nanomaterials or bulk materials :

Three-dimensional (3D) nanomaterials or bulk materials are nanomaterials that are not confined to the nanoscale in any dimension or dimensional range. All dimensions of a 3D material are outside the nanometer range or greater than 100nm, but the bulk material is made up of individual blocks that are in the nanometer scale (1–100nm), therefore, 3D nanomaterials have three arbitrary dimensions above 100nm. It includes nanoparticles dispersion, arrays of nanowires and nanotubes, nanotetrapods, nanoflowers, nanoprisms, and nanocombs and multi-nano layers in which the 0D, 1D, and 2D structural elements are in close contact and form interfaces. [1,16, 17].

Classification of nanomaterials based on their pore dimensions :

Microporous, mesoporous, and macroporous nanomaterials are three groups of nanomaterials that are classified based on the length of their pore diameter dimensions.

Micro porous materials :

Microporous materials are materials having narrow pores with diameters less than 2nm. They comprise small molecules such as gases or linear molecules. They show slow diffusion kinetics and high interaction properties. Naturally occurring clay materials are examples of microporous materials [1].

Mesoporous materials :

Mesoporous materials have pores with a diameter range larger than 2nm but smaller than 50nm large enough to hold some bigger molecules. MCM-41, MCM-48, and SBA-15 are mesoporous materials [1].

Macroporous materials

Macroporous materials are materials with pores having a diameter greater than 50nm to host very large molecules, such as polyaromatic systems or small biological molecules. Carbon microtubes, porous gels, and porous glasses are examples of macroporous materials [1].

Approaches for the synthesis of nanoparticles :

Describing the different possibilities for the synthesis of nanostructures, mainly two approaches are adopted: Top-down and Bottom-up approach.

Top-down : The top-down approach is essentially the breaking down of bulk material into nano-sized particles. The following techniques are utilized to achieve a top-down approach:

- ♦ **Mechanical milling :** The mechanical milling process uses balls inside containers and may be carried out in various mills, typically planetary and shaker mills. It is a practical approach to create materials at the nanoscale from bulk materials. For example, nanoalloys based on aluminium, nickel, magnesium, copper, and a variety

of other nanocomposite materials may all be created mechanically [20].

- ♦ **Electrospinning** : A technique for creating fibers called electrospinning draws charged threads from polymer melts or solutions up to fiber sizes of a few hundred nanometers. Coaxial electrospinning was a significant advancement in the field of nanotechnology. Core-shell and hollow polymer, inorganic, organic, and hybrid materials have all been developed using this technique [20].
- ♦ **Laser ablation** : This technique produces nanoparticles by striking the target material with an intense laser beam. Due to the high intensity of the laser irradiation being used in the process, the source material or precursor vaporizes, causing the production of nanoparticles. Laser ablation is an environmentally friendly technique for producing noble metal nanoparticles [20].
- ♦ **Sputtering** : Microparticles of a solid material are expelled off its surface during the phenomenon known as sputtering, which occurs when the solid substance is assaulted by intense plasma or gas particles. The sputtering method is intriguing because it is more affordable than electron-beam lithography and the composition of the sputtered nanomaterials is similar to the target material with fewer contaminants [20].
- ♦ **Sonication** : The most crucial step in the creation of nanofluids is sonication. After the mixture has been magnetically stirred in a magnetic stirrer, sonication is performed in an ultrasonication path, ultrasonic vibrator, and mechanical homogenizer. Probe sonication is highly effective for processing nanomaterials [20].
- ♦ **Lithography** : Lithography typically uses a concentrated beam of light or electrons to create nanoparticles. It is an affordable and easy-to-apply method [20].

Bottom-up : The bottom-up approach refers to the build-up of nanostructures from the bottom, atom-by-atom or molecule-by-molecule by physical and chemical methods which are in a nanoscale range (1 nm to 100 nm) using controlled manipulation of self-assembly of atoms and molecules. Following are some common methods utilized to achieve a bottom-up approach :

- ♦ **Chemical vapor deposition (CVD)** : Through a chemical process involving vapor-phase precursors, a thin coating is created on the substrate surface during CVD. Precursors are deemed appropriate for CVD if they exhibit sufficient volatility, high chemical purity, and strong evaporation stability. This technique is useful for creating 2D nanoparticles [20].
- ♦ **Sol-gel process** : A wet-chemical approach called the sol-gel method is widely utilized to create nanomaterials. Metal alkoxides or metal precursors in solution are condensed, hydrolyzed, and thermally decomposed. The result is a stable solution or sol. The gel gains greater viscosity as a result of hydrolysis or condensation. The particle size may be seen by adjusting the precursor concentration, temperature, and pH levels. It may take a few days for the solvent to be removed,

for Ostwald ripening to occur, and for the phase to change during the mature stage, which is necessary to enable the growth of solid mass. A few advantages of the sol-gel process are the uniform quality of the material generated, the low processing temperature, and the method's ease in producing composites [20].

- ♦ **Co-precipitation** : It is a solvent displacement technique and is a wet chemical procedure. Ethanol, acetone, hexane, and non-solvent polymers are examples of solvents used. One key benefit of this method is the natural ability to produce high quantities of water-soluble nanoparticles through a straightforward process [20].
- ♦ **Hydrothermal** : This method utilises a wide temperature (ambient to high temperature) range for the production of nanoparticles. At higher temperature ranges, the nanomaterials produced by hydrothermal synthesis could become unstable [20].

Inert gas condensation/molecular condensation: Large quantities of metal nanoparticles are produced using this method. Making fine nanoparticles using the inactive gas compression approach has been widespread, which creates nanoparticles by causing a metallic source to disappear in an inert gas. For example, Copper metal nanoparticles are created by vaporizing copper metal inside a container containing argon, helium, or neon [20].

Biological synthesis using microorganisms : Microbes use metal capture, enzymatic reduction, and capping to create nanoparticles. The use of microorganisms (especially bacteria, algae, and fungi) for the synthesis of metallic nanoparticles is environmentally friendly, fast and economical [20, 21].

Biological synthesis using plant extracts : Plant extracts are combined with a metal salt solution at room temperature to create nanoparticles. This method has been used to create nanoparticles of silver, gold, and many other metals. A leaf extract from *Polyalthia longifolia* was used to create silver nanoparticles; the average particle size was around 58 nm [20].

Biosynthesis of nano particles is a green scale down or bottom-up approach and involves redox (reduction/oxidation) reaction. The microbial enzymes or the phyto-chemicals present in plant extracts having anti-oxidant or reducing capabilities are usually responsible for reduction of metal compounds into their respective nanoparticles.

Properties of nanoparticles :

Nano materials can exhibit a wide range of unique properties (physical, chemical and biological) depending on their size, shape, and composition that do not exist in their larger-dimension counterparts of the same materials. There are three major physical properties of nanoparticles, and all are interrelated : (1) they are highly mobile in the free state, (2) they have enormous specific surface areas, and (3) they may exhibit what are known as quantum effects. Nano particles can exist as suspensions, colloids, or dispersed aerosols depending on their chemical and electromagnetic properties. When particle size is made to be nanoscale, properties such as melting point, fluorescence, electrical conductivity, magnetic permeability,

and chemical reactivity change as a function of the size of the particles. Two principal factors cause the properties of nanomaterials to differ significantly from other materials: increased relative surface area and quantum effects. As a particle decreases in size, a greater proportion of atoms are found at the surface compared to those inside. Nanoparticles have a much greater surface area per unit mass compared to larger particles. In tandem with surface-area effects, quantum effects can begin to dominate the properties of matter as size is reduced to the nanoscale [22].

Physicochemical properties of nanoparticles :

Optical property :

Noble metal nanoparticles (Cu, Ag, and Au) are known for their unique optical properties exhibited near UV and visible spectral wavelength range. Gold nanoparticles (AuNPs) are extensively used in optical imaging due to their unique plasmonic properties, hence Gold nanoparticles-assisted bioimaging is mainly used in direct visualization, monitoring of biomolecular events and physiological processes, and *in vivo* deep-tissue imaging [23].

Mechanical property :

Mechanical properties refer to the mechanical characteristics of a material under different conditions, environments, and various external forces. The mechanical properties of nanomaterials generally are strength, brittleness, hardness, toughness, fatigue strength, plasticity, elasticity, ductility, rigidity, and yield stress. Most inorganic, non-metallic materials are brittle and do not have significant toughness. On the other hand, organic materials are flexible and do not necessarily have brittleness and rigidity properties. Due to surface and quantum effects, nanoparticles display different mechanical properties compared to bulk materials. For example, conventional FeAl powder which is composed of microparticles (larger than 4 μm), is brittle, while ultrafine FeAl alloy powder displays a good combination of strength and ductility as well as enhanced plasticity [23].

Thermal property :

Heat transfer in nanoparticles primarily depends on energy conduction due to electrons as well as photons (lattice vibration) and the scattering effects that accompany both. The major components of the thermal properties of a material are thermal conductivity, thermoelectric power, heat capacity, and thermal stability. Nanoparticles size has a direct impact on the electrical and thermal conductivity of nanoparticles. As the nanoparticles size decreases, the ratio of particles' surface area respective to its volume increases hyperbolically. Since the conduction of electrons is one of the two main ways in which heat is transferred, the higher surface-to-volume ratio in nanoparticles provides a higher number of electrons for heat transfer compared to bulk materials. For example, the addition of CuGold nanoparticles to ethylene glycol enhances the thermal conductivity of the fluid by up to 40% [23].

Magnetic property :

Considering the technical advancements in Magnetic nanoparticles design and *in vivo*

studies, Magnetic nanoparticles-based drug delivery has received much attention in the field of nanomedicine. A magnetic drug delivery system is generally comprised of iron oxide nanoparticles with the drug of interest, and they are delivered to the tumor site with the aid of an external magnetic field. Among the various Magnetic nanoparticles, iron oxide is the most preferred material as it is biocompatible, biodegradable, and most importantly, possesses super paramagnetic (SPM) behavior. One of the interesting size-dependent phenomena of nanoparticles is super paramagnetism. Super paramagnetic nanoparticles display high magnetization only in the presence of a magnetic field, and once it is removed, they do not retain any magnetization [23].

Catalytic property :

Nano-catalysis, i.e., the use of nanoparticles as catalysts, is a quickly evolving field within chemical catalysis. Significantly enhanced or novel catalytic properties such as reactivity and selectivity have been reported for nanoparticles catalysts compared to their bulk analogs. The catalytic properties of nanoparticles depend on the size, shape, composition, interparticle spacing, oxidation state, and support of the nanoparticles. The dependency of catalytic activity on the size of nanoparticles is well studied. The relation is an inverse one, i.e., the smaller the nanoparticles the more catalytically active they are. The shape is also known to affect the reactivity and selectivity of the nanoparticles. For the oxidation of CO by Au nanoparticles, hemispherical nanoparticles were found to be more active than spherical ones. The utility of bio-nanomaterials in the biomedical field has attracted much attention owing to their biocompatibility and non-toxicity and other remarkable properties in the fields of biosensors, material science, aerospace engineering, pharmacology, etc. [23].

The beauty of nanocatalysis is that nanoparticles, nanomaterials and nanocomposites catalysts are highly tunable. A small change to their size, shape, composition, doping with heteroatoms or metals, or changes of the supporting material can alter their physicochemical and biological properties significantly and to the desired level.

Chapter 2 contains more information in this regard.

Characterization of nanomaterials :

Nanostructures show difficulty in being analysed properly because of their small size and exhibit properties different from their bulk analogs at the molecular level. A combination of analytical techniques is employed for the characterization of nanoscale materials to determine their elemental composition, crystal structure, surface area, and other properties including physicochemical, electronic, optical, and mechanical properties. Based on the selectivity, precision, cost, simplicity, and non-destructive nature, nanoparticles are analyzed in depth by microscopy-based techniques e.g. Transmission Electron Microscopy (TEM), High-resolution Transmission Electron Microscopy (HRTEM), and Atomic Force Microscopy (AFM) giving details of their size, and morphology, the magnetic properties of nanoparticles are determined using superconducting quantum interference device (SQUID), Vibrating-sample magnetometry (VSM), Ferromagnetic Resonance (FMR), and X-ray Magnetic Circular

Dichroism (XMCD) and other techniques like X-ray, spectroscopy and scattering techniques to study about their structure, optical properties, and specific physical properties, etc. [23].

More information in this regard is available in chapter 3.

Applications of nanotechnology :

Nanotechnology produces materials of various types at the nanoscale level that has a positive impact on the development of several fields, especially where conventional approaches may reach their limits. Nanotechnology offers, for the first time, tools to develop and set up new industries based on cost-effective and cost-efficient economies, and circular economies thus contributing to sustainable economic growth. Some of them are health and medicine, transportation, energy, and the environment. Remarkable progress has been made in different fields that have expanded in different directions to observe things from micro to nano, to even smaller scale sizes. Nanotechnology and nanoscience are of fundamental importance to industrial applications and medical devices, such as diagnostic biosensors, drug delivery systems, and imaging probes. For example, in the food industry, nanomaterials have been exploited to increase drastically the production, packaging, shelf life, and bioavailability of nutrients. Nanomaterials are being used to build a new generation of solar cells, hydrogen fuel cells, and novel hydrogen storage systems capable of delivering clean energy to countries still reliant on traditional, non-renewable contaminating fuels. However, the most significant advances in nanotechnology fall in the broad field of biomedicine and especially in cancer therapeutics because of their great potential to offer innovative solutions to overcome the limitations derived from traditional chemotherapy and radiotherapy approaches [24].

In medicine and health care :

Nanomedicine is the term used to refer to the applications of nanotechnologies in medicine and health care. Nanomedicine is the most promising use of nanorobotics. Nanotechnologies exhibit significant potential in the field of medicine, including their entry into imaging techniques and diagnostic tools, drug delivery systems, tissue-engineered constructs, implants, and pharmaceutical therapeutics, and have advanced treatments of several diseases, including cardiovascular diseases, cancer, musculoskeletal conditions, psychiatric and neurodegenerative diseases, bacterial and viral infections, and diabetes. For example, nano-oncology is a very attractive application of nanoscience and allows for the improvement of tumor response rates in addition to a significant reduction of the systemic toxicity associated with current chemotherapy treatments [25]. Further, nanotechnology helps resolve the problem of focused treatment administration, reduces the danger of side effects and maximizes therapeutic effectiveness.

In energy sustainability :

Sustainable energy production, transformation, and use are very much needed to maintain the readily and cheap access to energy to the growing and increasingly demanding world population while minimizing the impact on the environment. Nanotechnology is critical to overcome some of the technological limitations of various alternatives to non-renewable

energies. The utilization of nanotechnologies for the enhancement of electrical energy stores like batteries and super-capacitors turns out to be downright promising. Nanotechnologies can improve the capacity and safety of lithium-ion batteries decisively, for example through new ceramic, heat-resistant, and still flexible separators and high-performance electrode materials [26].

Environment :

Nanotechnology plays an important role in environmental remediation by degrading dyes, cleaning up organic chemicals that pollute groundwater, cleaning volatile organic compounds (VOCs), generating less pollution during the manufacture of materials. Researchers are developing nanostructured filters that can remove virus cells and other impurities from water, which may ultimately help create clean, affordable, and abundant drinking water. A nanofabric paper towel which can absorb 20 times its weight in oil, can be used for oil-spill clean-up operations [27]. Nanotechnology can transform the technology into sustainable, green, and clean alternatives to deal with global warming and climate change consequences.

In organic synthesis :

A wide range of applications of various supported metal nanoparticles in catalysis include Au, Ag, Pt, Cu, Cd, Ni, etc. in the form of reduced metals and compound form as heterogeneous catalysis. nanoparticles have the potential to improve the efficiency, selectivity, and yield of catalytic processes. Higher selectivity of the nanoparticles towards reaction proceeds through less waste and fewer impurities which could lead to safer technique and reduced environmental impact. In recent years, Au nanoparticles have attracted immense interest in different fields of science due to their unique features such as high X-ray absorption coefficient, ease of synthetic strategy, enabling precise control over the particle's physicochemical properties, strong binding affinity to thiols, disulfides, and amines, tunable, optical and distinct electronic properties. Zhan and coworkers used Au nanoparticles as catalysts for the 4-nitrophenol reduction reaction [28].

As sensors in electronic devices :

Nanoscale particles and films can be used to create sensors that can detect trace amounts of gases, toxins, or biomolecules and can be integrated into wearable or implantable devices for health monitoring. Nanomaterials constitute the driving force for advanced research in many fields, including nanosized energy conversion, environmental sustainability, catalysts, electronic devices, pervasive sensors, biomedical engineering, and more [29].

In food :

Nanotechnology has introduced alternative techniques to food processing in terms of both improving physicochemical qualities and increasing nutrient stability and bioavailability. Nanotechnology's application in food systems has resulted in a large variety of novel products with better food quality attributes such as texture, taste, sensory properties, stability, etc. It is used extensively in food preservation, food additives, and food packaging as an antimicrobial

compound. In comparison to traditional packaging methods, nanotechnology has several advantages, including improved mechanical barrier, heat-resistant properties, and biodegradability. Nanomaterials can be utilized to detect food deterioration using nanosensors due to their increased antibacterial properties [30].

In military applications :

It comes as no surprise that a majority of nanoscience research is sponsored by the military and thus, focused on military applications. For example, wound dressings coated with certain nanomaterials allow for control of the release of drugs and proteins for a specified period, which can ultimately accelerate wound healing. Nanotechnology has also allowed the development of durable nonwoven fabrics that can impart water absorbance or repellence, fire and thermal resistance, and antimicrobial treatment, to name a few. Nanosensors can be used to detect harmful chemicals and biological weapons, damages to military equipment, nature, and the magnitude of the possible risk when explosives are detected. Developments of nanosized drones that could be used in surveillance, remote detonations, communication, etc. These drones can have long battery lives with nano-sensors that allow facial and object detection [31].

In agriculture :

Due to several contributing factors such as increasing population, climate change, pollution, and growing demands for water and energy, there is a need for the enhancement of global food production and distribution which in turn would require an enormous amount of resources. The introduction of nanotechnology in the agricultural research field and food security over the recent years, has preferentially offered promising solutions for boosting crop productivity for sustainability. Also, the use of nanomaterials like nano pesticides, nano biosensors, and nanofertilizers has many feasible applications thereby improving soil health and leading to newer possibilities. Nanomaterials are ubiquitous and form the fundamental component of fertilizers, medicine, and other industrial products in the form of nanocapsules, nanoemulsions, and nanogelets. The beneficial effects of medications have however been surpassed by potential risks associated with them. The viability of copper-based nano pesticides (Cu, CuO, Cu (OH)₂) for instance, prompts various international organizations to take into account the probable risks of pest management and further investigate the durability and bioavailability of nano pesticides [32,33].

In communication systems :

The use of nanotechnology in communication systems, owing to their special features enables the production of integrated circuits and sensors that are efficient, inexpensive, less power-consuming yet smaller in size enabling new opportunities for health care technologies, medical diagnostics, services, environment monitoring, and logistics facilitating the quality of human life. The advancement of several wireless products like electronics, antennas, storage, etc. possible due to nanotechnology which in recent years has evolved into a potential topic of interest with active research areas in many sectors, including electronics, medicine,

engineering, and sciences. It can provide effective solutions in the field of sensing, cost-effective computation, and human-machine interface in telecommunication engineering. The exchange of information based on new molecules transmitting and receiving data in the nanoscale has been a suitable alternative to electromagnetic waves as a means of communication leading to the generation of a nanonetwork. The use of nanoscale electrical systems is one of the principal applications of nanotechnology in nanomachines which can further be implemented into wireless communication systems such as microprocessor control chips, radio frequency integrated circuits, and large-scale integration digital processing bearing essential implications in signal transmission and power consumption [34, 35].

In polymer composites :

Polymeric nanocomposites are an important class of materials widely used in packaging, transportation, research, industrial purposes, and daily life. They can be easily fabricated into desired shapes and sizes which enables one to control their properties like catalytic activity, optical and magnetic behavior, etc. resolving the everyday challenges on a global level. There are two main sub-classes of polymers- natural (silk, wool, protein) and synthetic (polyethylene, nylon, Teflon) polymers. Polymer nanocomposites reinforced with nanomaterials (nanofillers) usually show multiple advantages in mechanical, optical, and electric properties compared to their counterparts. Owing to their lightweight and inexpensive nature, polymeric nanocomposites contribute potentially to industrial and research environments [36].

For waste disposal and management :

The synthetic applications of nanotechnology can be seen as one of the most important emerging tools in the environmental protection industry particularly in waste treatment. Due to their small size, they can easily be adsorbed and undergo better reactivity towards persistent contaminant groups, such as heavy metals (e.g. chromium), organic (e.g. chloroethylenes), and inorganic compounds (e.g. nitrates) effectively than conventional methods. The treatment of waste materials mainly depends on the selection of nanomaterials (metals and carbon-based) for specific contaminants under suitable conditions. Nanotechnology can also be employed in liquid waste management as well as in-situ and ex-situ waste management technologies. However, they may be associated with toxicological risks when inhaled, ingested, or incorporated into the skin. Therefore, there is a need for scientists to study and understand their impact on the environment keeping in mind their high cost of production [37].

More information in this regard is available in chapters 4-18.

In silico studies of nanoparticles :

Nanoparticles potentially serve as therapeutics, particularly in the field of drug delivery systems. They provide a combined knowledge of the carrier transportation process among biological systems. This is attributed to the list of shortcomings of various experimental sections at the atomic level. Computational and AI-based experiments *via* predictive modeling and deep learning help alleviate some of the common restrictions of conventional approaches. In silico approaches can understand the interactions in nanoparticles-membrane based on

minimal data and predict changes attributed to physicochemical and in vivo pharmacokinetic characteristics of given nanoparticles. The development of high-performance computation, data science, simulation, new algorithms, and theory, and computation is associated with experiments for the discovery and innovation in nanoscience and nanotechnology. In the present scenario, computational approaches together with nanomedicine have limited applicability and need more attention based on the design and engineering of nanoparticles [38]. In silico nanotoxicology can provide valuable insight into the toxicity of nanomaterials [39].

Toxicity of Nanoparticles :

The very first question that comes to our mind is the possible toxicity of nanomaterials as we go for their applications. It is asked because as we have learned that when we shrink matter to such small sizes, the matter that is harmless in macro-scales can become fatal at nanoscales and can enter into the ecosystem by many routes. Nanoparticles can easily enter cells and target some biomolecules as the sizes of nanoparticles are comparable with the size of certain biomolecules like protein globules, and DNA etc. and thickness of cell membranes. Further, owing to their high stability, they remain in the environment for a longer time. This persistency is influenced by the size, charge and the chemical composition of the nanomaterials.

During the passage via land, nanoparticles in solid wastes, effluents, discharges, or accidental spillages can contaminate soil and water (both surface and ground) and finally aquatic systems. Further, wind or rainwater runoff also help the process. Oral consumption of nanoparticles that have migrated from packaging is a major source of human exposure to nanoparticles. Therefore, couple of studies have been conducted. There are some reports that support the toxicity of nanoparticles. According to some studies, ingestion of nanoparticles has been linked to protein denaturation, stimulation of oxidative stress responses, DNA damage, as well as other biological consequences. Oral exposure to nanomaterials, particularly solid nanoparticles, has been linked to gastrointestinal and secondary organ damage in several investigations. Determining the degree of exposure and developing efficient techniques to investigate the toxicity and food safety of nanoparticles is a significant pragmatic issue that must be addressed [40, 41]. However, our body also has some natural processes to handle such situations. Nanoparticles in the blood can also be filtered out by the kidneys and excreted in urine. Long-term results are scanty, but it is wise to assume that unexcreted nanoparticles will accumulate in organs they can reach if exposure continues over long periods, which may be fatal.

Other potential risks posed by nanomaterials include explosions and catalytic effects.

Nanotoxicity, or nanotoxicology is the term used with regard to the study of potential adverse health/environment effects caused by engineered nanomaterial exposure. The nanomaterials who have some toxic/hazardous effects are known as nanotoxins.

Efforts are there to reduce toxicity of some nanaoparticles. e.g. via coating of core shell nanoparticles.

Limitations of Nanoparticles :

Nanoparticles are prone to agglomeration, resulting in the formation of large clumps that are difficult to redisperse. In addition, nanoparticles do not always retain their unique size-related properties when they are incorporated into a composite material or when exposed to external stimuli such as heat or light. Poor solubility in water, non-biodegradable, and high cost for large-scale production are other drawbacks associated with nanoparticles. The strong attractive interactions between particles can result in the agglomeration and aggregation of nanomaterials, which negatively affects their surface area and their nanoscale properties. Agglomeration can be prevented by increasing the zeta potential of nanomaterials (increasing the repulsive force) optimizing the degree of hydrophilicity/hydrophobicity of the nanomaterial, or by optimizing the pH and the ionic strength of the suspension medium.

Toxicity, if any may also be a limitation of nanoparticles. However, if that toxicity can be used for some useful purposes, then it's not a limitation.

Prospects :

In terms of medicine, there is a lot of scope for functional and consistently developing future enterprises involving drug delivery systems and cell targeting to increase their efficiency, resourcefulness, and effectiveness. Nanotechnology is an emerging science, that is expected to have rapid and strong future development, considering the probability for the detection and diagnosis of various health-related issues. In future, nanotechnology will play an important role in the development of personalized medicine. Personalized medicine simply means the prescription of specific therapeutics best suited for an individual. It is usually based on pharma cogenetic, pharma cogenomic, and pharma coproteomic information but other individual variations in patients are also taken into consideration. Apart from refinements in molecular diagnostics, an important basis of personalized medicine, nanobiotechnology also helps in the discovery of biomarkers that are crucial for the development of personalized medicine. A good example of the application of nanobiotechnology for personalized medicine is that of cancer. In the case of cancer the variation in behavior of cancer of the same histological type from one patient to another is also taken into consideration. Personalization of cancer therapies is based on a better understanding of the disease at the molecular level and it is expected that nanotechnology will play an important role in this area.

More information in this regard is available in chapter 19.

Conclusion :

Undoubtedly, Nanoscience and Nanotechnology are inherently trans disciplinary fields of science. As evident from the subsequent chapters, nanomaterials have become essential in various industries, including energy, sensor and electronic devices, health care, the environment, agriculture, fishery, forensic, composites to name a few. Due to their extraordinary and fascinating properties, nanomaterials offer a good contender for many smart applications. Although the emerging field of nanotechnology has piqued the public's interest at large, nanotechnologies have also resulted in extensive discussions and debates regarding their safety and any possible health factor associated with their use. New challenges

arise with the use of nanomaterials, specifically in predicting, understanding, and governing the possible health risks. Therefore, safe and eco-friendly approaches are on urgent demand. Despite of all these, nanoscience is all about the ultra-small, but it has the potential to have an enormous impact on our lives. We have already been using and enjoying many unique and superior products enhanced with nanotechnology. Subsequent chapters will highlight some of the areas where the progress has been a lot more limited than people think. Still, it holds promise to revolutionise fields such as medicine impact on the survival of the human race, or help solve some of the world's difficult environmental problems for a better and sustainable future, which makes the future of nanoscience and nanotechnology most exciting.

The integration of nanotechnology in agriculture holds promise for a transformative second green revolution, fostering sustainability across the agricultural spectrum. To navigate this transformative potential responsibly, there is a need to underscore a strategic research and development paradigm throughout the agricultural value chain. Central to this initiative is the imperative to prioritize environmentally and socially responsible technological advancements, anticipate regulatory requisites, and garner stakeholder acceptance. Within the realm of optimism surrounding nanotechnology in agriculture, it is imperative to acknowledge and address societal, environmental, and ethical considerations. Potential violations on human rights, privacy implications, and the potential for adverse environmental effects demand careful scrutiny. Furthermore, the economic impacts on farmers, consumers, and associated industries require nuanced evaluation, recognizing the spectrum of potential outcomes, both positive and negative. Nevertheless, a comprehensive investigation into the application of nanotechnology in agriculture yields encouraging results across various critical parameters in most findings. These findings contribute positively to the discourse, suggesting a harmonious compatibility between nanotechnology and pivotal facets of agricultural and environmental integrity.

In a nutshell, nanoscience and nanotechnology require awareness among scientific workers and common people, adequate knowledge on the properties of nanomaterials, clear thinking with no ambiguity and pragmatism to pinpoint the most critical and burning issues backed by sound and solid science to address them. In analogy to the quote, “small is beautiful”, nanoscience and technology also refers to the belief that the size of an object is not the sole determinant of its value, but it's the property and targeted use that will determine it's value.

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Physical Properties of Elements and Materials at Nano Level

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Abstract :

This chapter explores how elements and materials transform when size is reduced to the nano level, revealing interesting and useful properties. We discuss here the origins of these unique nano scale features, focusing on quantum confinement effect, surface effect and reduction of imperfections. Important aspects of optical, electrical, mechanical, and magnetic properties of nano materials are highlighted with appropriate examples.

Keywords : Surface effects, Quantum confinement, Optical property, Electrical property, Mechanical property, Magnetic property

Introduction :

At the nanoscale, elements and materials display properties distinctly different from their bulk counterparts. Reduction in size induces dramatic changes in optical, magnetic, mechanical, and electrical properties, as well as catalytic and biological activities among other factors. Because of size dependence of many such properties, nanostructured materials can be tuned considerably simply by adjusting the size, shape or extent of agglomeration. Huge charge of interest on nano materials has been seen among researchers over the last few decades due to substantial modification of properties at the nano level [1-3]. To exemplify this property modification, it is commonplace to reference the example of gold nano particles [4]. While in bulk, gold appears yellow, a 10 nm Au particle appears red. Melting point goes down drastically as the size is lowered. Although gold is reputed as noble, its 2-3 nm nanoparticles (NPs) are very good catalysts. Moreover, Au nano particles demonstrate considerable magnetic property.

In general, the list of unique properties emerging at the nanoscale is quite extensive, encompassing high surface area, excellent mechanical properties (hardness, strength, elasticity), altered melting points, enhanced heat capacity, heightened thermal and electrical conductivity, modified magnetic behavior, novel optical properties, catalytic activity, solubility variations, color changes and so forth [5]. As a consequence of these property modifications at the nanoscale, diverse applications of nanomaterials have emerged.

Origin of the property modifications :**Surface Effects :**

At the bulk of matter, an atom experiences bonding from all directions. Conversely, an atom at the surface has a reduced number of bonded neighbors. Due to the diminished neighboring atoms and unsatisfied bonds, surface atoms exhibit higher energy compared to bulk atoms. The smaller the size of a particle or object, the greater the proportion of atoms at the surface, resulting in a higher average binding energy per atom. Thus, on the nanoscale, particles offer a more active surface. This, coupled with a significantly enhanced surface-to-volume ratio, contributes to the notable influence on their properties [6].

With decrease in size of the material to nano level, surface to volume ratio increases to a great extent, giving rise to various surface-dependent properties. This has significant implications for melting and other phase transition temperatures [6]. Melting point of the surface layer of a nano particle is low compared to that of the bulk. At the edges and corners, where atoms have even fewer neighboring atoms, stronger bonds are formed with foreign atoms and molecules. A similar scenario, with fewer neighbors, is observed in the narrow pores of nano materials, leading to a substantial reduction in the solubility of salts in pore-confined water, as well as alterations in the melting and critical points of fluids.

Surface effects exert a profound influence on the thermodynamics of nano materials [7]. Phase transitions, which is a collective phenomenon, become less well-defined at the nanoscale due to the lower number of surface atoms. The Gibbs phase rule loses applicability as phases and components become less distinguishable. Small clusters exhibit behavior more akin to molecules than bulk matter. Numerous thermodynamic concepts may break down, particularly when the system of interest comprises a single isolated cluster with a small number of atoms.

Quantum size effects :

Molecular Orbital Treatment: In bulk materials, atomic orbitals overlap, giving rise to molecular orbitals (MOs) that are delocalized throughout the entire material. In a bulk material with N atoms, each atom contributes one atomic orbital, resulting in N MOs. The large value of N in bulk materials causes the MO energy levels to be closely spaced, forming a continuous band of energy states. The band with the Highest Occupied Molecular Orbital (HOMO) is termed the valence band and the one with the Lowest Unoccupied Molecular Orbital (LUMO) is called the conduction band. Due to the presence of limited number of atoms in NPs, discrete energy levels for the resulting MOs are observed [8]. As the particle size decreases, the energy levels become more distinct, resulting in a quantized electronic structure. A band gap represents an energy range, where no MO is found to be populated. So, the presence of HOMO-LUMO gap means that the MO with lowest energy of the conduction band lies directly above the MO with highest energy of the valence band. This HOMO-LUMO gap in NPs is dependent on the particle dimensions. As the particle size diminishes, the HOMO-LUMO gap widens, causing electronic transitions to demand higher energy [9]. With decreasing particle size, electrons and electron holes draw closer, increasing the energy required for their activation, ultimately resulting in a blueshift in light

emission. The quantum confinement effect enables the fine-tuning of electronic and optical properties by manipulating the size of the NPs. For instance, in semiconductor quantum dots, adjusting the particle size allows for the tunability of the emission color of light [9, 10]. The HOMO–LUMO gap of semiconductor particles, and consequently their absorption and fluorescence wavelengths, becomes size-dependent. However, the manifestation of these effects strongly depends on the nature of the semiconductor material [11]. The **Figure 1** illustrates the alteration in electron energy levels and band gap between a nanomaterial and its bulk state.

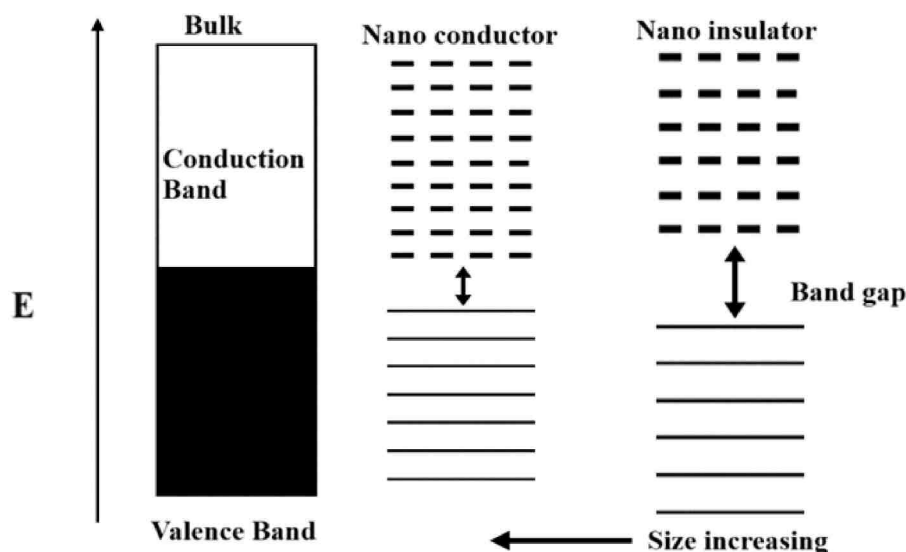


Figure 1 : Variation of size leading to discrete energy levels and large band gap at nano level.

Particle in a box model treatment :

The most precise way to explain how the HOMO-LUMO band gap energy changes with particle size is by employing quantum mechanical models, such as the “particle in a box” model. The behavior of conducting electrons within a nanoparticle can be approximated using the three-dimensional model of a “particle in a box” in which the size of the box is given by the size of the particle [8, 12]. This model establishes a mathematical connection between energy states and size of the particles. Decreasing the volume or dimensions of the available space results in an increase in the energy of the states. For instance, one can apply this model to Quantum dots (QDs). Quantum dots are like tiny spaces where electrons and holes move freely within, much like a particle in a box. However, there are a few differences we need to consider. In QDs, there are two particles - the electron and the hole - instead of just one. Also, unlike the square shape in the particle in a box model, QDs are spherical. So, we replace the box length (L) with the radius (R) to account for this change. Additionally,

we need to use the effective masses of the electron and hole (m_e^* and m_h^*) because of their interaction with the crystal lattice. One obtains an expression for the confinement energy of electrons in quantum dots, considering all these factors into account:

$$E_n = \frac{nh^2}{8m_e^*R^2} + \frac{nh^2}{8m_h^*R^2}$$

Reduced imperfections :

The behavior of materials is often governed by presence of defects, and nanocrystals are no exception. Nanomaterials always have some degree of imperfections. However, nanomaterials exhibit a notable reduction in imperfections compared to their bulk counterparts [13]. Nanoscale structures and materials demonstrate a propensity for a self-purification process during thermal annealing. This tends to migrate the impurities and intrinsic defects toward the surface. This enhanced perfection significantly influences the properties of nanomaterials. In comparison to their bulk counterparts, the mechanical properties of nanomaterials experience considerable enhancement owing to increased crystal perfection or diminished crystallographic defects [14].

Nano-Scale Properties :

Optical Properties :

Optical properties are the most interesting and useful properties that onset at the nano level. The common optical characteristics of nanomaterials such as absorption, transmission, reflection, and light emission, differ markedly from those displayed by the corresponding bulk material. The optical properties of nanomaterials may be tuned by variation of parameters such as size, shape, surface characteristics [10]. By doping a nanomaterial or by interacting the nano particles with a particular surrounding environment optical property can also be manipulated [15]. The shape of tiny metal NPs can significantly affect how they interact with light. For example, with the variation of size, semiconductor NPs, like CdSe, can change their light-related characteristics [16]. For metal NPs, changing their size doesn't have a big impact on their optical properties, like color. However, if their shape is altered by making them into nanorods instead of spheres, it can dramatically change their optical properties. For example, a 25-nm diameter spherical gold nanoparticle looks green, but a 100-nm diameter one appears orange. Similarly, a 100-nm diameter spherical silver nanoparticle looks yellow.

Optical properties of QDs :

Quantum dots (QDs) are tiny particles made of semiconductor materials [17]. These NPs have a unique ability to absorb and emit light at specific wavelengths, influenced by their size and shape, thanks to quantum confinement effects [18,19]. By adjusting the size and composition of quantum dots, one can fine-tune the wavelengths they emit, spanning from ultraviolet (UV) through the visible to the near-infrared regions of the light spectrum. For instance, changing the size of CdSe quantum dots from 2 nm to 8 nm alters their

emission, with smaller particles emitting blue light and larger particles emitting red light [19]. Quantum dots, acting as fluorescent materials, have advantages over organic fluorescent dyes. They can easily modify their optical properties by varying particle size, resist photo bleaching under intense illumination, offer comparable or higher quantum yields than organic dyes, and can be excited at greater distances from their emission peak [20]. This characteristic provides a large effective Stokes shift, offering flexibility in imaging and selecting excitation sources to avoid interference in biological samples.

These remarkable properties make quantum dots valuable in various applications [17], including photodetectors, solar cells, light-emitting diodes (LEDs), and even in everyday products like televisions. In the medical field, they are used for imaging purposes. Thus, the absorption and emission of light by semiconductors could be controlled via tuning the nanoparticle size in this range. However, in the case of metals, the mean free path of electrons is 10-100 nm and, due to this, electronic and optical effects are expected to be observed in the range of 10-100 nm. The colors of aqueous solutions of metal NPs can be changed via changing the aspect ratio. Aqueous solutions of Ag NPs show different colors at different aspect ratios. A red shift in the absorption band appears with an increase in the aspect ratio.

The optical properties exhibited by nanomaterials arise from various mechanisms, some key mechanisms are discussed here:

Quantum Confinement : This quantum size effect has already been discussed. Nanomaterials such as QDs and quantum wells may be treated as three-dimensional space where the electrons are confined leading to discrete electronic energy levels and tunable HOMO-LUMO gap. This quantum confinement effect results in size-dependent optical properties. As the size decreases, the HOMO-LUMO gap changes, leading to shifts in absorption and emission wavelengths [21]. Emission or absorption of light happens when there is transition of electron from the HOMO to the LUMO or vice versa. When an electron moves from a higher energy level to a lower one, it releases a packet of light called a photon. The wavelength of this light (λ) is determined by Planck's constant (h), the speed of light (c), and the energy difference between the electron's allowed energy levels (ΔE). If the energy difference (ΔE) is larger, the wavelength is shorter, making it shift towards the blue end. So, semiconductor nanomaterials absorb and emit light at specific wavelengths, influenced by both the size and shape of the particles due to quantum effects. By adjusting the size and composition of these nanomaterials, we can change the wavelengths of the light they emit, ranging from ultraviolet (UV) to visible and even near-infrared regions of the light spectrum.

Surface Plasmon Resonance :

When the size of a particle is smaller than the wavelength of the incoming radiation, the surface plasmon resonance phenomenon comes into being, influencing and controlling the optical characteristics of nanomaterials [22]. When a small spherical metallic nanoparticle is illuminated with light, the oscillating electric field induces coherent oscillations among its conduction electrons. This phenomenon is illustrated in **Figure 2**. When the electron

cloud undergoes displacement relative to the nuclei, a restoring force emerges due to Coulomb attraction between electrons and nuclei. This results in the oscillation of the electron cloud in relation to the nuclear framework. The frequency of this oscillation is governed by factors such as the electron density, effective electron mass, and the shape and size of the charge distribution. The collective oscillation of electrons is referred to as the dipole plasmon resonance of the particle, sometimes designated as “dipole particle plasmon resonance” to distinguish it from plasmon excitation occurring in bulk metal or metal surfaces. Higher order modes of plasmon excitation, such as the quadrupole mode, can also occur, wherein half of the electron cloud moves parallel to the applied field, while the other half moves antiparallel. When the wavelength of the incident light coincides with the oscillation frequency of electrons, strong absorption or scattering of light occurs, leading to the manifestation of vibrant colors in the particles [23]. The intensity of surface plasmon resonance is directly linked to the quantity of excited electrons and the dielectric constant of the surrounding medium. Metals like gold, silver, and aluminum can sustain surface plasmon modes, where the free electrons within the material resonate at a frequency determined by the particle’s composition, size, and shape [24].

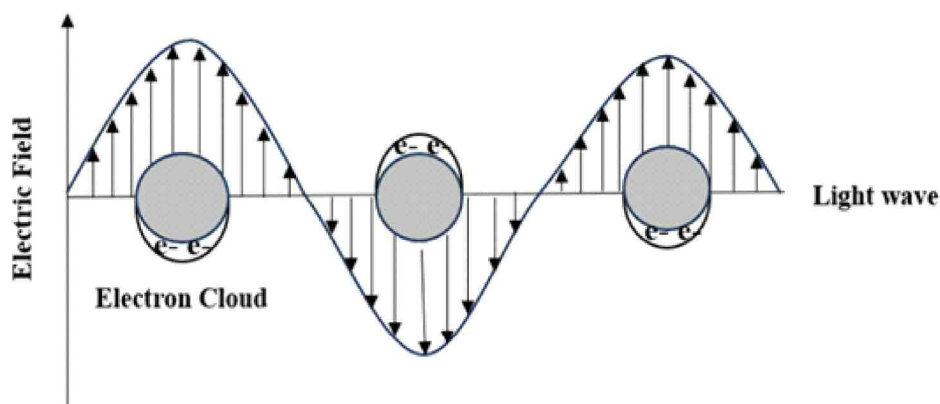


Figure 2 : Plasmon oscillation for a spherical nano particle showing the displacement of the conduction electron cloud relative to the nuclei

Electrical properties :

Features like conductivity or resistivity fall under the electrical properties category. Just like optical or magnetic properties, these features are noticed to undergo interesting changes at the nanoscale. Electrical properties of nanomaterials primarily depend on the movement of charge carriers. When material size is reduced to nano scale, like other properties, electrical properties also undergo changes drastically. When material dimensions are reduced to the nanometer scale, electrons are left to move in a limited space, quantum confinement effect comes into play. As a consequence of this quantization effect, a metal which is conducting at the bulk stage, may turn into a nonconductor at the nano scale [25]. For example, at the nano scale the famous electrical conductor copper metals can lose its

conductivity, whereas extraordinary electrical conductivity is exhibited by graphene attained from graphite [26].

The dielectric properties, resistivity and electrical transports of nano materials are also markedly influenced by factors other than size such as (i) increase in the number of surface atoms, (ii) increased crystal perfection or diminished crystallographic defects near the grain boundaries (iii) doping impurities [27]. Lowering of electrical conductivity when we approach to nano scale size is mainly governed by the HOMO-LUMO gap. For semiconductor nanomaterials, with a decrease in particle size, HOMO-LUMO gap usually increases leading to decrease in electrical conductivity. For example, at 2-3 nm range, gold particles lose conducting property [28]. Electrical conductivity in a material depends on how close the energy levels (MOs) are to each other. For the particles of nano dimension, with only a few atoms, the number of energy levels decreases drastically. This makes the gaps between energy levels bigger, thereby decreasing electrical conductivity of the nanoparticle. This is true for 0D nanomaterials, which are tiny particles with no dimension.

For 1D or 2D nanomaterials, like nanowires or nanotubes, the electrical behavior can be quite different [29, 30]. The electrical characteristics of nanomaterials cannot be easily inferred from those of their bulk counterparts. For instance, materials that exhibit no super conductivity in their bulk form may demonstrate superconducting behavior when configured as nanowires. Some nanowires exhibit super conductivity exclusively at their surfaces. When non-super conducting materials are transformed into nanowires and connected to superconducting electrodes, they can acquire superconducting properties. As the diameter of a nanowire or nanofiber decreases, the role of surface characteristics becomes increasingly significant due to a higher ratio of surface to bulk atoms in the nanowire. When we make nanowires thinner while keeping the voltage constant, the number of electron wave modes contributing to electrical conductivity decreases in quantized steps. When nanowires get really small, below 20 nm, the conductivity suddenly increases because of increased surface scattering for electrons and phonons, more surface area, and a high density of electronic states. The conductivity of nanowires with diameters smaller than 3 nm is notably influenced by crystal orientation and surface oxidation [31]. The electrical resistivity of these nanowires is impacted by surface scattering and oxidation. Surface oxidation gains significance as the ratio of surface atoms to bulk atoms in a nanowire becomes substantial, particularly in the case of small-diameter nanowires. Small modifications in the state of a nanowire, such as bending or stress, can influence its electrical properties. Nanowires can also exhibit insulating, semiconducting, metallic, or superconducting behavior, depending on factors such as composition, radius, and crystallinity.

Morphological properties at the nanoscale play a crucial role in determining the electrical properties of carbon nanotubes. Specifically, the chirality of carbon nanotubes is essential in determining whether they exhibit metallic or semiconducting behavior. In carbon nanotubes that conduct electricity, only one electron wave mode is observed, facilitating the flow of electrical current. The conductivity of carbon nanotubes is altered by changes in cross-sectional area and the application of shear forces, such as twisting. Additionally, the electrical conductivity of multi-walled carbon nanotubes differs from that of single-walled carbon

nanotubes of the same dimensions [32].

Mechanical Properties :

Traditionally, the mechanical properties of a material encompass its observable mechanical characteristics under various environmental and conditional settings, subjected to different external forces [33]. Mechanical properties of traditional materials usually consist of ten parts viz. hardness, strength, toughness, brittleness, fatigue strength, elasticity, plasticity, rigidity, yield stress and ductility [34]. Excellent mechanical properties are observed in case of nanomaterials due to surface and quantum effects compared to bulk material. Three aspects which gain much importance in the study of mechanical properties are influence of porosity, influence of grain size and superplasticity. When NPs are added to a common material, these particles will refine the grain to a certain level thereby forming an intergranular or an intragranular structure. As a result, the grain boundary gets improved and the mechanical properties of the materials are enhanced [35-37]. As NPs are small, addition of these to any matrix can fill the pores of the matrix thereby reducing its porosity and increasing its relative density. As such, mechanical properties get improved.

Many researchers worked on the modifications of the matrices by incorporating NPs to enhance the mechanical properties. Typically, study of mechanical properties of NPs usually deals with bulk metallic and ceramic materials, filled polymer composites, polymer-based nanocomposites filled with platelets, particle filled polymers, carbon nanotube-based composites [35, 38, 39]. Two types of materials seem to have grabbed much interest in the study of mechanical properties of nano materials. The first category of materials are polymers which contain NPs or nanotubes to improve their mechanical properties. The second category of materials are severely plastic-deformed metals, which exhibit amazing properties. Incorporation of polymers with NPs, nanorods or nanotubes leads to significant improvement in their mechanical properties. Such kind of developments mainly depend on the type of incorporates and the method in which the incorporation is performed. However, if the incorporates form aggregates, the advantages of a nanoparticulate incorporates may be lost. Particulate-incorporated polymer-based nanocomposites exhibit a broad range of failure strengths and strains. This depends on the shape of the incorporates, particles or platelets as well as on the degree of agglomeration. Potentially, the best composites are those which are incorporated with nanofibers or nanotubes, but sometimes such kind of composites have least ductility. Again, use of carbon nanotubes can produce composite fibers with tremendously high strain and strength at rupture. Among various exciting nanocomposites, polymer ceramic nanocomposites where the ceramic phase is platelet-shaped is very interesting. In nature, this type of composite is preferred and it is also found in the structure of bones.

Magnetic properties :

Magnetism is a basic physical property of material. A magnetic field is involved in this process which allows a particle to attract or repel other particles. Magnetism basically originates from the magnetic dipoles which are related to the electron spin and orbital moment. In a crystal lattice, each of these magnetic dipoles can couple with the adjacent

ones by exchange interaction which leads to an overall collective alignment that originates the macroscopic magnetic effect [40]. Three parameters measure the magnetic strength and magnetization of a material : (i) the external field needed to reduce the magnetization back to zero known as the coercive field, (ii) the maximum value of magnetization that a material can reach with increasing field known as the saturation magnetization and (iii) the residual magnetization of the material at zero applied field known as the remanent magnetization [40]. Depending upon the magnetic response of a particle *i.e.* the strength of attraction or repulsion of any particle by a magnetic field, particles are categorized as ferromagnetic, ferrimagnetic, antiferromagnetic, paramagnetic, diamagnetic etc. [41,42]. Among these, ferromagnetism is the cause for most of the permanent magnetism in particles. These categories are characterized basically by the orientations of the electrons in the presence of the magnetic fields. Besides these, another kind of magnetism is observed which is known as super paramagnetism. This type of magnetism is mainly exhibited by the small NPs showing ferrimagnetic or ferromagnetic properties.

Super paramagnetism generally illustrates the magnetic behavior of a material in which all the spins are always collinear to each other and its anisotropy energy is too low to keep them aligned along a particularly stable magnetocrystalline direction. As such, a super paramagnetic material is always parallelly aligned to one another, but together they move in a random fashion and the spins align along the same direction if an external magnetic field is applied along a certain direction [40].

There exist many metals which are not at all magnetic in bulk form. Key parameters for determining magnetic properties in bulk materials are basically crystallographic structure, composition, vacancy defects and magnetic anisotropy [43, 44]. However, as they enter in a nanoscale size, they can exhibit magnetic behavior with appropriate modifications due to their size and shape [45]. To understand this, examples of bulk Au and Pt can be taken. These are non-magnetic at bulk stage, but they can be made magnetic in nano scale by simple methodologies like interaction with other chemical species known as capping of NPs. The Au NPs exhibit ferromagnetic property by capping with suitable molecules and the charge localized at the surface of the particle leads to ferromagnetic behavior. Au NPs with surface and core with a diameter of 2 nm exhibit ferromagnetic and paramagnetic behavior respectively. On application of an external magnetic field on these NPs, they can interact with the magnetic fields as magnetic sensitivity of the particles is established. Overall, the magnetic features of NPs are controlled by their finite-size as well as their surface effects. Quantum confinement of electrons is basically responsible for inducing finite-size effects. On the other hand, surface effects can be caused by breaking of the symmetry of the crystal structures at the boundary of each particle. This can also happen by different magnetic as well as chemical structures of the NPs core and outer shell.

Many NPs having magnetic properties are synthesized. Mostly mono component Fe, Ni, Co-based magnetic nanostructures; metal alloy NPs highly promising with super magnetic featured NPs such as Iron-Palladium (FePd) and Iron-Platinum (FePt); metal carbide magnetic nanostructures such as ironcarbides (Fe_5C_2 , Fe_3C , and Fe_2C); metal oxide magnetic nanostructures etc. are reported in literature [46].

Conclusions :

When the material size is brought to nanoscale, a plethora of unique properties in elements and materials appear which significantly differ from their bulk counterparts. Reduction in size onsets profound changes in optical, magnetic, mechanical, and electrical attributes, as well as catalytic and biological activities. The versatility of nanostructured materials lies in their tunability through adjustments in size, shape, or agglomeration. The property modifications at the nanoscale have open the door to diverse applications of nanomaterials.

Surface effects become prominent as materials shrink to the nanoscale, resulting in increased surface-to-volume ratios and surface-dependent properties. Quantum confinement effects, observed due to the limited number of atoms in NPs, lead to discrete energy levels, creating a quantized electronic structure. The HOMO-LUMO gap varies with decreasing particle size, influencing electronic transitions.

The reduction of imperfections at the nanoscale significantly influences nanomaterial properties. Optical properties, driven by quantum confinement and surface plasmon resonance, differ markedly from bulk materials. Electrical properties undergo drastic changes at the nanoscale, with quantum confinement affecting the conductivity of materials. Mechanical properties improve as NPs fill matrix pores, reducing porosity and increasing relative density. Magnetic properties at the nanoscale also exhibit interesting behaviors, paving the way for innovative magnetic devices.

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Characterization of Nanoparticle and Nanomaterials

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Abstract :

Nanoparticles and nanomaterials have emerged as focal entities in various scientific and technological domains, owing to their unique and size-dependent properties. This abstract summarizes the essence of a comprehensive characterization approach employed to unravel the intricate features of these petite structures. The study encompasses a diverse array of analytical techniques, including microscopy, advanced imaging modalities, and spectroscopic methods, to discern the physicochemical attributes of nanoparticles and nanomaterials.

Keywords: Nanoparticle, Nanomaterial, Physicochemical characteristics, Characterization techniques

Introduction :

Nanoparticles and nanomaterials represent an active research area and gaining prominence in technological advancements due to their tunable physicochemical characteristics over their bulk counterparts. Nanoscience, nanotechnology, nanomaterials or nanochemistry are only a few of the nano-containing terms that occur frequently in scientific reports, popular books, newspapers and have become familiar to a wide public, even of non-experts [1, 2].

Instrumentation techniques for characterization of nanoparticle and nanomaterials :

Nanoparticles and nanomaterials have distinct physicochemical properties in terms of size, shape, length, internal structure, surface properties, molecular weight, composition, chemical structure, substructure, solubility and stability, optical properties, magnetic properties and mechanical properties. These properties are critical factors to understand their complex structure-properties applications relationships, physiological behaviour. Thus, it is very imperative to precisely determine the properties of the nanoparticles and nanomaterials. There are many sophisticated and diverse methods available for characterization. The following section encompasses the main techniques employed for characterization of nanoparticles and nanomaterials.

(a) Particle size, shape, morphology characterization :

Particle size distribution and morphology are the most important parameters of

characterization of nanoparticles and nanomaterials. These are crucial parameters in determining their physicochemical properties like catalytic activity, bioactivity, their optical, magnetic and mechanical properties. Dynamic Light scattering (DLS), electron and force microscopies are among the most common techniques used to investigate the size, shape, size distribution, and aggregation state of nanomaterials.

(b) Dynamic Light Scattering (DLS) :

Dynamic light scattering (DLS), also known as photon correlation spectroscopy (PCS) or quasi-elastic light scattering, is one of the fastest and widely used light scattering techniques employed to measure nanoparticle's size in solution or suspension by a monochromatic light source. It is a non-invasive, non-destructive, low-cost technique, and its operation is relatively simple [3-5]. When exposed to a light source within the visible spectrum, a particle allows some of part of the light to transmit through it, while another part may be absorbed. If the particle size is sufficiently small relative to the wavelength (λ) of the incident irradiation ($<\lambda/20$), the irradiation will be scattered in all directions without altering its energy or the wavelength of the incident light. This phenomenon is recognized as Rayleigh scattering, or elastic scattering of light. DLS monitors fluctuations of the Rayleigh scattering generated by the Brownian motion of the nanoparticles with size much smaller than the wavelength of the incident light, at a fixed scattering angle. The two important assumptions of DLS are: (i) 'Random walk' or 'Brownian motion' of nanoparticles caused by random collisions of the nanoparticles with the solvent molecules; (ii) nanoparticles are spherical in shape, noninteracting and small in size. When a monochromatic light beam, such as a laser, illuminates a solution with randomly moving nanoparticles, the light undergoes scattering at a distinct frequency linked to the particle size. Analyzing the intensity fluctuation traces, composed of constructive and destructive interferences of the scattered light, enables the determination of particle size. This analysis is conducted using the motion-dependent autocorrelation function of Stokes-Einstein or the photon autocorrelation function. An essential aspect of Brownian motion in the context of the DLS technique is the observation that larger particles exhibit slower movement compared to smaller ones. This implies an inverse relationship between the translational diffusion coefficient (D) of the particles and their size, as dictated by the Stokes-Einstein equation:

$$D = k_B T / 6\pi\eta R_h$$

where R_h is the hydrodynamic radius of the particles, k is the Boltzmann constant, T is the absolute temperature in Kelvin, η is the viscosity of the dispersing medium, and D is the translational diffusion coefficient.

Two other important characteristics of nanoparticles that can be obtained from DLS experiment are polydispersity (PDI) and stability of the nanoparticles. There are many advantages of DLS as mentioned earlier. But, DLS has also several limitations. The technique finds limited applicability in the examination of samples with a heterogeneous size distribution. Moreover, DLS is unsuitable for accurately determining the size of nanoparticles with non-spherical geometries. This limitation arises from the earlier assumption of spherical particle shape of nanoparticles. And finally, the size derived from DLS is contingent upon

the particle concentration, with accurate sizing achieved in highly dilute samples. Additionally, the type and concentration of ions in the solution can influence the results [6-8].

(c) Scanning Electron Microscopy (SEM) :

Operating on the principles of electron scanning, it serves as a tool for discerning the shape, morphology, and dispersion of nanoparticles within a bulk or matrix. SEM is capable of resolving details down to 1 nm and magnifying up to 400,000x. The incident focussed electron beam, bearing a negative charge, interacts with the material's specific arrangement and electron clouds, generating diverse signals that mirror the topographic intricacies and atomic composition of the scanned specimen surface. This interaction prompts the emission of X-rays from surface atoms, elastically backscattered (or primary) electrons, secondary inelastic electrons, and Auger electrons. Of these, secondary electrons prove most valuable for capturing sample morphology and topography. High-resolution images, revealing details in the range of 1-5 nm, can be produced from these secondary electrons. SEM offers insights into the shape, size, and size distribution of nanomaterials. Typically, the particle size determined by SEM corresponds to the mean Feret diameter of the projected equivalent surface diameter (dS) of a disc with an area S. For spherical particle, dS is given by the following equation:

$$dS(\text{nm}) = 2\sqrt{(S/\pi)(\text{nm})}$$

There are some limitations of conventional SEM measurements. The sample preparation includes drying and contrasting to determine the size, size distribution and shape of the nanomaterials. It may cause shrinkage of the specimen and change the characteristics of the nanomaterials. However, the advent of novel microscopy techniques enables the determination of nanomaterial topography without necessitating sample drying. Techniques like environmental SEM or cryo-SEM offer alternatives to traditional methods [6, 8-10].

(d) Transmission Electron Microscopy (TEM) :

TEM is the most effective electron microscopy technique for characterization of nanomaterials that can provide direct high-resolution images and detailed qualitative/quantitative chemical information for the nanomaterials at a spatial resolution from the atomic level (1-100 nm) up to micrometre level. In TEM, a high-energy electron beam is transmitted through an ultra-thin specimen. Upon entering the thin specimen, electron-electron interactions between the beam and the sample transforms the incident electrons into unscattered, elastically scattered, or inelastically scattered electrons. The scattered or unscattered electrons are then focussed a series of electromagnetic lenses, ultimately being projected onto a screen. It generates electron diffraction, amplitude-contrast images, phase-contrast images, or shadow images with variable darkness, depending on the density of unscattered electrons. TEM provide higher resolution images than SEM as it uses powerful electron beams, thus able to provide greater detail and information about crystallinity and granularity of the nanoparticles. Although TEM is a powerful imaging technique, its use is accompanied by certain drawbacks. The first drawback is that TEM measurements requires

high vacuum and ultrathin sample preparation. The need for ultrathin sample preparation increases the possibility of sample structure alternation. In addition to it, exposure to powerful electron beam can destroy the sample. Finally, similar to SEM, it may present biased statistical size analysis as a result of the examination of a small area of the sample over a certain period of time. TEM can be coupled with other analytical techniques like energy-dispersive X-ray spectroscopy (EDX), electron energy loss spectroscopy (EELS), and energy-filtered transmission electron microscopy (EFTEM) for qualitative and quantitative chemical analyses of the tested sample [6, 8, 11-12].

(e) Atomic Force Microscopy (AFM) :

Atomic force microscopy (AFM) or scanning probe microscopy (SPM) is a mechanical imaging technique used for nanoscale analysis, providing insights into the three-dimensional (3D) topography of a sample. Much like TEM and SEM, AFM provides valuable information about the shape, size, structure, and degree of dispersion of nanomaterials. The resolution achieved is similar to TEM, ranging from 0.1 to 1.0 nm in the x-y plane and is 0.01 nm in the z-direction. The AFM setup consists of a cantilever with a sharp mechanical tip (probe) at its end, which is used to scan the surface of a sample. AFM measurement is based on measuring interacting forces between the fine probe and the samples. When AFM scans the samples, the tip of the cantilever is positioned close enough to the surface such that it can interact with the force fields associated with the surface. The cantilever gets deflected according to Hook's law:

$$F = -k\Delta z$$

where F is the force, k is the spring constant of the cantilever, and z is the bending distance of the cantilever.

AFM can scan samples in contact, non-contact, and oscillating modes, contingent on the proximity between the probe and samples. It is primarily used for analysing various properties of nanoparticles, encompassing topography, elasticity, friction, adhesion, electrical properties, and magnetism. Additionally, AFM proves also helpful in determining particle size distributions and imaging complex arrays of nanoparticles. AFM has the capability to image a variety of nanomaterials at the subnanometer scale under environmental conditions or even immersed in fluids. However, one disadvantage of AFM is that the size of the cantilever tip is larger than the dimensions of the nanomaterials scanned, resulting in an overestimation of the dimensions of the sample [6, 8, 13-14].

Composition and Chemical structure characterization :

(a) Nuclear Magnetic Resonance (NMR) Spectroscopy : Nuclear magnetic resonance (NMR) spectroscopy is a non-destructive/non-invasive technique which can provide information about the structure, composition, purity, molecular weight, dynamics, and diffusion properties of nanomaterial. NMR can operate both in the liquid and the solid-state, in one-dimensional (1D), two-dimensional (2D), and multidimensional (nD) scenario. NMR can be used to determine the composition of block copolymer-based micellar nanoparticles. It is also very promising for characterization of the structure

and physical properties of dendritic nanoparticles, carbon nanotubes, nanocrystalline hydroxyapatite, zinc oxide, and polymeric nanoparticles within inorganic–organic composites. Moreover, 2D NMR techniques like ^1H - ^1H COSY (correlation spectroscopy), ^1H DOSY (diffusion-ordered spectroscopy) can characterize different types of organic molecules bound to nanoparticles. DOSY also finds applicability in determining the size of nanoparticles and can estimate the amount of encapsulated small molecules within nanoparticles [6, 8, 15-17].

- (b) **X-ray diffraction (XRD)** : XRD is the most extensively employed technique for nanoparticle characterization. XRD provides insights into the crystalline structure, phase nature (single or multiphase), lattice parameters, and particle size (calculated through the Debye-Scherrer formula) of nanoparticles. Applicable to both solid and liquid samples, it can determine compositions of nanoparticle by comparing peak positions and intensities with reference patterns. The mean crystalline size (D) of the particles can be determined from the XRD line broadening measurement using the Scherrer equation:

$$D = K\lambda/(\beta\cos\theta)$$

where D is the mean crystallite size, λ is the wavelength of the incident X-ray beam, β is the full width at half of the maximum intensity of the reflection peak, and K is the Scherrer constant, θ is the Bragg (diffraction) angle. K signifies the shape of the nanoparticles, assuming a value of 0.9 for spherical particles, while displaying variability for anisotropic particles. XRD is useful for characterization of nanoscale crystallites with diameters below 100-200 nm. But, for very small particles with a diameter below 2-3 nm the peak broadening becomes so pronounced that the peak width cannot be measured accurately or even sometimes detected. XRD is less informative for anisotropic nanoparticles and those with a nonuniform size distribution. In such cases, it is advisable to interpret XRD data with additional analyses, such as TEM [6, 8, 17-18].

- (c) **X-ray photo electron spectroscopy (XPS)** : X-ray photoelectron spectroscopy (XPS), also known as X-ray photoemission spectroscopy is a potent non-destructive surface-sensitive technique that allows the investigation of surface chemistry of nanoparticles and nanostructured materials. XPS utilizes X-rays to eject electrons from the core orbitals of elements. An X-ray source, typically $\text{AlK}\alpha$ or $\text{MgK}\alpha$, interacts with the sample surface, leading to the emission of a core level electron with a fixed binding energy. The binding energy values are roughly proportional to the square of the atomic number of elements and may be affected by the chemical environment of the element, causing small shifts in the position of the XPS peaks in the spectrum, known as chemical-shifts. Thus, the binding energy spectrum allows for the identification of elements in nanoparticle sample and determination of their oxidation states. Meanwhile, quantitative elemental information is obtained by measuring peak heights or the integrated area under a specific elemental peak [6, 8, 19-20].
- (d) **Fourier-transform infrared spectroscopy (FT-IR)** : Fourier transform infrared spectroscopy (FTIR) is a technique based on the measurement of the absorption of

infrared radiation with wavelengths within the mid-infrared region ($4000\text{--}400\text{ cm}^{-1}$). A recorded FT-IR spectrum gives the position of bands related to the strength and nature of bonds, and specific functional groups, providing thus information concerning molecular structures of nanoparticles and interactions present in nanomaterials. FT-IR can give insights into the successful anchoring of NP surfaces with inorganic or organic functional groups. It also offers information about the NP–stabiliser interaction and confirmation of the stabiliser type [6, 8, 21–22].

- (e) **Thermo gravimetric analysis (TGA)** : TGA can provide valuable information about mass and composition of nanomaterials. It can determine type, purity, and quantity of organic ligand present on nanomaterials. In this technique, a nanomaterial sample undergoes heating, causing components with different degradation temperatures to decompose and vaporize, while a change in mass is meticulously recorded. The TGA device records both the temperature and mass loss, and determines the initial sample mass, the type and quantity of organic ligands associated with the nanoparticles [6, 8].

Surface charge characterization (Zeta potential measurement) with electrophoretic light scattering (ELS):

The surface charge of nanoparticles is considered as one of the crucial parameters influencing the properties of nanomaterials. It plays an important role in determining the dispersion stability and aggregation state of these materials. Nanoparticles with a high positive or negative surface charge exhibit a tendency to repel each other, forming a stable colloidal dispersion. Conversely, nanoparticles with low surface charge often undergo agglomeration and precipitation. Measurements of zeta potential provide a means to determine the surface charge of nanoparticles in a colloidal solution.

The zeta potential (ζ) can be determined by measuring the electrophoretic mobility of nanoparticles in an electrical field that is. It is calculated from the Henry equation:

$$\mu_{\text{e}} = (2\epsilon\zeta / 3\eta) f(ka)$$

where μ_{e} is the electrophoretic mobility, ϵ is the dielectric constant, η is the absolute zero-shear viscosity of the medium, $f(ka)$ is the Henry function, and ka is a measure of the ratio of the particle radius to the Debye length.

The magnitude of the zeta potential is helpful in predicting the colloidal stability of nanoparticles in solution. Zeta potential values can range from +100 to -100 mV. Generally, nanoparticles with zeta potential values exceeding +30 mV or falling below -30 mV are deemed strongly cationic and anionic, respectively, exhibiting high levels of colloidal stability. Conversely, nanoparticles with zeta potential below +25 mV or above -25 mV are prone to agglomeration due to various interparticle interactions, including van der Waals, hydrophobic, and hydrogen bonding interactions. Finally, nanoparticles with a zeta potential value between +10 and -10 mV are considered approximately neutral.

Among the available methods for calculating zeta potential, electrophoretic light scattering (ELS) stands out as the most commonly employed technique, popular for its high sensitivity, accuracy, and versatility. Conventional ELS faces difficulty in handling turbid

samples, as the incident light struggles to penetrate them. Additionally, the zeta potential is not only influenced by nanoparticles but is also impacted by the surrounding environment in which the particles are suspended, encompassing factors such as solution pH and ionic strength [6, 23].

Optical properties:

Nanoparticles and nanomaterials often exhibit optical properties that are very sensitive to their size, shape, agglomeration state, and concentration. Ultraviolet-visible (UV-Vis) and fluorescence spectroscopies are among the highly valuable optical techniques for investigating the optical properties of nanomaterials. These methods allow us to investigate crucial features of nanomaterials without significant modification or permanent damage, given their noncontact and non-invasive nature.

- (a) **UV-Vis Spectroscopy** : UV-Vis spectroscopy, also known as electronic absorption spectroscopy, is widely used for nanomaterial characterization. It proves effective in exploring the size and oxidation state of carbon nanotubes, as well as determining the size and shape of metal nanoparticles. The bandwidth, intensity, and wavelength peak of the absorption spectrum, particularly the surface plasmon resonance of metal nanoparticles, are contingent upon their composition, size, shape, and aggregation state. In fact, as the nanoparticle size increases, the absorption spectra of metal nanoparticles undergo a shift towards longer wavelengths. The electronic spectra of semiconducting nanoparticles are intricately connected to the energy bandgap (ΔE) of the nanoparticles, with this relationship being expressed by the following equation and tied to the size of the nanoparticles:

$$\Delta E = n^2 h^2 / 8ma$$

where n is the principal quantum number, h is the Planck's constant, m is the effective mass of an electron, and a is the radius of the nanoparticle.

UV-Vis spectroscopy is also valuable for monitoring the stability of nanoparticles in solution. It aids in determining the extent of nanoparticle aggregation, as evidenced by a reduction in the intensity of the original extinction peak, often accompanied by broadening of the band or the emergence of a secondary band at longer wavelengths as the particles agglomerate [6, 24-25].

- (b) **Fluorescence spectroscopy** : Fluorescence spectroscopy, also known as fluorometry or spectrofluorometry, serves as a complementary technique to electronic absorption spectroscopy, analyzing the fluorescence from a sample. Fluorescence spectroscopy proves useful in characterizing fluorescent nanoparticles and quantum dots. Semiconductor nanocrystals exhibit the quantum confinement effect, leading to enhanced light absorption, luminescence, and a shift in emission to higher energies as nanoparticle size decreases. Consequently, fluorescence spectroscopy facilitates the determination the size of Quantum Dots (QDs) by measuring the emitted colour from nanoparticles with the same building block but of varying sizes [6, 18].

Magnetic properties:

The magnetic properties of nanoparticles are influenced by their size, size distribution, shape, chemical composition, and aggregation state. Two primary factors that dominate the magnetic characteristics of nanoparticles are: (a) finite-size effects, encompassing single-domain or multidomain structures and quantum confinement of electrons, and (b) surface effects. Surface effects arise from symmetry breaking in the crystal structure at the particle's surface due to factors such as oxidation, dangling bonds, presence of surfactants, surface strain, or variations in chemical and physical structures present in the internal core or surface shell of the nanoparticles.

- (a) **Measurement of magnetisation of nanomaterials :** The magnetic properties of nanoparticles are commonly explored by measuring magnetization (M) as a function of the magnetic field (H) in an M-H hysteresis loop. Such measurements are usually conducted using magnetometers like the superconducting quantum-interference-device magnetometer (SQUID) or the vibrating sample magnetometer (VSM). The distinctive features of these curves are linked to the material type, size, and shape of the magnetic nanoparticles [6].
- (b) **Ferromagnetic resonance (FMR) spectroscopy:** Ferromagnetic resonance (FMR) is yet another sensitive and potent spectroscopic technique employed for measuring the magnetic properties of nanostructured materials. It can provide important characteristics of nanomaterial such as the Curie temperature, total magnetic moment, relaxation mechanism, elementary excitations etc. [18].
- (c) **Mössbauer spectroscopy:** Mössbauer spectroscopy is an absorption technique often employed to investigate the magnetic properties of nanomaterials. This method probes a nucleus with gamma rays generated from radioelements. Three types of nuclear interactions are typically probed: (a) isomer shift (or chemical shift), (b) quadrupolar splitting, and (c) hyperfine splitting (Zeeman splitting). The hyperfine splitting originates from the interactions between the nucleus and the local magnetic field. Hence, Mössbauer spectroscopy can offer insights into the magnetic order of nanoparticles, as indicated by the hyperfine magnetic field. It can also provide information about the superparamagnetic behavior of the sample, as the collapse of the hyperfine field plays a crucial role in determining the blocking temperature of magnetic nanomaterials [26-27].
- (d) **Magnetic Force Microscopy (MFM) :** MFM, a special mode of AFM is a surface detection technique that employs a magnetic probe to characterize magnetic materials. In MFM, a magnetically coated tip scans the sample surface. As the tip magnetically interacts with the magnetic domains of the sample, it produces an MFM signal. MFM can provide insights into the magnetic behaviour, topography and the static magnetic configuration of individual magnetic nanostructures. It is also extensively used to probe the position of magnetic nanoparticles [28-29].

Measurement of mechanical properties by nanoindentation :

The traditional techniques used for the mechanical characterization of materials at the macroscale are not suitable at the micro or nanoscale. Nanoindentation, also referred to as instrumented indentation, stands as an efficient depth-sensing technique. It is usually employed to determine the mechanical properties of thin films and coatings, nanoparticles, and fibers as well as patterned structures. Modern nanoindentation instruments demonstrate remarkable precision and yield valuable information from a small sample volume through the measurement of indenter load and displacement. Nonetheless, results from analogous experiments on similar materials in different laboratories may differ from one another for various reasons: (a) the deformation processes during indentation are complex, (b) measurement and data analysis methods lack standardization, and (c) use of several experimental parameters can significantly impact the results [30].

Conclusion :

In essence, the characterization of nanoparticles and nanomaterials stands as a testament to human ingenuity, stretching the limits of our comprehension and paving the way for innovative solutions to the challenges of tomorrow. The voyage goes on, inviting researchers to delve deeper into the intricacies of the nanoworld and unlock the full potential of these remarkable entities.

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Importance of Nanotechnology in Energy Sustainability

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Abstract :

Developing renewable energy sources is one of the biggest technological challenges of the 21st century. In this regard, the significance of both energy storage and conversion system is striking. The application of nanotechnologies to improve electrical energy storage devices such as super capacitors (SCs) and metal-air batteries (MABs) as well as the conversion of primary energy sources into electricity such as fuel cells (FCs) is incredibly promising. This is because the electrode reactions in such alternative energy devices have slow reaction kinetics and hence require electrocatalysts to accelerate the reactions. Owing to its excellent electrical, optical, magnetic and mechanical properties, nanomaterials find use in many areas, including electrochemical catalysis. This chapter overview the importance of nanomaterials, mainly for the reactions involved in FCs, MABs and SCs.

Keywords : Fuel cells, metal-air batteries, nanoparticles, oxygen evolution reaction, oxygen reduction reaction, super capacitors

Introduction :

The world's energy consumption will inevitably rise due to rising living standards and population growth. Undoubtedly, the most significant technological problem of our time is how to meet the energy demands of a growing global population in a way that is both environmentally and strategically viable [1]. Two third of the renewable growth comes from renewable sources, primarily solar, photovoltaic and wind power. However, these renewable sources are not reliable. For instance; wind turbines may be inefficient in calm weather, solar panels may be inefficient in overcast weather and renewable energy sources may generate excess energy, occasionally overloading the system. Because of the high demands from a wide range of applications, research on energy conversion like fuel cells (FCs), solar cells, etc. and storage such as super capacitors (SCs), metal air batteries (MABs), etc. have been explored in recent decades. The main purpose of such technologies is to gather energy from several sources, convert it and store it for later use. The four primary categories of energy technologies are electrical, mechanical, chemical and electrochemical. Because the

electrochemical method can be used for both conversion and storage, it is highly favored [2]. The electrode reactions associated with the alternative energy technologies have slow kinetics. To increase the efficiency of energy conversion and storage devices, high-performance electrode materials must be designed and developed. In this regard, nanotechnology comes into play for the advancement of such devices for commercial applications.

Nanocatalysis is one of the most fascinating subfields of nanoscience. The main objectives of nanocatalysis are to promote, control, steer, and enhance chemical processes by altering the catalyst's size, shape, reaction center, chemical composition and dimensionality [3]. Different types of nanocatalysts include carbon nanoparticles (NPs), metal oxide NPs and metallic NPs. Metal oxide NPs are widely used in industry as an active component, promoter or support. Their distinctive physical and chemical properties can be attributed to the high density of defect sites and their sizes in the nanometer range. Metallic NPs are typically defined as metal fragments that form nanoclusters with sizes in the nanometer range. The very high catalytic activity and selectivity of the metallic NPs make them suitable for catalyzing electrochemical reactions, redox reactions, coupling reactions, etc. [4]. Thus, the design of electrocatalysts (ECs) in the nano scale range is crucial since electrochemical reactions have many roadblocks. This chapter illustrates the significance of nanomaterials or nano-ECs in energy storage and conversion devices with special emphasis on FCs, metal-air batteries (MABs) and SCs. A brief discussion on the working principle of such devices, their types and components, the reactions kinetics involved as well as the current development of nano-ECs are also included in this chapter.

Electrocatalysts (ECs) :

A particular type of catalyst that functions at the electrode surfaces or on solid/liquid interfaces is known as an EC. It can take part in an electrochemical reaction to speed up the rate of oxidation and reduction processes. ECs have the ability to accelerate the transfer of electrons between the electrode and reactants, as well as facilitate an intermediate chemical transformation that can be characterized by an overall half-reaction [5-8]. When it comes to practical application, the activity, cost, stability and durability of EC becomes the highest priority. Apart from these, good chemical stability in a range of electrolytes, high electronic conductivity and a high surface area catalyst support are also essential for synthesizing ECs. For example, graphene nanosheets act as a promising support material. Likewise, metal oxide NPs are significant electrocatalytic materials due to their highly distinctive optical, electrical and molecular characteristics, ease of functionalization and strong resistance to alkaline corrosion in an electrochemical environment [4, 6].

Fuel Cell: Its Types and Basic Working Principle :

FC is an electrochemical device that produces electrical energy from chemical energy without burning fuel. The byproducts that FC generates are water and heat. They are open systems, requiring a constant supply of fuel from outside the cell for the cathode and the anode to function as the charge transfer media. FCs are classified according to the type of

electrolyte used. Different types of FCs include proton exchange membrane FC, (PEMFC), alkaline FC (AFC), direct methanol FC (DMFC), molten carbonate FC (MCFC), phosphoric-acid FC (PAFC) and solid oxide FC (SOFC), which were categorized based on the electrolyte used. **Figure 2(a)** represents the various types of FCs and their basic components [4,9]. The basic working principle of FC was discovered by Swiss scientist Christian Friedrich Schonbein in 1838. The two electrodes, namely; the cathode and the anode are connected via an electrolyte on its either side. The anode is continuously supplied with H_2 fuel, while the cathode is continuously supplied with oxygen from the air in typical hydrogen (H_2) FC, as shown in **Figure 2(b)**. The positive and negative ions are produced by the decomposition of H_2 fuel at the anode terminal. The intermediate electrolyte membrane serves as an insulator for electrons and only allows positive ions to move from anode to cathode. In order for the system to stabilize, these electrons need to recombine on the other side of the membrane, due to which the free electrons travelled to the cathode side via an external electrical connection. At the cathode, the positive and negative ions recombine with the oxidant to generate pure water or depleted oxidant [10].

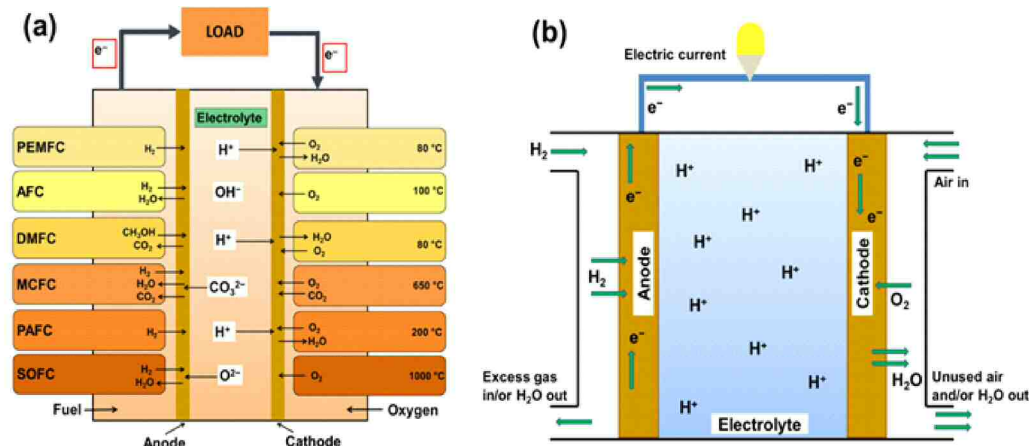
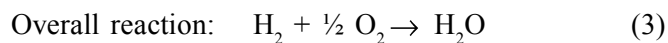
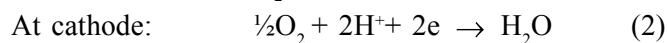


Figure 2 : (a) Various types of FCs along with their components and (b) working principle of a H_2 FC

Electrode Reactions in Fuel Cells :

The reactions involved in a typical H_2 FC are :



The electrode half-reaction that takes place in the cathode of a FC is known as the oxygen reduction reaction (ORR). Another important reaction that finds interest in URFCs, water-splitting devices and MABs is the oxygen evolution reaction (OER). Both of these

reactions have slow rate and require catalysts for progressing [11, 12]. ORR either proceeds via a direct pathway or an indirect pathway involving four or two electrons, respectively. The various pathways of ORR in acidic and alkaline media are summarized in **Table 1** [13,14].

Table 1: Pathways followed by ORR in acidic and alkaline media

Medium	Direct Pathway	Indirect Pathway
Acidic	$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$ $E^\circ = 1.229 \text{ V (vs. SHE*)}$	Step 1: $\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}_2$ $E^\circ = 0.695 \text{ V (vs. SHE)}$ Step 2 : $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$ $E^\circ = 1.775 \text{ V (vs. SHE)}$
Alkaline	$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ $E^\circ = 0.401 \text{ V (vs. SHE)}$ SHE = Standard Hydrogen Electrode	Step 1: $\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{HO}_2^- + \text{OH}^-$ $E^\circ = 0.076 \text{ V (vs. SHE)}$ Step 2: $\text{HO}_2^- + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 3\text{OH}^-$ $E^\circ = 0.878 \text{ V (vs. SHE)}$

ORR via “direct” pathway is a 4e process. The “indirect” ORR pathway is a two-step process that generates hydrogen peroxide (H_2O_2) and HO_2^- ion as intermediates. The intermediates are then reduced to give the final products. For effective ORR, however, the “direct” 4e pathway is favorable since there is no generation of intermediate ions.

Electrocatalysts for Fuel Cells :

The kinetically sluggish element *i.e.*, ORR in FCs has been a challenge for the researchers worldwide to develop high-performance ECs. The excellent electrocatalytic activity of platinum/carbon (Pt/C) or platinum-based materials made it the state-of-the-art catalyst for ORR in both alkaline and acidic medium. However, there are certain shortfalls while using Pt/C as EC such as its high price, insufficiency, CO poisoning effect, detachment, agglomeration, low durability and low stability, thereby limit its widespread commercialization [15,16]. Thus, to reduce Pt loading continuous efforts had been made by undertaking various approaches, such as, development of ultra-low Pt loading catalysts by dispersion of Pt NPs, atomic Pt atoms, nanoclusters, etc., on high surface area carbon supports; alloying Pt NPs with transition metals and inner transition metals; developing core-shell NPs, replacement of Pt by transition metal oxides (TMOs), chalcogenides, phosphates, nitrides, heteroatom-doped transition metal and so on [17]. Experimental and theoretical studies proved that TMOs are promising ECs for ORR, because of their ability to mix with other materials, presence of variable oxidation states, low cost, availability, resistance to alkaline corrosion and its crystalline nature that prevents agglomeration of NPs [18, 19]. Some research has indicated that the absence of bridging bonds between the metal oxide and carbon lattice slow down the electron transfer at the interface and affects electrocatalytic activity, demanding

the use of doped carbon or mixing TMOs with post-TMO NPs. Few examples of ECs that displayed remarkable ORR activity are CeO_2 -modified $\text{Fe}_2\text{O}_3/\text{C}$ [20], gold NPs on the surface of cuprous oxides (Cu_2O) [21], oxygen-deficient NiCo_2O_4 [22], porous Co_3O_4 nanorods [23], porous Co_3O_4 nanorods anchored on graphene nanosheets [24], cobalt iron carbide NPs encapsulated by nitrogen-doped graphene nanosheets [25], etc.

Metal-Air Batteries: Its Components and Basic Working Principle :

MABs are the systems where energy storage is based on electrochemical charge/discharge reactions that occur between a positive “air electrode” (cathode) and a negative “metal electrode” (anode). It means electricity is produced due to a redox reaction taking place between the metal (anode) and oxygen (cathode). The nature of the electrolyte used in MABs can be aqueous, non-aqueous, solid-state or hybrid [26] as shown in **Figure 3**.

An electrolyte with an aqueous or non-aqueous base is used to provide the ion transport mechanism between the electrodes. Generally, studies are conducted on the metal electrodes of Zn, Al, Li, Na and Mg. Because highly active metals like Li, Na and K are unstable in aquatic environments. Due to their relative inactivity, Al, Mg, Fe and Zn must use alkaline aqueous electrolytes during the process. Another part of MAB which is utilized to separate the two distinct electrolytes is the separator. It prevents the short circuit generated by the metal dendrites and also inhibits several mass transport activities that take place between the electrodes. The metal anode is made up of first-row transition metals with high electrochemical equivalency, such as Fe and Zn, or alkali metals (usually Li, Na, and K) or alkaline earth metals [27]. The working principle of MAB involves the conversion of oxygen to hydroxide ions and the transformation of metals or alloys into metallic ions at cathode and anode, respectively. The metal is oxidized in the anodic reaction, while, oxygen from the atmosphere diffuses through the porous carbon electrode. The overall reaction is catalyzed by a suitable catalyst. High energy densities result from the anodic material occupying the majority of the cell volume in the M-Air concept.

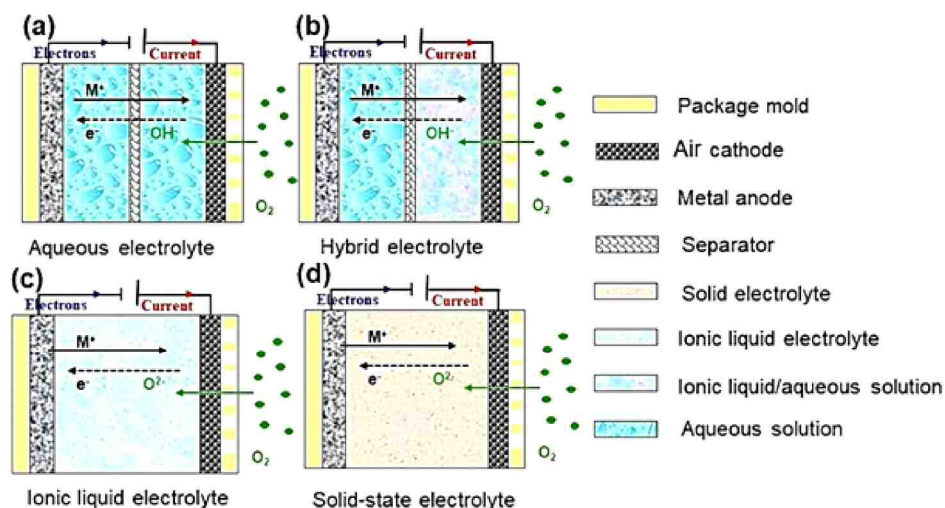


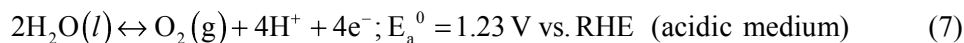
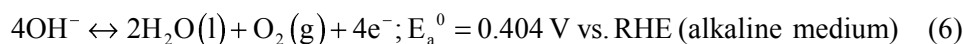
Figure 3 : Different electrolytes of MABs. Reproduced from ref 29.
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Electrode Reactions in Metal-Air Batteries :

ORR and OER are the two major electrochemical reactions involved respectively in the discharge and charge processes of MABs for its operation [28, 29]. The reactions taking place in the cathode and anode are as follows:



ORR in MAB follows the same pathways like that in FCs. Like ORR, OER is the limiting factor that obstructs MABs from practical use because a total of four electrons are required for the generation of O_2 . The kinetics involved the transfer of one electron at a time via multiple steps. In an alkaline medium, OH^- groups are oxidized to O_2 and H_2O , while, O_2 and 4H^+ are formed from two molecules of water in an acidic medium, during OER [30]. The anodic half-reactions at 25°C and 1 atm are shown below:



Therefore, in order to decrease its over potential and increase its rate, efficient ECs are required. Higher over potential will result in reduced energy storage efficacy and output performance. Other significant parameters that influence the reaction rate are high mass transfer, low-rate capability, poor round-trip efficiency and low Coulombic efficiency.

Nanomaterials for MABs :

The benchmark catalysts for OER, IrO_2 and RuO_2 , are costly, scarce and unstable in commonly used electrolytes that oxidize to their higher oxidation states despite having good activity, similar to that of Pt/C in ORR. The discharge and charge processes of MAB can be facilitated by using bifunctional ORR/OER ECs. In the free-state, noble metals like Au, Pt, Ag, etc., typically have low affinity to combine with O_2 , and under conditions of red heat, they may react without changing its composition. Recently, there has been an increase in global research interest in TMOs with various stoichiometric or even non-stoichiometric compositions [31]. For instance, it has been found that carbon supported TMOs of the form M_xO_y ($\text{M}=\text{Mn, Fe, Co, Ni, etc.}$) are very active as bifunctional ECs for MABs. Again, perovskites and spinel-oxides also act as active bifunctional ORR/OER materials. For instance; spinel-oxide based N-doped reduced graphene anchored morphology-controlled Co_3O_4 (cubic, octahedron, and polyhedron) nanocrystals and perovskite based Fe-doped $\text{PrBa}_{0.5}\text{Sr}_{0.5}\text{Co}_{1.5}\text{Fe}_{0.5}\text{O}_{5+\delta}$ nanofibres decorated by Han et al. [32] and Bu et al. [33] respectively, show excellent ORR/OER performance. Various types of nano-ECs for MABs can also be developed by tailoring the morphology, interface engineering, doping with heteroatoms and creating oxygen vacancies and defects. **Figure 4** displays various morphologies of nano-ECs such as $\text{Mn}_3\text{O}_4/\text{MnO}$ NPs, LaCoO_3 catalysts with nano rod-like structure, 3D ordered mesoporous structure Co_3O_4 and hollow cobalt oxide NPs [29].

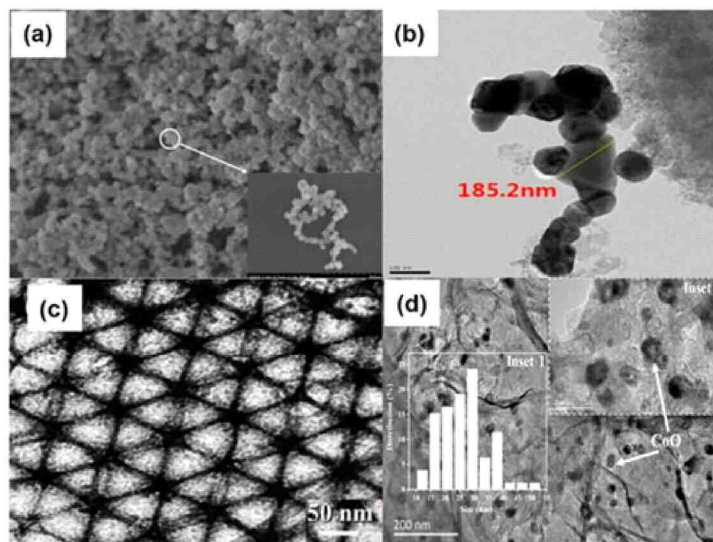


Figure 4 : (a) SEM images of $\text{Mn}_3\text{O}_4/\text{MnO}$ NPs, (b) TEM images of LaCoO_3 catalysts, (c) TEM image of honeycomb-like 3D ordered mesoporous spinel Co_3O_4 (d) TEM image of hollow cobalt oxide NPs. Reproduced from ref 29. Copyright The Authors, some rights reserved; exclusive licensee [MDPI]. Distributed under a Creative Commons Attribution License 4.0 (CC BY)

Super capacitors (SCs): Basic Working Principle and its Types

SCs are a kind of electrochemical energy storage devices with high specific capacitance and power density. These systems are capable of releasing large amounts of energy efficiently in a short period of time. SCs provide higher power densities than batteries and higher energy densities than dielectric capacitors, placing them in the middle of the energy-power range. SCs also have a greater temperature range, environmental friendliness, capacity to charge quickly materials, improved safety, increased dependability, and maintenance-free operation [34, 35]. **Figure 5(a,b)** represent the types of SCs along with its basic working principle and storage mechanism. As shown in **Figure 5(b)**, a SC has numerous components which are separators, polarizing electrodes, electrolytes and current collectors. In this application, the separator of the battery fulfills the same function as the battery itself. Furthermore, it isolates the two electrodes so that during a short circuit, they do not come into contact with one another [36]. The fundamental idea behind SC energy storage is electric double-layer capacitance created by charge separation at the interface between the bath solution and the electrolyte. Basically, SCs can be classified into two types based on their operating principle: electric double-layer capacitors (ELDC) and pseudo-capacitors.

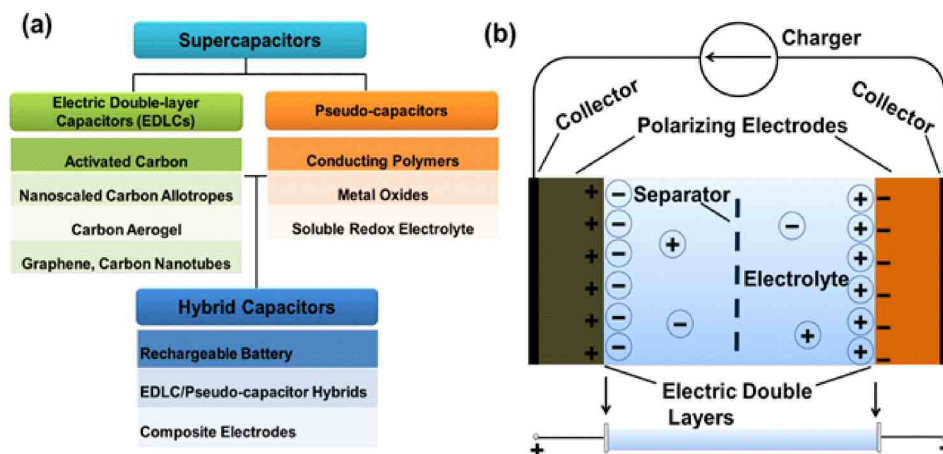


Figure 5 : (a) Types and (b) basic working principle and storage mechanism of SCs

ELDC is made up of two-plate capacitors. When the SC is being charged, electrons travel from the negative electrode to the positive electrode via the external loop, positioning the charge accumulator at the interface between the electrode and electrolyte. The ions and electrons go in the opposite direction during the discharge procedure. However, no charge transfer processes take place in ELDC. In pseudo-capacitors, a redox reaction takes place on the electrode, producing charges and facilitating the transfer of charges between the two layers [37]. The electrolytes for SCs can be aqueous-based, organic-based or ionic liquids. Typically, 6MKOH or 1 MH_2SO_4 aqueous electrolytes are used. In order to obtain higher performance, hybrid SCs are used which combine the characteristics of both EDLCs and pseudo-capacitors [34].

Nanomaterials for SCs

In order to serve as a SC, a material must possess both increased surface area and conductivity, both of which can improve the SC's efficiency. Carbon-based materials were successfully used as the electrode material for SCs because they can improve the capacitive double-layer charge and also increases the contact between the electrolyte and deposited materials [37]. Graphene is one of these materials that have shown to be an effective electrode material for SCs and has outstanding energy transfer characteristics. Graphene's enormous specific surface area, which is related to its layered nanostructure, is one of the key reasons that made it a suitable electrode material for SCs. Graphene in various forms (such as fiber, quantum dots, thin films, etc.) has been effectively used as an electrode for super capacitors. 35 But these carbon materials often have poor specific capacitance.

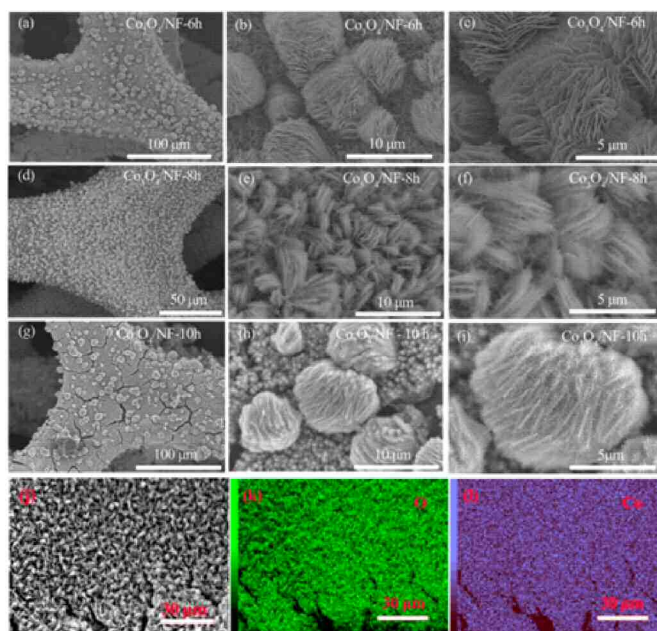


Figure 6 : SEM images: (a–c) Co_3O_4 /nickel foam (NF)—6 h; (d–f) Co_3O_4 /NF—8 h; (g–i) Co_3O_4 /NF—10 h; (j–l) element mapping images of Co_3O_4 nanoflakes. Reproduced from ref. 39. Copyright The Authors, some rights reserved; exclusive licensee [MDPI]. Distributed under a Creative Commons Attribution License 4.0 (CC BY)

Scientists are focusing more on pseudo-capacitor materials (such conducting polymers and TMOs/hydroxides) rather than pure carbon materials. Noble TMOs like RuO_2 and IrO_2 are unsuitable for use in super capacitors in the commercial sector due to its high cost and limited supply, apart from having many other advantages. Likewise, low-cost TMOs (e.g., MnO_2 , Co_3O_4 , NiO , etc.,) are good candidates, but have comparatively reduced cycling stability and low power density. This can be overcome by combining metal oxides/hydroxides with conductive materials and to create porous nanostructures. The former can quicken the kinetics of the process; the latter can lessen the stress caused by the electrode's expansion and contraction, offer more active sites for charge transfer reactions and shorten the diffusion and transfer pathways of electrolyte ions [38]. **Figure 6(a–i)** and **(j–l)** respectively shows the SEM images of Co_3O_4 /nickel foam (NF) with different heating times and their corresponding elemental mapping images [39].

Conclusions :

Energy conversion devices reduce the dependence on traditional energy sources and improve energy utilization. Similarly, energy storage system stores the excess and ensures a constant power supply during low energy production. The role of nanotechnology in

enhancing the efficiency and performance of energy conversion and storage devices is highly substantial. NPs are large-surface materials with a lot of active sites. Because of their incredibly small size, NPs create advantageous transfer pathways by shortening the electron/ion diffusion path. The morphology and dimensions of materials are crucial in enhancing electrochemical performance, due to which the synthetic materials at the micro and nanoscale are receiving more and more attention. It has been observed that there is a shift in the trend of using ECs from precious metals and metal-oxides to non-precious metals and oxides/hydroxides with various surface modifications, which declines the cost of such devices for commercialization. Nevertheless, there are many challenges that need to be addressed yet for smooth functioning and marketing of such devices.

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Functional Nanomaterials As Sensors In Electronic Devices

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Abstract:

Recent strides in the synthesis and fundamental comprehension of nanomaterial properties have notably advanced the domain of gas, chemical, and biological sensors grounded in nanomaterials. This article addresses an extensive spectrum of nanomaterials, encompassing nanoparticles, nanometal structures, nanoprobcs, carbon nanotubes, graphene, quantum dots, molecularly imprinted nanostructures, smart nanomaterials, magnetic-based nanomaterials meticulously designed for diverse gas, chemical, and biological sensing applications. The discourse extends to exploring novel perspectives in sensing methodologies, including nanoscaled electrochemical detection, functionally amplified optical assays employing nanomaterials, as well as colorimetric, fluorescence, and electrochemiluminescence techniques.

Keywords : nanomaterials, semiconducting materials, cost-effective materials, sensors.

Introduction :

Nanotechnology relies on nanostructured materials as its fundamental building blocks, essential for crafting intricate devices with specific functionalities [1]. Nanostructured materials, owing to their inherent quantum size and shape effects [2], find diverse applications across various domains such as electronics [3], optoelectronics [4], information processing [5], catalysis [6], biomedical science [7], environmental science [8], energy conversion and storage [9], advanced defense technologies, and more. The development of innovative nanostructured materials is significantly influenced by chemistry [10].

Functional materials on the other hand are materials that possess specific properties or functionalities that make them suitable for particular applications or purposes [11-20]. These materials are designed or engineered to exhibit specific behaviors or characteristics that enable them to perform a desired function when incorporated into a device, system, or structure [21].

Functional materials play a crucial role in various industries, including electronics, energy, healthcare, and more [22]. Some common types of functional materials are Semiconductors used in integrated circuits in computers, smartphones, and other electronic devices relying on semiconductors [23]. Superconductors exhibit zero electrical resistance and the expulsion of magnetic fields when cooled below a critical temperature [24]. They allow for the efficient transmission of electrical currents without loss [25]. Used in Magnetic resonance imaging (MRI) machines [26], maglev trains [26], and sensitive scientific instruments benefit from the unique properties of superconductors [27]. Piezoelectric Materials generate an electric charge in response to mechanical stress or vice versa [28]. This property is exploited in sensors [29], actuators [30], and energy harvesting devices [31]. Piezoelectric materials are used in sensors for pressure [32], acceleration [33], and vibration detection [34]. They are also employed in ultrasound devices and energy harvesting systems [35]. Ferroelectric Materials exhibit spontaneous electric polarization that can be reversed by an external electric field [36]. They are utilized in memory storage and capacitors [37]. Ferroelectric materials are used in non-volatile ferroelectric random-access memory (FeRAM) and certain types of capacitor [38]. Smart materials can change their properties in response to external stimuli such as temperature, light, or electric fields [39]. They can be adaptive or responsive to specific conditions [40]. Used in shape memory alloys, which return to a predetermined shape upon heating, are an example of smart materials [41]. They find applications in biomedical devices [42], robotics [43], and aerospace [44].

Functional nanomaterials on the other hand have emerged as critical components in the development of advanced sensors for electronic devices [45]. These nanomaterials exhibit unique properties and behaviors at the nanoscale, allowing for enhanced sensitivity [46], selectivity [47], and responsiveness in sensing applications [48]. Carbon Nanotubes (CNTs) are cylindrical structures made of carbon atoms [49]. They possess excellent electrical conductivity and mechanical strength [50]. CNTs are widely used in gas sensors due to their high surface area [51]. Changes in resistance occur when gas molecules adsorb onto the CNT surface, enabling the detection of gases such as ammonia and carbon dioxide [52]. A CNT-based gas sensor can be employed to monitor air quality in smart homes [53], detecting harmful gases and triggering ventilation systems [54]. Graphene is a single layer of carbon atoms arranged in a hexagonal lattice [55]. It is known for its high electrical conductivity [56] and large surface area [57]. Graphene-based sensors are employed in various fields, including environmental monitoring and healthcare [58]. For instance, graphene can be used to detect specific biomolecules by modifying its surface with functional groups that selectively bind to target molecules [59]. A graphene-based biosensor can be integrated into a wearable device to monitor glucose levels in real-time for individuals with diabetes [60]. Metal oxide nanoparticles, such as zinc oxide (ZnO) or tin dioxide (SnO₂), exhibit semiconductor behavior and sensitivity to gases [61]. Metal oxide nanoparticles are commonly used in gas sensors [62]. For example, ZnO nanoparticles can detect low concentrations of volatile organic compounds (VOCs) through changes in electrical conductivity [63]. In an industrial setting, a ZnO-based sensor can be employed to monitor the presence of hazardous gases, providing early warning and ensuring worker safety [64]. Quantum dots are semiconductor nanoparticles

with size-dependent optical and electronic properties [65]. Quantum dots are utilized in fluorescence-based sensors [66]. Changes in their fluorescence can be indicative of environmental conditions, such as pH or the presence of specific ions [67]. Quantum dots can be integrated into a sensor for water quality monitoring, where variations in fluorescence indicate the presence of contaminants like heavy metals [68]. Magnetic nanoparticles exhibit unique magnetic properties and can be manipulated by external magnetic fields [69]. Magnetic nanoparticles are employed in magnetic field sensors [70]. Changes in their magnetic behavior can be utilized to detect environmental parameters or changes in the physical state of materials [71]. Magnetic nanoparticle-based sensors can be incorporated into electronic devices to monitor structural integrity [72]. For instance, in civil engineering, they can detect stress or strain in buildings and bridges [73].

Functional nanomaterials offer a versatile platform for sensor development, enabling the creation of highly sensitive and selective electronic devices [74]. These nanomaterials exhibit unique properties and behaviors at the nanoscale, allowing for enhanced sensitivity, selectivity, and responsiveness in sensing applications [75].

Thus the myriad nanomaterials encompass a diverse range of functional materials, including metals, metal oxides, ionic compounds, ceramics, semiconductors, insulators, organics, polymers, biological materials, bioorganisms, and more [76-78]. Each functional material can take on numerous nanostructure forms [79]. Carbon stands out as a noteworthy example, presenting all dimensionalities with 0D fullerene (hollow bucky ball) [80], 1D carbon nanotubes (CNTs) [81-83], 2D graphene [84], and 3D graphite nanostructures. Beyond carbon, an abundance of nanomaterials, spanning various dimensions of metals [86], metal oxides [87], semiconductors [88], organics [89], polymers [90], biomaterials [91], and their composites [92], have been extensively documented.

Diverse types of nanostructured materials can be created through a variety of methods. Broadly speaking, there are three primary approaches for producing nanomaterials: top-down, bottom-up, and a combination of both [93].

Nanomaterials play a crucial role in a multitude of nanotechnology domains, spanning nanophotonics [94], nano-electromechanical systems (NEMS) [95], nanoelectronics [96], optoelectronics [97], bioelectronics [98], biochemistry [99], biomedicine [100], electrochemistry [101], nanomechanics [102-104], nanobiotechnology [105], nanochemistry [106], and more. This adaptability results in a diverse range of applications, such as nanosensors [107], advanced energy storage devices [108], and nanoactuators, quantum-effect lasers, photonic band gap devices [109], catalysts [110], photocatalysts [111-112], molecular electronic devices [114], nano fuel cells [115], nano drug delivery systems [116] (Figure 1). Notably, sensors are witnessing significant growth due to their high demand in real-world applications, including automobiles, communication, consumer electronics, industrial sectors, and biomedicine. Sensors can be classified into various categories, including mechanical, thermal, optical, magnetic, gas, chemical, and biological [117].

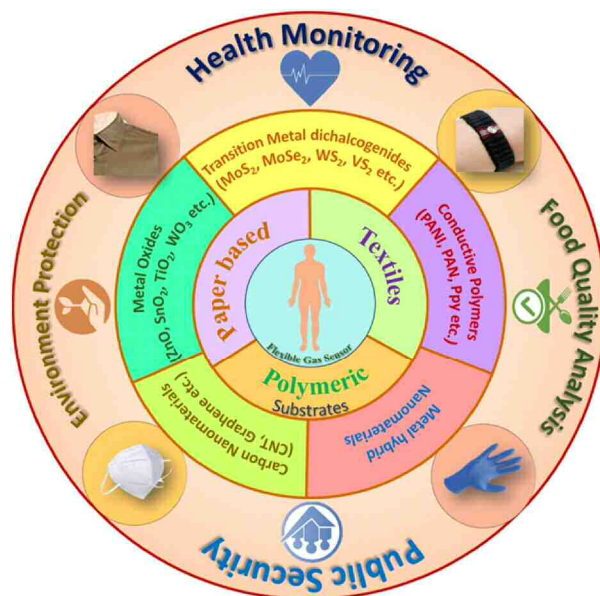


Figure 1: Use of metal nanoparticles in different fields

Various sensor types, specifically designed for gas, chemical, and biological applications, leverage the significant advantages offered by the high surface-to-volume ratio property of nanomaterials [118]. Typically, gas, chemical, and biological sensors comprise a sensing material responsive to changes in gas, chemical, or biological analytes, coupled with a transducer converting these alterations into electrical signals. Gas sensors can be classified based on sensing mechanisms, including chemoresistive, surface acoustic wave (SAW), quartz crystal microbalance (QCM), chemiluminescent, optical absorption, and dielectric types [119, 120]. Gas sensor applications span a wide spectrum, addressing toxic gases, flammable gases and volatile organic compounds (VOCs) [121].

Similarly, chemical sensors can be categorized based on sensing platforms, encompassing electrochemical, ion-sensitive field effect, chemiluminescent, optical, and mass spectrometric types [122]. Chemical sensing applications are diverse, covering a range of liquid-phase chemicals, including acids, bases, solvents, inorganic substances, and organic analytes (Figure 2) [123].

Likewise, commonly employed biosensing platforms include electrochemical, fluorescent, surface plasmon resonance (SPR), QCM, and microcantilever technologies [124]. Biosensing applications span a broad array of biologically relevant materials found in living organisms, such as glucose, cholesterol, uric acid, DNAs, RNAs, cells, proteins, organelles, and more.

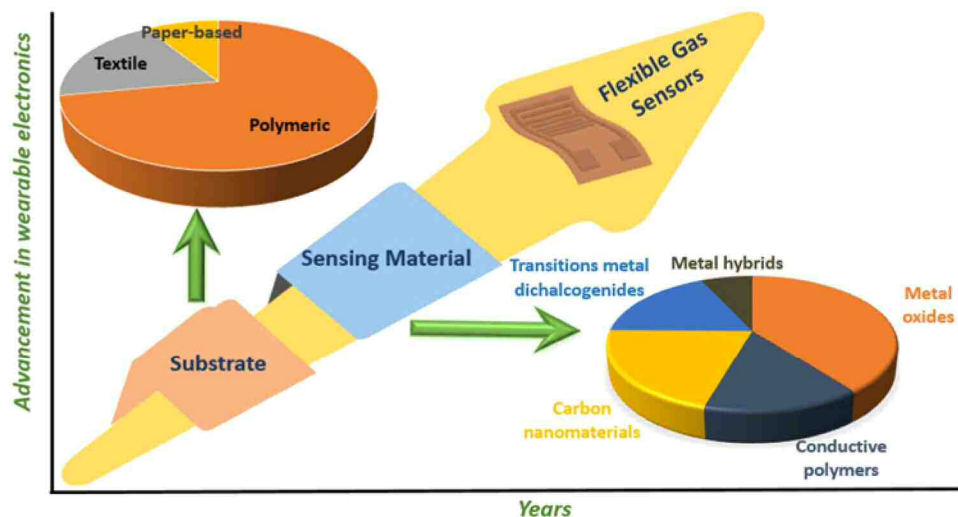


Figure 2 : Metal nano as sensors

In recent times, notable advancements have been made in the realm of nanomaterial-based sensors. A diverse array of nanostructured materials and composites has emerged across various sensing platforms, employing numerous preparation methods to cater to different sensing applications. For example, there has been extensive exploration of high sensitivity chemoresistive gas sensors utilizing metal oxide nanostructures like SnO_2 nanowires, ZnO nanotetrapods, and TiO_2 nanorods [125]. Additionally, considerable attention has been directed towards highly sensitive electrochemical biosensors. These biosensors leverage the synergistic combination of biofunctional materials such as enzymes, antibodies, and DNAs, along with innovative electrode materials such as carbon/metal/conductive polymer/metal-oxide nanostructures. Furthermore, nanocomposites like CNTs, graphene, gold nanoparticles, CNTs/polyaniline, CNTs/ ZnO , CNTs/gold nanoparticles graphene/polythiophene, and similar materials have sparked significant interest. This article comprehensively covers recent progress in selected metal oxide nanostructures, encompassing a broad spectrum of gas, chemical, and biological sensing applications, organized in a structured manner.

This chapter will specifically examine two predominant classes of nanostructured semiconducting metal oxide sensors. These classes comprise n-type semiconductors, including zinc oxide, titanium dioxide, tungsten oxide, tin dioxide, and vanadium oxides, as well as materials exhibiting p-type characteristics, exemplified by copper oxides and tin monoxide. The discourse will intricately explore the sensing properties, production methodologies, and techniques associated with these sensors, with a focal point on their application-oriented aspects. Furthermore, a comprehensive overview of diverse sensors, spanning gas, humidity, UV, and biosensors, integrating these nanomaterials, will be provided. The underlying mechanisms governing responses to gas, humidity, and UV share similar concepts. While

the subject remains contentious, the widely accepted mechanism for these sensors involves changes in the electrical conductivity or resistivity of semiconductor metal oxide materials. In gas sensors, the detection of gases results from a shift in the equilibrium of surface chemisorbed oxygen reactions caused by the presence of a target gas. This creates extrinsic surface acceptor states that immobilize conduction band electrons in the near-surface region of an n-type semiconductor material. In the case of p-type semiconductor materials, chemisorbed oxygen leads to the formation of an accumulated surface layer, thereby altering their conductance. Under ambient conditions, oxygen molecules are adsorbed onto the surface of n-type semiconductor materials, capturing inner free electrons from these materials.

Synthesis of metal oxides nanoparticles:

Navaleet.al have synthesized titanium dioxide nano particles using a simple hydrothermal method that helps in large scale synthesis of anatase titanium dioxide nano particles (TiO_2) having precursor material titanium glycolate and is utilised for the manufacturing of low-cost high-performance acetone (CH_3COCH_3) gas sensors. The nano particles were investigated using XRD, SEM, TEM analysis for crystallinity, phase-purity, and surface morphology. The fabricated TiO_2 shows chemiresistive properties that reveals a superior selectivity towards acetone with a maximum response of 15.24 (1000ppm) @270°C compared to other target gases [126].

Singh et.al have discussed the synthesis of zinc oxide nano rods and nano particles using simple methods through chemical process and reported their comparative study on ethanol sensors. The reaction temperature played a significant role in the selective synthesis of morphologically distinct nano structures. Zinc oxide nano particles were deposited as a thick films to act as gas sensors and their comparative response to ethanol vapours were focused at various concentration and temperature. The researcher reported that the formation of nano particles and nano rods was dependent on reaction temperature. The sensitivity of nanoparticles to ethanol vapours is exceptionally higher than that of nano rods under similar conditions [127].

Zhang et al. have made a rational fabrication of tin dioxide nano materials with superior design and highly desirable towards gas sensing. SnO_2 nanoparticles have three nanostructures with different morphologies, particles, core-shell spheres and facet exposed crystals are developed and further applied to trace amounts of volatile organic compounds, (VOCs). Porous tin dioxide core-in-hollow-shell sphere-based sensors exhibited a higher sensitivity than SnO_2 particle [128].

Ali et al. discussed the ZnO nanoparticle synthesis ranging from zero dimension to one dimension that can perform as an excellent gas sensor. The synthesized nano materials have been investigated using XRD, SEM, TEM, PL. X-ray diffraction (XRD) analysis proved the hexagonal Wurtzite structure of ZnO nanoparticle. However, the Photoluminescence spectroscopies (PL). TEM result confirms the non-agglomeration of zero dimension and one dimension nanoparticle. However, zero-dimension ZnO exhibits better acetone gas sensing performance compared to one-dimension ZnO under an optimum operating temperature of 250°C [129].

Application:

Novel biosensing devices have incorporated metal oxide materials, leveraging their superior electron-transfer kinetics and robust adsorption capability. These attributes create favorable microenvironments for the immobilization of bio-molecules, resulting in heightened electron transfer and improved biosensing characteristics. The biosensing process with metal oxide materials relies on the binding of biomolecules to these materials through either physical adsorption or chemical binding. The physical adsorption of biomolecules is influenced by various factors, including surface morphology, reaction medium, and net surface charge, particularly in the context of weak interactions such as van der Waals, electrostatic, and physisorption. Nevertheless, short-range forces like charge, steric, depletion, and solvent interactions can also impact the biointerface. Establishing an effective biointerface with metal oxide materials ensures an accelerated electron transfer rate and aids in maintaining the stable biological activity of biomolecules [130]. Additionally, it has been observed that incorporating metal oxides onto diverse surfaces can enhance the adhesion of negatively-charged bacteria due to their positive charge and hydrophobicity. This finding holds particular significance for the development of high-performance biosensing devices.

Semiconductor metal oxides play a prominent role in the realm of electronics and optoelectronics, finding extensive use in transistors, circuits, panel displays, and more recently, in printed and paper electronics. Despite their established presence in these applications, sensing technology has undergone significant evolution in recent years, continually advancing to ensure human well-being, quality, and safety across various domains, including food, air, and environmental protection. Materials such as ZnO, TiO₂, WO₃, CuO/Cu₂O, SnO/SnO₂, and VO₂/V₂O₅ exemplify this trend, as they have been seamlessly integrated into diverse sensor types, spanning gas, humidity, UV, and biological sensors. These materials exhibit diverse nanostructures at the nanoscale, including nanowires, nanospheres, or nanosheets, directly influencing their performance in the ultimate sensing applications.

TiO₂, a n-type semiconductor, typically exists in an amorphous state or manifests in three crystalline phases: the tetragonal phases of anatase, rutile, and an orthorhombic phase known as brookite [131]. Among these, rutile stands out as the most stable phase, while anatase and brookite are considered metastable, undergoing transformation to rutile when subjected to heat [132]. The phase transformation sequence from the amorphous state involves conversion to anatase between 300 to 500°C, followed by further transformation to rutile within the temperature range of 600 to 1000°C [133].

In the tetragonal structures of rutile and anatase, there are six and twelve atoms per unit cell, respectively. In both phases, each Ti atom coordinates with six O atoms, and each O atom coordinates with three Ti atoms. The TiO₆ octahedron exhibits slight distortion, with two Ti-O bonds being somewhat longer than the other four, resulting in some of the O-Ti-O bond angles deviating from 90° (with greater distortion observed in anatase compared to rutile). The crystal structures of rutile and anatase consist of chains of TiO₆ octahedra, sharing four edges in anatase and two in rutile. In the case of brookite, the distorted TiO₆ octahedra shares three edges. Additionally, brookite's orthorhombic cell comprises eight formula units, exhibiting interatomic distances and O-Ti-O bond angles akin to rutile and

anatase, although it features six distinct Ti-O bonds [134]. TiO_2 , characterized by a wide energy band gap, typically exhibits optical band gaps of 3.0 eV for rutile and 3.2 eV for anatase. In the case of brookite, the optical band gap ranges from 3.13 to 3.40 eV [135]. Nevertheless, for nanostructured materials, manipulation of TiO_2 's optical band gap is achievable by introducing structural defects or doping with non-metal and metal elements, offering opportunities to narrow the band gap.

Numerous methods have been documented for the fabrication of TiO_2 nanostructures or thin films, encompassing the hydrothermal and solvothermal synthesis [136, 137], sol-gel method [138], thermal evaporation [139], wet-chemical techniques [140, 141], sputtering [142], electrodeposition [143], microwave irradiation [144], and various others. Diverse and distinctive structures have been reported for TiO_2 nanomaterials, particularly in the form of nanoparticles, spanning nanowires, nanorods, nanotubes, nanobelts, nanowhiskers, nanospheres, and more.

Concerning sensors, a top-tier sensor is anticipated to exhibit heightened sensitivity, a high detection limit, rapid response/recovery times, and prolonged stability and life cycles. These attributes are intricately linked to the properties of the sensing material, such as grain size and microstructure, which exert a direct influence on the sensor's sensitivity [145]. In this context, nanostructured materials, particularly nanostructured TiO_2 , have found extensive application in sensors over the years. Nanostructured TiO_2 materials, known for their enhanced chemical stability facilitating low-temperature operation, possess inert characteristics and resilience against harsh atmospheric conditions. Notably, they are cost-effective in terms of production and compatible with wet-chemical synthesis routes [146].

TiO_2 has gained widespread usage as a photoactive layer in gas sensors, where TiO_2 -based gas sensors typically operate on the chemoreceptive principle. This working mechanism aligns with the conductance change observed in typical n-type semiconductor gas sensor materials, attributed to the adsorption/desorption processes of oxidizing and reducing gases [147].

TiO_2 nanowire sensors exhibited remarkable selectivity and high sensitivity, detecting NO_2 down to 100 ppm at room temperature, with response and recovery times of 10 s and 19 s, respectively. Additionally, these nanowires demonstrated commendable repeatability and selectivity, even in the presence of interfering gases like NH_3 , H_2 , and CH_4 [148]. Hydrothermally synthesized TiO_2 nanostructured films, comprising various morphologies such as nanoparticles and nanotubes, were found to possess high sensor responsiveness to toluene, particularly when composed of nanotubes [149].

It is well-established that well-ordered porous structures in metal oxides significantly enhance gas sensing efficiency by increasing the material's surface area, thereby improving its interaction with gaseous species [150]. TiO_2 spongy layers, exhibiting dual-scale porosity at both meso and nano scales, have been reported. These layers demonstrated a sensitivity of 44 ppm to ethanol at 250°C, with a notable characteristic of independence from ambient humidity and a rapid response time of approximately 10 s [151].

In the realm of humidity sensors, TiO_2 is renowned for its exceptional humidity

sensitivity, owing to its inherently hydrophilic nature resulting from surface defects, specifically Ti^{3+} defect sites or oxygen vacancy sites, which avidly adsorb water molecules from the atmosphere [152]. However, conventional TiO_2 -based humidity sensors often exhibit limited sensitivity due to factors such as a low surface/volume ratio, high resistance, or an inefficient charge transfer process in single-phased TiO_2 materials. To address these limitations, various strategies have been devised.

Ultrathin 2D TiO_2 nanosheets, characterized by a high specific surface area and surface oxygen vacancy defects, have demonstrated outstanding humidity sensing performance. These nanostructured sensors exhibited substantial impedance variation across the humidity range of 11% to 95%, coupled with a short response time (3s), recovery time (50s), and a minimal hysteresis of approximately 4.6% [152]. Lin et al. [153] introduced a humidity sensor based on graphene/ TiO_2 that, at relative humidity levels between 12% and 90%, showcased a sensing response ($S = 151$) and a humidity hysteresis value of less than 0.39%. In another study, flexible sensors incorporating TiO_2 and conducting polymers were developed, exhibiting a hysteresis (within the range of 30–90% RH) of 2% RH, with a response time of 30 s and a recovery time of 45 s [154].

Doping TiO_2 has also been explored to enhance humidity sensors. Zare et al. [155] explored the impact of silver doping on TiO_2 for improved humidity sensitivity, while Buvailo et al. [156] demonstrated the use of LiCl-doped TiO_2 -based nanomaterial to produce conducting thin film sensors.

In recent developments, TiO_2 has emerged as a promising alternative to the widely used silicon-based UV sensors [157]. This is attributed to TiO_2 's high photoactivity and stability under UV irradiation, owing to its distinctive band gap properties [158, 159]. Utilizing the hydrothermal method, highly oriented rutile TiO_2 nanorod arrays were synthesized and assessed as UV sensors, achieving a noteworthy photocurrent of $12.87 \mu\text{A cm}^{-2}$ under exposure to 365 nm UV light [160]. In a different study, anodic TiO_2 anatase nanotube arrays were evaluated as UV sensors, demonstrating a remarkable responsivity of 13 A W^{-1} at $\lambda = 312 \text{ nm}$, along with a rapid response featuring a rise time and decay time of 0.5 and 0.7 s, respectively [161]. Epitaxial TiO_2 thin films, produced through RF magnetron sputtering on LaAlO_3 single crystal substrates, showcased a maximum photoresponse of 3.63 A W^{-1} at 310 nm and an ultrahigh response speed of approximately 90 ns [162]. Additionally, self-powered UV sensors integrated with nano-branched TiO_2 arrays exhibited an increase in photosensitivity from 0.03 to 0.22 A W^{-1} , demonstrating exceptional spectral selectivity and swift response [157].

The utilization of metal oxides in sensing applications, particularly in biosensors, has experienced rapid growth in recent years. Biosensors, a dynamic field within analytical chemistry, are estimated to exhibit a substantial annual growth rate of around 60% [163]. This surge is propelled by the healthcare industry, food quality and safety concerns, as well as environmental monitoring. In the realm of TiO_2 -based biosensors, several studies have concentrated on generating 1D nanostructures forming continuous arrays [164,165]. The adoption of TiO_2 nanotube/nanorod arrays is attributed to their expansive internal surface area, negative surface charge, and high refractive index, facilitating the incorporation of

biomolecules and enhancing analyte sensitivity [164].

Zinc oxide has been the subject of extensive research since 1935, primarily due to its integration into various applications, with a particular focus on ZnO-based electronic and optoelectronic devices. ZnO is categorized as an n-type semiconductor, possessing a wide band gap ranging from 3.2 to 3.4 eV and a substantial exciton binding energy of 60 meV at room temperature [166]. Under room temperature and pressure conditions, ZnO crystallizes in the hexagonal wurtzite structure, where each anion is surrounded by four cations at the corners of a tetrahedron. Various ZnO nanostructures have been grown, as reported by multiple authors in recent years. Diverse synthesis methods, including chemical bath deposition, electrospinning, electrodeposition, laser-assisted flow deposition (LAFD), and hydrothermal/solvothermal synthesis — performed through conventional or microwave-assisted heating — have been employed [167]. Depending on the synthesis method, precursors utilized, their concentrations, solution pH, or the chosen solvents, zinc oxide may exhibit distinct nanostructures, spanning from nanorods and nanofibers to nanoneedles, nanowires, nanoplates, nanostars, tetrapods, and nanoflowers.

Metal oxide semiconductors are widely employed as gas sensors in both environmental monitoring and industrial applications, enabling the development of compact and cost-effective devices [168]. ZnO, in particular, exhibits the capability to detect a diverse range of gases, including CO₂, H₂S, NO₂, NO, NH₃, C₃H₈, and CH₄ [169]. The detection of carbon dioxide is particularly crucial, showcasing the utility of various ZnO nanostructures in detecting NO and NO₂ gases. Notably, Sadek et al. [170] utilized ZnO nanobelts to detect NO₂ gas at concentrations as low as 0.51 ppm, while Cho et al. [171] employed ZnO nanorods to successfully detect NO₂ gas concentrations at 1 ppm.

An alternative approach to enhance the performance of ZnO gas sensors involves doping the semiconductor metal oxide with additional materials. Dilonardo et al. [172] introduced Au nanoparticles as dopants in ZnO nanostructures, successfully detecting NO₂ gas at a concentration as low as 0.2 ppm. The sensor exhibited a total recovery time of 30 seconds after the gas was removed. Additionally, Li et al. [173] employed a ZnO/NiO nanoheterojunction with a porous morphology for ethanol detection. Li achieved the detection of 100 ppm of ethanol at 200°C. The use of a heterojunction introduces a dual-factor sensing mechanism. As ZnO is an n-type semiconductor, the presence of ethanol induces interaction with O²⁻ and O⁻ at the crystal surface, resulting in increased conductivity.

SnO₂ has also been combined with other metal oxides to create UV photodetectors/sensors. In a study by Xie et al. [174], UV photodetectors were developed based on CuO/SnO₂ p-n nanoscale heterojunctions. The responsivity for a single SnO₂ nanodevice was measured at 1.9 A W⁻¹, while for the p-n heterojunction, it significantly increased to 10.3 A W⁻¹. The wavelength-dependent photocurrent-to-dark current ratio for the CuO/SnO₂ photodetector at 290 nm was estimated to be approximately 592nm. In another approach, ZnO-SnO₂ nanowire arrays were synthesized using a near-field electrospinning method for flexible ultraviolet photodetectors. These sensors demonstrated outstanding photo response properties under 300 nm ultraviolet light illumination, including high I on/I off ratios (up to 103), as well as good stability and reproducibility [175]. Additionally, a separate study

detailed the fabrication of fully transparent photodetectors from electro spun ZnO-SnO₂ heterojunction nanofibers.

The exploration of WO₃ in the form of films for gas sensing has been investigated. In a study referenced as [176], oriented WO₃ nanoflakes array films were employed as gas sensors. The WO₃ thin film sensor demonstrated responses of 2.8-9.8%, 12.8-68.9%, and 26.2%-85% to H₂S concentrations ranging from 0.1 to 10 ppm at various operating temperatures (100°C, 200°C, and 300°C, respectively). Notably, the WO₃ thin film sensor exhibited a heightened response to H₂S, which increased with higher concentrations of H₂S gas. Another study [177] demonstrated sulfur dioxide gas sensors utilizing WO₃ screen-printed thick films composed of nano powders. These sensors exhibited responses to 1-10 ppm SO₂ at temperatures between 200-300°C, showing a strong dependence on WO₃ morphology. Furthermore, nanoparticle WO₃ films served as gas sensors for ethanol and H₂S, operating at temperatures between 150 and 250°C and successfully detecting concentrations as low as 200 ppb of ethanol and 20 ppb of H₂S.

WO₃ also holds promise for integration into UV photodetectors/sensors, thanks to its substantial indirect energy band gap of 3.3 eV. Recently, WO₃ nanostructures have been employed to address reported limitations in WO₃-based UV sensors, particularly slow response times exceeding 1 minute [178]. Utilizing WO₃ nanosheets as high-performance UV photodetectors/sensors, impressive optoelectronic performance was observed, featuring high sensitivity (293 A W⁻¹), a rapid response speed (40 ms), substantial on/off ratios (2000), and elevated external quantum efficiency (997%) [179]. Li et al. [180] presented WO₃ nanowires synthesized on carbon papers for use in UV photodetectors/sensors. These hierarchical WO₃ sensors exhibited response and decay times of approximately 3s and 20s, respectively. In another study, hexagonal WO₃ nanowires were evaluated as UV photodetectors, showcasing a conductivity increase of over two orders of magnitude under UV illumination at 312 nm, with a photoconductivity gain of 4.6×10^3 [181]. Reference [182] demonstrated the production of WO₃ nanobelts via electrospinning for UV photodetector applications. The sensor's photocurrent increased to approximately 12 nA and rapidly returned to its initial value (12 pA) upon turning off the light, indicating high sensitivity of the photodetector with a photo-dark current ratio of up to 1000. Remarkably, a responsivity and external quantum efficiency of up to 2.6×10^5 A W⁻¹ and 8.1 × 10⁷%, respectively, were achieved. Shao et al. [183] reported on 3D WO₃ nanoscale structured materials synthesized through hydrothermal synthesis.

Conclusion:

Various nanostructures, including nanowires, nanotubes, core-shell configurations, nanosheets, nanofibers, nano cubes, and nanospheres, represent a promising avenue for the development of novel and advanced sensor materials. Addressing the primary limitations of current sensors, several strategies have been explored to enhance material sensitivity and overall sensor performance. These approaches involve doping, combining with other metal oxides to create nanocomposites, or incorporating carbon-based materials. A dedicated section on FET-based sensing highlights the notable improvements in sensor performance, particularly

in sensitivity, selectivity, and response/recovery time, attributed to the extra gate electrode in comparison to two-terminal sensors. Nevertheless, achieving large-scale production of oxide nanostructure FET sensors on flexible substrates requires significant advancements in establishing reliable methods for obtaining controlled densities of high-quality nanostructure networks on these substrates.

Looking ahead, the scientific community is directing efforts toward innovative synthesis strategies capable of precisely tuning metal oxide structures at the nanoscale, along with their intrinsic properties. This is exemplified by the evolution of UV sensors featuring single 1D nanostructures, resulting in high-performance devices. These nanostructures are anticipated to reduce defect density, facilitating carrier transport within the nanostructures, thereby enabling high external quantum efficiency and fast response times. The miniaturization of sensors while maintaining or increasing sensitivity and selectivity is a major contemporary research focus, along with the development of flexible and cost-effective systems. Utilizing inexpensive, lightweight, abundant, recyclable, and environmentally friendly substrates, such as polymer and paper-based materials, emerges as a dependable approach to produce the next generation of sensing devices. These devices are expected to be highly adaptable, disposable, eco-friendly, sustainable, and recyclable smart products. Furthermore, these substrates facilitate large-scale device production and can be seamlessly integrated with printing methods.

In the near future, flexible sensing devices are anticipated to be seamlessly incorporated into opto-electronic smart devices like solar cells, enabling the emergence of a wide range of low-cost and disposable consumer products in our daily lives. The flexibility and conformability of this new generation of sensors are poised to contribute to the Internet of Things (IoT) revolution, enhancing well-being, satisfaction, and comfort for end-users.

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Carbon Nanomaterials: Synthesis, Properties and Applications

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Abstract :

Carbon is one of the most fascinating elements of the periodic table. Because of its catenation property carbon exists in many allotropic forms. The allotropes of carbon that exist in nano dimensions are broadly called carbon nanomaterials and include graphene, carbon nanotubes, nanodiamonds, Carbon Fibers and many more. Carbon nanomaterials have the gained attention in recent times due to their unique properties ranging from high electrical conductivity, flexibility, specific surface area, carrier mobility, optical transparency etc. They have a wide range of applications spanning from electrodes for energy storage, environmental remediations, catalysts, sensors, smart electronics etc. This chapter highlights the synthetic methods and properties of carbon nanomaterials and discusses their applications, particularly in the field of energy.

Keywords: Carbon nanomaterials, Supercapacitor, Li-ion battery, environmental remediations, specific surface energy.

Introduction :

Carbon-based materials or carbon nanomaterials are composed of carbon atoms. Carbon nanomaterials are an extensive family of carbon allotropes [1-3]. Carbon nanostructures have different shapes i.e. tube-shaped [e.g. carbon nanotubes (CNT)], horn-shaped [e.g. nano horns], spherical or ellipsoidal [e.g. fullerenes]. Technical applications of carbon nanomaterials are in various fields such as micro- and nanoelectronics, gas storage, conductive plastics, composites, displays, antifouling paints, textiles, batteries, gas biosensors, catalysis, electronics, energy storage, biomedicine, and therapy [4-6]. Carbon nanomaterials have several properties, including hardness and softness, an insulator, semiconductor, conductor, thermal insulation, and thermal conductivity [7-9]. Furthermore, full light absorption as well as full light transmission, heat resistance, radiation resistance, chemical resistance, surface and interface

properties etc., are also witnessed in carbon nanomaterials [10-15]. Carbon is the most promising material located on earth. It can be detected in nature in the form of graphite, diamond and coal. Specifically, the use of carbon allotropes and derivatives such as carbon nanotube (CNT), graphene, porous carbons, carbon nanofibers, fullerenes, nanodiamonds and carbon-based quantum dots have a powerful approach to solve the problem in the environment [16-20]. Carbon nanomaterials are used in many different fields such as water treatments, environmental remediation, catalysis, sensors, drug delivery, cancer therapy and energy storage [21-27].

In this chapter, we will concentrate on various synthesis processes, properties and applications of carbon nanomaterials elaborately.

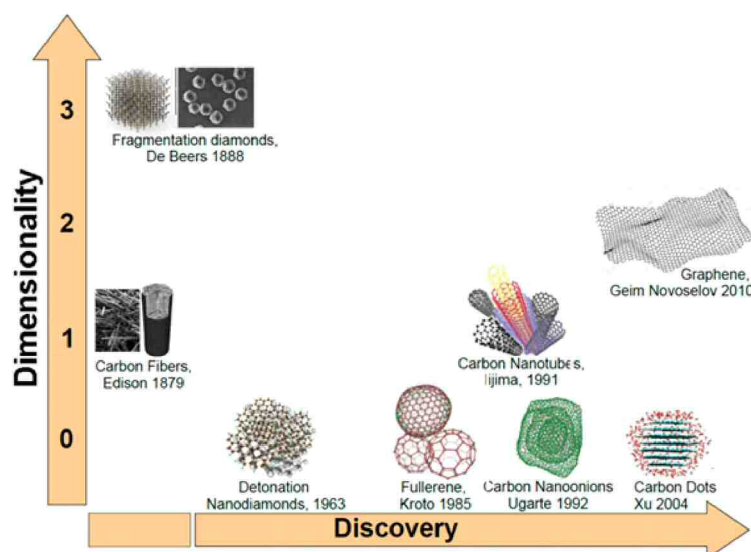


Figure 1: Carbon nanostructure dimensions and discovery time

Types of carbon nanomaterials:

(a) Zero-dimensional (0-D) Carbon nanomaterials :

- ♦ **Fullerenes :** Fullerene is a class of carbon materials having a hollow spherical cage-like geometry with sixty or more carbon atoms. Robert F. Curl, Jr., Richard E. Smalley, and Sir Harold W. Kroto discovered a new type of knowledge in material science i.e. fullerenes in 1985. Fullerenes have contributed to the research field in recent times. Researchers developed different synthesis processes of fullerene and its derivatives as shown in **Table 1** [28,29].
- ♦ **Carbon-Encapsulated Metal Nanoparticles :** Carbon-encapsulated metal nanoparticles are a subclass of zero-dimensional Carbon nanomaterials which has the shape of a core-shell structure in the nanoscale. It is a polyhedral metallic core-shell structure covered by a multilayer graphitized carbon shell. The carbon layers isolate the particles

magnetically from the external environment and protect them against corrosion and magnetic interaction. Researchers developed different synthesis processes of Carbon-encapsulated metal nanoparticles and its derivatives as shown in **Table 1** [30].

- ♦ **Nanodiamond** : Diamond is an allotropic form of carbon. They have sp^3 hybridized carbon atoms and are the hardest material. Due to their excellent hardness, chemical corrosiveness, thermal expansions and conductivity, electrical insulation and biocompatibility nanodiamonds have gained tremendous attention [31-35]. Researchers developed different synthesis processes shown in **Table 1** and derivatives of nanodiamonds.
- ♦ **Onion-Like Carbons** : Onion-like carbons have three to eight closed graphitic shell structures with a hollow core. The outer diameter of onion-like carbons is in the range of 20-100 nm and has quasi-spheroidal shape. The most significant property of these nanocarbons is their surface specificity [36]. The different synthetic procedures of onions like carbon and their derivatives are summarized in **Table 1**.

(b) One-Dimensional (1-D) Carbon Nanomaterials :

- ♦ **Carbon Nanofibers** : It is composed of stacked and curved graphene layers from a quasi-1-D filament which has cylindrical or conical nanostructures. It can be synthesised through hydrothermal, solvothermal, microwave irradiation, chemical vapour deposition etc. [37-38]. Researchers developed different synthesis processes of this class of material as shown in **Table 1**.
- ♦ **Carbon Nanotube (CNT)** : In 1991, Ijima discovered a new branch of knowledge in material science i.e. CNT. The carbon nanotubes are divided into two categories: single-walled carbon nanotubes (SWCNT) and multiwall carbon nanotubes (MWCNT). A SWCNT consists of a single sheet of graphene that has been rolled up into a tube, while MWCNTs consist of multiple graphene sheets rolled up into tubes (concentric tubes) shown in **Figure 1** [39-41]. Researchers developed different synthesis processes of this material as shown in **Table 1**.

(c) Two-Dimensional Carbon Nanomaterials :

- ♦ **Graphene**: Graphene is an example of two-dimensional hexagonal honeycomb crystal lattice. It is made up of a one-atom-thick planar sheet of sp^2 -bonded carbon atoms. Graphene has excellent conductivity, large surface area, long-term stability, and great processing flexibility arising from the arrangement of carbon atoms in a hexagonal framework [42-49]. Researchers developed different synthesis processes of graphene as shown in **Table 1**.
- ♦ **Carbon Nano walls**: Carbon Nano walls (CNWs) are a subclass of two-dimensional carbon nanomaterials. It consists of vertically aligned graphene sheets standing on the substrates, with large surface areas and sharp edges. The size of CNWs ranges from a few nm to a few tens of nm [50]. Researchers developed different synthesis processes for this class of materials as shown in **Table 1**.

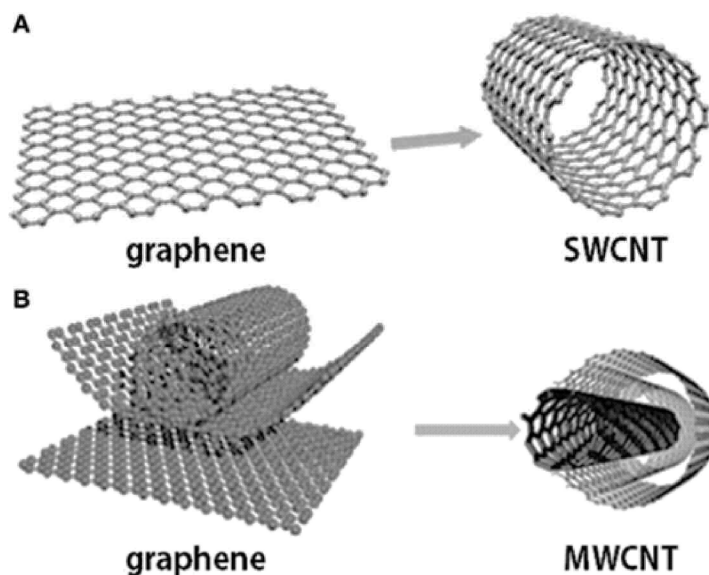


Figure 2: Formation of SWCNT and MWCNT from graphene

Synthesis Method of The Carbon Nanomaterials:

Carbon nanomaterials (Fullerenes, Nanodiamonds, Carbon nanofibers, Carbon nanotubes, Graphene, and Carbon Nano walls) were synthesised using various methods such as hydrothermal process, solvothermal process, microwave process, CVD etc. [51-53].

- (a) **Hydrothermal Process :** Hydrothermal process is a widely used synthetic method for their low cost and high yield. Water is abundant in nature, and has amazing qualities, and when hydrothermal conditions exist, it serves as a catalyst for the synthesis of the desired product. The fact that water is volatile and can be quickly separated from the substance makes it an excellent solvent [36].

Table 1: Different Synthesis Methods of the Carbon Nanomaterials

Sl. No.	Carbon nanomaterials	Synthesis method
1	Fullerene	Laser vaporization, Arc discharge, Combustion, Microwave plasma Pyrolysis, Flat flames
2	Carbon-Encapsulated Metal Nanoparticles	Arc-discharge, high-temperature heat treatment, mechanical milling, Co-carbonization, Pulse-laser irradiation
3	Nanodiamond	MPCVD, Hot-filament, CVD, Pulse laser ablation, Electro irradiation, High-energy X-ray diffraction

4	Onion-Like Carbons	Arc-discharge, High-electron irradiation, CVD, RFP Thermal annealing of diamond nano-particle
5	Carbon Nanofibers	Traditional vapour growth, Catalytic combustion, PECVD, Hot filament-assisted sputtering, Ultrasonic spray pyrolysis, Ion beam
6	Carbon Nanotube	Arc-discharge, Laser-ablation, CVD, and other methods
7	Graphene	Mechanical exfoliation, Epitaxial growth, Chemical exfoliation
8	Carbon Nano walls	MWPECVD, RFPECVD, HWCVD, EBEPECVD

- (b) **Solvothermal Process** : Another process is the solvothermal approach which is similar to the hydrothermal process, with solvents like ethanol, methanol, etc. Different solvents can be used to create unusual morphologies such as nanorods, microflowers, and plate-like morphologies. Solvothermal heating involves heating the reaction mixture in a closed vessel to a temperature greater than the boiling point of the solvents, similar to hydrothermal heating. It has been found that the shape of materials is significantly influenced by temperature and the surfactant [40].
- (c) **Microwave Process** : The microwave irradiation method is a unique one-step method owing to its facile and fast nature. In the process of microwave synthesis, a closed container is placed into the microwave oven and upon microwave irradiation the inside material rotates and migrates due to their dielectric property. After absorbing the microwave, the material quickly gets an extremely high temperature within a few seconds. Under the microwave irradiation heat is produced that support the reaction. The reaction rate can be enhanced with an increase in a collision that occurs between reactants within a confined space. Microwave irradiation is the fastest heating process and provides eco-friendly pathway that create porous morphology and high surface area material [33,45].

Properties and Application of The Carbon Materials:

- ♦ **Fullerenes** : Fullerenes is a category of zero-dimensional carbon nanomaterials. Fullerene can be used in medicine, drug deliveries, and cancer therapy (melanoma). Fullerene is used in superconductivity, and thermal conductivity due to its specific properties [28].
- ♦ **Nanodiamonds** : An extremely wide range of applications has been made possible by nanodiamonds, including most notable applications such as polishing materials, and additives to motor oil, fibre, industry, and electrical conductors. In recent years, there have been several researches conducted on the application of nanodiamonds in medical science, particularly in the production of chemotherapy drugs [32].
- ♦ **Carbon nanotubes** : CNTs can exhibit such a wide range of chemical and physical properties, that they can be used in a wide range of different fields (such as

nanotechnology, electronics, optics, materials science, and architecture). Based on structural applications of CNTs, they can be used in industries such as clothes, sports equipment such as stronger and lighter tennis rackets, bikes, various kinds of balls, combat jackets, concrete with hightensile strength, polyethene, the possibility of the space elevator, synthetic muscles in medical science and sports, high tensile strength fibres, in the build of bridges, ultrahigh-speed flywheels and applications in fire protection etc. Also, CNT can be used in chemical nanowires, conductive films, electric motor brushes, magnets, optical ignition, their applications to produce light bulb filament (as an alternative for tungsten filaments), the applications related to their fine superconductivity properties, in display screens such as field emission displays (FEDs) applications in transistor industrial, and the electromagnetic antenna. CNTs are used in air pollution filters, biotech containers, hydrogen storage, water filtration, and mechanical applications such as using them as faster oscillators, nanotube membranes, carbon nanotube actuators, infrared detector, radiometric standard, and their application as thermal radiation for space satellites based on the chemical application of CNT [54-56].

- ♦ **Graphene :** Graphene is the most important nanomaterial, due to its properties of high thermal conductivity, and high surface area, high power density. It can be applied in chemical and biochemical sensing. On the other hand, based on low resistivity and also very low thickness it has a great role in the mechanical fields, electrical conducting, and transparent films which are necessarily applicable in the field of electronics such as producing touch screens and photovoltaic cells. Graphene and modified graphene are used in a different field such as energy storage, sensors, and photocatalytic applications [57-58].

Application of the Carbon Material :

Energy Storage and Conversion Applications :

- (a) **Supercapacitor (SC):** Supercapacitors (SCs), also called ultracapacitors or electrochemical capacitors stimulate extensive attention with novel features such as rapid charging-discharging rate, favourable power density, extended cycle life and high dynamic of charge propagation. Supercapacitors are classified into three categories: electric double-layer capacitors (EDLCs), pseudo-capacitors and hybrid supercapacitors (the EDLCs and the pseudo-capacitors are combined into a single device). Electrode materials are one of the most promising components in SCs and the overall performance of supercapacitors very much depends on the active electrode material used. Carbon nanotubes and graphene either in pristine forms or in the composite state with transition metal sulfide, and metal oxide, are applied as electrode materials in supercapacitors [55, 59].
- (b) **Lithium-Ion Batteries (LIBs) :** Rechargeable lithium-ion batteries (LIBs), which were first made commercially available by Sony Corporation in the 1990s, have dominated the personal smart electronics market for over 20 years, and they are anticipated to be used in electric cars (EVs) and hybrid electric vehicles (HEVs). With significant commercial success, LIBs are well-developed and often utilized in a wide range of

applications, from small personal mobile devices to medium-sized automobiles (including hybrid, electrical, and mopeds). The grid-scale energy storage applications of these devices, however, call for cost-effective and performance qualities, which LIBs lack. Unfortunately, the use of LIBs in large-scale energy storage systems (LESSs) has been constrained by the scarcity, unequal distribution, and rising costs of lithium and cobalt precursors. LIB is made up of anode, cathode, separator and electrolyte. Carbon nanomaterials, such as carbon nanofibers (CNFs), fullerene, graphene and carbon nanotubes (CNTs), have several applications in lithium-ion batteries (LIBs). They can help improve the stability, safety, and performance of LIBs. They can also be used as anode materials to address chemo-mechanical issues [60-62].

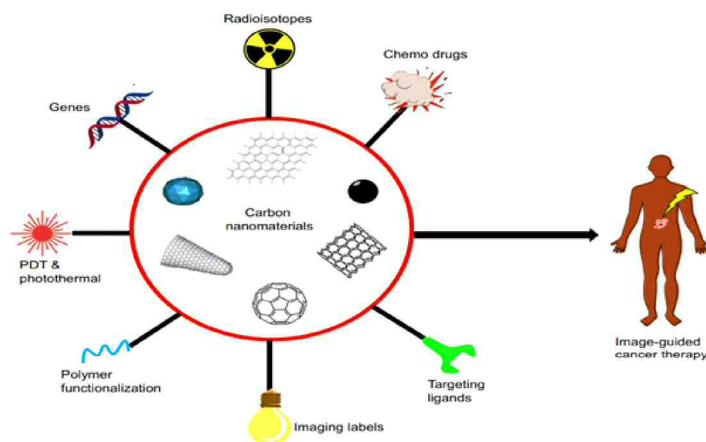


Figure3: Application of carbon nanomaterials

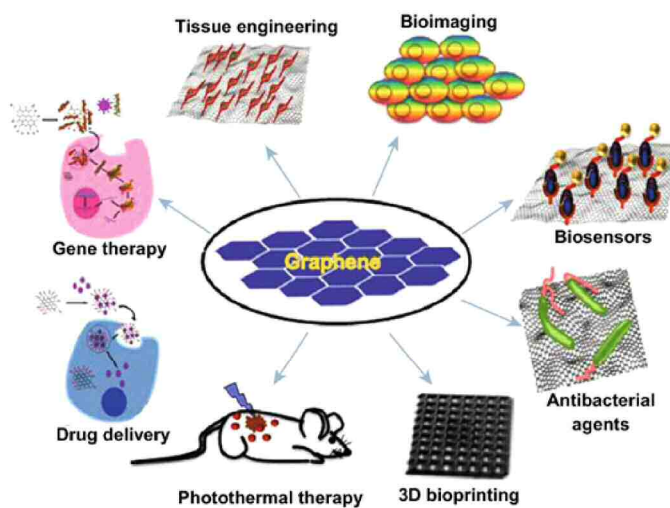


Figure4: Application of graphene

- (c) **Sodium-Ion Batteries (SIBs)** : Sodium-ion batteries (SIBs) are the most important rechargeable batteries due to their large-scale energy storage systems and low production cost. Carbon nanomaterials [Carbon dots and graphene etc.] have been used in sodium-ion batteries (SIBs) as anodes. Based on carbon nanotube (CNT)-based nanomaterials, SIBs have received great attention due to their superior features i.e. large specific surface area, high conductivity, excellent flexibility and chemical stability [60-62].

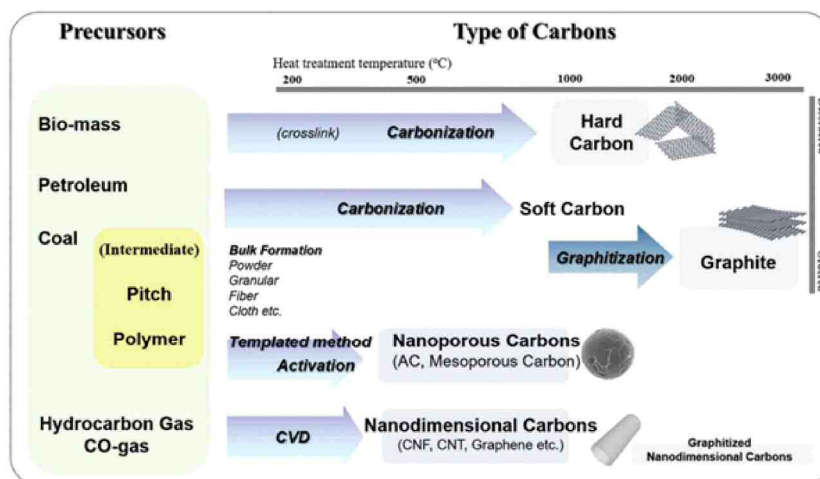


Figure 5: Application of carbon nanomaterials into sodium-ion batteries

Concluding Remarks :

Carbon nanomaterials offer a wide range of applications and exhibit fascinating properties that are tunable as desired. These materials can be synthesized by different synthetic procedures most of which are cost-effective, less robust and can be produced on a large scale. Because of its synthetic procedures and versatile properties extensive research has been devoted to carbon nanomaterials. Yet there are still some research gaps left. Carbon nanomaterials are believed to change the energy storage scenario in the near future. As a sustainable ecofriendly material, carbon nanomaterial is expected to replace many other materials indifferent fields in the coming days.

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Nanotechnology in Polymer Composites

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Abstract:

Nanotechnology has revolutionized the field of polymer composite with its powerful presence. Introduction of nanomaterial to the polymer composite enhances various physicochemical properties in the composite. Polymer nanocomposites, which consist of a polymer matrix with dispersed inorganic nanoparticles, have shown improved performance in various areas such as mechanical strength, electrical conductivity, and thermal stability. Now polymer nanocomposite has been extensively used in advance application like aerospace, energy storage, wearable technology, self-healing material etc. Depending on the types of nanoparticles within polymer matrix, polymer nanocomposite can be divided into various groups. Each group of composite has unique application. Processing of polymer nanocomposite mainly consist of two steps blending and moulding. Nanotechnology unlocks a vast potential for next-generation polymer composites, shaping a more efficient, sustainable, and technologically advanced future.

Keywords : Polymer nanocomposite, Nanoparticles, Clay nanocomposite, Carbon nanocomposite, Grapheme nanocomposite.

Introduction to Polymer Nanocomposite and Early History :

In modern era of scientific research, polymers and polymer composites holds a key position due to its versatility. Polymer composites are widely used in various fields such as agriculture sector, automobile sector, furniture and fixtures etc. The major advantages possess by the polymer composites over the other traditional materials are strength, weight, durability etc. Incorporation of nanomaterial into the polymer composite further improves the properties of the composite. This class of compound is known as Polymer nanocomposite.

Polymer nanocomposites (PNC) have a rich history that spans over several decades, marked by significant advancements and breakthroughs in material science and engineering. The concept of incorporating nanoscale fillers into polymer matrices to enhance their properties

can be traced back to the mid-20th century with the first work published in a patent by the National Lead Company in 1950. The market debut of commercially successful nylon nanocomposites occurred in 1989, pioneered by Toyota central R & D Laboratories in Japan.

However, the term “nanocomposite” gained prominence in the 1990s, coinciding with the rapid development of nanotechnology [1,2].

The early history of polymer nanocomposites can be linked to the pioneering work of researchers such as Paul Flory, who laid the foundation for understanding polymer behavior at the molecular level. The introduction of nanotechnology provided a new avenue for manipulating materials at the nanoscale, leading to the exploration of novel polymer-based nanocomposites.

One of the key milestones in the history of polymer nanocomposites was the development of clay-polymer nanocomposites in the 1990s. This marked a turning point in the field, as researchers demonstrated that the addition of small amounts of layered silicates, such as montmorillonite, could significantly improve the mechanical, thermal, and barrier properties of polymers [3,4].

Subsequent years saw a surge in research focused on various types of nanofillers, including carbon nanotubes, grapheme, metal oxides, and nanocellulose, among others [5-9]. The evolution of polymer nanocomposites has been driven by interdisciplinary collaboration between chemists, physicists, materials scientists, and engineers. This convergence of expertise has led to the development of advanced manufacturing techniques, characterization methods, and theoretical models to better understand and optimize the behavior of polymer nanocomposites.

Impact of Nanotechnology in current polymer research :

Nanotechnology has revolutionized various industries, including the field of polymer composites. With the integration of nanotechnology, polymer composites have seen significant improvements in terms of their mechanical, thermal, and electrical properties. Nanotechnology is a rapidly advancing field that involves manipulating materials at the nanoscale level to achieve unique properties and enhanced performance. One of the most promising applications of nanotechnology is in the development of polymer nanocomposites.

Polymer nanocomposites containing nanoparticles as fillers within the polymer matrix, has become a prominent area of research in material science. These nanocomposites utilize different types of nanoparticles, such as carbon nanotubes, grapheme, clay nanoparticles, and metal oxide nanoparticles, to enhance the overall performance of the polymer matrix. By incorporating nanoparticles into polymer composites, researchers have been able to achieve enhanced physicochemical properties. Because of the high degree of contact between nanofillers and polymers, also referred to as the “nano-effect,” the addition of nanofillers with the same filler loading into the polymer matrix significantly alters the electrical and mechanical properties of polymer composites.

These nanofillers, with their exceptional physical properties, have the potential to improve the strength, stiffness, and toughness of polymer composites. Additionally, nanotechnology

has also been utilized to improve the thermal properties of polymer composites [10]. For instance, the addition of thermally conductive fillers such as metal oxide nanoparticles can significantly increase the thermal conductivity of polymer composites [11].

This improvement in thermal conductivity is crucial, as it allows for better heat dissipation and thermal management, making the polymer composites suitable for applications that require efficient heat transfer, such as electronic devices and automotive components [12]. Furthermore, the integration of nanotechnology into polymer composites has led to advancements in electrical conductivity [13]. For example, the incorporation of carbon nanotubes or silver nanowires into a polymer matrix has shown remarkable improvements in electrical conductivity [14, 15].

Classification of Polymer nanocomposites :

There are several types of polymer nanocomposites, each distinguished by the nature of the polymer matrix and the nanofiller used. Here are some common types:

Clay nanocomposite : This is one of the earliest and most extensively studied types of polymer nanocomposites. A clay-based polymer nanocomposite is a material that combines a polymer matrix with clay nanoparticles to create a new material with enhanced properties. The clay nanoparticles are typically layered silicates, such as montmorillonite (MMT), which have a high aspect ratio and a large surface area. The process of creating a clay-based polymer nanocomposite involves dispersing the clay nanoparticles within the polymer matrix using methods such as melt blending, solution mixing, or in situ polymerization. The addition of clay nanoparticles can enhance the physicochemical properties of the polymer, such as increased strength, stiffness, and toughness [16]. The barrier properties, including gas barrier and flame retardancy, are also improved. **Figure 1** represents TEM micrograph of a typical Montmorillonite clay reinforced polymer composite with Polyvinyl chloride [17]. The delaminated clay layers of MMT are visible in **Figure 1.b)** while the agglomeration of MMT can be observed in **(Figure 1.c)**.

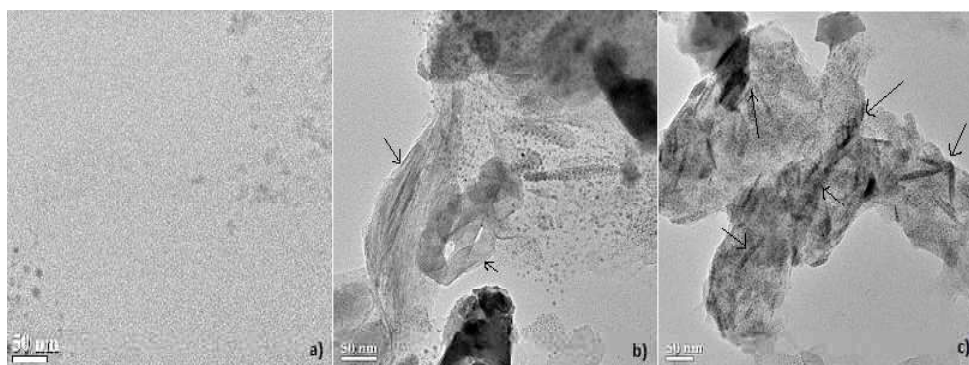


Figure 1 : TEM image showing Montmorillonite composite with PVC

Carbon Nanocomposites : Polymer carbon nanocomposites are a class of materials that have gained significant attention due to their exceptional properties and wide-ranging applications. Various types of polymer carbon nanocomposites have been developed to harness the unique properties of carbon-based nanocomposite for specific applications [18]. The world of materials science is abuzz with the potential of carbon nanocomposites, a class of materials where carbon in its various nanoforms takes center stage. These remarkable materials combine the unique properties of carbon with those of other materials, creating composites with strength, flexibility, conductivity, and functionalities that surpass conventional materials. Some of the most widely studied types of polymer carbon nanocomposites include those with thermoplastic polymer matrices, such as polyethylene, polystyrene, and polypropylene, as well as thermosetting polymers like phenolic resin [19,20]. Some of the most common types include:

- (a) **Carbon Nanotube (CNT) Composites :** Carbon Nanotube (CNT) These composites consist of carbon nanotubes dispersed within a polymer matrix. CNT composites can be embedded in polymers for enhanced strength and conductivity, making them ideal for lightweight aircraft parts, energy storage devices, and even bulletproof vests [21, 22]. Polymer composites have been prepared with almost all forms of CNT material such as Singled walled CNT, multi walled CNT and Doubled walled CNT. CNT-polymer composites offer a world of possibilities with their diverse properties and potential applications.

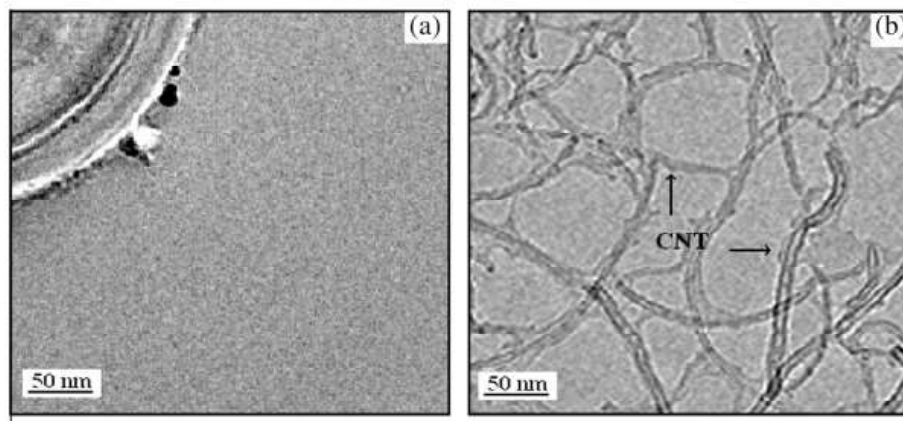


Figure 2 : Showing Dispersion of MWCNT in vacuum impregnation of melamine formaldehyde furfuryl alcohol copolymer composite [23]

- (b) **Graphene Nanocomposites :** Graphene, a single layer of carbon atoms arranged in a two-dimensional honeycomb lattice, is incorporated into a matrix material to form graphene composites. These composites exhibit excellent mechanical, electrical, and thermal properties, making them suitable for applications in electronics, energy storage, and structural materials [24, 25].

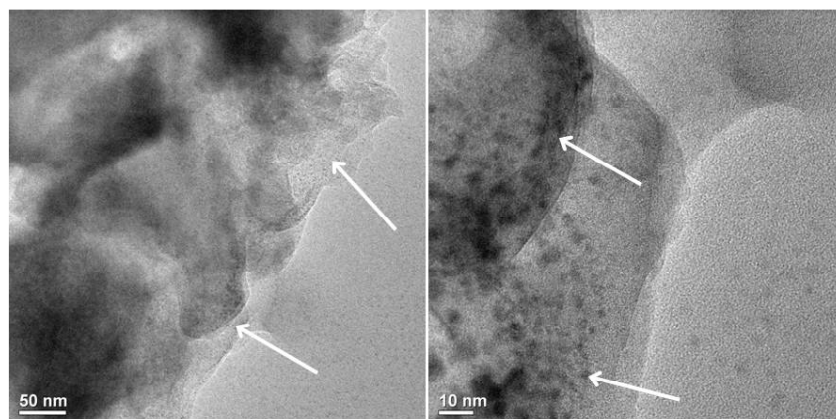


Figure 3 : shows the TEM micrographs of the composite containing 0.5 phr GO at two different magnifications in PVC based nanocomposite [26]

- (c) **Carbon Nanofiber Composites :** Carbon nanofibers, which are cylindrical nanostructures with diameters on the order of nanometers, are used to reinforce a matrix material in carbon nanofiber composites. These composites offer high strength, stiffness, and electrical conductivity, and are commonly used in aerospace, automotive, and energy applications [27].

Carbon Nanoparticle Composites :

These composites incorporate various carbon nanoparticles, such as fullerenes, carbon black, or carbon quantum dots, into a matrix material. They can exhibit unique properties, including enhanced electrical conductivity, thermal stability, and optical properties, making them suitable for applications in electronics, sensors, and energy storage.

- (a) **Metal Nanoparticle Nanocomposites :** This is a class of polymer nanocomposite where metal nanoparticles (e.g., silver, gold, or copper) incorporated into the polymer matrix. This type of nanocomposite offers enhanced antimicrobial properties, electrical conductivity, and catalytic activity and are commonly used in packaging materials, medical devices, and sensors [28].
- (b) **Nanocellulose Nanocomposites :** Here nanocellulose was used as reinforcing agent which is derived from natural cellulose fibers. Presence of nanocellulose in the polymer composite improves the mechanical properties and enhances the strength, stiffness, and thermal stability of the polymer matrix [29]. This class of composite is sustainable and biodegradable, making them suitable for eco-friendly applications.
- (c) **Quantum Dot Nanocomposite :** This is a type of minute semiconductor nanoparticles that exhibit unique optical and electronic properties due to their quantum confinement effect. Quantum dots are integrated into polymer matrices for applications such as optoelectronic devices, solar cells, and sensors [30]. The size and composition of quantum

dots can be adjusted to achieve desired optical properties.

MXene Base polymer nanocomposite:

MXene polymer composites have gained significant attention due to their unique properties and potential applications. MXenes are 2D nanomaterials that, when combined with polymers, can enhance various properties and add new functionalities to the resulting composites. Some of the applications of MXene polymer composites include EMI shielding, water filtration, energy harvesting, high light-to-heat conversion, excellent electromagnetic interference shielding, high charge storage, and anticorrosion [31, 32]. These composites have found applications in various fields such as chemical, materials, electrical, environmental, mechanical, and biomedical engineering, as well as medicine [33]. Polymers are favored for synthesizing composites with MXenes due to their versatility, compatibility, and cost [34]. The combination of the attractive properties of MXenes and the flexibility of polymers makes MXene/polymer nanocomposites promising for a wide range of practical applications.

Nanofluid :

Nanofluid polymer nanocomposites are a type of polymer nanocomposite that incorporate nanoparticles dispersed in a polymer matrix, along with a fluid component. Nanofluids have gained attention in thermal and heat transfer applications, and their incorporation into polymer matrices has been shown to enhance the thermal and mechanical properties of the resulting nanocomposites [35]. The use of nanoparticles in polymer nanocomposites has led to improved performance in various areas such as mechanical strength, electrical conductivity, and thermal stability [36]. Nanomaterials additive synthesis and preparation of nanofluids are a new generation of nanomaterials with outstanding performances in industrial applications [37]. The future of this field holds potential for building ordered arrays of nanoparticles in the polymer matrix and manufacturing nanocomposite circuit boards, further expanding the range of applications for these advanced materials [38].

Processing Method of PNC :

Polymer nanocomposites are materials that combine the properties of polymers with those of nanoparticles, resulting in materials with unique and often superior properties. The processing method used to create these materials can significantly impact their final properties and performance. The aim of all techniques is to produce nanocomposite materials with uniform dispersion and without aggregations. The processing of polymer nanocomposite comprises of mainly two steps. The first step is the blending of the polymer matrix and the nanoparticle along with other additives. In this step nanoparticle is incorporated into the polymer matrix.

Blending of Polymer and Nanomaterials :

Here are some of the most common mixing methods for polymer nanocomposites:

- (a) **Solution Mixing :** In this method, both the polymer and the nanofiller along with other additives are dissolved in a common solvent. The solution is then stirred or sonicated

to disperse the nanofiller throughout the polymer matrix. The solvent is then removed by evaporation or precipitation, leaving behind the polymer nanocomposite.

Advantages :

- Good dispersion of the nanofiller
- Can be used for a wide variety of polymers and nanofillers

Disadvantages :

- Requires the use of a solvent, which can be expensive and environmentally hazardous
- Can be limited by the solubility of the polymer and nanofiller
- May require additional processing steps to remove the solvent

- (b) **Melt Mixing :** In this method, the polymer, nanofiller and the additives are mixed together in the molten state. The mixture is then sheared or extruded to disperse the nanofiller throughout the polymer matrix.

Advantages :

- Does not require the use of a solvent
- Can be used for high-molecular-weight polymers
- Can be used for continuous processing

Disadvantages :

- May require high processing temperatures
- Can be difficult to achieve good dispersion of the nanofiller
- May degrade the polymer or the nanofiller

- (c) **In Situ Polymerization :** In this method, the nanofiller is added to the monomer mixture before polymerization. The monomer is then polymerized around the nanofiller particles.

Advantages :

- Can lead to strong interfacial bonding between the polymer and the nanofiller
- Can be used to tailor the properties of the nanocomposite

Disadvantages :

- Can be limited to certain types of polymers
- May require specialized equipment

- (d) **Layer-by-Layer (LbL) Assembly :** In this method, the polymer and the nanofiller are alternately deposited onto a substrate in thin layers. The layers are held together by electrostatic or other intermolecular forces of nanocomposite.

Advantages :

- Can be used to create very thin films with controlled properties

- Can be used for a wide variety of materials

Disadvantages :

- Can be a slow and tedious process
 - May not be suitable for all applications
- (e) **Ultrasonication** : This method uses high-intensity ultrasound waves to disperse nanoparticles in a polymer matrix. This can be done in a solvent or in the molten state. Ultrasonication can help to break up agglomerates of nanoparticles and improve their dispersion in the polymer matrix.
- (f) **Electrospinning** : In this method, a solution of polymer and nanofiller is charged and sprayed through a high-voltage needle. The resulting jet of solution dries and solidifies into thin fibers. This method can be used to produce nanocomposites with a high surface area and unique morphologies.

Molding Process of Blended mixture of PNC :

Molding methods are commonly used to shape and form polymer nanocomposites into desired shapes and dimensions. Here are some molding methods used for polymer nanocomposites:

- (a) **Injection molding** : In this method, molten polymer nanocomposite material is injected into a mold cavity under high pressure. The mold cavity is shaped according to the desired shape of the final product. After cooling and solidification, the molded part is ejected from the mold. Injection molding is suitable for mass production of complex-shaped polymer nanocomposite parts with high precision and dimensional accuracy.
- (b) **Compression molding** : In this method, a mixture of polymer nanocomposite material and a curing agent is placed in a mold cavity and subjected to heat and pressure. The heat and pressure cause the material to flow and fill the mold cavity, and the curing agent initiates crosslinking or polymerization reactions to solidify the material. Compression molding is suitable for producing simple-shaped parts with uniform thickness.
- (c) **Transfer molding** : This method is similar to injection molding, but the molten polymer nanocomposite material is first transferred from a heating chamber into a mold cavity using a plunger or screw. The material is then cured under heat and pressure to form the final product. Transfer molding is suitable for producing complex-shaped parts with good surface finish and dimensional accuracy.
- (d) **Extrusion molding** : In this method, molten polymer nanocomposite material is forced through a die with a specific shape. The extruded material solidifies as it exits the die, forming a continuous profile or shape. Extrusion molding is suitable for producing long, continuous products such as pipes, sheets, and films.
- (e) **Blow molding** : In this method, a parison of molten polymer nanocomposite material is extruded and then inflated inside a mold cavity using compressed air or gas. The inflated material conforms to the shape of the mold cavity and solidifies, forming a

hollow product such as a bottle or container.

The choice of molding method for a particular polymer nanocomposite will depend on factors such as the desired shape and dimensions of the final product, the production volume, and the properties of the polymer nanocomposite material.

Applications :

The incorporation of nanoparticles into the polymer matrix leads to improved mechanical, thermal, electrical, and optical properties of the resulting material. The size of the nanoparticles plays a crucial role in determining the properties of the PNCs. The smaller the size of the nanoparticles, the larger the surface area, which leads to enhanced properties [39].

The unique characteristics, like enhanced strength, electrical conductivity, and thermal stability, unlock a vast spectrum of applications across various industries. Here's a detailed exploration of the diverse applications of polymer nanocomposites:

1. Automotive Industry :

- **Light weighting** : Replacing traditional metals with polymer nanocomposites in car parts reduces weight, leading to improved fuel efficiency and reduced emissions.
- **Enhanced Durability** : Nanocomposites offer superior scratch resistance, making them ideal for car interiors and exteriors.

2. Electronics and Electrical :

- **Semiconductors** : Nanoparticles can tailor the electrical properties of polymers, making them suitable for transistors and other electronic components.
- **Electrostatic Discharge (ESD) Protection** : Nanocomposites shield sensitive electronics from harmful static shocks.
- **Batteries** : Polymer nanocomposites with high ionic conductivity are being explored for next-generation battery electrodes.

3. Aerospace and Defence :

- **Radar Absorbing Materials (RAM)** : Specific nanocomposites absorb radar waves, making aircraft and military vehicles stealthier.
- **Lightweight Aircraft Parts** : The high strength-to-weight ratio of nanocomposites enables the development of lighter and more fuel-efficient aircraft.
- **Bulletproof vests** : Nanocomposites can be used to create stronger and lighter bulletproof vests for improved protection.

4. Medical and Healthcare :

- **Drug Delivery** : Nanoparticles can be used to target specific cells or tissues for controlled drug delivery, improving treatment efficacy and reducing side effects.
- **Tissue Engineering** : Scaffolds made from polymer nanocomposites mimic the natural extracellular matrix, promoting tissue regeneration.

- **Medical Implants** : Biocompatible nanocomposites are used in artificial bones, joints, and other implants for enhanced durability and biocompatibility.

5. **Packaging and Food Safety :**

- **Biodegradable Packaging** : Nanocomposites can be used to create sustainable and biodegradable packaging materials, reducing plastic waste.
- **Active Packaging** : Antimicrobial nanoparticles incorporated into packaging can extend food shelf life and prevent spoilage.
- **Oxygen Scavengers** : Nanocomposites can absorb oxygen, creating an ideal environment for preserving food freshness.

6. **Energy and Environment :**

- **Solar Cells** : Polymer nanocomposites with improved light absorption and charge transport properties are being developed for efficient solar cells.
- **Fuel Cells** : Nanocomposite membranes can enhance the performance and durability of fuel cells.
- **Water Purification** : Membranes made from certain nanocomposites can effectively filter pollutants and contaminants from water.

These are just a few examples of the countless applications of polymer nanocomposites. As research and development progress, we can expect even more innovative uses of these versatile materials in the future, revolutionizing various sectors and shaping a more sustainable and efficient world. It's important to note that the specific applications of polymer nanocomposites depend on the type of polymer and nanoparticles used, their properties, and the fabrication techniques employed. The ongoing research in this field holds immense promise for further advancements and breakthroughs in diverse fields.

Future scope of Polymer nanocomposite :

Polymer nanocomposite has gained a lot of popularity due to their usefulness in various fields. In modern scientific era it is considered as one of the wonderful materials. Polymer nanocomposites hold immense promise for the future, offering a revolutionary new class of materials with vastly improved properties compared to traditional polymers. Their potential applications span diverse sectors, from medicine and energy to aerospace and electronics. Here are some key areas of future scope for polymer nanocomposites:

- (a) Improved mechanical properties** : The future of polymer nanocomposites lies in enhancing their mechanical properties such as strength, stiffness, and toughness. This can be achieved by developing new types of nanoparticles and optimizing their dispersion within the polymer matrix.
- (b) Nanocomposites for aircraft interiors** : Nanocomposites based on poly (furfuryl alcohol) (PFA), a bio-based and renewable polymer, have been developed for aircraft interiors. They have shown good fire resistance, mechanical performance, and peel strength, which are important for the safety and durability of aircraft components².

- (c) **Sustainable and eco-friendly nanocomposites :** The future scope of polymer nanocomposites also includes the development of sustainable and eco-friendly materials by using bio-based polymers and renewable nanoparticles. This could lead to the creation of biodegradable and recyclable nanocomposites. Developing methods for recycling and upcycling nanocomposite materials will further enhance their sustainability and reduce waste.
- (d) **Smart and responsive materials :** Polymer nanocomposites can be engineered to respond to external stimuli such as temperature, pH, or light. This could lead to the development of smart materials with self-healing, shape memory, or sensing capabilities.
- (e) **Biomedical applications :** Nanocomposites can be used to create biocompatible implants, drug delivery systems, and tissue engineering scaffolds. Their tailored properties can mimic natural tissues, enhance healing, and target specific cells for drug delivery.
- (f) **Energy storage and conversion :** Nanocomposites can improve the efficiency and lifespan of batteries, supercapacitors, and solar cells. Their ability to conduct electricity and store energy makes them ideal for next-generation energy technologies.
- (g) **Lightweight and high-strength materials :** Nanocomposites can be significantly lighter and stronger than metals, making them ideal for applications in aerospace, automotive, and construction industries. Their ability to withstand extreme temperatures and pressures also opens doors for use in extreme environments.
- (h) **Functionalization for specific applications :** Polymer nanocomposites can be tailored for specific applications by functionalizing the nanoparticles with additives or surface treatments. This could lead to the development of nanocomposites with enhanced electrical, thermal, or barrier properties.

Overall, the future of polymer nanocomposites is bright. With continued research and development, these materials have the potential to revolutionize various industries and contribute to a more sustainable and efficient future.

Conclusion :

Nanotechnology has significantly influenced the field of polymer composites by improving various properties. The incorporation of nanoparticles into the polymer matrix enhances stiffness, strength, and toughness of the composites. Nanotechnology continues to drive advancements in polymer composites, opening doors for the development of innovative materials with enhanced properties and performance. Today, polymer nanocomposites continue to be a subject of intense research and development, with a focus on sustainability, multifunctionality, and scalability. The integration of nanotechnology with polymer science has opened up new frontiers for creating high-performance materials with tailored properties, paving the way for diverse applications in industries ranging from electronics and packaging to construction and energy.

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Role of Nanotechnology in Medicine

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Abstract :

Nanotechnology has extensive application as nanomedicine in the medical field. Some nanoparticles have possible applications in novel diagnostic instruments, imagery and methodologies, targeted medicinal products, pharmaceutical products, biomedical implants, and tissue engineering. Nanomedicine is a relatively new field of science and technology. Brief discussion on role of nanotechnology medicine is included here. The different types of nanoparticles used in the medical field are discussed in the chapter.

Keywords : Nanomedicine, nanotechnology, drug delivery, regenerative medicine, theragnostic, tissue engineering.

Introduction :

Nanotechnology is now regarded as a novel technique in medical research [1]. In healthcare area, nanotechnology opens new area in the life sciences industry. Nanotechnology has possibility in manipulating things at the atomic level to change many parts of medical treatment, such as diagnosis, monitoring for diseases, operating equipment, regenerative medicine, developing vaccines, and drug delivery. It also opens the way through sophisticated research instruments to develop drugs to improve treatments for various ailments. Nanotechnology can be utilized for medication to particular cells in the body, thereby reducing the risks of failure and rejection. Thus nanodrug delivery systems can not only transport encapsulated or grafted small chemotherapeutic drugs, with a size of less than a dozens of nanometres, but also deliver them inside cells once they have penetrated them. Such systems can also be decorated with fragments of antibodies on their surface to target specific tissues, thus improving the specificity of the drug delivery.

Various disease like diabetes, cancer, Parkinson's disease, Alzheimer's disease, cardiovascular diseases and multiple sclerosis as well as different kinds of serious inflammatory or infectious diseases constitute a large number of serious and complex illnesses which are posing a major problem for the mankind. Nanotechnology in medicine makes use of nano materials and nanoelectronic biosensors. Nanomedicine is defined as the medical application of nanotechnology. Nanotechnology in medicine may be helpful in early detection and

prevention, improved diagnosis, proper treatment and follow-up of diseases. Gene sequencing has become more efficient with the invention of nano devices like gold nano particles, these gold particles when tagged with short segments of DNA can be used for detection of genetic sequence in a sample. With the help of nanotechnology, damaged tissue can be reproduced or repaired. These so called artificially stimulated cells are used in tissue engineering that may revolutionize the transplantation of organs or artificial implants. Biosensors with advanced features can be developed with the help of Carbon nano tubes. This technology may be used to develop sensors for cancer diagnostics. Though CNT is inert, it can be functionalized at the tip with a probe molecule. Nanotechnology has made excellent contribution in the field of stem cell research. Nanodevices can be used in stemcell research in tracking and imaging them. Stem cells can be modulated by mixing of nanocarriers with biological molecules [2].

Nanotechnology in Drug Delivery:

Drug delivery is one of the important considerations in drug development and the therapeutics. New technologies are applied for constructing innovative formulations and delivering them. The focus is on targeted drug delivery. This is important for delivery of biopharmaceuticals and treatment of diseases such as cancer and neurological disorders. In the pharmaceutical industry, there is potential to provide new formulations and routes of drug delivery [3].

The suitability of nanoparticles for use in drug delivery depends on a variety of characteristics, including size and porosity. Nanoparticles can be used to deliver drugs to patients through various routes of delivery. Nanoparticles are important for delivering drugs intravenously so that they can pass safely through the body's smallest blood vessels, for increasing the surface area of a drug so that it will dissolve more rapidly, and for delivering drugs via inhalation. Porosity is important for entrapping gases in nanoparticles, for controlling the release rate of the drug and for targeting drugs to specific regions.

It is difficult to create sustained-release formulations for many hydrophobic drugs because they release too slowly from the nanoparticles used to deliver the drug, diminishing the efficacy of the delivery system. Modifying water uptake into the nanoparticles can speed the release, while retaining the desired sustained-release profile of these drugs. Water uptake into nanoparticles can be modified by adjusting the porosity of the nanoparticles during manufacturing and by choosing from a wide variety of materials to include in the shell. Potential nano drugs will work by very specific and well-understood mechanisms; one of the major impacts of nanotechnology and nanoscience will be in leading development of completely new drugs with more useful behavior and less side effects.

Nano medicines used for drug delivery, are made up of nano scale particles which can improve drug bioavailability. For maximizing bio availability both at specific places in the body and over a period of time, molecular targeting is done by nanoengineered devices such as nanorobots. The molecules are targeted and delivering of drugs is done with cell precision. *In vivo* imaging is another area where Nanotools and devices are being developed for *in vivo* imaging. Using nano particle images such as in ultrasound and MRI, nanoparticles are used as contrast. The nanoengineered materials are being developed for effectively treating illnesses

and diseases such as cancer. With the advancement of nanotechnology, self-assembled biocompatible nanodevices can be created which will detect the cancerous cells and automatically evaluate the disease, will cure and prepare reports. The pharmacological and therapeutic properties of drugs can be improved by proper designing of drug delivery systems, by use of lipid and polymer based nano particles. The strength of drug delivery systems is their ability to alter the pharmacokinetics and bio-distribution of the drug. Nanoparticles are designed to avoid the body's defense mechanisms can be used to improve drug delivery. New, complex drug delivery mechanisms are being developed, which can get drugs through cell membranes and into cell cytoplasm, thereby increasing efficiency. Triggered response is one way for drug molecules to be used more efficiently. Drugs that are placed in the body can activate only on receiving a particular signal. A drug with poor solubility will be replaced by a drug delivery system, having improved solubility due to presence of both hydrophilic and hydrophobic environments. Tissue damage by drug can be prevented with drug delivery, by regulated drug release. With drug delivery systems larger clearance of drug from body can be reduced by altering the pharmacokinetics of the drug. Potential nanodrugs will work by very specific and well-understood mechanisms; one of the major impacts of nanotechnology and nanoscience will be in leading development of completely new drugs with more useful behavior and less side effects. Thus, nanoparticles can be used successfully and efficiently for targeting and distribution, further research can be done on nano toxicity so that its medical uses can be increased and improved [4].

Nanotechnology in the treatment of Cancer :

To fight against cancer, nanotechnology is regarded as one of the most promising breakthroughs in medication supply. Nanoparticles are likely to transport standard cancer medicines to tumours with fewer adverse effects and allow for the targeted killing of cancer cells. With nanoparticles permitted chemotherapeutic administration, more cytotoxic medicine may be delivered at the targeted location, and the toxicity is decreased.

On the other hand, nanoparticles can be of great use in imaging. Nano particles, such as quantum dots, with quantum confinement properties, such as size-tunable light emission, can be used in conjunction with magnetic resonance imaging, to produce exceptional images of tumor sites. As compared to organic dyes, nano particles are much brighter and need one light source for excitation. Thus the use of fluorescent quantum dots could produce a higher contrast image and at a lower cost than organic dyes used as contrast media. But quantum dots are usually made of quite toxic elements.

Nanoshells of 120 nm diameter, coated with gold were used to kill cancer tumors in mice by Prof. Jennifer at Rice University. These nanoshells are targeted to bond to cancerous cells by conjugating antibodies or peptides to the nano shell surface. Area of the tumor is irradiated with an infrared laser, which heats the gold sufficiently and kills the cancer cells[5].

Cadmiumselenide nanoparticles in the form of quantum dots are used in detection of cancer tumors because when exposed to ultraviolet light, they glow. The surgeon injects these quantum dots into cancer tumors and can see the glowing tumor, thus the tumor can

easily be removed.

Nano particles are used in cancer photodynamic therapy, where in the particle is inserted within the tumor in the body and is illuminated with photolight from the outside. The particle absorbs light and if it is of metal, it will get heated due to energy from the light. High energy oxygen molecules are produced due to light which chemically react with and destroy tumors cell, without reacting with other bodycells. Photodynamic therapy has gained importance as a non-invasive technique for dealing with tumors.

The applications of various nanotechnology in cancer therapy may be summarized as:

- Carbon nano tubes, 0.5-3 nm in diameter and 20-1000 nm length, are used for detection of DNA mutation and for detection of disease protein biomarker.
- Dendrimers, less than 10 nm in size are useful for controlled release drug delivery, and as image contrast agents.
- Nanocrystals, of 2-9.5 nm size cause improved formulation for poorly-soluble drugs, labeling of breast cancer marker $_{\text{Her2}}$ surface of cancer cells.
- Nano particles are of 10-1000 nm size and are used in MRI and ultrasound image contrast agents and for targeted drug delivery, as permeation enhancers and as reporters of a poptosis, angiogenesis.
- Nanoshells find application in tumor-specific imaging, deep tissue thermalablation.
- Nanowires are useful for disease protein bio-marker detection, DNA mutation detection and for geneex pression detection.
- Quantum dots, 2-9.5 nm in size, can help in optical detection of genes and proteins in animal models and cell assays, tumor and lymphnode visualization[1].

Nanotechnology for Study of the Nervous System :

Diseases of the nervous system are an important part of medicine. Inspite of all the advances in neurology, particularly in the last decade of the twentieth century, there are serious deficiencies in our understanding of the path mechanism of several neurological disorders as well as our ability to diagnose and treat these disorders. Nanobiotechnology will have an impact on improving our understanding of the nervous system and developing new treatment for disorders of the nervous system. Electrical recording from spinal cord vascular capillary bed has been achieved demonstrating that the intravascular space may be utilized as a means to address brain activity without violating the brain parenchyma. Working with platinum nanowires and using blood vessels as conduits to guide the wires, researchers have successfully detected the activity of individual neurons lying adjacent to the blood vessels. This can provide an understanding of the brain at the neuron-to-neuron interaction level with nonintrusive, biocompatible and biodegradable nanoprobes. This technique may one day enable monitoring of individual brain cells and perhaps provide new treatments for neurological diseases. Because the nanowires can deliver electrical impulses as well as receive them, the technique has potential as a treatment for Parkinson's disease. It has already been shown that

patients with Parkinson's disease can experience significant improvement from direct stimulation of the affected area of the brain. But the stimulation is currently carried out by inserting wires through the skull and into the brain, a process that can cause damage to the brain tissue. By stimulating the brain with nanowires threaded through blood vessels, patients can receive benefits of the treatment without the damaging side effects. The challenge is to precisely guide the nanowire to a predetermined spot through the thousands of branches in the brain's vascular system. One solution is to replace the platinum nanowires with new conducting polymer nanowires. Not only do the polymers conduct electrical impulses, they change shape in response to electrical fields, which would allow the researchers to steer the nanowires through the brain's circulatory system. Polymer nanowires have the added benefit of being 20-30 times smaller than the platinum ones used in the reported laboratory experiments. They are biodegradable and therefore suitable for short-term brain implants[3].

Nanotechnology in operative dentistry:

Nanotechnology aims at the creation and utilization of materials and devices at the atomic, and molecular level, supra molecular structures, and in the exploitation of unique properties of particles of size 0.1 nm to 100 nm. Nano filled composite resin materials are believed to offer excellent wear resistance, strength, and ultimate aesthetics due to their exceptional polishability and luster retention. In operative dentistry, nanofillers constitute spherical silicon dioxide (SiO_2) particles with an average size of 5-40 nm. The real innovation about nano fillers is the possibility of improving the load of inorganic phase. The effect of this high filler load is widely recorded in terms of mechanical properties. Micro hybrid composites with additional load of Nano fillers are the best choice in operative dentistry. It is expected that in near future, it would be possible to use a filler material in operative dentistry, whose shape and composition would closely mimic the optical and mechanical characteristics of the natural hard tissues [6].

Nanotechnology in Ophthalmology:

Some applications of nanotechnology to ophthalmology include treatment of oxidative stress; measurement of intraocular pressure; theragnostic; use of nanoparticles for treatment of choroidal new vessels, to prevent scars after glaucoma surgery, and for treatment of retinal degenerative disease using gene therapy; prosthetics; and regenerative nano medicine. Treatments for ophthalmic diseases are expected by applications of nanotechnology. A novel nanoscale dispersed eye ointment (NDEO) for the treatment of severe evaporative dry eye has been successfully developed [7]. The excipients used as semisolid lipids were petrolatum and lanolin, as used in conventional eye ointment, which were coupled with medium-chain triglycerides (MCT) as a liquid lipid; both phases were then dispersed in polyvinyl pyrrolidone solution to form nanodispersion. A transmission electronmicrograph showed that the ointment matrix was entrapped in the nanoemulsion of MCT, with a mean particle size of about 100nm. The optimized formulation of NDEO was stable when stored for six months at 4°C, and demonstrated no cytotoxicity to human corneal epithelial cells when compared

with commercial polymer-based artificial tears[1].

Nano technology in Tissue Engineering:

Tissue engineering is an interdisciplinary field, which applies the principles of engineering and the life sciences to the development of biological substitutes that restore, maintain, or improve tissue function. Tissue engineering is an emerging field between traditional medical devices and regular pharmaceuticals. It faces many challenges and is also a field that is extremely interdisciplinary requiring the efforts of physicians, cell biologists, material scientists, chemical engineers, and chemists. Apart from the use of nanoparticles for diagnostic and therapeutic purposes, nanotechnology has applications for the development of tissue engineering as indicated by some of the studies in life sciences. Potential areas for application of nanobiotechnology in regenerative medicine include CNS injury, neurogenerative disorders, diabetes, cartilage injuries, and bone fractures.

The response of cell motility and metabolism to changes in substrates has been thoroughly studied in the past decade. Size, structure, geometry, integrin-binding, and other factors have all been investigated. Various techniques have been employed to create micro patterned surfaces of different materials to study cell behavior. In the presence of patterned stripes of bovine serum albumin and laminin, Schwann cells aggregate preferentially on the laminin regions. Osteogenic cells have been cultured in 3D nanohydroxyapatite/collagen matrix, which is precipitated in such a manner that hydroxyapatite crystals are uniformly distributed in a matrix of collagen, seemingly ideal for bone construction.

Microfluidic devices enable the study of methods for patterning cells, to topographical control over cells and tissues, and bioreactors. They have not been used extensively in tissue engineering but major contributions are expected in two areas. The first is growth of complex tissue, where microfluidic structures ensure a steady blood supply, thereby circumventing the well-known problem of providing larger tissue structures with a continuous flow of oxygen as well as nutrition and removal of waste products. The second, and probably more important function of microfluidics, combined with micro/nanotechnology, lies in the development of in vitro physiological systems for studying fundamental biological phenomena[3].

Nanotechnology in Vaccine Delivery

Nanoparticles are used in DNA Vaccines. Nanoparticles are promising as delivery systems for DNA vaccines. Nanoemulsions or nanoparticle aerosol vaccines are also under development. Bacterial spores can be used for vaccine delivery. The spore coat can act as a vehicle for heterologous antigen presentation and protective immunization. Use of bacterial spores will solve the problem of stability and transport in the developing countries as they do not require refrigeration and the administration is needle free. Proteosomes can be used as Vaccine Delivery Vehicles. Components of the immune system recognize particles more efficiently than soluble proteins. Proteosomes serve as vaccine delivery vehicles by virtue of their nanoparticulate nature, forming vesicles and vesicle clusters comparable to the size of small viruses. Proteosome vaccine vesicles and vesicle clusters may range in size from 20 to 800 nm, depending on the type and amount of antigen formulated with the Proteosomes.

The hydrophobic nature of the Proteosome porin proteins also contributes to vaccine delivery capabilities by facilitating interaction of the vaccine particles with, and uptake of the vaccine by, cells that initiate immune responses. The fact that Proteosomes are effective nasal vaccines is considered to be particularly related to this enhanced recognition and uptake afforded by Proteosomes' particulate and hydrophobic nature. This technology is being applied to develop vaccines for influenza, allergy, plague and respiratory syncytial virus[3].

Ethical, Legal, and Social Implications of Nanomedicine:

Nanotechnology's impact in the past was mostly in engineering, communications, electronics, and consumer products. Now that nanotechnology is being applied to human medicine, there is awareness of the consequences, good or bad, that are still not well understood. Communication with the public is important at this stage. This will involve education of the public and consideration of their opinion. Legal aspects of nanobiotechnology in healthcare are discussed in a separate article [8]. Ethical aspects are important for all new technologies and nanobiotechnology is no exception. Although nanotechnology has not raised any new ethical issues, it is worth while to keep this consideration in mind while developing and applying nanobiotechnologies to medicine. On the legal front, experts do not believe that structures set up specifically to deal with nanomedicine are needed right now. However, they suggest monitoring existing regulatory systems to ensure they address all nano medical products.

Conclusion :

Nanotechnology has revolutionized the health care management. Nanotechnology helps resolve the problem of focused treatment administration, reduces the danger of side effects and maximizes therapeutic effectiveness. Nano medicine is the most promising use of nanorobotics. Its applications cover several fields, such as the creation of vaccines, medication delivery, wearable equipment, diagnostic and imaging equipment, and antimicrobial products. It is possible to develop tailored mechanisms for medication administration, new diagnostic methods, and nanoscale medical devices. In future nanotechnology will play an important role in the development of personalized medicine. Personalized medicine simply means the prescription of specific therapeutics best suited for an individual. It is usually based on pharmacogenetic, pharmacogenomic, and pharmacoproteomic information but other individual variations in patients are also taken into consideration. Apart from refinements in molecular diagnostics, an important basic of personalized medicine, nanobiotechnology also helps in the discovery of biomarkers that are crucial for the development of personalized medicine. A good example of application of nanobiotechnology for personalized medicine is that of cancer. In case of cancer the variation in behavior of cancer of the same histological type from one patient to another is also taken into consideration. Personalization of cancer therapies is based on a better understanding of the disease at the molecular level and nanotechnology will play an important role in this area.

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Applications of Nanotechnology in Waste water Management: A Review

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Abstract:

In this comprehensive review, we focused on the application of nanomaterials in waste water treatment. The availability of water is crucial for every living creature since it is a vital component of life. Conversely, there is a serious drinking water crisis facing every corner of the globe. Water pollution is caused by a variety of gases, microbes, and other pollutants (chemicals and heavy metals) that are introduced into the water by rain, flowing water, etc. The numerous applications of nanomaterials in the elimination of diverse pollutants from contaminated water are discussed in this review article. Many types of nanomaterials have the potential to be extremely successful in treating contaminated water that contains metal toxins and other organic and inorganic contaminants. This is because of their special qualities, such as their larger surface area and ability to function at low concentrations. Technologies based on nanostructured catalytic membranes, nanosorbents, and nanophotocatalysts to remove pollutants from wastewater are eco-friendly and efficient, but they demand more energy and cost for purifying the wastewater. To be commercially viable, new, advanced wastewater treatment technology needs to be versatile, affordable, and effective.

Keywords: Nanocatalysts, nanomembranes, nanosorbents, nanomaterial applications, waste water treatment.

Introduction :

Water is an essential resource on Earth, and its availability in its purest form is critical for humans and other living species, as the notion of existence is unbelievable without it. Because of its potential qualities, such as its solubility power, water is also known as the universal solvent. Water contamination is currently the biggest challenge facing the entire world. There are several causes of this problem, including poor sewage treatment, industrial waste, problems with marine dumping, radioactive waste, certain agricultural practices, etc. [1, 2]. In addition to its detrimental effects on the ecosystem, water pollution can also cause air pollution, which poses a serious risk to human health. Water pollution also carries a

negative effect on the economic development and social perspectives of the concerned societies and countries. According to a recent UN assessment, the availability of fresh water and purified water is a global concern that will grow more difficult for everyone to access in the twenty-first century since contaminated water makes it unsafe for living things to survive. [3, 4]. When undesirable elements enter reservoirs or other bodies of water, the water becomes contaminated and unfit for human consumption. There are numerous chemical, physical, and mechanical ways of dealing with this growing issue. Additionally, scientists are still investigating many novel technologies to enhance low-cost water filtration techniques [5, 6]. Recently, the field of developing nanotechnology has offered the possibility of a low-cost, highly effective method of purifying water by removing impurities [7]. Nanomaterials have been effectively used in a number of fields in the past, including catalysis and medical science.

Experts discovered that nanomaterials, with their unique properties such as nanosize, large surface area, high reactivity, strong solution mobility [8], strong mechanical properties, porosity characteristics, hydrophilicity, dispersibility, and hydrophobicity [9], are a better option for treating wastewater when the world is facing serious drinking water issues. Various nanomaterials have been shown to be successful in eliminating organic and inorganic contaminants, harmful microorganisms, and heavy metals such as Pb, Mo, etc. [10]. The World Health Organization (WHO) currently reports that four billion cases of various health conditions are recorded annually owing to waterborne infections, and over 1.7 million deaths are attributed to water pollution. The various types of water contaminants are shown in **Table 1**, along with their causes and harmful consequences [11]. Recent advances in nanomaterials have included the development of nanophotocatalysts, nanomotors, nanomembranes, and nanosorbents (which have a high adsorption capacity and are useful for a variety of water treatment techniques). Certain imprinted polymers have also proven to be useful in the treatment of contaminated water. To summarize, it is thought that research on the use of nanomaterials in water filtration can provide useful insights. Consequently, in this review article, we have summed up the function that nanomaterials play in wastewater treatment to mitigate water crises. Although nanoengineered materials offer a major and potentially useful water treatment strategy that is easily adaptable, there are still certain flaws that require additional attention and are specifically outlined in this article. Additionally, we discussed the restrictions, benefits, drawbacks, and prospects for these nanomaterials in the future.

Table 1: Indicates different water pollutants with theirs sources and adverse effects.

Water Pollutants	Source of Pollutants	Effect of Pollutants
Pathogens	Viruses and bacteria	Water borne diseases
Agricultural pollutants	Agricultural chemicals	Directly affect the fresh water
Sediments and suspended solids	Land cultivation, demolition, and mining operations	Damaging fish spawning, affecting the aquatic environment of insects and fish

Water Pollutants	Source of Pollutants	Effect of Pollutants
Inorganic pollutants	Metal compounds, trace elements, inorganic salts, heavy metals, and mineral acids	Aquatic flora and fauna, public health problems
Organic pollutant	Detergents, insecticides, and herbicides	Aquatic life problems, carcinogenic
Industrial pollutants	Municipally polluted water	Caused water and air pollution
Radioactive pollutants	Different isotopes	Effects on bones, teeth, and skin
Nutrient pollutants	Plant debris, fertilizers	Effect on the Eutrophication Process
Macroscopic pollutants	Marine debris	Plastic pollution
Sewage and contaminated water	Domestic waste water	Waterborne diseases

Water Treatment Methodologies:

Nanophotocatalysts:

The word “photocatalysis” is composed of the Greek words “photo” and “catalysis,” which mean decomposition of compounds in the presence of light [12]. Photocatalysis can be used to describe a procedure that uses light (UV, visible light, or sunlight) to activate or stimulate the compounds. During the process of chemical transformation, a photocatalyst can alter the pace of the reaction without being directly involved. Additionally, the key difference between a photocatalyst and a conventional thermal catalyst is that the latter is activated by light energy in the form of photons, whereas the former is activated by heat [13]. Nanophotocatalysts are frequently employed in wastewater purification processes because of their increased surface ratio and shape-dependent properties, which contribute to the catalyst’s increased reactivity. Because of their unique surface characteristics and quantum effects, materials based on nanoscales react differently from materials based on bulk dimensions. Additionally, it helps to improve their optical, mechanical, magnetic, and chemical reactivity. Research has demonstrated that nanophotocatalysts can increase oxidation capacity by efficiently producing oxidizing species on the material’s surface, which aids in the efficient breakdown of contaminants from contaminated water [14]. The majority of the applications for nanoparticles, such as semiconductors, zero-valence-based metals, and some bimetallic types, are in the remediation of environmental contaminants, such as nitroaromatics, organochlorine insecticides, azo dyes, and chlorpyrifos [15]. Additionally, a number of studies have demonstrated in the literature that TiO_2 -based nanotubes can be used to remove organic pollutants from waste water, including azo dyes, Congo red, phenol aromatic base pollutants, toluene, dichlorophenol trichlorobenzene, chlorinated ethene, and others. However, SiO_2 , ZnO , TiO_2 , Al_2O_3 , and other metal oxide nanophotocatalysts are the most prevalent and important ones. Titanium dioxide (TiO_2) is an outstanding photocatalyst from all available

materials due to various factors such as its inexpensive cost, toxic-free property, chemical stability, and easy availability on Earth. In addition, TiO_2 is found in three natural states: rutile, brookite, and anatase. Anatase is currently regarded as an excellent nanophotocatalyst material [16]. This state can absorb ultraviolet light (below 387 nm) and has a bandgap of 3.2 eV. But other photocatalysts, such as ZnO, have also been developed to remove impurities from wastewater and have an effective recyclable nature. When considering composite nanomaterials, it was also tested to see how well the CdS/TiO₂ composite worked as a catalyst to degrade the reference chemical, dimethyl sulfoxide, in order to assess the photocatalytic performance of water treatment. The ferromagnetism of the iron-doped nanoparticles makes recycling and reuse easier. Similar to this, Pd-incorporated ZnO nanomaterials have been employed to remove *Escherichia coli* from wastewater because of certain qualities, including, strong photocatalytic reactivity [17]. Recent efforts have focused on modifying metal oxides with other elements, such as metals or metal ions, carbonaceous-based materials, dye sensitizers, and many others, to improve their photocatalytic performance under visible light irradiation. Moreover, the process of nanophotocatalysis can happen in two different ways: homogeneously or heterogeneously. Due to its vast range of uses in water cleansing and environmental-related applications, heterogeneous nanophotocatalysis is the most intensively investigated state in the modern era. A solid photocatalyst (such as a metal or semiconductor) and a fluid (both reactants and reaction products) must first form an interface in the case of heterogeneous photocatalysis [18]. When light-based semiconductor photocatalysts are utilized in conjunction with gaseous or liquid phases, the term “heterogeneous photocatalysis” is typically used. Applications based on heterogeneous photocatalysis heavily rely on scaled-up reactors built on advanced, more efficient designs. Primarily, reactor design involves optimizing mass transfer and effectively illuminating the nanocatalyst, especially in liquid-phase applications. Although light-emitting diodes and optical fibers can enhance photon transport, significant advancements in this area are currently absent. Furthermore, a great deal of work has been put into developing solar photo reactors. In terms of minimizing the issue at the business level, it is still not ideal. Furthermore, the current lack of significant commercial applications is due to a lack of effective photoreactor designs and the photocatalytic competence of photocatalysts. In spite of this, heterogeneous nanophotocatalysts offer intriguing benefits such as minimal chemical consumption, lack of additives, capacity to function at lower concentrations, and chemical stability (e.g., TiO₂ stability in aqueous medium) [19]. As a result, heterogeneous photocatalysis has recently achieved pre-industrial scale.

Advantages and Disadvantages of Nanophotocatalyst:

The mineralization of hazardous organic compounds at 25°C has been shown to be greatly aided by nanophotocatalysis, which has been shown to be a very successful and efficient means of detoxifying water. Additionally, the majority of nanophotocatalysts exhibit favorable characteristics, including reduced toxicity, affordability, chemical stability, ease of use, and superior photo active qualities at the nanoscale, or within the range of 1-100 nm [20]. In general, nanophotocatalysts provide a number of benefits, including low cost, eco-

friendliness, and stability (both chemical and physical). TiO_2 has strong photostability, but a lot of nanophotocatalysts show very low chemical stability because of photocorrosion, including zinc oxide, metal sulfide materials, copper-based materials, and many more. Photocatalysts undergo oxidation or reduction, depending on the materials they are made of. This changes their oxidation state by producing holes or electrons, which causes the catalysts to break down and lose efficiency. Consequently, the synthesis of a nanocomposite is necessary to attain stable photocatalysts for extended periods of operation.

The primary benefit of nanotechnology is the quantum-size effect, which increases the energy bandgap while decreasing particle size. Additionally, photodegradation is a technique that has many benefits, including full degradation, reusability, and low cost. Despite all of the advancements, there are still problems with nanophotocatalysts, such as toxicity and catalyst recovery from mixtures. The potential and uses of nanophotocatalysts at a higher level are restricted by these kinds of problems [21]. Therefore, in order to lessen toxicity while utilizing nanocomposite materials in the water treatment process, scientists are currently concentrating on additional nanocomposite materials. Therefore, it is recommended that researchers develop novel photocatalysts that can function in visible ranges for long-term results and encourage the doping of photocatalysts with various materials, such as graphene and its derivatives, to lessen toxicity. One important strategy, the employment of magnetic nanophotocatalysts in wastewater treatment, could be utilized to overcome the catalyst recovery drawback. When using magnetic nanophotocatalysts, catalyst recovery is possible via external magnetic fields. This allows for multiple nanocatalyst recycling processes and more efficient and naturally responsive water purification processes. **Figure 1** illustrates some of the novel uses for nanophotocatalysts. In addition, the method of using nanophotocatalysis to remove contaminants from water has been described as being highly successful.

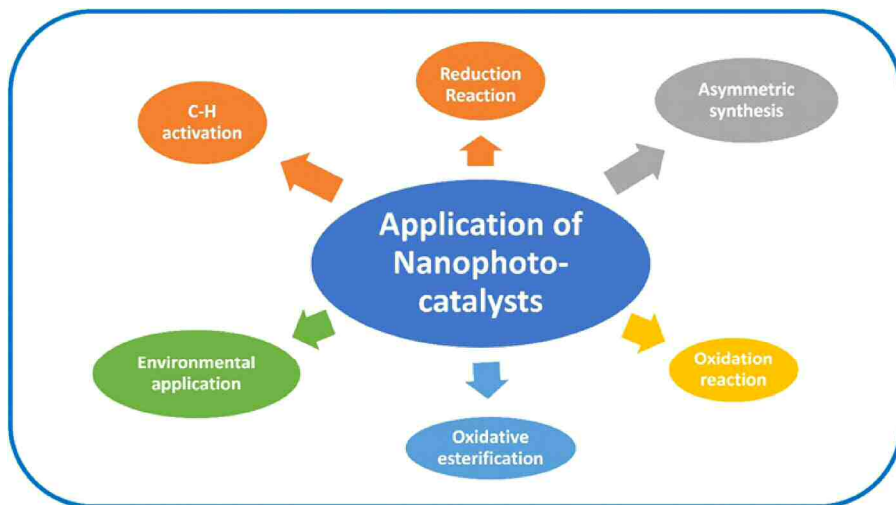


Figure 1: Applications of Nanophotocatalysts

Future Perspectives on Nanophotocatalysts:

In the subject of nanophotocatalysis, extensive research on nanomaterials is now underway, which has resulted in certain advancements in reactor design and developments pertaining to alterations in nanophotocatalysts. Despite the fact that nanophotocatalytic materials have advanced significantly, there are still some crucial questions about their properties that need to be answered. Limitations on mass transfer and increased photon consumption are the main obstacles to intensification [22]. By merging nanocomposites with nanophotocatalytic reactor structures, the concept of nanocomposites can be extended in its ability to solve the electron pair recombination problem. Microfluidic reactors are new, cutting-edge reactors that provide new opportunities for in-depth analysis of reaction and synthesis phase features. Reactors that operate with reactants on a microscale are known as microfluidic reactors. Microreactors are more suitable as a material than traditional reactors because of their huge surface-to-volume ratio, enhanced diffusion effect, high mass transfer coefficient factor, extremely stable hydrodynamics, and reduced Reynolds flow. Nevertheless, the large-scale application of photocatalysis to real wastewater remains challenging. Additionally, the production of important structures with enhanced functional and structural features, such as nanorods, nanospheres, nanoflowers, nanoflakes, and nanocones, could open up a new area of research. Measured synthesis techniques have the ability to yield multiple architectures of nanomaterials with desirable characteristics. New photocatalysts should be created in order to further investigate this area of study [19]. It is imperative to synthesize innovative nanophotocatalysts with great stability, affordability, and exceptional efficiency. Additionally, in order to fully utilize pollutant treatment, a number of strategies, including adsorption, photocatalysis, electrocatalysis, and other thermodynamic processes, should be combined in a rational manner. In the early decades, there was a lot of research done on the creation of nanocomposites with ZnO and TiO₂ for the treatment of water contaminants. The process of creating nanocomposites with carbonaceous, polymeric, and ceramic components is still in its early stages. It can produce perfect nanocomposites with enhanced characteristics. A number of primary technical issues that require careful investigation stand in the way of the use of heterogeneous photocatalysis for wastewater remediation. With prompt evaluation, it is possible to understand a substantial photocatalytic treatment that has superior solar-driven effectiveness.

Nanomembranes:

In order to remove undesirable nanoparticles from the aqueous phase, a special kind of membrane called a nanomembrane is created using various nanofibers. Reverse osmosis can be performed at a very fast elimination speed with a condensed fouling tendency by using this approach, which also functions as a pre-treatment step [23]. Numerous studies on membrane nanotechnology have been published with the goal of creating multifunctional membranes using various nanomaterial substances in membranes based on polymers. Reverse osmosis, ultrafiltration, nanofiltration, and other processes were carried out by water-porous membranes. A composite layer with a porous support is present in the membrane. Carbon-based material [graphene oxide/carbon nanotube (CNT)] distributed into a polymer matrix

is typically the most significant composite layer in practice. This offers noteworthy and encouraging advancements in water transport and fouling resistance. For instance, CNTs have antimicrobial qualities that can lessen biofilm formation, fouling, and the likelihood of mechanical failures [24]. When nanomaterials (such as zeolite, alumina, TiO_2 , etc.) are doped into polymers, ultrafiltration membranes with hydrophilicity and fouling resistance are formed on the membrane's surface. In order to create a polymeric membrane that inhibits the formation of biofilms on the membrane's surface and prevents bacterial attachment, antimicrobial materials such as silver metal particles are additionally doped with a polymer [25]. In addition to its ability to reduce membrane biofouling, it also functions to inactivate viruses. Through doping in surface modification, the creation and expansion of nanomaterial thin films are integrated into active thin film composites.

In nanomembranes, there are often two main problems: membrane fouling and clogging, which must be resolved. The prevention of fouling and clogging may be verified by adding super hydrophilic nanoparticles to a thin-film nanocomposite membrane. Additionally, because they have higher hydrophilicity, higher porosity, better fouling resistance, and a more homogenous nanopore, metal oxide nanoparticles (Al_2O_3 , TiO_2), antimicrobial nanoparticles (nano silver, CNTs), and aquaporin-based membranes are very helpful materials to overcome the membrane clogging and fouling issue. Typically, the kind, amount, and size of nanoparticles determine how they influence a membrane's permeability and selectivity. Furthermore, a variety of biological membranes with great permeability and selectivity are found [26]. Because of their many advantageous qualities, including high homogeneity, optimization, fast reaction times, ease of handling, and high order of response, nanomembranes are also utilized in wastewater treatment. To prepare the nanocomposite membrane for the breakdown of organic contaminants, certain nanophotocatalysts can be added. TiO_2 -incorporated nanomembranes and films, for instance, can be utilized to decompose organic contaminants and deactivate certain microorganisms. Advances in nanotechnology, particularly in the field of nanomaterials, have led to the creation of a number of nanostructured catalytic membranes with unique characteristics, such as increased foul resistance, elevated selectivity, and a rapid disintegration rate [27]. Numerous methods were employed to create these kinds of nanomaterials with multifunctional properties through synthesis. The use of nanostructured catalytic materials in pore membranes, such as enzymatic and iron-catalyzed free radical reactions, illustrated a positive advancement in this technology. As a result, oxidative processes can be used to purify water and remove pollutants without the need for harmful chemicals. In order to demonstrate the effectiveness of nanostructured catalytic membranes in industrial and commercial settings, a hydrogen peroxide reaction involving immobilized membrane nanoparticles (ferrihydrite and iron oxide) was conducted in order to produce free radical ions that could be used to remove organic pollutants based on chlorination from actual groundwater. In numerous additional environmental applications, the development of nanostructured materials is still highly beneficial [28]. Numerous studies on the immobilization of metallic nanoparticles on membranes (chitosan, cellulose acetate, polysulfones, etc.) for the purpose of dichlorination and the degradation of a toxic substance with novel properties like surface reduction, high reactivity, absence of agglomeration, and nanoparticle hindrance

have been reported. Pd nanoparticles have a novel sort of interaction that helps to increase the effectiveness of water treatment. Palladium acetate and polyetherimide are both employed to make nanocomposite films. Under several circumstances, in situ and ex situ techniques are also utilized to create nanomaterials inside the matrix with precursor film. This presents numerous chances to create nanomaterials with adjustable characteristics. Furthermore, numerous noteworthy catalytic membranes with excellent permeability, great resistance to fouling, and high selectivity have been made possible by advancements in nanotechnology. In the field of wastewater treatment, contemporary techniques include hybrid and bottom-up approaches to strengthen its multi-functionality features [29].

Advantages and Disadvantages of Nanomembrane:

When it comes to drinking water, the primary goal of membrane technology is to remove harmful particles from the water. The water safety level was also measured using the nanomembrane filters. The advantages of using a nanomembrane over more conventional filtration techniques are that with traditional filtration techniques, calcium and magnesium needed an additional ion to balance out the process, so Na^+ ions are typically used as an exchanger; with nanomembranes, this is not necessary [30]. The constraints of nanomembranes typically result in a reduction in their efficiency when compared to other traditional methods. The first is nanomembrane fouling, which develops after a few membranes are used. It is the main problem that makes this strategy more costly and ineffective. The occurrence of fouling issues in nanomembranes is solely dependent on the working conditions. In certain cases, inappropriate working conditions, such as excessive temperature, pressure, or optimization, are also thought to be the cause of membrane fouling. Membrane stability is the second significant restriction. The stability of the nanomembrane could not be maintained for very long. Eventually, stability starts to lose effectiveness and stops producing the same high-quality results as before. Therefore, in order to get great outcomes, the nanomembrane will need to be changed. Nevertheless, this will lead to a number of other problems, including high costs and the possibility of impurities throughout the process [31]. The fundamental chemical resistance that is used for material cleaning determines its stability. Less reliability, a sluggish operation procedure, less selectivity, expensive maintenance, and a decline in working efficiency over time are the drawbacks. The lack of thorough research is due to a few prevalent drawbacks. **Table 2** provides a summary of several types of nanomembranes, their benefits, drawbacks, and uses [32].

Table2. Advantages, disadvantages, and applications of nanomembranes

Different types of nanomembrane	Advantages of nanomembrane	Disadvantages of nanomembrane	Application
Nanofiber membranes	Excellent porosity, tailor-made, bactericidal, good permeate efficiency,	Pore blocking, conceivably discharge of nanofibers	Ultrafiltration, prefiltration, filter cartridges, water handling, and separate filtration devices

Different types of nanomembrane	Advantages of nanomembrane	Disadvantages of nanomembrane	Application
Nanocomposite membranes	High hydrophilicity, better water permeability, high fouling resistance, and good thermal and mechanical stability	Resistant substances used to discharge nanoparticles	Entirely dependent on composites
Aquaporin-based membranes	Improved ionic selectivity and better permeability	Poor mechanical stability	Ultrafiltration, less pressure, and salination
Self-assembling membrane	Homogeneous nanopore membranes	Laboratory-scale availability	Ultrafiltration
Nanofiltration membranes	Charge-based repulsion, comparatively less pressure, and better selectivity	Membrane blocking	Color, reduction of hardness, odor

Future Perspectives on Nanomembranes :

By employing size exclusion and diffusion in solution, nanomembranes remove organic, inorganic, and virus-based contaminants from water supplies. While the effectiveness of these nanomembranes has been demonstrated in research laboratories, there remains a hurdle in scaling them up to lower costs and making them suitable for industrial use. The commercialization goal requires a concerted effort by manufacturing firms and research institutes to overcome this obstacle to productive upscaling. To prevent membrane fouling, it is necessary to first increase the selectivity of the nanomembrane and then enhance its resistivity. Polymers based on surface grafting, such as zwitterionic, could be a promising option for creating next-generation membranes. However, fouling resistance to the inner wall pore would not be reported by surface grafting [33]. The next crucial step is the installation of additional suitable anti-fouling convertors into the membrane matrix. Enhancing polyamide membranes' high sensitivity to various oxidants, such as chlorine and ozone, is also crucial. In addition, greater emphasis must be paid to the manufacture of multifunctional membranes in order to improve innovation on an industrial scale. To produce more affordable and efficient nanomembranes for water treatment, a number of issues still need to be resolved.

Nanosorbents:

Wide-ranging characteristics of nanosorbents, such as their strong sorption capacity, make them more effective and suitable for water treatment [34]. Although these nanosorbents are quite uncommon in the marketplace, researchers and specialists are working hard to make them in greater quantities and on a commercial scale. Carbon-based nanosorbents, such as carbon black, graphite, and graphene oxide, are the most often described types. Moreover, polymeric and metal/oxide nanosorbents are known to exist. In order to lessen the

impact of toxicity during the wastewater treatment process, composites of various materials, such as Ag/polyaniline, Ag/carbon, C/TiO₂, etc., are extremely important. Depending on how they are synthesized, carbeneous materials like carbon nanotubes (CNTs) with a cylindrical form can appear as single-walled or multiwalled nanotubes. CNTs have quantifiable adsorption sites and sustained surfaces because of their large surface area. Because the surface of CNTs is hydrophobic, it needs to be stabilized to avoid aggregation that would reduce the number of surface-active sites. As a result, it is an effective material for adsorbing contaminants. In a similar vein, dendrimers and other polymeric nanoadsorbents work well to remove heavy metals and organic contaminants from wastewater [35]. For instance, the dendrimer-ultrafiltration system assisted in the reduction of copper ions. They merely underwent a pH shift to renew and exhibit biocompatibility, biodegradability, and a toxic-free atmosphere. Additionally, around 99% of dyes and other organic contaminants have been removed. Zeolites, which have an absorbent structure that allows several nanoparticles, such as copper and silver ions, to be implanted, are another significant type of nanosorbent [36]. Zeolites have the benefit of regulating the concentration of metals and acting as an antimicrobial agent. Additionally, magnetic nanosorbents are an essential component of water treatment and a special tool for eliminating various organic contaminants. Magnetic filtering is another method used to eliminate certain organic contaminants. Magnetic nanoparticle coating of ligands results in the synthesis of magnetic separation nanosorbents. Regenerating these nanosorbents has been described using a variety of techniques, including ion exchange, cleaning agents, magnetic forces, etc. These renewed nanosorbents have the potential to be more commercially viable and cost-effective. Carbon still possesses certain health concerns. Studies that have been published have shown that surface changes, chemical stabilizers, and nanoadsorbent shape all affect hazardous effects [37]. Therefore, it is imperative to focus on synthesizing more stable morphology (size and form) in order to address toxicity concerns and health dangers. Additionally, the excellent biodegradability, good biocompatibility, and nontoxicity of the bioadsorbents make them interchangeable with chemically generated nanosorbents. The scientific community recommends graphene oxide because it is a relatively new substance that may be used as a nanosorbent to remove pollutants, and because of its superior qualities, it can produce better results than other materials. **Table 3** lists a few known nanosorbents along with their purposes.

Table 3: Most commonly used nanosorbents and their functions

Different Nanosorbents	Treatment Function
Carbon-based nanosorbents	Nickel ions present in water
Graphite Oxide	Removal of dyes
Regenerable polymeric Nanosorbents	Organic pollutants and inorganic contaminants of waste water
Nano clay	Hydrocarbons, Dyes
Nano-metaloxides	Different heavy metals

Nano aerogels	To remove uranium from drinking water
Nanoiron oxides	To eliminate the hormones and toxic pharmaceuticals from water
Polymer fiber	To remove arsenic and other toxic metals

Advantages, Disadvantages, and Future Perspectives of Nanosorbents:

Since nanoparticles have such extraordinary physiochemical capabilities, there has been a lot of interest in their potential as sorbents in explaining ecological challenges like wastewater purification. These characteristics set them apart from more standard sorbents in a number of disciplines. An ideal sorbent would have a large surface area, an excellent rate of adsorption, and short adsorption times. Like nanosorbents, nanoparticles attracted attention due to their nanoscale, which allowed for a rapid and high rate of adsorption. Additionally, nanoparticles are nanosorbents that can be utilized as a separating medium in water decontamination to remove organic and inorganic-based pollutants from contaminated water [38]. A survey of the literature reveals that significant attempts have already been made to treat wastewater, with nanomaterials being utilized as sorbents to achieve effective outcomes. For the goal of commercializing nano-size sorbents for water decontamination, a number of obstacles must be fully overcome. These obstacles include production scalability, excellent adsorption measurements, selectivity, material stability, material operational duration, etc. Toxic ions and chemicals from wastewater must be controlled; hence, there is a great need for an active approach to wastewater treatment and the creation of novel nanosorbents [39]. The possibilities for the future are promising as the scientific community works to improve adsorption mechanisms and enhance understanding. These days, scientists, advisors, and government representatives are worried about the health hazards associated with wastewater pollution and are working hard to come up with a convincing case for why industrial-scale usage of nanomaterials is appropriate.

Conclusive Remarks and Future Perspectives

Our planet is unique and superior to other worlds because of water. However, there is enough clean water accessible around the globe to meet all current and anticipated water demands. Drinking water resources are insufficient in a number of places to meet domestic developmental, economic, and basic demands. The lack of fresh water in these places to meet sanitation and water needs for humans is undoubtedly detrimental to both human and other living beings health. The scientific community and research institutions need to figure out how to get rid of these restrictions. Furthermore, the world is having a difficult time accomplishing that because of a changing and unclear future climate, a rapidly growing population that is propelling urbanization and globalization, as well as expanded community and financial growth. It successfully tackles these obstacles, which require investigating every aspect of water management. Due to the terrible state of the water and the global need for fresh water, the use of nanomaterials for the treatment of water pollutants is growing quickly in the current day. Innovative, progressive techniques for water treatment are

desperately needed, especially to certify high-quality water for drinking by eliminating micro- and macro-contaminants and boost industrial production improvements with adaptable water treatment methods.

Nanotechnology has demonstrated remarkable success in managing water purification issues and is poised for further development. Although nanomaterial approaches such as nanosorbents and nanostructured catalytic membranes are very effective, time-efficient, energy-efficient, and environmentally friendly, they are not yet widely employed for commercial wastewater purification due to their high cost. Because of their fast rate of response, nanomaterials exhibit great efficiency. However, some shortcomings still need to be worked out. Additionally, appropriate recommendations to address this issue should be prepared by some research institutes and international research communities in order to limit the health risk [39]. But there is a lot of room for water revolutions using nanoengineered materials, especially when it comes to seriously biodegradable contaminants, point-of-use approaches, and decentralized water treatment technology. In addition, it is imperative to synthesize modified nanomaterials that are efficient, user-friendly, eco-friendly, and effective. The commercialization of these technologies for wastewater treatment and their associated costs must also be taken into consideration. The various uses of nanomaterials can offer a huge opportunity to supply drinking water to every country on the globe.

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Nanotechnology in Agriculture

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Abstract :

Nanotechnology, presents innovative solutions to pressing challenges such as diminishing factor productivity and environmental degradation associated with conventional agricultural practices. Nano-fertilizers and nano-pesticides, distinguished by their high surface area and reactivity, introduce a sophisticated “smart delivery system.” This system enhances precision targeting, diminishes reliance on agrochemical inputs, and augments nutrient use efficiency, thereby fostering sustainable agricultural practices. Nanotechnology also opens new avenues to identify and quantify different biomolecules by using nanoprobe, nanomembrane for irrigation water purification and nano soil moisture sensor. Beyond crop-centric applications, nanotechnology extends its impact to agri-allied sectors like animal husbandry and fisheries. As nanotechnology progressively integrates into agricultural frameworks, emphasis on strategic research and development, environmental stewardship, and inclusive stakeholder engagement becomes imperative for realizing its transformative potential. The present research in Agriculture using nano products in shows a harmonious coexistence of nanotechnology with agricultural and environmental integrity, delineating a promising trajectory toward a sustainable second green revolution. Despite the prevailing optimism, a judicious consideration of societal, environmental, and ethical ramifications is essential, necessitating vigilant scrutiny to mitigate potential violations on human rights, and adverse environmental consequences.

Keywords : Nanotechnology, Agriculture, Sustainability, Environmental stewardship, Smart delivery system

Introduction :

Nanotechnology has brought a revolution in agricultural science in the form of nano technology-aided products like nano-fertilizers, nano-pesticides (including insecticides, fungicides, bactericides, herbicides etc.), nano pest repellents [2], nanomagnets, nanomembranes [1-5], nano-based soil moisture monitoring [6] etc. All these products and services can increase the productivity (production/unit area) of crops and cropping systems

while maintaining the ecological balance [7]. It is seen that in recent years agricultural factor productivity has been declining [8] as a result of injudicious and imbalanced chemical fertilization, the use of high-analysis fertilizer and the depletion of organic matter in soil [9]. There is a wide gap between mining and addition of nutrients. In some states like Assam addition is less than mining [10] except in tea gardens and vegetable belts while in other states like Punjab and Haryana, excessive applications of nitrogenous and phosphatic fertilizers pollute the groundwater [11] and cause eutrophication in aquatic ecosystems [12]. To overcome this problem, we have to reduce the dependency on large-scale application of conventional chemical fertilizers. By cutting large-scale chemical fertilization a farmer can not only save money for organic amendments but also can reduce environmental pollution. No doubt that conventional fertilizers boost crop production, but their excessive and careless application, especially without organic manures, disrupts soil mineral balance and fertility and therefore factor productivity decreases over time [13]. Apart from this, fertilizers harm soil structure and mineral cycles, and damage beneficial soil microorganisms [14]. To address this, there's a persistent need to optimize chemical fertilizer use and reduce environmental risks. A solution lies in "nanotechnology" which can provide us such smart materials that can precisely release agro-chemicals whether nano-nutrients or pesticides that can hit the specific target zone with small quantities but provide bigger positive impact on crop production while preserving natural environment and the ecosystem as a whole [15].

Advantage of nano technology approach :

There are a lot of advantages using nano technology approach in field of agricultural sciences. It has been found that nano particles have high surface area and therefore highly reactive than macro scale particles [16]. Nano particles are reported to be very effective catalyst for plant and microbial metabolism [5]. Nano particles can easily penetrate inside the cell because of their tiny size [17]. Therefore, a small quantity of chemical fertilizer or pesticide do the same job which can be achievable by application of very high dose of conventional chemical fertilizers and pesticides. By doing so, nano molecules indirectly also reduce pollution of the environment that may arise due to heavy dependency of chemical intensive farming and side by side also improve nutrient use efficiency [18]. Nano technology also opens new avenues to identify and quantify different biomolecules by using nanoprobe and nano sensors [19]. With the use of nanotechnology, biosystems may be better understood and transformed. It also has a significant influence on subcellular dynamics, regeneration processes, genome description [20], and food characterization [21]. Therefore, nano technology is a buzzing platform for synthesis of effective agrochemicals not only in crop production but also for animal husbandry [22] and fishery sector [23] which help in increasing farmers income in an all-inclusive way. For example, phosphorus use efficiency through conventional chemical fertilizer is hardly 15-20 percent. However, if we apply nano P, fertilizer use efficiency was reported to increase up to 58-65 percent [24]. Similarly, nitrogen use efficiency of nano urea is claimed to be 85-90 percent which is 30-35% only in case of conventional urea [25]. Therefore, Govt. of India is also promoting nano urea so that they can stop import of urea to fulfil the domestic demand by 2025 [26]. On the other hand, use

of nano P can not only increase the P use efficiency but also can save foreign exchange that otherwise would have been spent for purchasing of high-grade rock phosphate and phosphoric acid required as raw material for producing phosphatic fertilizers [27]. Applications of nano-induced microbial gum improve soil aggregation, carbon sequestration, and moisture retention [28]. Again, nano pesticides utilize nano materials to enhance the efficiency, targeted delivery, and overall effectiveness of the pesticide compounds through improve penetrations, targeted delivery and control release [29]. Nano pest repellents are formulations that use nanotechnology to create substances capable of repelling pests, such as insects or rodents with precision targeting, reduced environmental impact and enhanced durability and, therefore, have good scope for post-harvest management of pests [30]. Environmentally friendly and fairly priced, magnetic nanoparticles (MNPs) have several advantages. They have therefore been applied in agriculture as antimicrobial agents [31], in nanoelectronics, and in the identification and removal of poisons [32]. Nano membranes are ultra-thin membranes, usually less than 100 nanometers in thickness. These complex structures are very successful in purifying irrigation water because they have nanopores or nanochannels that selectively permit the passage of some chemicals while blocking others [33]. This technology is very much suitable for treating saline, sodic or brackish water or iron contaminated water when used for irrigation [34]. These membranes have enhanced filtration efficiency, longevity and durability. Besides, they are cost effective and less energy consuming. Furthermore, the highly precise and exact monitoring of soil moisture levels is made possible by nanoparticles. Because of their tiny size, they may more easily permeate the soil, guaranteeing that the data accurately depicts the moisture level. By offering constant, real-time monitoring, nanoparticles lower the possibility of over-or under-irrigation. Based on precise, current information, farmers may modify their water consumption, improving crop health and resource use efficiency [35]. Nanotechnology is gaining momentum in veterinary sector too. For instance, Nanozinc oxide can improve farm animals' and birds' immunity, growth rate, and ability to reproduce. It can also reduce the incidence of diarrhea in young pigs [36]. Thus far, studies have indicated that the application of nano-minerals in animal immunology, reproduction, and production holds potential [37]. However, there have been some negative reports too [38]. In wider gamut of agriculture nano technology also finds place fishery sector for several applications such as water purification, sterilizing fish ponds, feeding fish with nano-feed, and managing aquatic diseases [39]. Nano technology is also used to produce nano sensors. Soil contaminants, pests, soil nutritional content, and plant stress due high or low temperature, pressure and drought can be detected by these sensors [40].

What is quantum dot?

Quantum dots, semiconductor nanocrystals endowed with exceptional quantum properties, have proven instrumental in the realm of biological research. Their capacity for in vitro and in vivo labelling of living cellular entities, through injection or protein conjugation, has facilitated unprecedented insights into biomolecular dynamics [41].

Why Nano-particles behave differently?

Nano materials behave differently mainly because of their high surface area to volume ratio [42]. Specific targeting is possible due to their tiny size. Nano molecules may be engineered to get desired surface characteristics [43]. Difference between nano and conventional fertilizer is discussed in Table 1. Thus, nano technology provides us a “smart delivery system” because of all these unique properties such as high surface area, control release and high solubility [44].

Table 1 : Comparison between Conventional and nano-fertilizers

Sl No	Parameter	Conventional-fertilizer	Nano-fertilizer
1	Surface area	Less	High
2	Solubility	Less	More
3	Targeted delivery	No	Yes
4	Bioavailability	Less	More
5	Nutrient use Efficiency	Less	More
6	Controlled release	Mostly not except for some slow-release fertilizers	Yes
7	Loss due to leaching and volatilization	More	Very less

Techniques for applying nano chemicals: Plants grown *in vitro* are fed either hydroponically — without soil — in nano nutrient-rich water or aeroponically — fed with nano-nutrient-rich fluid that is regularly sprayed into the roots of the plants using pump and sprinkler system [45]. In *in vivo* methods nano chemicals can either be applied as soil application [46] or foliar application [47].

How nano materials are produced:

Nanotechnology research requires a hyper-clean laboratory (Class 10,000, 1000 or 100) depending on the process involved. Nanoparticles can be synthesized by several methods *viz.* Physical Methods, Chemical Methods, Biological Methods and Aerosol Technique.

- **Physical Synthesis of Nanoparticles :** For preparation of nano particles such nano-P by using physical methods we require a High Energy Ball Mill or Pot Mill to produce nanoparticles with an average size of 28 and 70 nm, respectively [48].
- **Chemical Synthesis of Nanoparticles :** Nano particles can easily be synthesized using chemical methods too. Nano Zinc oxide powder can be produced by reacting Zinc acetate with potassium hydroxide in presence of methanol [49]. Detail procedure is

described in Fig 1. Nano P or nano Zn fertilizer can be produced by reacting nano-Zn or nano-rock phosphate powder with Oleic acid in o-Xylene. Details is described in Fig 2.

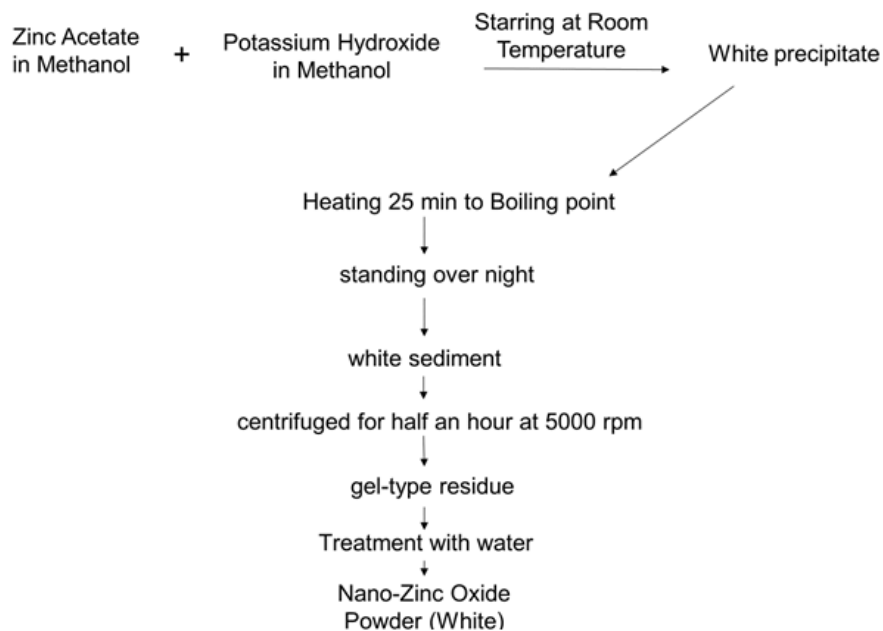


Fig 1 : Chemical Synthesis of nano particles [49]

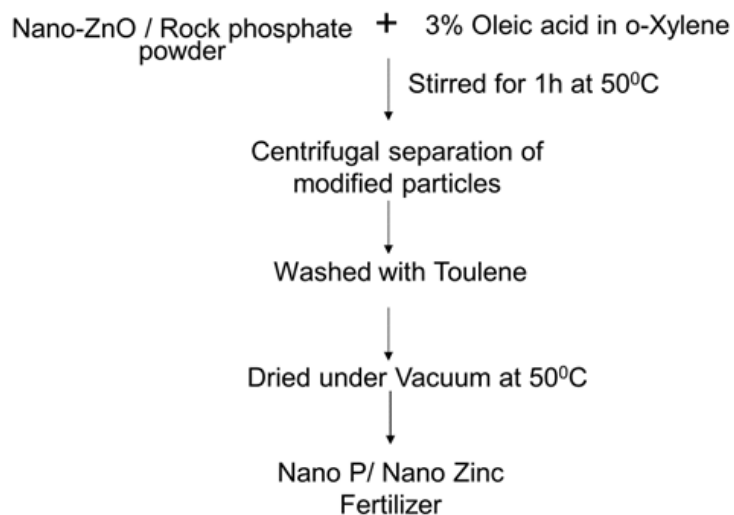


Fig 2 : Preparation of nano fertilizer

- **Vinyl (Pyrrolidene) Technique :** This is a technique is used for solution phase synthesis of colloidal particles. The water-soluble polymer compound polyvinylpyrrolidone (PVP), usually referred to as polyvidone or povidone, is derived from the monomer N-vinyl pyrrolidone [50]. PVP is available in a range of molecular weights and corresponding viscosities, thus it may be selected according to the desired application characteristics [51]. The hydroxyl groups of PVP are good reductant for the aqueous synthesis of nanoparticles made of Lead, Silver, platinum and Gold with varying shapes vis. circular, hexagonal and triangular [52].

Synthesis of nano particles by microorganisms :

The generation of nanoparticles using microbiological synthesis has become a fast-expanding field of study in nanotechnology worldwide, with a variety of biological entities being used on a regular basis creating an impute substitute for traditional techniques. Nanoparticles may be created from simple prokaryotes to eukaryotic creatures, including higher plants [53]. For example, to isolate nano particles by fugus at first specific fungus from soil are isolated using serial dilution technique. Selected fungal strains are cultured to obtain fungal balls and separated from broth by centrifugation. extracellular fungal proteins are isolated and added to the solid/suspended compounds in a flask/test tubes at room temperature for synthesis of nano particles.

Aerosol method :

To produce nano particle by aerosol method evaporation of a targeted metal is done in a low-density gas. Nanoparticles are formed in vapour by homogeneous nucleation. Produced nano particles are then isolated using thermophoresis (a transport force occurs due to temperature gradient and movement of particles occur from hot gas to the cold wall) technique and condensed [54].

Characterization of nano particles :

Nano particles are characterized by the following instruments/techniques.

1. Particle size analyzer : Dynamic Light Scattering (DLS)
2. Transmission Electron Microscope (TEM)
3. Scanning Electron Microscope (SEM)
4. Atomic Force Microscope (AFM)
5. Fourier Transform Infrared Spectroscopy (FTIR)
6. X-ray Diffraction (XRD)
7. Energy Dispersive X-ray Spectroscopy (EDS, EDX, EDXS, or XEDS),
8. Lithography
9. Inductively Coupled Plasma Mass Spectrometer (ICP-MS)
10. Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES)

Conclusion :

The integration of nanotechnology in agriculture holds promise for a transformative second green revolution, fostering sustainability across the agricultural spectrum. To navigate this transformative potential responsibly, there is a need to underscore a strategic research and development paradigm throughout the agricultural value chain. Central to this initiative is the imperative to prioritize environmentally and socially responsible technological advancements, anticipate regulatory requisites, and garner stakeholder acceptance. Within the realm of optimism surrounding nanotechnology in agriculture, it is imperative to acknowledge and address societal, environmental, and ethical considerations. Potential violations on human rights, privacy implications, and the potential for adverse environmental effects demand careful scrutiny. Furthermore, the economic impacts on farmers, consumers, and associated industries require nuanced evaluation, recognizing the spectrum of potential outcomes, both positive and negative. Nevertheless, a comprehensive investigation into the application of nanotechnology in agriculture yields encouraging results across various critical parameters in most findings. These findings contribute positively to the discourse, suggesting a harmonious compatibility between nanotechnology and pivotal facets of agricultural and environmental integrity.

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Nanotechnology and Food Industry

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Abstract :

Nanotechnology is an area of emerging interest and opens up a whole universe of new possibilities for the food industry. Recent innovations in nanotechnology have transformed a number of scientific and industrial areas including the food industry. The food market demands technologies, which are essential to keep market leadership in the food processing industry to produce fresh authentic, convenient and flavourful food products. Applications of nanotechnology have emerged with increasing need of nanoparticle uses in various fields of food science and food microbiology, including food processing, food packaging, functional food development, food safety, detection of food borne pathogens, and shelf-life extension of food and/or food products. Nano food packaging materials may extend food life, improve food safety, alert consumers that food is contaminated or spoiled, repair tears in packaging, and even release preservatives to extend the life of the food in the package. Nanotechnology provides a vast range of opportunities for the development of new products and applications in food system. Functional foods, nutraceuticals, bioactive, pharma foods, etc. are very recent example of it. Lowering of the cost of food additives is a milestone of using nano food additives.

Keywords : Nanoparticles, food safety, food preservation, functional food, food nutrition, nano-processed food products

Introduction :

Over the past few decades, nanotechnology has increasingly been considered as to be attractive technology that has revolutionized the food sector. The created nanomaterials possess one or more external dimensions, or an internal structure, on the scale from 1 to 100 nm that allowed the observation and manipulation of matter at the nanoscale. It is observed that these materials have unique properties unlike their macroscale counterparts due to the high surface to volume ratio and other novel physiochemical properties like color, solubility, strength, diffusivity, toxicity, magnetic, optical, thermodynamic, etc. [1]. Nanotechnology has brought new industrial revolution and both developed and developing countries are interested in investing more in this technology [2]. Therefore, nanotechnology offers a wide range of opportunities for the development and application of structures,

materials, or system with new properties in various areas like agriculture, food, and medicine, etc.

The rising consumer concerns about food quality and health benefits are impelling the researchers to find the way that can enhance food quality while disturbing least the nutritional value of the product. The demand of nanoparticle-based materials has been increased in the food industry as many of them contain essential elements and also found to be non-toxic [3]. They have been also found to be stable at high temperature and pressures [4]. Nanotechnology offers complete food solutions from food manufacturing, processing to packaging. Nanomaterials bring about a great difference not only in the food quality and safety but also in health benefits that food delivers. Many organizations, researchers, and industries are coming up with novel techniques, methods, and products that have a direct application of nanotechnology in food science [5].

The applications of nanotechnology in food sector can be summarized in two main groups that are food nanostructured ingredients and food nanosensing. Food nanostructured ingredients encompass a wide area from food processing to food packaging. In food processing, these nanostructures can be used as food additives, carriers for smart delivery of nutrients, anti-caking agents, antimicrobial agents, fillers for improving mechanical strength and durability of the packaging material, etc. whereas food nanosensing can be applied to achieve better food quality and safety evaluation [6]. In this review, we have summarized the role of nanotechnology in food science and food microbiology and also discussed some negative facts associated with this technology.

Nanotechnology in Food Processing :

The nanostructured food ingredients are being developed with the claims that they offer improved taste, texture, and consistency [7]. Nanotechnology increasing the shelf-life of different kinds of food materials and also help brought down the extent of wastage of food due to microbial infestation [8]. Nowadays nanocarriers are being utilized as delivery systems to carry food additives in food products without disturbing their basic morphology. Particle size may directly affect the delivery of any bioactive compound to various sites within the body as it was noticed that in some cell lines, only submicron nanoparticles can be absorbed efficiently but not the larger size micro-particles. An ideal delivery system is supposed to have following properties: (i) able to deliver the active compound precisely at the target place (ii) ensure availability at a target time and specific rate, and (iii) efficient to maintain active compounds at suitable levels for long periods of time (in storage condition). Nanotechnology being applied in the formation of encapsulation, emulsions, biopolymer matrices, simple solutions, and association colloids offers efficient delivery systems with all the above-mentioned qualities. Nano polymers are trying to replace conventional materials in food packaging. Nanosensors can be used to prove the presence of contaminants, mycotoxins, and microorganisms in food [9].

Nanoparticles have better properties for encapsulation and release efficiency than traditional encapsulation systems. Nanoencapsulations mask odors or tastes, control interactions of active ingredients with the food matrix, control the release of the active

agents, ensure availability at a target time and specific rate, and protect them from moisture, heat [10], chemical, or biological degradation during processing, storage, and utilization, and also exhibit compatibility with other compounds in the system [11]. Moreover, these delivery systems possess the ability to penetrate deeply into tissues due to their smaller size and thus allow efficient delivery of active compounds to target sites in the body [12]. Various synthetic and natural polymer-based encapsulating delivery systems have been elaborated for the improved bioavailability and preservation of the active food components (Table 1). Further, the importance of nanotechnology in food processing can be evaluated by considering its role in the improvement of food products in terms of (i) food texture, (ii) food appearance, (iii) food taste, (iv) nutritional value of the food, and (v) food shelf-life. It is a matter of fact that surprisingly nanotechnology not only touches all the above-mentioned aspects but has also brought about significant alterations in food products providing them novel qualities.

Texture, Taste, and Appearance of Food :

Nanotechnology provides a range of options to improve the food quality and also helps in enhancing food taste. Nanoencapsulation techniques have been used broadly to improve the flavor release and retention and to deliver culinary balance [13]. Zhang et al. (2014) used the nanoencapsulation for the highly reactive and unstable plant pigment anthocyanins which have various biological activities. Through, encapsulating cyanidin-3-O-glucoside (C3G) molecules within the inner cavity of apo recombinant soybean seed H-2 subunit ferritin (rH-2) improved the thermal stability and photostability. This design and fabrication of multifunctional nanocarriers for bioactive molecule protection and delivery. Rutin is a common dietary flavonoid with great important pharmacological activities but due to poor solubility, its application in the food industry is limited. The ferritin nanocages encapsulation enhanced the solubility, thermal and UV radiation stability of ferritin trapped rutin as compared to free rutin [14]. The use of nanoemulsions to deliver lipid-soluble bioactive compounds is much popular since they can be produced using natural food ingredients using easy production methods, and may be designed to enhance water-dispersion and bioavailability [15].

As compared to larger particles which generally release encapsulated compounds more slowly and over longer time periods, nanoparticles provide promising means of improving the bioavailability of nutraceutical compounds due to their subcellular size leading to a higher drug bioavailability. Many metallic oxides such as titanium dioxide and silicon dioxide (SiO₂) have conventionally been used as color or flow agents in food items [16]. SiO₂ nanomaterials are also one of the most used food nanomaterials as carriers of fragrances or flavors in food products [17].

Nutritional Value:

A majority of bioactive compounds such as lipids, proteins, carbohydrates, and vitamins are sensitive to high acidic environment and enzyme activity of the stomach and duodenum. Encapsulation of these bioactive compounds not only enables them to resist such adverse

conditions but also allows them to assimilate readily in food products, which is quite hard to achieve in non-capsulated form due to low water-solubility of these bioactive compounds. Nanoparticles-based tiny edible capsules with the aim to improve delivery of medicines, vitamins or fragile micronutrients in the daily foods are being created to provide significant health benefits [18,19]. The nanocomposite, nano-emulsification, and nanostructuration are the different techniques which have been applied to encapsulate the substances in miniature forms to more effectively deliver nutrients like protein and antioxidants for precisely targeted nutritional and health benefits. Polymeric nanoparticles are found to be suitable for the encapsulation of bioactive compounds (e.g., flavonoids and vitamins) to protect and transport bioactive compounds to target functions [20].

Preservation or Shelf-Life :

In functional foods where bioactive component often gets degraded and eventually led to inactivation due to the hostile environment, nanoencapsulation of these bioactive components extends the shelf-life of food products by slowing down the degradation processes or prevents degradation until the product is delivered at the target site. Moreover, the edible nano-coatings on various food materials could provide a barrier to moisture and gas exchange and deliver colors, flavors, antioxidants, enzymes, and anti-browning agents and could also increase the shelf-life of manufactured foods, even after the packaging is opened. Encapsulating functional components within the droplets often enables a slowdown of chemical degradation processes by engineering the properties of the interfacial layer surrounding them. For example, curcumin the most active and least stable bioactive component of turmeric (*Curcuma longa*) showed reduced antioxidant activity and found to be stable to pasteurization and at different ionic strength upon encapsulation [21].

Nanotechnology in Food Packaging :

A desirable packaging material must have gas and moisture permeability combined with strength and biodegradability [22]. Nano-based “smart” and “active” food packaging confer several advantages over conventional packaging methods from providing better packaging material with improved mechanical strength, barrier properties, antimicrobial films to nanosensing for pathogen detection and alerting consumers to the safety status of food [23].

Application of nanocomposites as an active material for packaging and material coating can also be used to improve food packaging [24]. Many researchers were interested in studying the antimicrobial properties of organic compounds like essential oils, organic acids, and bacteriocins [25] and their use in polymeric matrices as antimicrobial packaging. However, these compounds do not fit into the many food processing steps which require high temperatures and pressures as they are highly sensitive to these physical conditions. Using inorganic nanoparticles, a strong antibacterial activity can be achieved in low concentrations and more stability in extreme conditions. Therefore, in recent years, it has been a great interest of using these nanoparticles in antimicrobial food packaging. An antimicrobial packaging is actually a form of active packaging which contacts with the food

product or the headspace inside to inhibit or retard the microbial growth that may be present on food surfaces [26]. Many nanoparticles such as silver, copper, chitosan, and metal oxide nanoparticles like titanium oxide or zinc oxide have been reported to have antibacterial property [27], Figure 1.

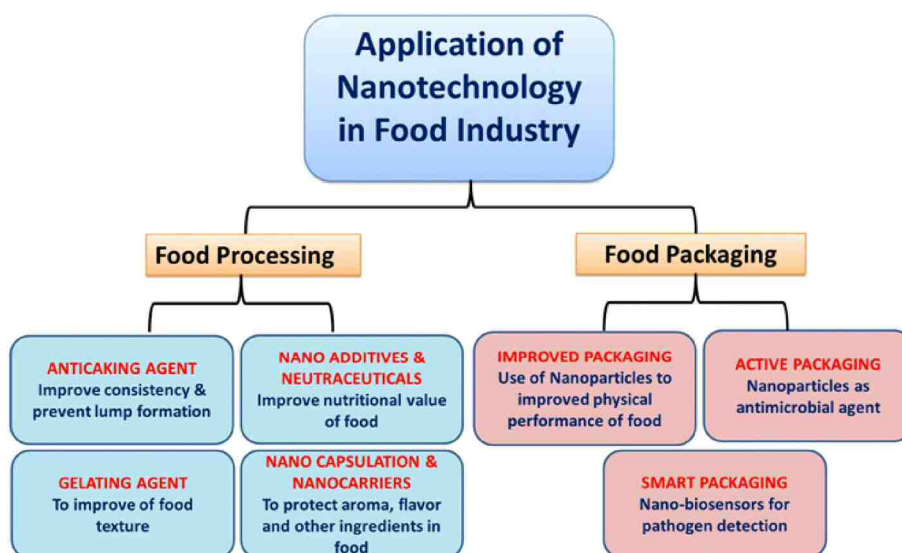


Figure 1. Schematic diagram showing role of nanotechnology in different aspects of food sectors.

Table 1. Different Nano techniques to encapsulate and delivery of functional ingredients.

Nanotechnique	Characteristic feature	Examples	Reference
Edible coatings	To preserve the quality of fresh foods during extended storage	Gelatin-based edible coatings containing cellulose nanocrystal Chitosan/nanosilica coatings Chitosan film with nano-SiO ₂ Alginate/lysozyme nanolaminate coatings	Fakhouri et al., 2014 Shi et al., 2013 Yu et al., 2012 Medeiros et al., 2014
Hydrogels	Can be easily placed into capsules, protects drugs from extreme environments, and to deliver them in response to environmental stimuli such as pH and temperature	Protein hydrogels	Qui and Park, 2001
Polymeric micelles	Solubilize water-insoluble compounds in the hydrophobic interior, high solubility, low toxicity	PEO-b-PCL [poly(ethylene glycol)block-poly(caprolactone)] polymeric micelles Methoxy poly(ethylene glycol) palmitate polymeric micelles	Ma et al., 2008 Sahu et al., 2008
Nanoemulsions	(i) Greater stability to droplet aggregation and gravitational separation; (ii) Higher optical clarity; and, (iii) increased oral bioavailability	β -Carotene-based nanoemulsion β -Carotene-based nanoemulsion	Kong et al., 2011 Yuan et al., 2008
Liposomes	Since liposome surrounds an aqueous solution inside a hydrophobic membrane, it can be used delivery vehicles for hydrophobic molecules (contained within the bilayer) or hydrophilic molecules (contained in the aqueous interior)	Cationic lipid incorporated liposomes modified with an acid-labile polymer hyper-branched poly(glycidol) (HPG)	Yoshizaki et al., 2014
Inorganic NPs	They display good encapsulation capability and their rigid surfaces allow controlled functionalization	Mesoporous silica nanoparticles	Tang et al., 2012

As compared to larger particles which generally release encapsulated compounds more slowly and over longer time periods, nanoparticles provide promising means of improving the bioavailability of nutraceutical compounds due to their subcellular size leading to a higher drug bioavailability. Many metallic oxides such as titanium dioxide and silicon dioxide (SiO_2) have conventionally been used as color or flow agents in food items [28]. SiO_2 nanomaterials are also one of the most used food nanomaterials as carriers of fragrances or flavors in food products [29].

Nanosensors for Pathogen Detection :

Nanomaterials for use in the construction of biosensors offers the high level of sensitivity and other novel attributes. In food microbiology, nanosensors or nanobiosensors are used for the detection of pathogens in processing plants or in food material, quantification of available food constituents, alerting consumers and distributors on the safety status of food [30]. The nanosensor works as an indicator that responds to changes in environmental conditions such as humidity or temperature in storage rooms, microbial contamination, or products degradation [31]. Various nanostructures like thin films, nanorods, nanoparticles and nanofibers have been examined to their possible applications in biosensors [32]. Thin film-based optical immunosensors for detection of microbial substances or cells have led to the rapid and highly sensitive detection systems. In these immunosensors, specific antibodies, antigens, or protein molecules are immobilized on thin nano-films or sensor chips which emit signals on detection of target molecules [33]. A dimethylsiloxane microfluidic immunosensor integrated with specific antibody immobilized on an alumina nanoporous membrane was developed for rapid detection of foodborne pathogens *Escherichia coli* O157:H7 and *Staphylococcus aureus* with electrochemical impedance spectrum. Nanotechnology can also assist in the detection of pesticides, pathogens and toxins [34] serving in the food quality tracking–tracing–monitoring chain.

Biosensors based on carbon nanotubes also gained much attention due to their rapid detection, simplicity and cost effectiveness and have also been successfully applied for the detection of microorganisms, toxins, and other degraded products in food and beverages [35]. Toxin antibodies attached to these nanotubes causes a detectable change in conductivity when bound to waterborne toxins and therefore are used to detect waterborne toxins. Further, the use of electronic tongue or nose which consists of the array of nanosensors monitor the food condition by giving signals on aroma or gases released by food items. The quartz crystal microbalance (QCM)-based electric nose can detect the interaction between various odorants and chemicals that have been coated on the crystal surface of the QCM. Many studies on small molecule detection have used quartz crystal surfaces that have been modified with different functional groups or biological molecules, such as amines, enzymes, lipids, and various polymers [36].

Safety Issues :

Besides a lot of advantages of nanotechnology to the food industry, safety issues associated with the nanomaterial cannot be neglected. Many researchers discussed safety

concerns associated with nanomaterial giving emphasis on the possibility of nanoparticles migrate from the packaging material into the food and their impact on consumer's health [37]. Although a material is being considered as GRAS (generally regarded as safe) substance, additional studies must be acquired to examine the risk of its nano counterparts because the physiochemical properties in nanostates are completely different from that are in macrostate. Moreover, the small size of these nanomaterials may increase the risk for bioaccumulation within body organs and tissues [38]. For example, silica nanoparticles which are used as anti-caking agents can be cytotoxic in human lung cells when subjected to exposure [39]. There are a lot of factors that affect dissolution including surface morphology of the particles, concentration, surface energy, aggregation, and adsorption. A model to study the migration of particles from food packaging has been developed by Cushen et al.. They studied the migration of silver and copper from nanocomposites and observed that the percentage of nanofiller in the nanocomposites was one of the most crucial parameters driving migration, more so than particle size, temperature, or contact time. Since every nanomaterial has its individual property, therefore, toxicity will likely be established on a case-by-case basis [40]. Further, regulatory authorities must develop some standards for commercial products to ensure product quality, health and safety, and environmental regulations.

Conclusion :

Over past years the popularity of the uses of structures on the nanometer scale in the food sector is increasing, therefore, interest and activities in this research area have greatly focused. As nanobiotechnology steps forward, devices or material based on this technology become smaller and more sensitive. Its applicability in the areas of food packaging and food safety are well known. Additionally, promising results have been achieved in food preservation using nanomaterial where they might protect the food from moisture, lipids, gases, off-flavors, and odors. They offer excellent vehicle systems to deliver bioactive compounds to the target tissues. Although the advances in nanotechnology are paving new paths day by day, there still persist many challenges and opportunities to improve the current technology and also issues about the consequences of nanotechnology that must need to be addressed in order to alleviate consumer concerns. The transparency of safety issues and environmental impact should be the priority while dealing with the development of nanotechnology in food systems and therefore compulsory testing of nano foods is required before they are released to the market.

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Nanopesticides in Agriculture: Management of Insect Pests

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Abstract :

With the escalating cost of insect pest management and increasing concern about the adverse effects of the pesticide residues, there is an urgent need to explore other avenues of alternate pest management strategies. Nano pesticides for plant protection is an emerging research field, offers new methods to design active ingredients amid nanoscale dimensions. Besides the broad categories of active ingredients (AIs) of synthetic insecticides, pheromone and natural resource-based molecules having pesticidal activity can also be used using NPs as a carrier as an alternate of traditional pest control agents. Nanopesticide based formulations have a potential and bright future for developing more effective and safer pesticide or biopesticides.

Keywords : Nano-pesticides, nano-particles; IPM, insect pest, Nano bio-pesticides

Introduction :

Pesticides are increasingly being created using nanotechnology to reduce the hazardous effects of pesticides and make their application to plants and cereals simpler. The use of nanopesticides is growing in popularity among scientists and extension workers [1]. According to Kah *et al.* [2] and Rani and Sushil [3], nanopesticides are as effective as conventional pesticides in controlling pests. Bio-efficacy of the nano-formulations in the field and its potential impacts on ecosystems may differ from laboratory tests if nano-specific quality assurance and adequate controls are lacked [2]. Studies have shown that nano-formulations may reduce drift and volatile losses during application [4]. Several Plant-derived products (PDP) based biopesticides have been synthesized as nano-formulations or nanoparticles [5-6]. This will reduce chemical loads and treatment costs while increasing crop yield potential [4]. PDP nanopesticides have not yet been compared to other market products. Pesticides are increasingly being manufactured using nanotechnology to reduce the hazardous effects of pesticides and make their application safe and efficient on agricultural crops. The popularity of the use of nano pesticides is growing among scientists and extension workers [1]. According

to Kah *et al.* [2] and Rani and Sushil [3], nano pesticides are as effective as conventional pesticides in controlling pests. In the absence of quality assurance and adequate controls, the bio-efficacy of nano-formulations in the field and their potential impacts on ecosystems may differ from laboratory findings. [2]. Studies have shown that, nanoformulations may reduce drift and volatile losses during application [4].

Nanotechnology is the field of emerging science which plays very important character to resolve all these problems which are associated with the future of agriculture. NPs are small, uniform pieces of common materials that are smaller than 100 nm [7]. NP research is a part of nanotechnology, which has moved into modern biological and medical fields and could soon change everything from the size of electronics to diseases diagnosis tools.

The Nanoscale materials have multifunctional capabilities in nature which provide the following compromising properties like, programmed, specific to target, time controlled and self-operating systems. As a result of absorbed nanoparticles, the plant increases its stress and disease resistance capability. By using nano biosensors, it is very easy to detect the pathogens in early stage and nanoparticles plays very important role in killing of pathogens. Smart crops could be developed by the assistance of Phyto-nanotechnology [8].

To take appropriate control measures, economic threshold level (ETL) is considered as one of the crucial requirements, indicates the density of population at which different control measures for the management of pest should be adopted to avert increasing pest population from touching the economic injury level (EIL) (Fig. 1). In agriculture, the EIL is very high for those pests, which infest the new shoots as their damage directly affects the yield and in case of the pests, which infest the other parts of the plant, the immediate EIL is low but their damage affects slowly to the growth of the new shoots as well as the health of the plant [9].

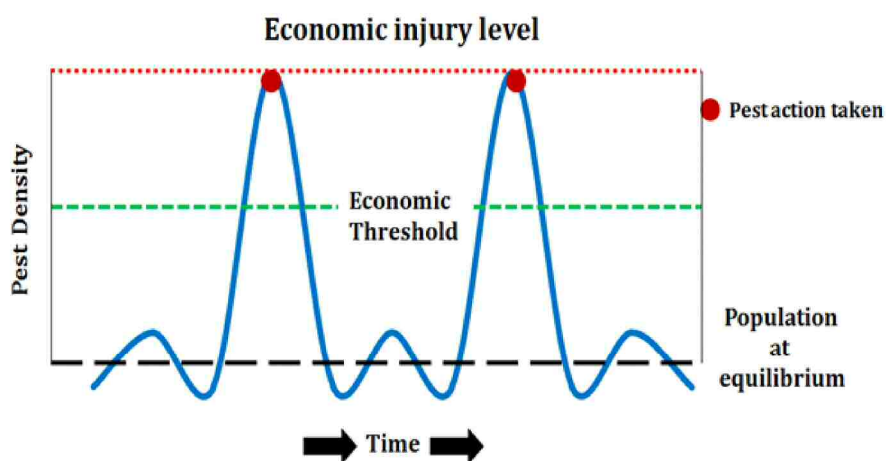


Fig. 1: Showing Economic injury level (EIL) and Economic Threshold Level (ETL) of insect pests. Source: Deka *et al* [10].

Management of pest in conventional methods :

Based on the ecological characteristics of a fields and the cropping system, a tentative IPM system comprising all possible control methods has been proposed (Fig. 2) [9-12].

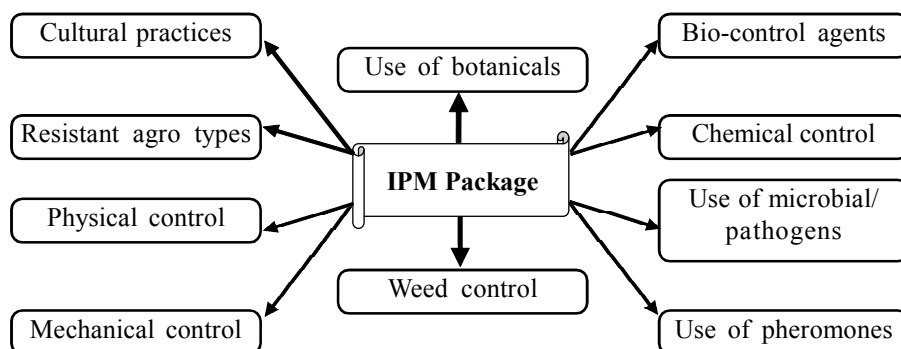


Fig. 2: IPM components for insect pest management.

Source: Deka *et al* [10].

Why nano pesticides for insect pest management :

The term “nano pesticides” describes the application of nanotechnology to plant protection. This is a novel area of study that offers innovative ways to formulate and transport active chemicals at nanoscale dimensions. This area of study encompasses a wide range of topics, such as the fundamental understanding of how insects interact with nanoscale materials, the formulation of AI into accessible pesticides and nano-emulsions, and the creation of novel nanopesticides that employ nanomaterials as active pesticide agents or nanocarriers to deliver them [13]. It is anticipated that much research on nanopesticides will address the limitations of currently available pest control methods and provide revolutionary nano-based formulations that penetrate the target (pest), stay stable and active in the environment, and don’t harm non-target organisms [14].

According to Auffan *et al.* [15], a nanoparticle (NP) is an ultrafine particle with dimensions ranging from 1 to 100 nm and unique properties not shared by non-nanoscale particles of the same chemical makeup. In addition to toxicity, the primary characteristics of nanoparticles (NP) that make them suitable for use as nanopesticides are their shape (rods, spherical, irregular, tubes), size, surface to volume ratio, and crystal phase (amorphous, crystalline). When compared to conventional pesticides, NP-based pesticides typically exhibit several advantages, including increased formulation stability and solubility in water, the removal of hazardous organic solvents, the capacity to release artificial intelligence (AI) more gradually, improved mobility, and insecticidal activity [16]. According to recent research, a variety of materials with different chemical compositions — such as carbon, silicates, polymers, metal oxides, lipids, semiconductor quantum dots, ceramics, emulsions, proteins, and dendrimers — were synthesized to produce NP [17].

NPsto improve pesticide formulations to control insect pest :

Recent studies also showed different plant synthesized NPs and their efficacy against a wide range of economically important insect pests, including, mosquitoes [14], beetles [18], moths [19], hard ticks [18], lice [20], louse flies [21], etc. Hence, most of the formulation of pesticides are available in oil-in-water (O/W) emulsions or emulsifiable concentrates (ECs) and are sparingly water-soluble [22]. EC formulations typically use a combination of surfactant emulsifiers or highly costly, combustible organic solvents that are more hazardous. However, the O/W emulsions need a high energy input for emulsification because they are based on a combination of non-ionic and polymeric surfactants [23]. New formulations based on micro and nano-emulsions are being introduced to counter these drawbacks of conventional insecticides [24]. Numerous formulations based on micro-emulsion are available for the management of a wide range of pests and diseases in various agroecosystems, such as plant growth regulators and systemic fungicides [25]. Micro-emulsions have advantages over conventional formulations, including improved stability and tank mix compatibility, reduced flammability and toxicity to handlers, and increased efficacy due to better uptake and penetration from surfactants with high solubilizing power [22]. Additionally, micro-emulsions are steadier than nano-emulsions, which require high energy input for scale-up for commercial production and difficult for handlers to produce on-site. On the other hand, micro-emulsions demonstrated several drawbacks, including greater surfactant concentrations, extremely low AI levels, and a restricted selection of appropriate surfactants [26]. The development of such appropriate release systems could improve the pesticide's performance and efficiency while reducing the negative environmental effects. These limitations could be partially resolved by different formulation concepts based on nano-suspension or nano-dispersion, where AIs of nanocrystals creates nano-dispersions containing similar properties of solutions [27]. NPs are typically engineered to transport a particular biomolecule to an organ, tissue, or even a single cell. By employing a nanocarrier, these products can be precisely delivered to insects or plants and easily enter their cells [28]. Numerous inorganic nanomaterials with unique chemical and physical characteristics, including as silica, metal oxides, metal, semiconductors, and carbon-based materials, have been created for delivery and tracking applications [29]. Nano-delivery vehicles can improve plant growth, production, and seed vigor in addition to shielding crop plants from pests and diseases. They may even be utilized to manipulate genes [30]. The NPs penetrate plant cells by forming new pores through ion channels or by attaching to a carrier protein. These processes can undergo major alterations as a result of the NPs' actions [31]. Although plant cell walls are remarkably capable of allowing NPs to enter the plant [32], there have been some improbable effects of NP application that have been brought to light. As a result, research on the phytotoxic effects of NPs [33] and their impact on plant development [34] is still being conducted. The ability of NPs to readily enter an insect's cell, mitochondria, or nucleus was also demonstrated by electron microscope research on the impact of NPs on plants and insects. This finding raises the possibility of using NPs as a pesticide for targeted delivery [32].

Pesticide delivery system using nanoparticles as nanocarriers :

The “pesticide delivery system” (PDS) idea was derived from the drug delivery idea in medicine, which uses NPs to deliver medications to the intended organs [35]. PDS is intended to minimize adverse effects on non-target organisms while delivering the AI to a designated target for a predetermined amount of time and concentration in order to achieve the intended biological efficacy [36]. Controlled distribution is crucial to ensuring that the right amounts of pesticides are released in the right amounts at the right times [35]. The use of NPs as nanocarriers is encouraged by their unique properties, which include a wide surface area, quick mass transfer to the insect’s body (target), high effective loading capacity, and ease of attachment for different pesticide compounds. Following encapsulation, pesticide molecules exhibit more gradual release over time, necessitating fewer administrations. At the same time, NPs impede the pesticides’ loss of potency due to degradation. Several ideas, including encapsulation, adsorption, entrapment inside the NP, and covalent attachment reconciled by different ligands, are used for loading pesticide molecules on NPs [37] (Fig. 3).

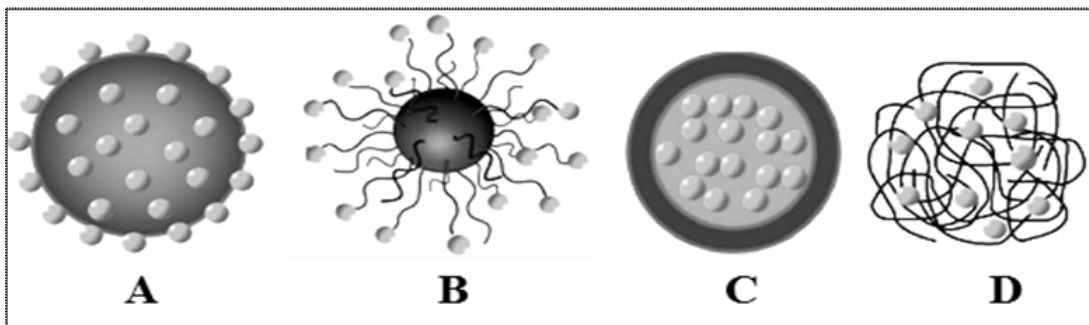


Fig. 3: Schematic representation of NPs for delivery of pesticides, A: adsorption on NP, B: attachment on NP by different linkers; C: encapsulation inside polymeric hydrophobic or hydrophilic core (polymer micelles); and D: entrapment inside polymeric nanoparticle (source: Deka *et al* [10].

It is possible to achieve control and slow-release properties of the molecules based on the way the AI bonds to the material, the nanocarrier’s properties of degradation, and the surrounding circumstances. Nanocarriers such as synthetic silica, polymer, titania, silver, alumina, and copper are typically employed for the delivery of pesticides. The application of the nanoencapsulation method suggests less exposure to the environment and safer pesticide use. Because of their structural flexibility, which allows them to form NPs of various shapes and sizes, and their capacity to form pores when loading biomolecules, solid and mesoporous silica NPs have demonstrated the greatest potential as agrochemical delivery agents among all the NPs currently available [38].

Nano bio-pesticides :

The selection of nano bio-pesticides necessitated considering a number of crucial factors, including ease of preparation, economic viability, effectiveness against a broad range of pests, safety for non-target organisms and the environment, non-toxicity, lack of accumulation in the food chain, negligible or nonexistent residues, and ease of application [39]. Generally, the conventional techniques for the preparation of synthetic pesticides are combined with in order to increase efficacy, better solubility, a slower rate of release, and degradation [40]. Comparably, biological molecules with pesticidal qualities are combined with silver salt to generate nano bio-pesticides, where they act as reducing and capping agents [39]. The actions and consequences of naturally occurring and chemically generated nanoparticles on plants and insect pests are different. NPs have a wide surface area, which allows them to circulate easily inside the system of a lepidopteran insect and link with other substances swiftly [41]. Research has shown that plant secondary metabolites in nanocarrier materials induce dyspepsia, break down the water-protection barrier, and ultimately kill insects [40]. In addition to protecting the active secondary metabolites and facilitating the regulated release of the chemical, the compounds' nanosized makes them more effective when used in small amounts as biopesticides [40]. Both the insecticidal efficacy and the shelf life were greatly enhanced by combining a specific plant extract with nano-silica [42]. It has been shown that the majority of terpene chemicals have antifeedant action. Alpha-pinene and linalool, two terpene chemicals, are combined with nano-silica to create a nanoformulations that has a longer shelf life of almost six months and increased efficacy (antifeedant activity) against *Spodoptera litura* and *Achaeajana* [43]. Because they share some of the essential properties of conventional synthetic pesticides—permeability, stiffness, crystallinity, biodegradability, and thermal stability—nano biopesticides can be used against pests both directly and indirectly through vectors. AgNPs biosynthesized from the medicinal plant *Piper betle* leaf extract demonstrated lower toxicity when tested against *Daphnia magna* for ecotoxicological studies. This confirmed that biologically synthesized AgNPs are safer than chemically synthesized AgNPs due to the formation of a protein core around the NPs during synthesis [43]. *P. betle* leaf extract was used to biosynthesize platinum (Pt) and palladium (Pd) nanoparticles, which were claimed to have similar environmentally benign properties [44].

Advantages of the use of nano-pesticides over conventional pesticides :

By creating target-specific nano insecticides, nanotechnology provides a platform for creating innovative, environmentally acceptable pest control formulations. Target-specific nanopesticides typically reduce pollutants and unwanted residues in final product while also increasing the efficacy of pesticides. Given that nanopesticides are made with high polymer materials that are sensitive to temperature, light, humidity, soil pH, and enzymes, they have a sluggish release and protective capability. These formulations enhance droplet adherence on plant surfaces while restricting the enhancement of target pesticide molecules' dispersion and bioactivity through AI. For these reasons, nanopesticides are more effective at managing pests than traditional pesticides. Because of their small size, nonapeptides are effective and

environmentally acceptable. When sprayed in the field, they improve plant surface adherence of droplets, wettability, and target absorption speed. Nano pesticides enable a sustainable production of any crop product since they lower the amount of chemicals used, improve plant protection, and decrease harmful residues [45].

Potential of nano-pesticides for agricultural pest control :

Plant-derived products (PDP) known as metabolites (terpenoids, flavonoids, and alkaloids) are made up of a complex mixture of chemicals that have a variety of effects on insects, including being toxic, repellent, antifeedant, oviposition deterrent, and insect growth regulators. Their mechanisms of action also vary [46]. The majority of research on botanicals or PDP is limited to the lab and has not been verified under field situations [47]. Indeed, locally accessible plant products (water extract, crude oils, seed cakes, etc.) and commercial formulations (emulsifiable concentrations, wettable granules, solvent-based products, active ingredient (AIs) have been introduced and routinely used by marginal and small farmers in developing countries [46]. As successful commercial formulations, these PDPs have been proven [48]. Essential oils (EO) with AI and allelochemicals have been shown to be more effective than PDP extracts [49]. Farmers use chemical pesticides in agriculture because of the quick “knockdown effect” on insect life stages and the availability of ready-made formulations. However, most of the time, these pesticides do not reach the target insect and are lost in the air, leached out in soil, or dissolved in water [36]. In addition, overuse or misuse of dosages without adhering to the correct recommended mixtures and repeated pesticide applications cause a number of problems, such as the emergence of insect resistance and undesired residues in commodities, which can be harmful to non-target organisms, humans, and the environment. Sunlight exposure would cause PDP to break down quickly, leaving less residue in the environment, less impact on creatures that are not targets, and less persistence [46]. The superior qualities of PDP nanotechnology over conventional chemical pesticides are supported by a controlled system that releases small-sized molecules at the site of action, minimizing residual issues, optimizing AI action, raising target specificity, and enhancing AI’s physicochemical stability and effectiveness [50]. It is known that nanopesticides are less harmful to animals and non-target organisms, and they also have less of an effect on human health and the environment [50].

Current issues and challenges :

Bio-efficacy:

Due to the low bioavailability of nanocarriers, in order to achieve the desired degree of pest control, both nanocarriers and AI must be provided in sufficient quantities. Therefore, it is necessary to determine the components’ bioavailability and durability in PDP-based nano insecticides [51]. According to da Costa *et al* [52], commercial products more precisely, the powder formulations confirmed superior UV stability compared to the encapsulated nanoformulations. The farmers actually prefer the liquid formulas. Therefore, increasing stability and persistence in solid formulations presents a challenge to the makers. In order to address this, Vrignaudet *al.* [53] proposed encapsulating hydrophilic molecules in polymer-

based nanoparticles and evaluating the hazards to creatures other than the target while boosting the uptake of AI to improve the bio-efficacy of the same.

Toxicity to plants :

When plants are exposed to sunshine during their development, the genotoxic and cytotoxic effects of azadirachtin in nanoformulations are diminished to an undetectable level [54]. Liu and Xing [55] reported that several forms of nanomaterials, including nanotubes, Al, ZnO, Zn, and multi-walled carbon, did not exhibit any negative effects on ryegrass and five distinct vegetables during the germination of seeds and the termination of root elongation. Psquoto-Stgliiani [50] found that when oleic acid and nanocapsules were applied to corn plants independently, oleic acid had a detrimental effect on physiological parameters whereas nanocapsules had no phytotoxic effect.

Toxicity to non-target organisms :

According to Carvalho *et al.* [56], the addition of neem oil to polymeric nanocarriers such as beta-cyclodextrin and poly- ϵ -caprolactone resulted in a decrease in toxicity towards parasitoids of the whitefly *Bemisia tabaci* and a parasitic wasp, *Encarsia formosa*. According to Kookana *et al.* [57], the concentration, particle size, number, and distribution of nanopesticides attached to the active component determine how hazardous plant-derived metallic nanoparticles are to microorganisms and earthworms. Because plant-derived metallic nanoparticles have different effects on the metalcore and capped phytochemicals, the release of metallic nanoparticles by microbes into the terrestrial environment is still up for debate [58]). Psquoto-Stgliiani *et al.* [50] verified that soil bacteria were unaffected for up to 300 days following exposure to nanocapsules containing poly- ϵ -caprolactone treated with neem oil. Furthermore, after three months of exposure to the formulation, the biomass of earthworms in soil treated with nanoformulations increased [59]. In the aquatic environment, engineered nanomaterials can influence surface charge, surface coating, and other physicochemical features [58]. Understanding the environmental fate of surface-capped plant-derived compounds and their participation in nano-specific activities is crucial since nano insecticides might not be environmentally friendly [58].

Toxicity to animals:

While nanomaterials are harmless when large, they frequently turn harmful when they get smaller, especially the non-biodegradable components (mainly metals). Mammals, plants, and soils may be poisonous due to its ability to accumulate and activate the immune system. Additionally, it has the ability to change how detoxifying enzymes function [60]. Because the ingredients in nanotubes are also able to stick to the body surface of insect pests, particularly in their hairs, the aluminosilicate used in them can stick to leaf surfaces and increase the stability of the product. Eventually, the ingredients will enter the animal's body and affect its physical functions [61]. Apart from direct intake, dermal contact or inhalation also can lead to nanoparticles entering the human body. In pest control operators, chronic exposure and sub-acute causes inhalation toxicity, which is very common, and sometimes

it leads to major health risks.

Future Perspectives:

Only synthetic insecticides used to control insect pests are relevant for evaluating the harm to the environment under the current regulatory processes. Planning the standards for the nano insecticides is necessary since nanoformulations have a variety of qualities [51]. Thorough completion of the risk assessment methodology is necessary for future applications, since, unlike synthetic pesticides, the results of research conducted for commercial bulk production may not apply to nanopesticides [62]. While recommending nanopesticides for insect pest management, more research on cost-benefit ratios, cost of nanopesticides, farmer recognition, safe and simple application techniques, availability, etc. would be desirable. Numerous studies on a range of agricultural crop pests revealed a high degree of confidence in the potential and bright future for nano pesticide-based formulations, such as nano-dispersions, nano-emulsions, and nanoparticles, to develop safer and more effective pesticide/biopesticide formulations for controlling insect pests. On the other hand, due to the unknown, unstandardized, and unexplored potential toxicity of nanopesticides, this research will undergo a thorough inspection by national and international safety regulators. Shape and dimension changes can occur during synthesis due to technique changes, and using these materials carries some risk. For this reason, the synthesis of nanomaterials is a crucial step. Since there are no set rules for using these formulations on nanomaterials, it is necessary to investigate the compounds' nature of toxicity to insects and plants before to using them. This means that research pertaining to risk assessment are crucial. In-depth research using a combination of analytical techniques that can identify the surface, detect the size, type, or shape, and measure the adjuvant and active chemicals emerging from the formulations is required before nano pesticides become popular for controlling pests. Nano pesticides will increase the profitability and environmental friendliness of agricultural output by lowering the quantity of chemicals needed for insect pest management. Although the progress in research and development is very slow, there seems to be an intense future for nano pesticides for insect pest management.

Conclusions :

A number of well-known pests, including as nematodes, mites, and insects, cause significant crop loss every year. The efficient application of nanopesticides to protect plants is a developing field of study that offers novel approaches to the formulation and delivery of active chemicals in nanoscale dimensions. Insect pests can be controlled more effectively and safely with the use of nanopesticide-based formulations, such as nano-dispersions, nano-emulsions, and nanoparticles. This field has a bright future ahead of it. The use of nanopesticides effectively for managing insect pests appears to have a lot of potential.

Abbreviations :

Active ingredients:	AIs
Artificial intelligence:	AI

economic injury level:	EIL
economic threshold level:	ETL
Essential oil:	EO
Integrated Pest Management:	IPM
Nanoparticles:	NPs
pesticide delivery system:	PDS
Plant-derived products:	PDP

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Application of Nanotechnology in Sustainable Aquaculture

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Abstract :

Aquaculture has emerged as one of the world's fastest-growing food industries in recent years, helping to meet the growing demand for animal protein. Nanotechnology offers a lot of potential in the aquaculture industry. For example, nanotechnology can be used to provide more effective fish feed for aquaculture species, Delivery of Nutraceuticals, Antifouling and preservation, Water purification, Aquaculture production and diseases control, Removal of heavy metals, Monitoring and treatment of fish diseases, Feed for fish, Antimicrobial and antibiofilm, Antifouling and preservation, The use of nanoparticles in aquaculture has shown promise in improving water quality, aquatic animal feeding, drug administration, disease diagnostics, and management, but there have been few studies done in a greener manner, which is developing a new horizon in the aquaculture era. Due to an innovative cleaning system and specific plates that clean each drop of water in each direction of water movement, silver nanoparticles are used to clean freshwater inhabited fish. Nanotechnology can help aquaculture and shrimp culture by enhancing disease management, feeding formulations, and biofouling control. This chapter emphasizes on applications of nanoparticles in aquaculture and fisheries.

Keywords : Antimicrobial, Drug administration, Disease diagnostics, Fish feed, Nanotechnology

Introduction :

Nanotechnology touches many science and technology fields, but it is one of the most promising technologies. Nanosciences and nanotechnologies have made rapid advances in recent years, opening up new possibilities for a variety of industrial and consumer sectors, including agriculture and allied industries. Nanotechnology is fast emerging as the new science and technology platform for developing and transforming agri-food systems and improving poor conditions.

Terrestrial and aquatic animals provide different protein types. Nevertheless, aquatic

protein options are preferred because of their health benefits and crucial compositional characteristics. Most countries consider fish an essential part of their diets [1]. It accounts for about 17% of the global population's intake of animal protein [2]. Fish play a vital role in nutrition, food security, and livelihoods. Fish provide the highest-quality protein sources and a wide range of other nutrients especially essential amino acids and fatty acids that our body needs, in addition to vitamins and other vital elements such as iodine and selenium which are not found in other crops or meat [3]. The minerals contained in fish include iron, calcium, zinc, iodine (from marine fish), phosphorus, selenium, and fluorine. These minerals are extremely bioavailable after being absorbed in by the body. Fish is an excellent source of vitamin B complex, and liver oil has a significant amount of fat-soluble vitamins A, E, K, and D, besides other vitamins like vitamin E, vitamin K, and vitamin C [1]. Fish oil contains polyunsaturated fatty acids (PUFAs) that can't be synthesized by the human body [1]. The major building blocks of our nervous system are omega-3 fatty acids such as eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA). With a growing world population, the demand for fish and fish products is expected to increase as they are considered the cheapest and most available animal proteins compared to other animal protein sources. In poor countries relying on staple foods (such as maize, rice, and other cereals), fish are considered a crucial component of nutrition. 150g of fish protein supplies approximately 50-60% of the daily protein requirement of an adult [1] when correcting an imbalanced calorie/protein ratio. Fish, especially oily fish, lowers coronary heart disease (CHD) mortality risk. In spite of this, the aquaculture sector faces uncertainty when it comes to its sustainability, as ever-increasing waste from aquaculture has negative effects both on aquaculture productivity and on the aquatic environment. [4] argue that nanotechnology is quickly emerging as the new forum for science and technology for agri-food systems' growth and transformation.

Applications of Nanotechnology in Aquaculture (Figure 1) :

Nanotechnology holds great promise for transforming agriculture and related industries, such as aquaculture and fisheries. It can offer fresh instruments for fish biotechnology, genetics, reproduction, aquaculture, and aquatic health, among other fields. Numerous riddles pertaining to animal health, productivity, reproduction, illness prevention, and treatment may be resolved with the use of nanotechnology instruments such as nanomaterials, nanosensors, DNA nanovaccines, gene delivery, and smart drug delivery, among others (Figure 1).

Delivery of nutraceuticals :

Nanotechnology has a wide range of applications in aquaculture and seafood. There is limited information available concerning marine species impact. There has been evidence that iron nanoparticles cause young carp and sturgeon to grow faster. A diet supplemented with nano-selenium was also shown to increase fish weight, relative growth rate, and antioxidant status, as well as the activity of glutathione peroxidase and muscle selenium concentrations in crucian carp (*Carassius auratus gibelio*) [5-6].

In addition, fish growth and performance that have been tested with the above mentioned nano materials have shown that nano level delivery of nutraceuticals can boost fish growth. In fish farming, silver nanoparticle-coated water filters may prevent fungal infections in rainbow trout by reducing exposure to silver nanoparticles directly in the water. However, silver nanoparticles directly in water have been found to harm young trout. It can be noted that fish health can be controlled through nanotechnology applications in aquaculture systems. This includes nanotechnology-based delivery of veterinary products to fish food using porous nanostructures, and nanosensors for pathogen detection in fish farming systems. Therefore, in a wide range of pond-ecosystem environments, nanomaterials have shown great potential [7].

Antifouling and preservation :

It is important to note that the above technology contributes to antifouling of fishing nets and aquaculture nets, antibacterial substances for aquaculture tanks, as well as new packaging materials for shipping seafood products, and new equipment for detecting seafood products shelf-life. It is evident from the above statement that various nanotechnology processes can preserve food quality.

Feed for fish :

A few studies have been conducted in the aquaculture industry on the use of nanoparticles, such as the rapid growth of young carp (*C. auratus gibelio*) after they were fed iron and selenium [8-9]. Using iron nanoparticles, Russian scientists recorded that young carp and sturgeon grew faster (30% and 24%, respectively). According to research, different sources of selenium (nano-Se and selenomethionine) supplemented to a basal diet increased the final weight of the animal, the relative gain rate of the animal, and the antioxidant status of the animal (*Carassius auratus gibelio*). Furthermore, nano-Se was more effective than organic selenomethionine at increasing muscle selenium levels. Similar to these results, fish growth and efficiency were assessed to be higher when these nutraceuticals were delivered at a nanoscale [10].

Antimicrobial and antibiofilm :

Nanoparticles such as silver, zinc oxide, copper oxide, and iron oxide have antimicrobial and antibiofilm properties. By immobilizing nanoparticles on fish aquarium surfaces and concrete water tanks, disinfection can be achieved. Two fundamental methods can be applied to immobilize nanoparticles on surfaces/walls: (a) physical association to form a multilayered thin film of nanoparticles, and (b) chemical connections necessitating coating the platform with linkage molecules.

Water purification :

One of the most crucial components needed for sustainable aquaculture is water treatment. The greatest significant health risk in the world today is water pollution, which is increasing as a result of improper disposal of waste from cities, businesses, and agriculture

as well as improper use of antibiotics and other synthetic chemicals in fisheries. This form of water degradation has an indirect impact on human health as well as aquatic animals, whose intake might result in various food-borne diseases. Direct effects include a reduction in the availability of pure groundwater. In addition, the presence of heavy metals and microbes in these waterways results in fish mortality and growth retardation, which causes a significant financial loss for the fishing business. Nanotechnology is widely used in aquaculture for water treatment, which creates a safe and favorable environment for fish to spawn. From this angle, the scientific community supports adsorption and photocatalysis as the most cost-effective and efficient methods of water filtration (**Table 1**).

Table 1: Potential applications of nanotechnology in water/wastewater treatment [11].

Applications	Examples of nanomaterials	Some of novel properties
Adsorption	CNTs/nanoscale metal oxide and nanofibers	High specific surface area and assessable adsorption sites, selective and more adsorption sites, short intraparticle diffusion distance, tunable surface chemistry, easy reuse, and so forth.
Disinfection	Nanosilver/titanium dioxide (Ag/TiO ₂) and CNTs	Strong antimicrobial activity, low toxicity and cost, high chemical stability ease of use, and so forth.
Photocatalysis	Nano-TiO ₂ and Fullerene derivatives	Photocatalytic activity in solar spectrum, low human toxicity, high stability and selectivity, low cost, and so forth.
Membranes	Nano-Ag/TiO ₂ /Zeolites/Magnetite and CNTs	Strong antimicrobial activity, hydrophilicity low toxicity to humans, high mechanical and chemical stability, high permeability and selectivity, photocatalytic activity, and so forth.

Aquaculture production and disease control:

Nanotechnology can improve aquaculture production and shrimp culture by improving disease control, feeding formulations, and biofouling control. By coating or painting nanostructures with metal oxide nanoparticles such as ZnO, CuO, or SiO₂, it is possible to monitor invertebrates (mussels and barnacles) and algae (seaweeds and diatoms). In order to achieve this, it is necessary to create a highly effective antifouling surface and increase antifouling control efficiency [5,12].

Removal of heavy metals :

The removal of heavy metals can be achieved effectively by using ligand-based nanocoatings, which have a high absorption propensity and are cost-effective. The bifunctional self-assembling ligand can be regenerated in situ by using the nanocoating media previously used to treat it. Several metal layers are bonded to a substrate in order to purify water using crystal clear technology [13]. Today, nanomaterials are commonly used to remove heavy metals from water and wastewater due to their high reactivity and large surface areas.

Metal oxide nanoparticles, such as nanosized ferric oxides, aluminum oxides, manganese oxides, cerium oxides, magnesium oxides, and titanium oxides, have a high surface area and affinity for heavy metal adsorption in aqueous systems. A hot topic has been the development of new technologies to synthesize metal oxide nanoparticles as well as the development of more practical applications, such as composite materials and granular oxides, for the evaluation of their heavy metal removal under various experimental conditions, or for determining the underlying mechanisms of metal removal based on mathematical models or analytical techniques, such as XAS and NMM [14-15]. Tanget *al.* [16] conducted a detailed study of the effects of humic acid and fulvic acid on heavy metal removal from aqueous solutions using various nanomaterials, primarily iron-based nanomaterials, carbon-based nanomaterials, and photocatalytic nanomaterials.

Chitosan nanoparticles are used as adsorbents to remove heavy metals. Several recent studies have investigated the removal of heavy metals from clays such as kaolinite, bentonite, and montmorillonite by chitosan nanoparticles. This is because clays, including chitosan and chitin, are intrinsically capable of doing so. It has been reported in recent years that nanochitosan-clay composites are used for metal ion removal [17-18]. Chitosan-magnetite nanocomposites have also been shown to remove heavy metals from aqueous solutions [19-20].

Nano-vaccines :

Aquaculture is regarded as most adversely affected by disease time to time resulting tumbling development and sustainability of the system. The conventional method of delivery of vaccine has some negative aspects due to the poor immune response of fish. To solve this problem nanoparticle can be applied for effective release and greater protection. Chitosan and Poly Lactide Glycolid Acid (PLGA) can be used as nanoparticles in vaccination study. These substances can be effectively used not only for bacteria but also in destruction of viral pathogen by their mild inflammatory response in fish and shrimp culture practices. In case of mass vaccination where the delivery of vaccine is a crucial factor, nanocapsule containing nanoparticles can effectively used to retain bioavailability for longer duration of time.

Nanoparticles as Fish Growth Promoter :

Nanoparticles containing different molecules are studied as growth enhancer. From the report it is found that young carp and sturgeon shows fast growth when iron containing nanoparticles are implied. The selenium (Se) containing nanoparticles are also assumed to

promoter of high final body weight gain and improve the antioxidant property. This is found to be very potent in Crussian carps. Moreover, more growth is found at the nanolevel application of neutraceuticals to fish. Since neutraceuticals are very much expensive their consistency is very important. In this case the nanoparticles not only improves the consistency but also found to be improved the taste of the aquafeed.

Nano-Tagging and Barcoding :

The barcod can be detected by the application of nanoscale components such as Radio Frequency ID (Rfid). This tag can be scanned by nanoparticle from very far and can able to detect the presence of any other particles on it automatically. This type of tag can also be used in fish to detect swimming behaviour, feeding behaviour and body metabolism status. In the processing and export industry nanobarcod can be used effectively to observe various aspects of delivery system. Apart from these, nanosensors in processing industry can able to detect any pathogen and temperature charge in the different products to retain their good aesthetic value.

Monitoring and treatment of fish diseases :

Nanotechnology for fish disease treatment Nanotechnology enables the observation, measurement, manipulation, and production of things at the nanometer level. The nanometer (nm) is a SI unit of 10^{-9} duration or one-billionth of a meter distance [21-22]. As a result of their small size, form, surface area, conductivity, or surface chemistry, these novel materials have been developed with specific physical or chemical properties. They have been applied to textiles, electronics, engineering, and medicine [23]. Nanomaterials are materials ranging from 1 to 100 nanometers [24]. The term nanomaterial has been developed from the Greek word “nano,” meaning “dwarf.”

Nanotechnology can therefore be considered an effective means of preventing and monitoring diseases and pathogens, as well as multiplying aquaculture benefits. Fish health applications of nanotechnology include developing antibacterial and antifungal surfaces using porous nanostructures and nanosensors in aquaculture systems for detecting pathogens, as well as nanodelivery of veterinary products and fish medicines through fish foods [25]. There is a higher utilization level of nano-trace elements than conventional inorganic trace elements, which are typically very limited in usage due to the fact that nano-trace elements reach the animal body through direct penetration [2, 25-26].

It may be possible to control viruses, bacteria, and fungi through nanomaterials, since they operate on the same scale as particle-infecting viruses [2, 25-26]. The use of chitosan-based wrapping for vaccines as a nano-encapsulation carrier for the treatment of bacteria and viruses causing fish diseases has been demonstrated, since nano-encapsulated materials are durable and can withstand high temperatures and acidic environments. The use of these nanomaterials is becoming increasingly critical for effective agriculture, particularly in the development of pathogen-free fish seedlings and shrimp or prawn post-larvae [25].

Nanotechnology is used in the aquaculture and marine industries, including water purification, pond sterilization, nanofeed for fish nutrition, and aquaculture disease

management. In the past, nanotechnology was used to treat water and breed fish. As a result of the application of nanotechnologies to aquaculture of seawater shrimp, it was found that the nanosystems were capable of improving water quality, decreasing water exchange rates, and increasing shrimp survival and yield [27]. As a result of nanotechnology application to seawater shrimp aquaculture, the nanodevice was found to reduce water exchange rates as well as increase water quality and shrimp survival rates as well as yield. The results showed an increase in fish survival rate of 100% and a reduction in both water nitrite and nitrate. Nitrite decreased to 1/4 of the control group. Nanonet treatment was the most effective nanotechnology device among several nanotechnology devices. As a result of nanotechnology, the pH of water has also been increased and the efficiency of water has been greatly improved [28].

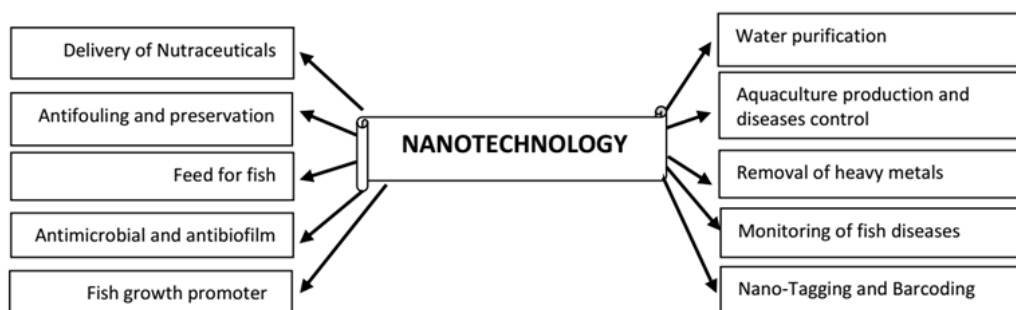


Figure 1: Application of nanotechnology in different fields of aquaculture

Future Perspective :

Nanotechnology certainly contributes to a significant role in aquaculture development and sustainability. So far, different nanotechnology-based systems have been employed to strengthen aquaculture and fishery pillars. However, there is a growing concern about the toxicity of NPs, excessive antibiotics and other synthetic compounds used in this discipline. Therefore, safe and eco-friendly approaches is inevitable. In this regard, recently natural bioactive compounds have attracted much attention, particularly curcumin which has shown a potent role in the fishery.

Aquaculture's efficiency is impacted by a multitude of factors, including the prevalence of sickness, synthetic pollution, and ecological concerns. The goal of scientists and researchers has always been to integrate aquaculture with any of the available technologies. At the same time, they considered that aquaculture combined with nanotechnology intervention would be the most effective way to address the aforementioned problematic factors. Aquaculture's future is protected by a variety of nanotechnology approaches and ideas, including nanosensors, nanoencapsulations, and many more, which help satisfy the demands of the expanding aquaculture industry.

It is imperative to synthesize safe and efficient vehicles for their encapsulation and

delivery. Pickering emulsion which is stabilized by solid particles (nutrients) instead of surfactants is used as a carrier for numerous compounds due to its non-toxic and long-term stable characteristics. But to date scarce evidence is available on utilization of Pickering emulsion in fishery thereby leaving a huge gap in fishery in terms of using this approach. Based on this information, we strongly recommend that future research be conducted in aquaculture and fishery to:

- Synthesize novel nanoparticle-based adsorbents and films for water purification from heavy metals and coliforms.
- Avoid excessive antibiotic and nanoparticle use with known toxicities.
- Identify new natural bioactive compounds that could be used against different diseases and as growth promoters.
- Incorporate nutrient-stabilized Pickering emulsions as delivery systems for the encapsulation and delivery of these compounds to enhance their efficacy and ensure their targeted and sustained release.

Conclusion :

Nanotechnology is still at an early stage in aquaculture. However, it may have the potential to solve most problems in aquaculture and fisheries with better technical innovation at different levels. Besides protein and fatty acids, fish also provides essential nutrients, like iodine and selenium, which cannot be found in other foods. People consider fish a valuable food source. According to research focusing on cultured fish species all over the world, they suffer from various infectious and non-infectious diseases as do most aquatic organisms because of water mixed with agricultural and industrial pollutants and intensive aquaculture. Furthermore, increased stress on fish leads to increased disease transmission. In recent years, fish diseases have attracted considerable attention. Despite the widespread expansion of fish culture and economic losses caused by infections in intensive culture systems, reliable information regarding the side effects of antibiotics and indirect treatments by synthesis and natural products is available. Classical methods of treatment and remedy have significant disadvantages. As a result, nanotechnology in aquaculture has become a comprehensive tool for solving many problems. These include disease diagnosis and treatment as well as water quality control, fish nutrition, environmental management, and others. Furthermore, this review discusses the role of selenium nanoparticles as an effective element in aquaculture. There is a promising path towards solving major aquaculture problems, such as disease control. This path may extend to diet formulation and other marine industries in the future, as demonstrated throughout this review. Despite being considered in aquaculture's infancy, nanotechnology may have the potential to mitigate most aquaculture and fisheries problems.

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Integration of Nanoscience in Forensic Science: Enhancing Sensitivity, Speed, And Accuracy

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Abstract :

This study aims to explore the contribution of nanoscience to forensic investigation, focusing on fingerprint enhancement, DNA analysis, forensic drug detection, trace evidence analysis, rapid toxin detection, forensic imaging, forensic document examination, and firearm and toolmark analysis. A comprehensive literature review is used in the study to investigate the applications of nanoscience in forensic investigation. The research design involves a qualitative analysis of scholarly articles, research papers, and scientific findings related to nanoscience and its integration into forensic science. The results underscore the various ways nanoscience can be applied in forensic investigations. The integration of nanoscience in forensic science has the potential to enhance the precision, sensitivity, and efficiency of forensic investigations, leading to more accurate and reliable results in criminal cases. The study emphasizes the significance of ongoing research and development in nanotechnology-based forensic techniques.

Keywords: Nanoscience, Forensic Science, Nano-Forensics, Fingerprint Enhancement, DNA Analysis, Toxin Detection

Introduction :

Nanoscience has numerous applications in medical, electronics, energy, materials science including the forensic science. Instead of incremental improvements to existing technologies, nanotechnology offers disruptive, gamechanging breakthroughs and innovations that can provide immediate answers and solutions to help our society, environment, and the planet [1].

Forensic science is also a multidisciplinary field that applies scientific principles and methods to the investigation of crimes and legal issues. It involves the collection, analysis, and interpretation of physical evidence to provide information for use in the legal system. Forensic scientists use their expertise in various scientific disciplines to uncover evidence, reconstruct events, and support criminal investigations. Advances in nanotechnology and

other scientific methods continually shape and improve the field of forensic science. The development of innovative polymers and nanomaterials, along with better instrumentation, has provided a sustainable answer to a variety of forensic difficulties [2]. Because of their different properties, the manufacturing of nanomaterials such as nanosensors, nanocodes, nano-films, nano-coatings, quantum dots, and composites has been enhanced. Nowadays, introduction of nanotechnologies in forensic science i.e. nano-forensics will significantly transform the processes of investigation by making them faster, more accurate, more effective, more sensitive, and easy to apply that explains the indisputable significance of this technologies [3,4]. The evolution of nanosensors replaces the large bulky instruments into a much smaller chip-based platform. This greatly aids in crime investigation and also to identify anonymous evidence in a shorter analysis method of investigation with higher accuracy and sensitivity for solving crimes [5].

Contribution of Nanoscience to Forensic Investigation :

Incorporating nanoscience into forensic science can improve the sensitivity, speed, and accuracy of numerous analytical techniques, hence boosting the capacities of forensic investigation in problem-solving. The following paragraphs highlight some of the important fields of forensic investigation that involve the application of current nanoscience approaches.

Nanoscience in fingerprint enhancement :

Fingerprints are one of only two human characteristics that are unique to an individual, the other being DNA. Fingerprint detection research is very useful in identifying criminals and giving evidence against them in court. Criminals are typically identifiable in fingerprint detection by the ridges and furrows of their fingers. Due to the simple sight of fingerprint ridges, latent fingerprint detection is the major method of detection used to identify criminals' fingerprints [6]. In the context of fingerprint analysis, nanotechnology can be applied in several ways to enhance the sensitivity, selectivity, and accuracy of fingerprint detection and identification. Here are some applications of nanoscience in fingerprint analysis.

Nanoparticle-based development agents :

Carbon black and aluminium flake have been used as latent fingerprint developers for many years because it was discovered that they worked well under specific conditions, but there has been little activity in developing engineered materials specifically designed for fingerprint development. The ideal fingerprint powder will adhere to the residues left by the finger, giving rise to the distinctive patterns that everyone recognises as a fingerprint, but will not adhere to anything else. To address this issue, nanotechnology is being used to engineer particles [7]. In fingerprint-detecting techniques, nanoparticles can be utilised as development agents. These nanoparticles can be engineered to adhere selectively to the components of the fingerprint residue, improving fingerprint visibility.

Nanomaterial-enhanced surfaces :

The use of nanomaterials, such as nanotubes or nanowires, on surfaces can enhance the

adhesion of fingerprint residues [8]. This can improve the quality and durability of fingerprint impressions, making it easier to analyze and compare prints. Nanostructured surfaces can be designed to interact more effectively with the chemical components of fingerprint residues. These substrates can increase the contrast and resolution of fingerprints, making it easier to distinguish and identify individual ridge patterns.

Nanotechnology in imaging techniques :

Nanotechnology has contributed to the development of advanced imaging techniques. For instance, scanning probe microscopy at the nanoscale allows for extremely high-resolution imaging of fingerprint details, enabling more accurate identification [9].

Quantum dots for tagging and labelling :

Quantum dots, nanoscale semiconductor particles, can be used as labels or tags in fingerprint analysis. Functionalized quantum dots can selectively bind to specific components of the fingerprint, making it easier to detect and analyze the details of the print [10].

Nanotechnology in spectroscopy :

Nanoscale materials have been integrated into spectroscopic techniques for fingerprint analysis. This allows for more precise identification of the chemical composition of fingerprint residues, aiding in the differentiation of prints from different sources [11]. Nanoparticles can be incorporated into various visualization techniques, such as fluorescent nanoparticles for enhancing contrast under certain lighting conditions. This can improve the visibility of latent fingerprints on different surfaces.

By integrating nanoscience into fingerprint analysis, forensic experts can potentially overcome challenges related to poor print quality, substrate variability, and environmental conditions, leading to more reliable and effective fingerprint identification methods. However, it's important to note that the field is continually evolving, and research in this area is ongoing to further improve the accuracy and efficiency of forensic fingerprint analysis.

Nanoscience in DNA analysis :

For crimes such as rape and murder, DNA identification is validated through the study of blood stains, sperm, and hair. Nanotechnology has made significant contributions to the field of forensic DNA analysis, offering improvements in sensitivity, speed, and accuracy. DNA collection and isolation is a critical step in the forensic analysis of crime scene materials. For forensic identifications, body fluids and skeleton remains require rigorous DNA polymerase chain reaction (PCR) separations. In forensic technology evaluations, iron nanoparticles, silica coated iron nanoparticles, and copper nanoparticles have been employed for DNA PCR separation [12]. Gold nanoparticles can improve thermal conductivity during PCR, leading to faster and more efficient amplification of DNA. In addition to application in PCR, applications of nanotechnology in DNA analysis within the forensic context can be summarized as follows :

- **Nanoparticle-assisted DNA extraction :** Nanoparticles can be functionalized to

selectively bind to DNA molecules, aiding in the extraction process. Magnetic nanoparticles, for example, can be used to isolate DNA from complex biological samples more efficiently.

- **Nanopore sequencing :** Nanopore sequencing involves threading DNA strands through a nanoscale pore, and changes in electrical currents as the DNA passes through can be used to determine the sequence. This technology allows for rapid and direct sequencing of DNA without the need for amplification [13].
- **Nanofluidic devices for DNA analysis :** Nanofluidic devices, which manipulate fluids at the nanoscale, can be employed for DNA analysis. These devices enable precise control of sample volumes, facilitating more efficient DNA separation, purification, and analysis [14].
- **Nanotechnology in DNA fingerprinting and DNA delivery :** Nanoscale materials and devices can be integrated into DNA fingerprinting techniques to enhance resolution and accuracy. This can improve the discrimination power of DNA profiling methods used in forensic investigations. Nanoparticles can be used to deliver DNA probes or markers to target cells, facilitating the tagging and identification of specific genetic sequences. This is particularly useful in situations where the collection of biological samples is challenging.
- **Nanoscale sensors for DNA detection :** Nanosensors can be designed to detect specific DNA sequences with high sensitivity. Functionalized nanoparticles can selectively bind to target DNA, leading to changes in electrical or optical properties that can be easily measured.

The integration of nanotechnology into forensic DNA analysis has the potential to revolutionize the field, providing forensic experts with advanced tools for more accurate and efficient DNA profiling and identification. As technology continues to advance, additional applications and improvements in nanotechnology-based forensic DNA analysis will likely emerge.

Nanoscience in Forensic drug detection :

Nanoscience plays a significant role in advancing forensic drug detection techniques. The application of nanotechnology in this field has led to the development of highly sensitive and specific methods for detecting and identifying drugs in forensic samples. Here are some ways in which nanoscience is employed in forensic drug detection:

- **Nanomaterials for sensing :** Nanoparticles can be functionalized to selectively interact with specific drug compounds. These functionalized nanoparticles can enhance the sensitivity and selectivity of detection methods [15]. Quantum dots are semiconductor nanocrystals that emit specific wavelengths of light. They can be used as labels for drug molecules, enabling highly sensitive detection in forensic samples.
- **Nanobiosensors and nanocomposites :** Carbon nanotubes and nanowires can be functionalized to selectively bind with target drug molecules. Changes in electrical

conductivity or other properties can then be measured to detect the presence of drugs. Nanocomposites, which are combinations of nanomaterials, polymers, and other substances, can be designed for specific drug detection applications, providing a high degree of sensitivity [16].

- **Lab-on-a-Chip Devices :** Nanoscale components in lab-on-a-chip devices enable precise control of fluids and enhance the efficiency of drug detection processes. These devices can analyze small volumes of samples rapidly and with high sensitivity [17].
- **Nanoparticle Imaging Agents :** Nanoparticles can be designed as imaging agents to visualize drug distribution in tissues and cells [18]. This is valuable for understanding the pharmacokinetics of drugs in forensic investigations

The integration of nanoscience into forensic drug detection methods has the potential to revolutionize the field by providing more accurate, rapid, and sensitive analyses. It allows forensic scientists to detect drugs at lower concentrations, making it particularly valuable in cases where trace amounts of substances need to be identified.

Nanotechnology in trace evidence analysis :

Nanotechnology has shown promise in advancing trace evidence analysis, which involves the identification and characterization of tiny or microscopic materials found at crime scenes. Trace evidence can include fibers, paint, glass, and other small particles that may be transferred during a criminal activity. Nanoparticles can be engineered to serve as tags or labels for trace evidence. These nanoparticles can be attached to specific materials, making them easier to detect and analyze under various imaging techniques [19]. Advanced imaging techniques at the nanoscale, such as scanning electron microscopy (SEM) or atomic force microscopy (AFM), allow for higher-resolution imaging of trace evidence. This helps forensic analysts better characterize and identify materials at the microscopic level. Nanomaterials, such as nanofibers or nanotubes, can be employed for efficient sample collection and extraction. These materials can have high surface areas and specific properties that enhance their ability to capture and retain trace evidence from surfaces [20]. Nanosensors can be designed to detect specific chemical compounds present in trace evidence. These sensors can be highly sensitive, allowing for the identification of even minute amounts of substances and aiding in the analysis of complex mixtures [21]. Nanotechnology can be applied in the development of paints and coatings with unique nanoscale features. These materials can serve as markers in trace evidence analysis, helping to link samples to specific sources [22]. Nanoparticles can also be employed in the analysis of gunshot residue. Specific nanoparticles can selectively bind to gunshot residue components, enabling a more accurate and sensitive detection of firearm discharge residues [23]. Nanomaterials can be integrated into preservation methods to improve the stability and longevity of trace evidence. This is crucial for maintaining the integrity of samples over time, especially in challenging environmental conditions [24].

The application of nanotechnology in trace evidence analysis holds great potential to improve the precision, sensitivity, and efficiency of forensic investigations, contributing to

more accurate and reliable results in criminal cases. Continued research and development in this area are likely to lead to further innovations in forensic science.

Rapid detection of toxins or poisons in biological samples:

Nanosensors play a crucial role in the rapid detection of toxins or poisons in biological samples. These nanoscale devices offer high sensitivity, specificity, and speed, making them valuable tools for applications in environmental monitoring, food safety, and clinical diagnostics. Nanosensors can be designed as biosensors with specific recognition elements (such as antibodies or aptamers) attached to nanomaterial surfaces. When the target toxin binds to the recognition element, it induces a detectable signal, allowing for rapid and selective toxin detection [25]. Nanoparticles can be incorporated into immunoassays to enhance their sensitivity [26]. These nanoparticles can carry enzyme labels or other signaling molecules, amplifying the detection signal in the presence of toxins, thus improving the limit of detection. Carbon nanotubes (CNTs) can be functionalized to detect specific toxins. Changes in the electrical conductivity of CNTs upon toxin binding can be measured, providing a rapid and label-free detection method. Nanofiber-based sensors can be designed for toxin detection by incorporating specific recognition elements. These nanofibers provide a large surface area for interaction with toxins, allowing for rapid and efficient detection. Nanosensors are well-suited for integration into portable and point-of-care testing devices. These devices enable rapid, on-site detection of toxins, making them valuable tools for emergency response, field testing, and resource-limited settings [27].

The application of nanosensors in toxin detection not only improves the speed and sensitivity of analytical methods but also facilitates the development of portable and user-friendly devices for rapid on-site testing in various applications, from environmental monitoring to clinical diagnostics.

Nanotechnology in forensic imaging:

Nanotechnology has the potential to revolutionize forensic imaging by enhancing the sensitivity, resolution, and specificity of imaging techniques used in forensic investigations. Here are some applications of nanotechnology in forensic imaging :

- **Nanoscale Imaging Agents :** Nanoparticles can be engineered to act as imaging contrast agents. These nanoparticles, such as quantum dots or super paramagnetic iron oxide nanoparticles, can be functionalized to target specific tissues, cells, or biomolecules, improving the visibility of forensic evidence in imaging modalities like magnetic resonance imaging (MRI) or fluorescence imaging [18].
- **High-Resolution Scanning Electron Microscopy (HR-SEM) and Atomic Force Microscopy (AFM) :** Nanotechnology can contribute to improvements in scanning electron microscopy (SEM) by enhancing the resolution of images [28]. Advanced nanomaterials may be used as substrates or coatings to improve sample preparation and increase the quality of SEM images, providing detailed information about the microscopic structure of forensic samples. AFM, also a high-resolution imaging

technique, can benefit from nanotechnology by utilizing nanoscale tips for imaging surfaces. This enables forensic scientists to capture detailed three-dimensional information about the topography of samples, such as tool marks or surface textures.

- Nanostructure-Assisted Fluorescence, Infrared and Multimodal Imaging :** Nanostructure-assisted fluorescence, infrared and multimodal imaging are techniques that use nanomaterials to enhance the contrast, resolution and sensitivity of different imaging modalities, such as fluorescence, infrared, magnetic resonance, computed tomography, photoacoustic and photothermal imaging. Nanomaterials can act as imaging agents, probes, sensors, or platforms that can interact with light or other electromagnetic waves in the near-infrared (NIR) region, which is optimal for biological imaging due to its low absorption and scattering by tissues. Nanomaterials can also be functionalized with different biomolecules, such as antibodies, peptides, drugs, or genes, to achieve specific targeting, delivery, or therapy of diseases, such as cancer, inflammation, or infection. Fluorescent nanoparticles can be employed to enhance the contrast and sensitivity of fluorescence imaging. Functionalized nanoparticles can selectively bind to specific forensic markers, making them highly visible under fluorescence microscopy. Nanomaterials with unique infrared properties can be utilized to enhance infrared imaging techniques. This is particularly useful for the detection and characterization of specific materials, such as drugs or biological fluids, in forensic samples. Nanotechnology enables the development of multimodal imaging agents that can be detected by multiple imaging techniques. This integration of different imaging modalities provides complementary information, enhancing the overall diagnostic capability in forensic investigations [29].
- Nanoparticle-Based X-ray Contrast Agents :** Nanoparticle-based X-ray contrast agents are nanomaterials that can enhance the contrast and resolution of X-ray imaging modalities, such as computed tomography (CT), by increasing the X-ray attenuation of the target tissues or organs. Nanoparticle-based X-ray contrast agents can be made from various heavy metals, such as gold, silver, bismuth, gadolinium, and lanthanides, that have high X-ray absorption coefficients. They can also be functionalized with different ligands, such as antibodies, peptides, and drugs, to achieve specific targeting, delivery, or therapy [30]. These contrast agents can be designed to selectively accumulate in specific tissues or structures, aiding in the visualization of anatomical details or foreign objects.

The integration of nanotechnology into forensic imaging not only improves the capabilities of existing techniques but also opens up new possibilities for more accurate and detailed analyses in forensic investigations. Continued research and development in this field are likely to lead to further innovations and advancements in forensic science.

Nanotechnology in document examination :

Nanotechnology can be applied in document examination within the field of forensic science to enhance the analysis and authentication of documents. Here are some potential

applications of nanotechnology in document examination:

- **Nanoparticle-Based Inks and Nanoparticle Tracers for Ink Analysis :** Nanoparticles can be used to create inks with unique optical or chemical properties. These inks can be applied to documents for security purposes, making them difficult to replicate. The use of nanoparticle-based inks can enhance the traceability and authentication of documents. Nanoparticles can also serve as tracers in ink analysis. Different inks may contain unique nanoparticle compositions that can be identified using advanced analytical techniques. This can help in distinguishing between inks used in different parts of a document [31].
- **Nanoscale Imaging for Forgery Detection :** Advanced imaging techniques at the nanoscale, such as atomic force microscopy (AFM) or scanning tunneling microscopy (STM), can provide high-resolution images of document surfaces [18]. This enables forensic experts to detect subtle changes, alterations, or additions made at the nanoscale, aiding in the identification of forgeries. Nanotechnology allows for the incorporation of nanomaterials, such as quantum dots or nanobarcodes, into documents as security features. These features can be invisible to the naked eye but easily detectable using specialized equipment, making counterfeiting more difficult.
- **Nanoscale Tagging for Document Authentication :** Nanoscale tagging for document authentication is a technique that uses nanomaterials, such as nanoparticles, nanowires, or nanofibers, to create unique and unclonable patterns or codes that can be embedded in documents, such as banknotes, passports, or certificates, to prevent counterfeiting or forgery. Nanoscale tagging can provide high security and reliability, as the nanomaterials have diverse and tunable properties, such as optical, magnetic, or electrical, that can be detected and verified using different methods, such as microscopy, spectroscopy, or scanning [32].
- **Nanoparticle-Based Watermarks and Forensic Ink Dating :** Nanoparticle-based watermarks and forensic ink dating are two applications of nanotechnology in the field of document security and authentication. Nanoparticle-based watermarks are invisible patterns or codes that can be embedded in documents, such as banknotes, passports, or certificates, to prevent counterfeiting or forgery. Nanoparticle-based watermarks can be made from nanomaterials that have different optical, magnetic, or chemical properties, such as fluorescence, plasmon resonance, or catalysis, that can be detected and verified using different methods, such as microscopy, spectroscopy, or scanning [19]. Forensic ink dating is a technique that can determine the age of a document by analyzing the chemical composition and degradation of the ink used to write or print on it. Nanoparticles can be used as tracers or markers that can be added to the ink to provide a unique signature or code that can be identified and measured using different methods, such as chromatography, mass spectrometry, or Raman spectroscopy [33, 34].

The integration of nanotechnology in document examination enhances the forensic capabilities to detect forgeries, authenticate documents, and analyze various features at a more detailed level. As technology continues to advance, nanotechnology may offer even

more sophisticated tools for document forensics.

Integration of nanotechnology into firearm and toolmark analysis :

Firearm and toolmark analysis are the examination of the microscopic marks left by firearms or tools on objects. When a firearm is fired it leaves marks on the bullet and cartridge case. To determine if a bullet or cartridge case found at a crime scene was shot by a certain firearm, forensic firearm and toolmark examiners compare these marks. The marks made by screwdrivers, pry bars, and bolt cutters are compared in a manner similar to this. Nanotechnology has the potential to enhance the capabilities of firearm and toolmark analysis in forensic investigations [35]. This field involves the examination of ballistic evidence, such as bullets, cartridge cases, and toolmarks, to link them to specific firearms or tools. Advanced imaging techniques at the nanoscale, such as scanning electron microscopy (SEM) or atomic force microscopy (AFM), can provide higher-resolution images of toolmarks and firearm surfaces. This can assist in the identification of unique microscopic features and improve the accuracy of comparisons. Some of possible applications are highlighted below:

- **Nanomaterial Coatings for Toolmark Preservation :** Nanotechnology can be applied to develop coatings that enhance the preservation of toolmarks on surfaces. These coatings may protect microscopic features, preventing degradation over time and improving the quality of the evidence. Nanomaterial coatings are thin layers of nanoscale particles or structures that can modify the surface properties and characteristics of a substrate. Nanomaterial coatings can have various applications, such as enhancing the corrosion resistance, wear resistance, hardness, thermal stability, optical performance, and electrical conductivity of the substrate [36]. Toolmark preservation is important for the accuracy and reliability of toolmark analysis, as the toolmarks can be altered or degraded by various factors, such as environmental conditions, corrosion, wear, and tampering. Nanomaterial coatings can help to protect the toolmarks from these factors by forming a barrier layer on the surface of the tool marked object. Nanomaterial coatings can also enhance the contrast and visibility of the toolmarks, especially on dark or reflective surfaces, by changing the optical properties of the surface [37].
- **Nanostructured Materials for Bullet Identification :** Bullet identification is currently based on the subjective judgment of the examiner, who uses a comparison microscope to visually inspect the similarities and differences between the bullet marks. This method has been criticized for its lack of objectivity, reliability, and validity [38]. Therefore, there is a need for more scientific and quantitative methods to improve the accuracy and confidence of bullet identification. One of the potential uses of nanostructured materials in forensic science is to enhance the bullet identification, which is the examination and comparison of the bullets and their casings to determine the source and origin of the firearm [39]. Nanostructured materials with unique properties can be incorporated into bullets to create identification markers. These markers, which may be invisible to the naked eye, can serve as unique signatures to link a bullet to a specific firearm.

- **Nanoparticle Tracers for Gunshot Residue Analysis and Detection :** Nanoparticle tracers are nanoscale particles that can be added to the gunpowder or primer of a firearm cartridge to produce a unique and detectable signature of gunshot residue (GSR). Nanoparticle tracers can help to improve the forensic analysis and identification of GSR by providing more information and features for comparison, such as size, shape, composition, and fluorescence [40]. These tracers can be applied to firearms, ammunition, or hands to aid in tracking the presence of gunshot residue, enhancing the investigative process. Nanoparticles can be employed in the development of highly sensitive sensors for the detection of gunshot residue. This can contribute to more rapid and accurate analyses of crime scenes involving firearms.

Firearm and toolmark analysis is currently based on the subjective judgment of the examiner, who uses a comparison microscope to visually inspect the similarities and differences between the marks. This method has been criticized for its lack of objectivity, reliability, and validity [41]. Therefore, there is a need for more scientific and quantitative methods to improve the accuracy and confidence of firearm and toolmark identification. The integration of nanotechnology into firearm and toolmark analysis has the potential to improve the accuracy, efficiency, and reliability of forensic investigations in this field. Continued research and development in nanotechnology are likely to lead to further innovations that enhance the capabilities of forensic scientists in linking ballistic evidence to specific firearms or tools.

Conclusions :

The research review highlights the diverse applications of nanoscience in forensic investigation, including the use of nanoparticles in fingerprint enhancement, nanotechnology-assisted DNA analysis, forensic drug detection, trace evidence analysis, rapid toxin detection, and forensic imaging. The integration of nanoscience in forensic science has the potential to enhance the precision, sensitivity, and efficiency of forensic investigations, leading to more accurate and reliable results in criminal cases. The study underscores the significance of ongoing research and development in nanotechnology-based forensic techniques.

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Recent advancements in Nano Composites based on Graphene Oxide (GO) for the Removal of Heavy Metal Ions from Wastewater

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Abstract :

The global challenge of a shortage in fresh water resources, brought about by heavy metal contamination, has serious implications for both environmental protection and human health. Consequently, there is a pressing need to devise and create materials that are efficient, cost-effective, and environmentally friendly for detecting and removing heavy metal ions. In recent years, various advanced methods, such as nanomaterials, polymers, porous materials, and biomaterials, have been employed for the removal of toxic ions. These materials showcase real-time detection capabilities, excellent removal efficiency, anti-interference properties, quick response times, high selectivity, and low limit detection. This book chapter delves into recent investigations and techniques centred around graphene oxide nanocomposites for the adsorptive removal of heavy metal ions from wastewater.

Keywords : Nanocomposites, graphene oxide, heavy metal, toxicity

Introduction :

Water constitutes a fundamental requirement for all living organisms on Earth, and envisioning life without it is inconceivable. Our bodies, in fact, consist of 70% water. Despite its abundance in nature, water is a nonrenewable resource, and not all of it is safe for consumption. The swift advancements in industrialization, urbanization, and the use of chemical compounds in various sectors pose a threat to the quality of potable water [1]. The deterioration of water quality is often attributed to the presence of heavy toxic metals [2]. Such contamination can stem from either natural processes or human activities, with the latter being a major contributor [3-5].

Heavy metal contamination, originating from activities like metal plating, mining operations, metal finishing, welding, alloy manufacturing, and agricultural practices, poses significant environmental and health risks [6-7]. These metals are non-biodegradable and accumulate in living organisms, leading to severe health issues. For instance, the infamous

Minamata disease in Japan was a result of the distribution of harmful Hg (II) ions in aqueous solutions, causing mental and neurological disorders in humans. Cu (II) ions, found in contaminated water, interfere with proteins and enzymes in the body, resulting in gastrointestinal problems, osteoporosis, and various diseases, including Alzheimer's disease [8-9].

To address the issue of heavy metal contamination in water, several methods are employed, including filtration [10], adsorption [11], ion exchange [12], and membrane separation [13]. Among these, adsorption-based removal methods are highly favored due to their cost-effectiveness, simple operating systems, high efficiency, and adaptability for large-scale water and wastewater treatment operations [14-15]. Notably, graphene and its derivatives, graphene oxide (GO) and reduced graphene oxide (r-GO), have gained attention for their advanced thermal, mechanical, and optical properties in removing heavy metals from wastewater. This discussion focuses specifically on the utilization of graphene oxide-based nanomaterials for the effective removal of heavy metal ions from water.

Sources of contamination and human toxicity:

Heavy metals are elements naturally occurring in nature, characterized by higher atomic weight and density than water. Due to their elevated toxicity, non-biodegradable nature, and persistent characteristics, they are classified as pollutants. The proliferation of heavy metals in the environment, air, soil, and water is attributed to diverse industrial, domestic, agricultural, technological, and medical applications. Pb^{2+} , Cd^{2+} , Cr^{3+} , Cr^{6+} , Ni^{2+} , Fe^{2+} , Fe^{3+} , Hg^{2+} , Co^{2+} , Cu^{2+} , As^{3+} , and Zn^{2+} stand out as prevalent heavy metal ions found in water contamination. Industries, including tanning, chrome plating, coal, metalworking, plastic processing, food production, ceramic applications, and agrochemical waste, contribute significantly to water contamination with heavy metals, releasing ions such as As (III), Cd (II), Pb (II), Cr (VI), Ni(II), Hg (II), and Cu (II) before proper treatment [16]. Domestic waste discharge into rivers is another source of heavy metal ion contamination, along with agricultural practices involving fertilizers, herbicides, pesticides, and excessive salt use [17]. The water cycle is further polluted by fertilizers, nutrients, and pesticides containing Cd (II), Pb (II), Cr (VI), As (III), Hg (II), Ni (II), Cu (II), and Zn (II) [18]. Dumping waste in landfills is a significant contributor to water pollution, as hazardous leachate containing Cd (II), Cr (VI), Cu (II), Pb (II), Ni (II), and Zn (II) ions can contaminate both surface and groundwater. Additionally, natural phenomena such as volcanic eruptions, forest fires, rock weathering, biogenic processes, and marine salt release can introduce heavy metals into water sources.

Heavy metals found in water contamination pose toxicity to all living organisms. Even in trace amounts, elements such as As, Cd, Pb, Hg, and Cr can be detrimental to human health. These metals enter the human body through food and water, disrupting essential biological processes. Toxic metals like Cr (VI), Pb (II), and As (III) negatively impact the body by inhibiting enzyme activities [19]. Cr (VI) specifically damages proteins, DNA, and RNA, reducing the intercellular space of cell membranes [20]. The health consequences of heavy metal exposure range from mild to severe, including symptoms like loose intestines, anxiety, lung disease, fatigue, kidney problems, stomach issues, skin infections, neurological problems, and the potential for malignant growth [21]. The infamous Minamata disease in

Japan resulted from Hg (II) poisoning, illustrating the severe impact heavy metals can have on the human body [22]. Additionally, heavy metals like Cd(II) contribute to Itai-Itai disease and cause reproductive, cardiovascular, pulmonary, and gastrointestinal disorders [23]. Pb (II) damages the central nervous system [24], while high toxicity of As (II) can lead to skin diseases, cardiovascular diseases, diabetes, and even cancer [25]. Ni (II) toxicity is associated with various health issues such as dermatitis, asthma, and respiratory tract cancer [26]. **Table 1** illustrates the primary sources and adverse impacts of these metallic ions, along with the permissible limits recommended by the World Health Organization (WHO).

Table 1: Sources and side-effects of heavy metal ions [27-39]

Heavy Metal Ions	Permissible limit in water (mg L ⁻¹) (WHO)	Sources	Side Effects (Humans)
Pb (II)	0.05	Paint, gasoline, smelting, lead containing pipes, battery, pigments, arm industry and printing of books.	Inhibit enzyme activity, damages kidney and nervous system, responsible for miscarriage in pregnant woman, male infertility and are carcinogenic.
Cd (II)	0.005	Fossil fuel, agricultural activity, cigarette smoke and sewage sludge.	Causes reproductive, cardiovascular, pulmonary, and gastrointestinal disorders and cancer
As (III)	0.05	Wood preservatives, alloying, semiconductors, glassware and pesticides.	Responsible for diabetes, neurological disorder, cardiovascular disease, lung and bladder cancer.
Cr (III) and Cr (VI)	0.05	Steel industries, construction materials, transportation, electroplating, paint, dyes, plastics, leathers and photographs.	Damage Protein, DNA and RNA, Causes vomiting, nausea, indigestion and hemorrhage.
Fe(II) and Fe(III)	0.1	Rock weathering, minerals, sewage, land fill leachate, acid mine drainage and engineering industries.	Responsible for diabetes, heart problem and retard growth in children.
Cu(II)	1.5	Rock weathering, animal manure, fossil fuel burning, bubble bursting, and aerosols.	Causes depression, anxiety, allergies, anorexia and fatigue
Zn(II)	5.0	Electroplating, vegetable fat producing industries, paint, galvanizing plants and porcelain enameling.	Causes dizziness and stomach pain.

Heavy Metal Ions	Permissible limit in water (mg L ⁻¹) (WHO)	Sources	Side Effects (Humans)
Ni(II)	0.1	Petroleum, metallurgy, petroleum procedure, nickel-cadmium batteries and plating.	Causes gastrointestinal problems, respiratory tract problems, heart diseases and cancer.
Hg(II)	0.001	Volcanic eruptions and naturally occurs as mercury salts.	Create reproductive, gastrointestinal, immunity problem. Damage nervous, excretory and cardiovascular system.
Co(II)	0.01	Mining, manufacturing units of electronic devices, petroleum industries and electroplating.	Cause reproductive, cardiovascular, pulmonary, and gastrointestinal disorders

GO based nanocomposites for adsorptive removal of different heavy metal ions:

Graphene-based materials have garnered significant attention for the adsorption-based removal of heavy metal ions. The exceptional properties of graphene oxide (GO) and its derivatives, such as a large surface area, sufficient functional groups, and robust mechanical strength, render them effective materials for wastewater decontamination containing heavy metal ions. Composite materials comprising GO and various nanomaterials have been acknowledged for their outstanding capability to remove heavy metal ions. Table 2 presents noteworthy recent investigations focusing on the application of graphene oxide-based nanocomposites for the adsorptive removal of heavy metal ions.

Table 2: Nanocomposites based on graphene oxide utilized for the adsorptive removal of various heavy metal ions

GO based Nano Composites	Heavy Metal Ions	% Adsorption/ Absorption Capacity	References
GO			
Chitosan, Sodium Alginate and GO @ Fe ₃ O ₄ gel beads	Cr(VI)	92.8%	Ref 40
MnO ₂ /Fe ₃ O ₄ /graphene oxide	Cu ²⁺ , Cd ²⁺ and Pb ²⁺	55.96, 86.28 and 189.04mg/g respectively	Ref 41
and MnO ₂ /Fe ₃ O ₄ /graphene	Cr(VI)	193.1 mg g ⁻¹	Ref 42
Poly (N-vinylcarbazole) PVK-GO polymer	Pb ²⁺	887.98 mg g ⁻¹	Ref 43

GO based Nano Composites	Heavy Metal Ions	% Adsorption/ Absorption Capacity	References
Polyamidoxime (PAO) brush grafted graphene oxide (GO)	Pb(II), Ag(I), Cu(II), and Fe(III),	116.7 mg g ⁻¹ , 258.6 mg g ⁻¹ , 192.2 mg g ⁻¹ and 167.9 mg g ⁻¹ respectively	Ref 44
Graphene oxide/TiO ₂ (GO/TiO ₂) GO-Si Nanocomposite	Rhodamine B (RB), Cd(II) and Zn(II) ions	11.299, 14.326 and 11.326 mg/g, respectively.	Ref 45
Nano-Fe ₃ O ₄ /triisopropanolamine functionalized graphene oxide composites (TI/GO@Fe ₃ O ₄)	Pb ⁺² and Hg ²⁺	347.2 and 266 mg/g, respectively	Ref 46
ZnO Nanorod-Reduced Graphene Oxide (ZnO NR-rGO)	Pb ⁺²	461 mg/g	Ref 47
Terpyridine-graphene oxide hybrid (GO-Tpy and GOh-Tpy)	methylen blue (MB), methyl orange (MO) and rhodamine B (RhB) dyes, Cu(II) and Co (II) ions	93.9, 88.08 and 75.3% respectively for dyes and 98.6 90.1% respectively for the ions	Ref 48
Graphene oxide/PVC composite papers functionalized with p-Phenylenediamine (PVC/PPD/GO)	Ni(II), Zn(II) and Co(II)	462, 421 and 336 mg g ⁻¹ respectively	Ref 49
Reduced graphene oxide supported ZnO/CdS heterojunction (CdS/ZnO6:5/RGO6)	Pb(II)	44.80 mg g ⁻¹	Ref 50
Few layered magnetic graphene oxide (FLMGO)	Cr(VI)	93.2%	Ref 51
Graphene Oxide/Polyvinyl Alcohol/Fe ₃ O ₄ Nanocomposite (GO/PVA/Fe ₃ O ₄)	Cd(II) and Cu(II)	401.14 and 1114.22 mg·g ⁻¹ respectively	Ref 52
Graphene oxide iron modified clinoptilolite-based composites GO-FeZ (GOFeZA and GOFeZB)	Co(II)	373.37 mg·g ⁻¹	Ref 53
Magnetic composite graphene oxide/Fe ₃ O ₄ /glucose-COOH (GO/Fe ₃ O ₄ /GC)	As ⁺⁵	557.86 and 554.64 μ g/g respectively	Ref 54
MnFe ₂ O ₄ /GO nanocomposite	U(VI)	390.70 mg g ⁻¹	Ref 55
Ultrathin Graphene Oxide-Based Nanocomposite	Pb ²⁺ and neutral red (NR)	98.8% and 94% respectively	Ref 56
In their investigation, Mandal and Chakraborty explored the elimination of Cr(VI) by	Na ⁺ , Mg ²⁺ , Ni ²⁺ and Pb ²⁺	99%, 98%, 82% and 79% respectively (rejection explored)	Ref 57

utilizing graphene oxide (GO) derived from graphite [40]. Their findings indicated that GO efficiently adsorbed Cr(VI) at a pH of 4 with an adsorption dose of 0.01g. The study unveiled that the adsorption process adheres to pseudo-second-order kinetics, and the reaction is characterized as endothermic and entropy-driven. Furthermore, their research highlighted that the reaction rate exhibited an upward trend with an increase in temperature within the range of 30 to 70 degrees Celsius. Interestingly, they observed that the removal efficiency of GO decreased as the initial concentration of Cr(VI) increased.

In the research conducted by Zhengguo Wu and co-workers on the removal of Cu^{2+} , Cd^{2+} , and Pb^{2+} from water using Chitosan (CMS), Sodium Alginate (SA), and $\text{GO@Fe}_3\text{O}_4$ gel beads [41]. It was observed that the adsorption capacity of the gel beads improved with the addition of GO. The study emphasized the significant influence of pH on the adsorption process, revealing that, with an initial concentration of 100 mg/L, the capacity of CMS/SA/ $\text{GO@Fe}_3\text{O}_4$ gel beads increased initially and then decreased as the pH varied from 1 to 7. The optimal adsorption of Cu^{2+} , Cd^{2+} , and Pb^{2+} occurred at pH values of 5, 6, and 5, respectively. The kinetic study indicated that the adsorption of Pb^{2+} ions was rapid within the first 4 hours, following a pseudo-second-order model, suggesting a chemical adsorption process. The maximum adsorption capacity of the gel beads for Pb^{2+} ions was observed at 30°C. In contrast, the adsorption of Cu^{2+} and Cd^{2+} ions increased with rising temperature, indicating an endothermic nature. Additionally, the research highlighted that the gel beads exhibited a higher capacity for Pb^{2+} compared to Cu^{2+} and Cd^{2+} ions. The study also assessed the reusability of the gel beads, revealing that the adsorption efficiency remained above 80% of its original capacity even after five cycles.

Yan Liu et al investigated the adsorption capabilities of $\text{MnO}_2/\text{Fe}_3\text{O}_4$ /graphene oxide and $\text{MnO}_2/\text{Fe}_3\text{O}_4$ /graphene [42]. Their findings indicated that the adsorption capacity of $\text{Fe}_3\text{O}_4/\text{GO}$ surpassed that of $\text{Fe}_3\text{O}_4/\text{rGO}$. Furthermore, $\text{MnO}_2/\text{Fe}_3\text{O}_4/\text{GO}$ (at pH 5) exhibited a higher adsorption capacity than $\text{MnO}_2/\text{Fe}_3\text{O}_4/\text{RGO}$ (at pH 5), suggesting that organic functional groups on GO, such as hydroxyl, carboxyl, and carbonyl groups, served as active centres for Cr^{6+} adsorption. The study highlighted the remarkable absorption capacity of $\text{MnO}_2/\text{Fe}_3\text{O}_4/\text{GO}$ within the pH range of 2 to 5, with initial concentrations ranging from 50 mg L⁻¹ to 300 mg L⁻¹. Analysing the adsorption kinetics, the researchers observed that an increase in temperature resulted in an enhanced adsorption capacity for $\text{MnO}_2/\text{Fe}_3\text{O}_4/\text{GO}$, and the adsorption process could be optimized following the pseudo-second-order kinetic model. Importantly, the study demonstrated that the adsorption capacities of $\text{MnO}_2/\text{Fe}_3\text{O}_4/\text{GO}$ for Cr^{6+} remained above 84% even after four cycles of reuse, indicating the reusability of these nanocomposites.

Yvonne Ligaya and co-workers investigated the adsorptive behaviour of a Poly (N-vinylcarbazole) PVK–GO polymer nanocomposite for the removal of Pb^{2+} from aqueous solutions [43]. The PVK–GO nanocomposite solution was created through a mixing process and characterized using XPS, ATR-IR, and FT-IR Spectrometer. The adsorption study encompassed various parameters such as dosage, contact time, pH, and concentration. Their findings indicated that, at pH 7 with a 90 minute contact time, the PVK–GO polymer nanocomposite exhibited a maximum adsorption capacity for Pb^{2+} of 887.98 mg g⁻¹. The

research demonstrated that the percentage removal of Pb^{2+} ions increased proportionally with the amount of GO in the PVK–GO polymer, suggesting the pivotal role of GO in removing heavy metal ions. The nanocomposite PVK–GO, comprising 90:10 wt% of PVK–GO, proved to be the most effective in removing heavy metal ions.

The synthesis and characterisation of polyamidoxime (PAO) brush grafted graphene oxide (GO) was done by Dejian Liu and co-workers [44]. The adsorption capacity of PAO brush grafted GO exhibited a pH-dependent behavior. At the optimum pH of 5, the PAO brush grafted GO demonstrated maximum adsorption capacities of 116.7 mg g^{-1} for Pb (II), 258.6 mg g^{-1} for Ag (I), 192.2 mg g^{-1} for Cu (II), and 167.9 mg g^{-1} for Fe (III). The effectiveness of PAO brush grafted GO was observed at an initial concentration of 100 ppm with a contact time of 30 minutes. The researchers suggested that the outstanding adsorption performance of PAO brush grafted GO could be attributed to the abundant functional groups made accessible to metal cations due to the polymer brushes swelling in water.

Joshi et al conducted an investigation into the synthesis, characterization, and applications of Graphene oxide/ TiO_2 (GO/TiO_2) for the adsorptive removal of (RB), Cd(II), and Zn (II) ions from water [45]. The GO/TiO_2 adsorbent underwent characterization using P-XRD, FT-IR, FE-SEM, and EDX methods. The optimal conditions for maximum adsorption efficiency of (RB), Cd(II), and Zn (II) ions were identified at a contact time of 60 minutes, a dosage of 1g, and a pH of 6, owing to the presence of very few or non-protonated binding groups available for adsorption. Their results demonstrated that the highest removal percentages for (RB), Cd(II), and Zn (II) ions were achieved at pH 6, measuring 81.38%, 90.99%, and 99.09%, respectively. Furthermore, the researchers explored the impact of temperature and concentration on the removal efficiency of these ions. They observed an increase in efficiency with rising temperature, reaching its peak at 60°C . Conversely, the removal efficiency of these ions decreased with increasing concentration.

The removal of Poisonous Pb^{2+} and Hg^{2+} from aqueous solution using GO-Si Nanocomposite were studied by Jayakaran Pachiyappan and co-workers [46]. The outcomes revealed that the ions achieved maximum adsorption at 347.2 and 266 mg/g, respectively, and exhibited a good fit with the Langmuir isotherm and pseudo-first-order kinetics. The optimal removal efficiency for both Pb^{2+} and Hg^{2+} ions was observed at a pH below 5. Additionally, the GO-Si Nanocomposite demonstrated the ability to be reused up to five times in water treatment.

Zhan-fang Cao and collaborators documented the effective elimination of Pb^{2+} ions through the use of Nano- Fe_3O_4 /triisopropanolamine functionalized grapheneoxide ($\text{TI}/\text{GO}@ \text{Fe}_3\text{O}_4$) nanocomposites [47]. Their study determined that the highest adsorption took place at 298K, with a pH of 5, utilizing a dosage of 10mg, and a contact time of 120 minutes. The adsorption maximum of $\text{TI}/\text{GO}@ \text{Fe}_3\text{O}_4$ nanocomposites exhibited a good fit with the Langmuir isotherm and adhered to pseudo-first-order kinetics.

Kuglaur Shanmugam Ranjith and colleagues conducted an investigation into the removal efficiency of ZnO Nanorod-Reduced Graphene Oxide (ZnO NR-rGO) by synthesizing and characterizing the nanocomposites [48]. They observed that ZnO NR-rGO nanocomposites

exhibited higher absorption efficiency for MB, RhB, and MO compared to pure ZnO NRs. Furthermore, the photodegradation efficiency of MB, RhB, and MO for ZnO NR-rGO, after 180 minutes of irradiation under visible and natural sunlight, was found to be seven times higher than that of pure ZnO NRs. Notably, the researchers found that 50mg of ZnO NR-rGO could remove 80% of pollutants within 90 minutes under visible illumination. Their investigation also demonstrated that ZnO NR-rGO possesses the capability to remove heavy metal ions, such as Cu (II) and Co (II), along with other hazardous materials from wastewater. They determined that the removal efficiency of ZnO NR-rGO for Cu (II) ions reached a maximum of approximately 98.6% at pH 8, while for Co (II) ions, it increased from 11.7% to 90.1% across a pH range from 2 to 10. The kinetic data for the adsorbent were well-fitted to the pseudo-second-order kinetic model and Freundlich isotherm.

Dawid Pakulski and collaborators investigated the adsorption characteristics of Ni(II), Zn (II), and Co (II) using the Terpyridine–graphene oxide hybrid (GO–Tpy and GOh–Tpy) [49]. Their research also explored the selective absorption of these ions from mixtures containing alkali metals [Na (I), K (I)] and alkaline Earth metals [Mg (II), Ca (II)]. The hybrid was thoroughly analyzed using XPS analysis, Raman spectroscopy, SEM, BET analysis, X-ray diffraction, and TGA analysis. According to their findings, the adsorption capacity of Ni(II), Zn (II), and Co (II) ions on GOh–Tpy reached its maximum in the pH range of 2–6, slightly decreasing by pH 7. The researchers noted that 1 g of GOh–Tpy could adsorb 6 to 8 mmol of divalent heavy metal ions, and the affinities of the metal ions towards GOh–Tpy followed the order $\text{Ni} > \text{Zn} > \text{Co}$. Moreover, the hybrid demonstrated excellent performance even in the presence of interfering I and II group cations. The adsorption of Ni(II), Zn (II), and Co (II) on GOh–Tpy was well-described by the Langmuir isotherm model and the pseudo-second-order kinetics model.

The adsorption of Pb (II) ion on graphene oxide/PVC composite papers functionalized with *p*-Phenylenediamine (PVC/PPD/GO-0.08) buckypapers were reported by Zaheen Ullah Khan and co-workers [50]. Their study reveals that the composite papers achieved the maximum removal efficiency of heavy metal ions at pH 6.5 and a temperature of 70°C, with an initial concentration of 0.01 mol L⁻¹, within a contact time of 120 minutes. The PVC/PPD/GO-0.08 buckypapers exhibited the highest Pb (II) adsorption performance, attributed to their elevated surface area, hierarchical porosity, remarkable wettability, chelating ability, and clear morphology.

Yitao Zhao and colleagues investigated the photocatalytic elimination of Cr(VI) ions from wastewater using Reduced graphene oxide supported ZnO/CdS heterojunction (CdS/ZnO6:5/RGO6) [51]. Their findings indicated that the composite exhibited optimal performance in reducing Cr(VI) when the ZnO/CdS molar ratio was 6:5, enhancing the formation of heterojunctions. They attributed the increased adsorption of Cr(VI) to the addition of RGO, which inhibited the aggregation of ZnO/CdS and provided more active sites for adsorption. Remarkably, within 30 minutes, the composite completely reduced 40 mL of 10 mg/L Cr(VI) under visible light at room temperature. The photocatalytic reduction rate constant of Cr(VI) by CdS/ZnO6:5/RGO6 was determined to be 1.7. Additionally, the efficiency of the photocatalytic reduction of Cr(VI) by the ZnO/CdS6:5/RGO6 composite

remained above 93.2% even after five uses.

The study on the mechanism of efficient adsorption of Cd(II) and Cu (II) ions from waste water by few layered magnetic graphene oxide (FLMGO) were done by Ting Guo and co-workers [52]. When comparing the adsorption efficiency of Fe_3O_4 , GO, and FLMGO, it was observed that the order of adsorption efficiencies for both Cd(II) and Cu (II) was $\text{FLMGO} > \text{GO} > \text{Fe}_3\text{O}_4$. The impact of pH on the adsorption of Cd(II) and Cu (II) ions by FLMGO was also investigated, revealing that the maximum adsorption for Cd(II) occurred at pH=7, while for Cu (II) it occurred at pH=8. Notably, $\text{Cu}(\text{OH})_2$ became the dominant species at a lower pH compared to $\text{Cd}(\text{OH})_2$. High adsorption efficiencies of 92.77% for Cd(II) and $1114.22 \text{ mg}\cdot\text{g}^{-1}$ for Cu (II) were reported at contact times of 5 and 7 minutes, respectively. It was observed that an increase in the adsorbent dosage led to a decrease in adsorption amount, while an increase in the initial concentration of the adsorbate resulted in an increase in adsorption amount. The presence of electrolytes had a minor effect on the adsorption of both ions on FLMGO, but the presence of humic acid decreased the adsorption of Cd(II) and had a limited influence on the adsorption of Cu (II). Additionally, an increase in temperature from 10 to 35°C resulted in an increase in the adsorption efficiency of both ions.

Thu Dieu Le and colleagues explored the application of graphene Oxide/Polyvinyl Alcohol/ Fe_3O_4 (GO/PVA/ Fe_3O_4) nanocomposite for removing Co (II) ions from water [53]. The composite was characterized using various analytical methods, including XRD, FT-IR spectrophotometry, UV-Vis spectrophotometry, and SEM. It was observed that approximately 60% of Co (II) ions were removed within the contact time range of 3 to 100 minutes. The optimal dosage of the adsorbent and temperature for Co (II) ion adsorption on the composite were reported to be 0.0123 g and 50°C , respectively. The highest adsorption of ions occurred at pH 5.2. The results further indicated that the sorption process was well-described by the Langmuir isotherm and pseudo-second-order kinetic model. Importantly, the GO/PVA/ Fe_3O_4 composite demonstrated the capability to be regenerated for at least 5 cycles, maintaining an adsorption efficiency of 80% after the 5th cycle.

Meltem Bilici Baskan and Seçil Hadimlioglu conducted an investigation into the removal of arsenate using Graphene oxide iron modified clinoptilolite-based composites (GO-FeZ) [54]. The results obtained from the study indicated that the arsenate adsorption capacity of GOFZA and GOFZB was highest within the pH range of 7-8. Notably, both composites were effective in reducing the residual arsenate concentration in water to below the 10 ppb MCL set by the WHO. At an initial arsenate concentration of $450 \text{ }\mu\text{g/L}$, the arsenate adsorption on GOFZA and GOFZB was reported as 557.86 and $554.64 \text{ }\mu\text{g/g}$, respectively. The sorption of arsenate by both composites was well-described by pseudo-second-order kinetics.

The synthesis, characterisation and adsorption capacity of Magnetic composite graphene oxide/ Fe_3O_4 /glucose -COOH(GO/ Fe_3O_4 /GC) nanocomposite were studied by Aili Yang et al [55]. They noted that the mentioned composite demonstrated a removal capacity of U(VI) exceeding 99%. Their observations revealed that the efficiency increased with an elevation in pH within the range of 4-7, reaching its maximum at pH 5. This optimal performance was achieved with an initial concentration of the adsorbent at 10 mg L^{-1} , a dosage of 0.15 g L^{-1} ,

and a contact time within 30 minutes. The presence of coexisting cations such as Na^+ , K^+ , Ca^{2+} , Mg^{2+} , and Al^{3+} had a minimal impact on the adsorption efficiency of the composite. The adsorption process was aptly described by the Langmuir model and pseudo-second-order kinetic model, respectively.

Khadijah MohammedsalehM Katubi and colleagues conducted an investigation into the removal of Pb^{2+} and Neutral Red (NR) dye from water using $\text{MnFe}_2\text{O}_4/\text{GO}$ nanocomposite [56]. Various analytical techniques, including X-ray diffraction, transmission electron microscopy, Fourier transform infrared spectroscopy, scanning electron microscopy, X-ray photoelectron spectroscopy, Raman spectra, and vibrating sample magnetometer, were employed to characterize the prepared nanocomposite. The removal efficiency increased from 39% to 98.8% for Pb^{2+} and from 63% to 94% for NR dye, respectively, with an increase in the adsorbent dose at an optimum pH of 6. The optimal contact times for Pb^{2+} ions and NR dye were reported as 120 and 30 minutes, respectively. The adsorption capacities were reported as 636.94 and 46.08 mg/g for the adsorption of Pb^{2+} and NR dye, respectively. These capacities were well-fitted to the Langmuir isotherm, and the nanocomposite demonstrated reusability after 5 cycles.

Lamellar membranes based on two-dimensional graphene oxide (GO) have found widespread applications in water purification. In a study conducted by Faheeda Soomro and colleagues, the removal of Ni^{2+} and Pb^{2+} using $\text{MoO}_2@\text{GO}$ and $\text{WO}_3@\text{GO}$ nanocomposite membranes was investigated [57]. The lamellar membrane exhibited a notably high water permeance of approximately $\sim 445 \pm 20 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$. Under a pressure of 1 bar and a concentration of 0.01M, the decline rate of ions through the membrane was observed in the order of $\text{Ni}^{2+} > \text{Pb}^{2+} > \text{Mg}^{2+} > \text{Na}^+$. The nanocomposite membrane emerges as a promising choice for water filtration, demonstrating a remarkable 99% rejection efficiency for Ni^{2+} ions.

Conclusion :

The release of heavy metal ions into aquatic systems poses a significant threat to the natural ecosystem and human health. Addressing this challenging issue, graphene-based nanocomposites prove highly effective in removing heavy metal ions from both water and wastewater through methods like adsorption or photocatalysis. Their advantages include high efficiency, reusability, and adaptability. To sum up, we have explored the distinctive features and practical applications of various graphene oxide-based nanocomposites in eliminating toxic ions from water and wastewater. This chapter is expected to be beneficial for researchers engaged in the field of wastewater treatment using nanocomposites.

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Nanotechnology for Wireless Communication Systems

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Abstract :

Communication systems are required to be compact and conformable due to which there are certain restriction on the size and structure of the various components of the system. Antennas which are an integral part of such wireless communication systems are required to be smaller and more efficient. Miniaturization in antennas is achieved by adjusting the parameter μ_r , which is complex permittivity and permeability of the substrate material used. Current antenna materials, such as low-temperature cofired ceramics and polymer dielectrics, have limitations in component downsizing due to their low permittivity and lack of magnetic characteristics. Magnetic nanoparticles have demonstrated a favourable contribution to high permeability behaviour, enabling microstrip antennas to be smaller without sacrificing performance. Another important aspect of wireless systems is Electromagnetic interference (EMI) which presents a serious problem for equipment designers, and the development of more efficient EMI shielding materials has been fuelled by the widespread usage of electronics. Magnetic nanoparticles, such as metallic element and alloy nano-powders, nanofiber, hybrid nanoparticles, and nanosized ferrite powders, have been used to create composites for RF attenuation applications.

Keywords : Antennas, EMI shielding materials, ferrite nano-composites

Introduction:

Advent of nanotechnology has led to a tremendous proliferation of novel materials bearing a wide range of applications. Electronic materials being one such among the spectrum, wherein, the potential of nanotechnology has been exploited to a large extent by researchers to attain unique electrical properties. Communication technology, which has grown exponentially since the last decade, has relied on such electronic materials to cater to requirements of the evolving wireless devices and products. Antennas are an essential part of any communication system and so are the electromagnetic shielding materials. The current chapter discusses about the application of nanomaterials in the domain of communication systems.

Antennas :

An antenna is, by definition, a device that converts an electromagnetic wave in free space from an RF signal that is traveling along a conductor. A characteristic of antennas is called reciprocity, which denotes that an antenna will remain the same whether it is transmitting or receiving. Most antennas are resonant devices, meaning they work well within a constrained frequency range. Antenna reception and transmission will be negatively impacted if it is not tuned to the same frequency range as the radio system to which it is attached. An antenna will emit radiation that is dispersed in space in a particular way when a signal is supplied into it.

Depending on the applicability, there are many types of antennas which vary in their shape and size. Microstrip patch antennas are one of such category of antennas which find application in many hand held and wearable communication devices. The inexpensive cost, conformability, and light weight of microstrip patch antennas (MPA) make them appealing. Active devices and printed strip-line feed networks can be incorporated with these antennas. In antenna engineering, this is a relatively young field.

Since the mid-1950s, it has been recognized that microstrip structures had certain radiation properties. When missiles needed conformal antennas in the early 1970s, this kind of antenna was put to use. Microstrip resonant patches, both rectangular and circular, have been widely used in numerous array topologies. The present revolution in electrical circuit shrinking, fueled by advancements in large scale integration, is a fundamental cause behind the recent advancements in microstrip antennas. Micro strip antennas built on photolithographic technology are considered an engineering marvel since traditional antennas are frequently large and expensive components of an electronic system.

A microstrip patch antenna, as depicted in **Figure 1**, is essentially a radiating patch on one side of a dielectric substrate and a ground plane on the other. The patch can be whatever shape you like and is usually composed of conductive materials like copper or gold. Typically, photo etching is used to create the radiating patch and feed lines on the dielectric substrate. The patch is typically square, rectangular, circular, triangular, elliptical, or some other common shape to simplify analysis and performance prediction. The length L of a rectangular patch is typically $0.3333 \lambda_0 < L < 0.5 \lambda_0$, where λ_0 represents the free-space wavelength. The patch is chosen to have a thickness of $t \ll \lambda_0$, meaning that it is extremely thin. Typically, the dielectric substrate's height, h , ranges from $0.003 \lambda_0$ to $0.05 \lambda_0$.

Usually, the substrate's dielectric constant (ϵ_r) falls between 2.2 and 12.

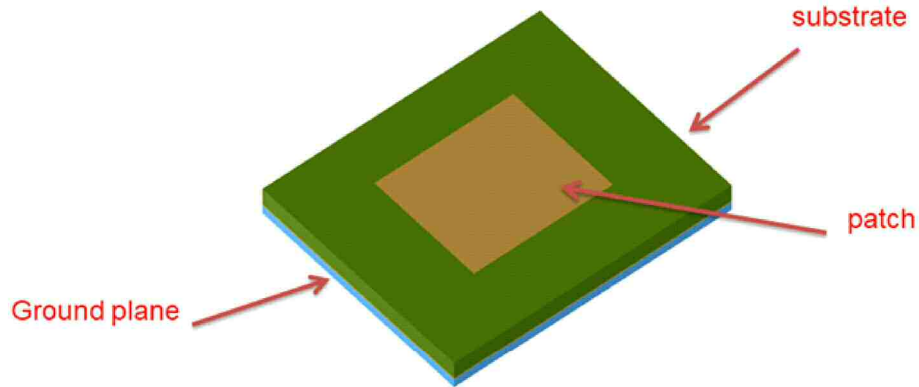


Figure 1: Structure of a simple microstrip patch antenna

The main source of radiation from microstrip patch antennas is the fringing field that exists between the patch edge and the ground plane. For optimal antenna functioning, it is preferable to have a thick dielectric substrate with a low dielectric constant. Such a combination offers higher radiation, greater bandwidth, and improved efficiency. But this arrangement results in a bigger antenna. Higher dielectric constant substrates, which are less efficient and have a narrower bandwidth, are required in the design of a tiny Microstrip patch antenna. As a result, it is necessary to balance the antenna's performance and size.

Miniaturization in antennas (planar antennas), denoted by factor n is given as, $n = \sqrt{(\epsilon_r \mu_r)}$, where ϵ_r is complex permittivity and μ_r is complex permeability of the substrate material used in the fabrication of the antenna [1].

Due to their poor qualities, such as low permittivity and lack of magnetic characteristics, existing antenna materials, such as low-temperature cofired ceramics (LTCCs) and polymer dielectrics, exhibit basic constraints in terms of component downsizing. The temperature and frequency dependency of characteristics, as well as dielectric loss, are typically associated with increased permittivity or permeability, rendering them unsuitable for use in high-performance radio frequency components [2]. The main limitations in using high ϵ_r substrates are (i) decrease in the radiation efficiency, (ii) degradation of radiation pattern (due to excitation of surface waves) and (iii) decrease in the impedance bandwidth. Another modification to achieve miniaturization is by alteration of the other parameter μ_r .

Research on magnetic materials at frequencies higher than 100 MHz has been extensive. They fall into four groups: bulk ferrites, anisotropic composite structures, microscale metal composites, and nanoscale metal composites.

Bulk ferrites characteristics such as, (i) high saturation magnetization values, (ii) dispersive magnetic loss and (iii) high external dc-bias field requirements for permeability alteration marks them incompatible for compact applications. In contrast, Nano sized ferrites

have shown suitability for miniaturized applications with their low saturation magnetizations and low eddy current losses. Composites made using nano ferrite inclusions in polymers can result in natural magneto-dielectric materials with low magnetic losses and enhances the gain. ϵ_r and μ_r , of the composite can be altered by varying the amount of the inclusions to achieve the desired performance. Hansen and Burke [3], gave a relation showing the dependence of zero-order bandwidth on ϵ_r and μ_r of the magneto-dielectric substrate material for an antenna

$$BW = \frac{96 \sqrt{\frac{\mu_r}{\epsilon_r}} \frac{t}{\lambda_0}}{\sqrt{2} [4 + 17 \sqrt{\mu_r \epsilon_r}]}$$

Where, t is thickness of the substrate and λ_0 is the free space wavelength. Hence, improvement in the impedance bandwidth can be achieved by addition of magneto-dielectric nano fillers.

Antenna dimensions can be significantly reduced by a factor of four when permeability and permittivity in magnetic nanocomposites are increased fourfold compared to typical polymer dielectrics. Materials can be engineered to have a lower permittivity-to-permeability ratio in order to obtain good impedance matching with free space by fine-tuning the formulation. Therefore, compared to high-permittivity dielectric substrates that offer the same downsizing from the permittivity alone, the antenna bandwidth on such magnetic substrates will be greater [4-6]. The increased radiation efficiency and suppression of backscattering brought about by the high permeability lead to even greater performance improvement [2].

Dispersed into a polymer matrix, magnetic nanoparticles have demonstrated a favorable contribution to a high permeability (μ) behavior, enabling microstrip antennas to be smaller without sacrificing performance. The literature states that most viscoelastic matrices, such as polydimethylsiloxane (PDMS) [7], polyvinyl alcohol [8], polystyrene [9], commercial Rogers® polymer [10, 11], chitosan/gelatin [12], galactomannan [13], collagen [13,14], and polyvinylidene fluoride [15], are dispersed with magnetic nanoparticles, which are typically Fe, Fe_3O_4 , CoFe_2O_4 , $\text{Y}_3\text{Fe}_5\text{O}_{12}$, $\text{NiZnFe}_2\text{O}_4$, $\text{MnxZn}_{(1-x)}\text{Fe}_2\text{O}_4$, and $\text{Ba}_{12}\text{Fe}_{28}\text{Ti}_{15}\text{O}_{84}$.

For instance, a Fe_3O_4 /PDMS magneto-dielectric composite could reduce dielectric and magnetic loss in the presence of a magnetic field and effectively increase bandwidth and yields with an adjustable dielectric property [10, 16-18]. Fe_3O_4 nanoparticles were in fact highly desirable when applied to magneto-dielectric substrates because of their super paramagnetic quality, which makes them easy to obtain at a low cost and minimizes losses from hysteresis and eddy current induction. Super paramagnetic iron oxides (SPIONs) with high magnetization are difficult to synthesis because decreasing their size causes an increase in oxidation, which in turn reduces magnetism [7].

An alternative magneto-dielectric (MD) substrate material is nano ferrites in a flexible

polymer matrix [19-24] (**Figure 2**).

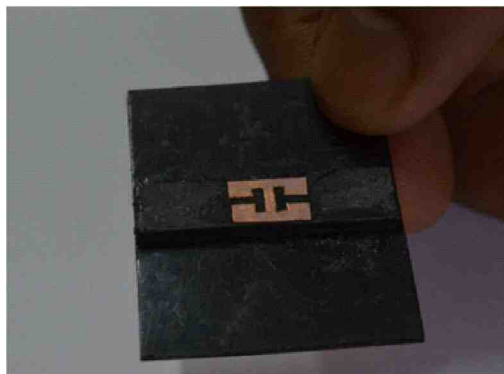


Figure 2: Patchantenna on a magnetodielectric nanocomposite substrate

Furthermore, by adjusting the number of magnetic inclusions in the polymer, the permittivity and permeability of magneto-dielectric (MD) materials can be adjusted to moderate values [20-26]. For instance, nickel ferrites, which have a 500°C Curie temperature, can withstand strong microwave energies without losing their magnetic characteristics. Zn added to NiFe_2O_4 stoichiometrically lowers the magneto-crystalline anisotropy, which lowers coercivity and magnetic losses [27-33]. These low loss nano composite materials [34] exhibit possibilities for excellent antenna substrates.

Emi Shielding Materials :

Electrical and electronic devices experience faults and inaccurate readings due to electromagnetic interference (EMI). It can wreak havoc with communications networks, data processing equipment, control systems, and instrumentation. In a world where electrical and electronic devices are essential, electromagnetic interference (EMI) presents a serious problem for equipment designers. The development of more efficient EMI shielding for a variety of applications has been fuelled by the widespread usage of electronics. The necessity to cover equipment increases with its sensitivity and criticality. The fundamentals of shielding equipment against electromagnetic interference (EMI) haven't changed in the last fifty years, but design advancements have made shields lighter and smaller, which is essential for today's sophisticated uses.

By using visible, infrared, microwave, radio, and ultraviolet wavelengths, the electromagnetic spectrum can be used for detection purposes. Advances in these detection techniques have been made possible by the theory of physical optics. In contrast, the theory of physical optics is utilized to limit reflections, emissions, and thus, detection through the use of camouflage or signature reduction. Reducing signatures is achieved by a mix of materials and optics. The use of electromagnetic waves in the gigahertz range in wireless

communication tools, local area networks, personal digital assistants, and other communication equipment has increased recently. As a result, efforts are being made to find appropriate materials that can absorb electromagnetic waves and prevent electromagnetic interference (EMI). The absorption qualities of a material for electromagnetic waves are mostly determined by the properties of its constituent materials, namely its permittivity (ϵ_r) and permeability (μ_r), which are dielectric and magnetic properties, respectively. Magnetic composite absorbers rely on a type of magnetic hysteresis effect that is produced when magnetic particles are added to a matrix, whereas dielectric composite absorbers rely on the ohmic loss of energy, which can be achieved by adding lossy fillers like carbon, graphite, or a metal particle/powder to a matrix [35]. These lossy fillers have strong thermal dissipation and a high absorption capacity of electromagnetic radiation.

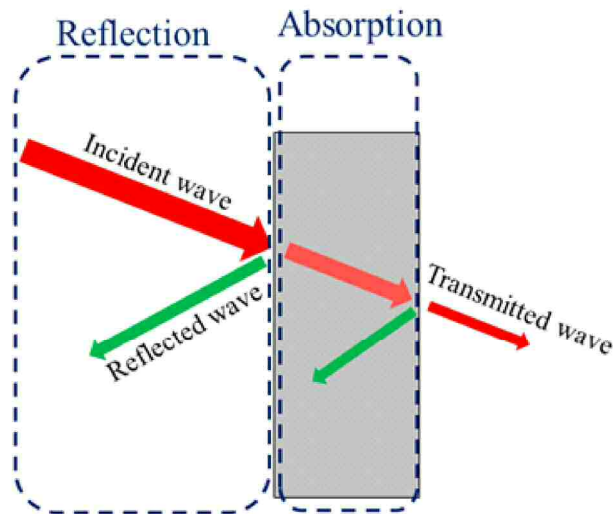


Figure 3: Mechanisms of EMI shielding

Further, composites for RF attenuation applications have been synthesized using magnetic nanoparticles such as metallic element and alloy nano-powders, nanofibre, hybrid nanoparticles of metallic alloys and oxides, and nanosized ferrite. Nano-powders have been produced using a variety of techniques, including high energy ball milling, chemical vapor deposition, dc arc discharge, and chemical precipitation (reduction) [19, 36-44]. Properties like intrinsic permeability of metallic alloy nano particles suggest the electromagnetic nature of such composites. Observations of the electromagnetic responses of these materials show two very distinct characteristics. First is the presence of different resonance peaks when the particle size enters the nano range from the micron range. Second is the shifting of the resonance peak to higher frequency range with the decrement of the particle size. Such behaviour has been explained on the basis of non-uniform resonance modes coexistence

originating from the exchange energy contribution towards the magnetization [36,45].

Ni nanofiber-based composites were also found to have enhanced microwave attenuation [37]. While the magnetic loss tangents of the two composites were almost equal, the dielectric loss tangent of the composite made with Ni nanofibres was larger over the microwave frequency range than the composite manufactured with Ni particles. The composite including Ni fibers exhibited superior microwave attenuation behaviour, as predicted [37].

A technique for creating low density ordered mesoporous carbon silica nanocomposites with a controlled Fe content was described by Zhou et al. [38]. This method's fundamental idea was to introduce pores to lower dielectric permittivity while retaining the required impedance matching. Low density would be another benefit of porous materials. According to Zou et al., a Fe-graphite oxide nanocomposite was reported, with typical Fe particle sizes ranging from 10–100 nm [46].

Additionally, ferrite can be reduced in H_2 to create FeCo-hexaferrite nanocomposite particles [40, 47]. Even though the nanocomposites permeability was higher than that of their parent ferrites, their permittivity was further enhanced [40]. Because of their low impedance matching, they are not the best options for microwave attenuation applications. Composites with Fe nanowires of diameter 70–200 nm and length of 20–50 mm, synthesized by chemical breakdown process, has shown improved dielectric and magnetic properties when compared to their microwire counterparts [48].

Recent advances in EMI shielding nanocomposites looks very promising for future EMI shielding applications. EMI shielding qualities are affected by the frequency-dependent features, morphological structure adjustment, and processing parameters. In spite of much technological progress in this field, there are many hindrances like poor thermal stability and limited absorption bandwidth that limits its applicability. Thus, research in these domains holds prospects for future studies.

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Nanotechnology for waste disposal and management

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Abstract:

Substantial growth of human population triggers swift urbanization, economic development, industrial expansion and enhanced living standards. However, this progress comes at the cost of numerous environmental challenges, with waste generation emerging as a predominant issue. This results in a deterioration of the health of both human and animals, which impacting the entire ecosystem. Consequently, it is essential for us to prioritize effective waste management. Nanotechnology is an evolving biotechnological innovation employed across diverse industrial sectors, including pharmaceuticals, food safety, electronics and various other domains. Due to their very small dimension (1-100 nm), nanomaterials exhibit an elevated propensity for oxidation/ reduction reactions and catalysis involving diverse organic and inorganic contaminants. Moreover, they display effective adsorption capabilities for various heavy metals. As a result, nanomaterials have emerged as innovative tool in the field of waste management. The primary focus of this chapter revolves around waste generation, its impact on human life and environment and the transformative role that nanotechnology has played as a revolutionary approach in waste management.

Keywords: Nanotechnology, Waste management, Nanocatalysts, Nanofilters, Nanosensors

Introduction :

The current landscape of waste disposal and management is riddled with intricate challenges that demand urgent attention and innovative solutions. One of the primary issues is the rapid urbanization outpacing the development of adequate infrastructure, resulting in improper waste disposal practices and heightened environmental risks [1]. The pervasive use of single-use plastics has given rise to a plastic pollution crisis, exacerbated by insufficient recycling facilities that struggle to cope with the sheer volume of plastic waste. Simultaneously, the surge in electronic devices has led to a burgeoning electronic waste problem, characterized by complex material composition and hazardous components that demand specialized disposal methods. The limited adoption of waste-to-energy technologies, attributed to high costs, public opposition, and environmental concerns, further complicates the quest for sustainable waste management. Illegal dumping and littering persist as pervasive

issues, straining existing waste management systems and posing risks to public health [2]. Overall, the challenge lies not only in waste reduction but in the development of a comprehensive strategy that includes improved recycling infrastructure, technological innovation, heightened public awareness, and international collaboration to foster sustainable waste management practices and mitigate the environmental impact of waste disposal.

Nanotechnology and its potential applications in addressing waste-related issues :

Nanotechnology, a field that manipulates matter at the nanoscale, holds immense promise in revolutionizing various aspects of waste management and addressing environmental challenges. At the nanoscale, materials display distinctive properties and behaviours, enabling the development of innovative solutions for waste-related issues. Nanotechnology's potential applications in waste management span across various domains. For instance, nanomaterials can be employed in advanced sensors for real-time monitoring of pollutants, providing a more precise and efficient means of tracking environmental contamination. Additionally, nanocomposites and nanocatalysts have shown promise in enhancing the efficiency of waste treatment processes, such as wastewater purification and air pollution control. Nanoparticles also offer opportunities for more effective and selective removal of contaminants from different waste streams [3].

Furthermore, nanotechnology plays a crucial role in the development of novel materials for packaging that can be more sustainable and environmentally friendly. As we navigate the challenges of waste disposal and environmental degradation, the integration of nanotechnology into waste management strategies holds the potential to usher in a new era of more efficient, sustainable, and environmentally conscious solutions.

Nanotechnology and solid waste management :

Solid waste materials refer to the byproducts generated by human activities to fulfil their needs, subsequently entering the environment. The burgeoning global population and shifts in lifestyles have amplified the production of these materials, posing a significant contemporary environmental challenge. Conventional technologies struggle to effectively eliminate these pollutants. Addressing this issue necessitates the integration of modern technologies, with nanotechnology emerging as a promising solution [4]. Nanomaterials, characterized by their minuscule size and unique properties, play a pivotal role in waste management. Several types of nanomaterials find extensive application in this domain, encompassing nanofilters, nanosensors, nanophotocatalysts, and nanoparticles. Nanofilters, equipped with nano-sized pores, exhibit enhanced filtration capabilities, enabling the removal of microscopic pollutants from water and air. Nanosensors provide real-time monitoring of pollutant levels, facilitating prompt responses to environmental threats. Nanophotocatalysts leverage nanoscale catalysts to accelerate chemical reactions, aiding in the degradation of pollutants in water and air [5].

Nanophotocatalyst and porous nanocatalysts :

The photocatalyst is a material that facilitates a chemical reaction in the presence of

sunlight without undergoing any change itself. These materials do not directly participate in oxidation and reduction reactions but provide the necessary conditions for these reactions to occur. Titanium dioxide exhibits almost all the characteristics of an ideal photocatalyst. When exposed to water, oxygen, and UV radiation, titanium dioxide generates free radicals that can effectively break down various harmful compounds into less toxic carbon compounds [6,7].

Titanium dioxide nanoparticles, with their higher surface-to-volume ratio, exhibit more favourable photocatalytic properties in comparison to larger particles. This material finds applications as a nanocrystalline microspheres, coating for fixed membranes, and membranes combined with silica to treat leachate produced from waste material landfills. Due to its hydrophilic nature, titanium dioxide can decompose most organic contaminants, making it effective in absorbing heavy metals from wastewater [8].

Porous nanocatalysts play a crucial role in the conversion of waste materials into ethanol through the gasification process. In this method, carbon compounds are transformed into syngas under high pressure and temperature within a controlled environment. Subsequently, porous nanocatalysts facilitate the conversion of syngas into ethanol. By creating suitable conditions for the interaction between the porous nanocatalysts and carbon monoxide molecules, the process promotes the synthesis of ethanol. This conversion represents a sustainable and environmentally friendly approach to utilize waste materials, providing an alternative source for ethanol production [9,10].

Nanofilters in waste management :

Toxic and nonbiodegradable materials present in waste, including substances such as arsenic, ammonia, xenobiotic organic compounds, heavy metals, and inorganic macro components, pose challenges for conventional biological and physicochemical methods to meet environmental standards. In addressing this issue, pretreatment, posttreatment, and modern technologies are being considered to effectively remove contaminants from the environment. Nanofiltration has emerged as a viable solution for waste treatment, eliminating the need for chemicals in the process, concentrating the produced waste, and reducing transportation and disposal costs. Nanofilters have demonstrated the capability to remove 60 to 70% of Chemical Oxygen Demand (COD) and 50% of ammonium in leachate. Furthermore, nanofilters prove effective in removing a broad spectrum of pollutants, including anions and cations, chromium, arsenic, pathogens and uranium from wastewater [11,12].

However, the successful implementation of nanofiltration technology requires the management of membrane fouling. Traditional approaches, such as membrane cleaning or replacement, are expensive and impractical. A more convenient solution involves the use of compounds like fullerenes to prevent biological fouling. Filtering membranes and pipes can be prone to the accumulation of bacteria and microorganisms over time, forming an organic film that obstructs the membranes. Fullerenes offer a preventive measure against bacterial fouling by inhibiting their attachment and growth, acting as membrane antifouling agents. As clot-busting drugs, fullerenes also play a role in preventing the clogging of pipes and filtration membranes. Coating pipes and membranes with these nanoparticles emerges as a

strategic and effective approach to mitigate biological fouling [13, 14].

Nanotechnology for wastewater treatment :

Despite approximately three-quarters of Earth's surface being covered with water, ensuring a safe and sufficient water supply remains one of the significant challenges faced by humanity. To enhance the efficiency of water treatment systems, the integration of modern tools capable of rapidly transforming contaminated water into potable water is essential. Nanotechnology has proven instrumental in mitigating the influx of pollutants arising from human activities into water and the environment. Key applications of nanotechnology in environmental contexts encompass the utilization of nanoparticles, nanoclays, nanofilters etc. for water treatment. Additionally, nanosensors play a crucial role in the rapid detection of water pollution [15,16]. Nanotechnology's impact on water treatment is multifaceted. Nanofilters demonstrate exceptional efficiency in removing contaminants from water, ensuring a higher quality of treated water. Nanoparticles contribute to advanced water purification processes, facilitating the removal of diverse pollutants and enhancing overall treatment effectiveness. Nanosensors, on the other hand, enable rapid and sensitive detection of water pollution, allowing for prompt response and intervention.

Nanoparticles :

Nanoparticles exhibit a significantly larger surface area compared to larger particles, enabling them to interact with various chemical groups and thereby increasing their affinity for desired compound [17,18]. Capitalizing on these characteristics, nanoparticles serve as effective tools for absorbing pollutants and treating contaminated water. Metal nanoparticles, in particular, are considered promising adsorbents with high capacities for adsorbing a variety of metal ions and anions [19]. Key metal nanoparticles used in water and wastewater treatment include silver, manganese oxide, lanthanum, gold, zinc oxide, and iron nanoparticles. Zero-valent iron (ZVI) nanoparticles, in particular, are prioritized in treatment processes due to their abundance, cost-effectiveness, non-toxicity, and rapid reactivity with pollutants. Iron nanoparticles, especially zero-valent iron (nZVI), represent a new generation of biological monitoring technologies, offering potential solutions to various environmental challenges. These nanoparticles decompose and oxidize polluting compounds such as carbon tetrachloride, trichloroethylene, PCBs and dioxins, transforming them into less harmful carbon compounds [20,21]. Neutral iron nanoparticles are employed for both non-situ and in-situ groundwater treatment. They can be directly injected into underground sources for in-situ treatment or utilized in filters for external applications. These bimetallic particles boast high activity, durable decomposition, and reduced production of toxic materials. However, concerns persist regarding the toxicity of catalyst metals and the potential deactivation of the catalyst surface due to the formation of a thick layer [22,23]. Nanostructured bimetallic compounds, such as iron-palladium, iron-silver, and zinc-palladium, find extensive applications in refining environmental pollutants [24]. These compounds showcase the versatility of nanoparticles in addressing pollution challenges, underscoring their potential to contribute to sustainable and efficient water treatment processes.

Nanofilters :

The selection of water treatment methods depends on the type and size of separable compounds, and various techniques, including microfiltration (100 nm), ultrafiltration (10 nm), nanofiltration (1 nm), and reverse osmosis (<1 nm), are available. Nanofilters are situated between ultrafiltration and reverse osmosis techniques with respect to their pore or membrane diameter. Ultrafiltration is employed when there is an excess of contaminants beyond permissible levels, while reverse osmosis is used to achieve higher purity than required, albeit with a higher cost [25].

Nanofiltration, compared to other methods, offers the advantage of treating contaminated water at an appropriate volume and desired quality while consuming less energy and allowing for higher water penetration. These filters can effectively separate various contaminants such as viruses, bacteria, organic pollutants, pesticides, and heavy metals from water. Unlike conventional chemical methods, nanofiltration for water softening does not rely on the use of chemicals, thereby minimizing adverse environmental effects. In water treatment, nanofilters not only separate bacterial and viral materials but also have the capability to completely destroy them beyond the filter wall. This capacity is also harnessed in air nanofilters, which utilize nanoparticles like zinc and silver coated on a filter to contact and destroy any living structures [26]. Nanofiltration operates on the basis of size, and filters with nano-sized pores effectively block the passage of even the smallest viruses. The addition of active materials such as silver nanoparticles, titanium dioxide, and ultraviolet light sources enhances the effectiveness of nanofiltration by destroying viruses, bacteria, and fungi trapped in the filter. As the size of the filter pores decreases, the filter's capacity for separation increases. Notably, nanofilters exhibit high selectivity in removing ions. By using nanofilters, essential minerals for human health are retained in the water, while harmful substances are effectively removed [27]. The selectivity power of nanofilters in preserving essential minerals sets them apart from other filtration methods.

Nanosensors :

Nanosensors are miniature devices designed to identify and react to stimuli on a scale as small as one nanometre. Employing nanotechnology is imperative for achieving measurements at the molecular and atomic levels. Nanoscale sensors exhibit exceptional sensitivity and possess selective functionality [28,29]. One practical application of nanosensors is in managing the unpleasant odor of water post-treatment. The odors emanating from wastewater treatment comprise various compounds, such as reduced sulphur or nitrogen compounds, organic acids, aldehydes, or ketones. The commercial sensors that are called electronic nose, have been developed to detect microorganisms and heavy metals in drinking water. They are also utilized to analyse the characteristics of odors resulting from a mix of steam collected in a closed container [30].

Nanotechnology and air pollution :

The rise of industrialization in human societies has resulted in a heightened production of detrimental air pollutants, including nitrogen oxides (NO_x), carbon monoxide (CO),

sulfur oxides (SO_x), and others. The current methods employed to manage these pollutants encounter certain constraints, with some being cost-ineffective and others generating hazardous byproducts. Moreover, many conventional methods struggle to efficiently remove very tiny contaminants from the environment. Addressing these challenges requires the development of new, low-cost solutions [31]. In this context, nanotechnology has emerged as a contemporary method with global applicability. The production of nanosensors, nanocatalysts, nanocomposites, nanofilters, and nanobiomaterials exemplifies the use of nanotechnology in environmental issues aimed at reducing air pollutants [32]. Nanotechnology offers innovative approaches to mitigating air pollution challenges. Nanosensors are designed to detect and monitor minuscule quantities of pollutants with high sensitivity and precision. Nanocatalysts play a crucial role in catalyzing chemical reactions that lead to the breakdown or conversion of pollutants into less harmful substances. Nanocomposites leverage the unique properties of nanoparticles to enhance the performance of materials used in pollution control. Nanofilters utilize nano-sized pores to effectively capture and remove tiny particles and contaminants from the air. Nanobiomaterials, combining nanotechnology with biological elements, offer novel solutions for environmental challenges. By harnessing the capabilities of nanotechnology, these advancements provide more efficient and sustainable alternatives for mitigating air pollution. The development and implementation of nanotechnology in environmental applications represent a promising avenue for addressing the limitations of traditional methods and finding effective, low-cost solutions to combat air pollution.

Nanocatalysts:

Nanostructured materials play a vital role as environmental catalysts in the purification of automobile exhaust gases and the filtration of air. While conventional ceramic or metal-based catalysts demonstrate good efficiency, their high cost has led to the exploration of nanostructured catalysts as more economical alternatives [33]. Research indicates that carbon nanotubes, in particular, are highly effective devices for adsorbing toxic pollutants like dioxin and other substances emitted from incinerator furnace chimneys. Dioxin, a hazardous byproduct of various industrial processes, poses a long-term contamination threat to air, soil, water, and living organisms due to its stability. Some dioxins are known to be carcinogenic, while most can disrupt the human immune system. Despite recent efforts to control the production of dioxins in many countries, their environmental dangers persist. Although carbon nanotubes are recognized as the most suitable means for adsorbing toxic pollutants like dioxins, their high cost poses a challenge. Researchers are actively working on developing cost-effective methods for producing carbon nanotubes. Moreover, the use of carbon nanotubes in computer displays has additional environmental benefits by reducing the consumption of heavy metals, contributing to a decrease in environmental damage [34,35]. This dual application of carbon nanotubes underscores their versatility and potential for addressing environmental challenges, both in air purification and in reducing the environmental impact of electronic devices.

Nanostructured membranes:

Filters play a crucial role among devices used to control air pollution, employing a porous structure through which gases pass while particles are retained in the filter. The filter membrane operates through three primary mechanisms to remove particles: direct contact between particles and the filter structure, the inertia force exerted when the gas changes direction, and the electric charge effect involving particles and the filter structure. The primary drawback of filters is the creation of a pressure drop that necessitates a significant amount of energy to overcome. However, nanofilters offer higher efficiency compared to conventional filters. With pores ranging from 1 to 10 nm, nanofilters can effectively remove various types of bacteria, organic contaminants and viruses [36]. Membranes based on carbon nanotubes exhibit an enhanced capacity for separating carbon dioxide from other gases. Carbon nanotubes can trap gases at a rate over 100 times greater than other gas separation technologies, making them suitable for large-scale separation processes. Unlike conventional membranes, which show an inverse connection between the volume of gas passing through and the quality of gas separation, carbon nanotube-based membranes do not suffer from these shortcomings. Nanomembrane technology holds significant potential for widespread use in the purification and separation of gases and pollutant vapours across various industries, preventing their release into the environment [37].

Nanosensors:

Continuous monitoring is a critical requirement in addressing air pollution, and nanosensors have emerged as a new technology capable of identifying and responding to physical stimuli at a nanometer scale. Nanosensors can be classified into two main categories: nanoparticles and nanostructured materials. Nanoparticles, function as receptors and visual-spatial sensors. Nanosensors constructed from materials like porous silicon have been developed to detect chemical and biological reactions [28,38]. The industrial sector poses the risk of noxious and toxic gas leaks, and conventional sensors used in industries often detect low concentrations of these gases with a delay. To address this, new technologies are needed to produce faster and more accurate sensors. Nanosensors, comprised of single-layer or multilayer nanotubes, can adsorb molecules of toxic gases and detect trace amounts of fatal gases [39]. Nanosensors have been specifically crafted to detect a range of gaseous compounds, such as nitrogen oxides, dioxin, sulfur dioxide, carbon dioxide, ammonia, and others. These nanosensors demonstrate superior accuracy and speed when compared to traditional sensors [40]. A novel application of nanosensors involves the concept of “smart dust.” Smart dust consists of ultrafine particles made of silicon that can function as very light nanocomputers. These particles are capable of wirelessly transmitting collected data to a central database. With a data transfer rate of about one Kbps or more, smart dust particles can remain suspended in the air for hours, deriving energy from the sun. Smart dust can transmit data such as temperature, pressure, humidity, and the concentration of chemicals in the air up to 20 km, providing continuous monitoring capabilities for controlling air pollution in specific areas [41].

Nanotechnology in recycling processes :**Nanocatalysts for recycling plastics :**

Plastics are specifically characterized as a subset of polymers, consisting of long-chain organic molecules with tailored properties. Approximately 80% of plastic products end up as waste and enter various environments, including land, oceans, and rivers. The waste plastics present in water or on land degrade over time, breaking down into smaller particles, including micro- and nano-plastics, through daily use [42]. These tiny plastic particles pose a threat to ecosystems by damaging nutrients, contaminating wildlife through entanglement and ingestion, and introducing toxins into the food chain. Ultimately, these contaminants can find their way back to humans [43-45]. In alignment with the principles of a circular economy, it is crucial to establish new protocols and technologies [46] to address the challenges associated with waste plastics management. Nanocatalyzed chemical upcycling of plastic wastes have gained much attention in recent times.

Nanocatalyzed pyrolysis :

Pyrolysis employs high temperatures to generate heat, facilitating the transformation of long-chain polymers into small molecules consisting of gaseous, liquid, and solid products. Catalytic pyrolysis is a process in which a mixture of waste plastics, such as polyethylene (PE) and polypropylene (PP), undergoes treatment in the presence of a Lewis acid catalyst or a solid acid catalyst. This chemical reaction involves catalyzed carbocation formation, isomerization, cyclization (and aromatization), and cracking through β -scission reactions [47]. The predominant outcomes of this catalytic pyrolysis are the generation of gaseous and aromatic products. The methodologies employed in the synthesis of nanocatalysts exert a significant influence on the ultimate pyrolysis performance. Wang et al conducted a comparative study on ternary metal oxides ($\text{Fe-Ce@Al}_2\text{O}_3$) nanocatalysts, which were prepared using coprecipitation, impregnation, and a direct mixing method, with a focus on investigating the pyrolysis of polycarbonate (PC) wastes [48]. Among the three synthesis methods, ternary metal oxides prepared by coprecipitation (FC@CP) exhibited notable characteristics, including a high surface area (104 m²/g), large pore volume (0.19 cm³/g), a substantial content of weak acid sites (0.21 mmol/g), and effective dispersion of Fe and Ce active species due to the ultrafine nanoparticle size of 7.5 nm. By virtue of these advantageous nanostructures, FC@CP demonstrated enhanced catalytic conversion of PC wastes into targeted monocyclic aromatic hydrocarbons when compared to its counterparts [48].

In the generation of hydrogen-rich syngas, Ru nanocatalysts have proven to be highly effective, as evidenced by significantly higher carbon conversions and elevated hydrogen yields observed in the gasification of polypropylene (PP) and polystyrene (PS) [49,50]. Bimetallic nanocatalysts generally exhibit superior performance compared to their monometallic counterparts, attributed to the synergy effect. For instance, finely dispersed $\text{Fe}_{0.64}\text{Ni}_{0.36}$ alloy nanoparticles (NPs) supported on carbon nanofibers/porous carbon surpassed the monometallic counterparts, demonstrating increased yield and improved selectivity in the production of hydrogen (H₂) and carbon monoxide (CO) during the pyrolysis-gasification

of biomass and plastic wastes [51].

Nanotechnology in E-waste recycling :

On a global scale, the accumulation of waste electrical and electronic equipment (WEEE) amounts to nearly 30-50 million tons annually, with an anticipated annual growth rate of up to 5% [52]. The conceptualization and practical utilization of end-of-life (EoL) waste products as a starting material for synthesizing new high-value and marketable commodities, such as nanoparticles (NPs), represent a potential strategy to stimulate industrial interest in adopting rigorous recycling efforts. This approach holds the promise of attractive economic returns, particularly with state subsidies, thereby encouraging a more sustainable and economically viable recycling process [53]. Recycling is a sustainable technology that has been widely employed for the recovery of metals and metal oxides from electronic waste (e-waste) [54,55].

Conclusion :

In conclusion, the investigation of nanotechnology for waste management and disposal represents a paradigm change in how we tackle the escalating problems associated with global trash. This chapter demonstrated the ability of nanoparticles to revolutionise a number of waste management-related issues. Applications for nanotechnology range from the accuracy of nano-based sensors in pollution detection to the catalytic power of nanomaterials in waste conversion. These demonstrate the effectiveness and efficiency of nanotechnology. This technology frontier offers opportunities for resource recovery and recycling at the molecular level in addition to promising a more environmentally friendly and sustainable waste management paradigm. To limit possible hazards, it is imperative to use caution while navigating this promising area. This involves assuring ethical considerations, acceptable research techniques, and strong regulatory frameworks. Essentially, this chapter illustrates the route to a future in which trash is not just an issue but also a chance for innovation, resource optimisation and environmental resilience.

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Nanotechnology and Gene Therapy

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Abstract :

In recent days treatment of genetic disorder becomes a hot topic of research for the genetic engineers and physicians. Gene therapy is the new branch of Medical science for the treatment of genetic disorder. Entry of nanotechnology in gene therapy further triggers the advancement of gene therapy. In this article attempt has been made to give a brief introduction of gene, gene therapy and application of nano technology in gene therapy.

Key words : Gene, Gene therapy, Nanoparticle, Nanotechnology,

Introduction :

Living beings are differentiated from non living things by some characters which are known as living characters. Some of these characters are governed by environment and rest of the characters are obtained from parental generation. The characters that transmitted from parents are known as the hereditary characters. Many scientists or philosophers have been trying to know- how these characters are transmitted to offspring from the parents and who are responsible for transfer of these characters, but till the mid- Nineteenth century no one was able to crack the mystery. It was Sir Gregor Johann Mendel who for the first time systematically studied and tried to explain the way how the hereditary characters are passed from parental generations to their offspring. In fact Mendel was the person to construct the foundation of Modern day Genetics and he is regarded as the father of Genetics. During 1956 to 1863 Mendel studied the pattern of transmission of hereditary characters from Parental generation to progeny in Pea Plants and formulated three theories which are popularly known as Mendel's laws of inheritance. Mendel concluded that there are some hereditary units who carry the hereditary characters generations after generations and he termed these units as factors. He said that for each and every character there is a factor. But Mendel knew nothing about physical and chemical properties of these factors. In 1909 Wilhelm Johannsen coined the term Gene to describe the factors proposed by Mendel. The meaning of Gene is generation. But before that in 1905 English Botanist William Bateson introduced the term Genetics which is dealt with the study of the Hereditary.

The Gene :

Conventionally genes are considered as basic unit of heredity. After the discovery of the structure and function of DNA (Deoxyribonucleic acid) scientist started to unfold different aspects of gene. DNA was initially identified by Swiss Chemist Friedrich Miescher in the late 1860 but the detail structure and function of DNA was worked out after a long gap by American biologist James Watson and English physicist Francis Crick in 1957 [1]. After the discovery of structure and function of DNA, gene is defined in a new way. Now the gene is defined as a discrete segment of DNA capable of producing at least one polypeptide chain. Genes are linearly arranged in a nucleo-protein structure called Chromosome. Boveri and Sutton postulated a theory where it is stated that Genes are located in a specific location of the Chromosomes. This theory is known as Chromosome theory of inheritance. Soon it was established that besides transmission of hereditary characters from one generation to next generation, genes are also responsible for expression of the different characters of a living body.

It is surprising to notice that all the characters of a living being are not expressed at the same time. If some characters are expressed in the child age, some others are expressed in teenage or some other characters are expressed after forty years of age. Again if some characters are expressed only in women then some others are only in men. Who controls all these mysterious actions? It is the gene who is doing all these things. Characters of a living being are expressed through a series of biochemical reactions. Each and every biochemical reaction again needs some enzymes. These enzymes are synthesized in ribosome of a cell with the direction of gene. In 1940 George Beadle and Edward Tatum first established the connection of gene with enzyme. They postulated a hypothesis which is popularly known as one gene one enzyme hypothesis. According to this hypothesis each gene encodes a single protein. A particular gene encodes a particular enzyme i.e. protein. This particular enzyme will catalyze a particular biochemical reaction and as a result of this biochemical reaction a particular character is expressed. Again a particular gene encodes a particular enzyme in a particular time frame for which there is a regulatory mechanism. This is how characters of a living organism are expressed.

Gene therapy :

It is already an established fact that all the characters of a living organism are controlled by genes. So, conclusion may be drawn that there are also some gene which are responsible for some undesired characters or defective characters or in a nut shell these genes are responsible for different genetic disorder. It is known to everyone that due to some metabolic irregularities or due to some microbial infections or due to some other reasons some time people suffer from illness. This kind of illness can be cured by applying appropriate drugs and by some surgery. But disorders developed by genetic influence cannot be treated by such conventional methods of treatment. In fact earlier it was impossible to treat genetic disorders. But with the development of modern Biotechnology nowadays different approaches have been made to correct some of the genetic disorders. But even today it is in experimental level. Scientists are being trying with their level best to make these approaches successful

and some good results have already been approved. One such approach is Gene Therapy. French Anderson in National Institute of Health (NIH) in 1990 for the first time became successful in clinical trial of gene therapy [2]. Two patients suffering from severe combined immunodeficiency (SCID) were cured using Novel gamma retro virus vector technology. Gene therapy is a technique where genetic disorders are corrected by altering the genetic makeup of a person. In this technology instead of using drugs or surgery, gene responsible for the disorder is replaced by a normal gene of same activity. Gene therapy is actually an aspect of Genetic Engineering or Recombinant DNA Technology. Genetic engineering needs sophisticated laboratory facilities and expertise of the worker in the field. Nowadays Physicians start to take the advantage of Nanotechnology in the field of Gene Therapy.

Application of Nanotechnology in gene therapy :

Nano is a word originated from the Greek language which means dwarf. Nano symbolize the smallness. Modern science defines the smallness of nano as one billionth of a meter. The technology deals with the dimensions smaller than 100 nanometres, is termed as nanotechnology. Nanotechnology is basically involves the production and application of Physical, Chemical and Biological systems at sizes ranging from individual atom or molecules to sub-micron dimensions [3]. In simple word nanotechnology deals with the production of nanoparticles. Nanoparticles are spherical, polymeric particles composed of natural or artificial polymers. The size of nanoparticles ranges between 10 and 500 nanometer. As a consequence of their spherical shape and high surface area to volume ratio, these particles have a wide range of potential applications [4]. Recently nanotechnology is also applied in gene therapy. In gene therapy, for transferring genetic material i.e. gene or DNA to a desired location some vectors are utilised. Basically circular DNA found in Bacterial Cell which are also known as Plasmids are used as Vector in Genetic Engineering or Recombinant DNA Technology. Culture of Bacterial colony, Isolation of plasmids from Bacterial cell and its processing as a vector needs utmost cautions and time. Besides, poor cellular uptake and instability of DNA in the physiological milieu limits its therapeutic potential [5]. In such condition, development of a vector which can protect the DNA and trigger the DNA efficiently in target cell is the need of the hour. Application of Nanotechnology in this field may help the scientists to overcome these difficulties and entry of Nanotechnology in Gene therapy has already been started. Applying the nanotechnology some nanoparticles have been developed as effective artificial vectors for gene therapy. Application of Polymer nanoparticle (PNPs), Dendrimers, Liposome, Magnetic nanoparticles, Gold nanoparticles has changed the scenario of progress of gene therapy. Most of these nanoparticles are synthesized from non toxic biodegradable and biocompatible polymers. Another successful and commercialized nano carrier in gene therapy is lipoplexes [6]. Lipoplexes are synthesized using cationic lipid such as quaternary ammonium lipids and cholesterol based cationic lipids. Lipoplexes can quickly form a self assembled structure that can penetrate the cell membrane and escape endosomal elimination. Gradually different nanoparticles are being developed by the scientists for effective use in gene therapy.

Conclusion :

The research and development of both nanotechnology and gene therapy is in embryonic stage and most of the findings are in clinical trial. Application of nanotechnology in gene therapy craving the route for the treatment of wide range of diseases such as cancer, cystic fibrosis, heart disease, diabetes, haemophilia, AIDS etc. At present the advantages of gene therapy is enjoyed by only a few people of the developed and developing countries. But it is expected that soon a day will come when nanotechnology will boost the development of gene therapy in such a way that every person society will be able to get the benefit of gene therapy.

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Unlocking the Future of the Nanotechnology Revolution: Insights from a Chemical Perspective

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Abstract :

The term ‘Nanotechnology’ is a ubiquitous reality of the 21st century that requires no introduction. It is one of the emerging principle areas of research that is interdisciplinary of chemistry and materials science, and in some instances merges with biology to generate novel and yet undiscovered properties that can be exploited to gain new opportunities. It is the science of reinventing the world, one atom at a time. The headway of nanoscience has been demonstrated by several facts. Altogether, nanotechnology holds tremendous promise for transformative developments in various areas, as well as new avenues for scientific discovery and technological innovation. However, as research progresses, the responsible development and applications of this budding technology will be crucial to realize its full potential in a wide range of disciplines. The scientific and technological tendencies enforcing the development of nanotechnology are presented in the several aforementioned articles. With a special emphasis in the field of chemistry, this article provides a comprehensive overview of potential future directions and challenges, emphasizing sustainable deployment in shaping the nanotechnology revolution.

Keywords: Nanotechnology, nanoscience, nanomaterials, future direction, sustainable deployment, interdisciplinary collaboration.

Introduction:

Following Feynman’s discovery of this new area of study, which piqued the curiosity of many scientists, a multitude of studies emphasized the enormous potential of nanotechnology, establishing it as one of the most intriguing areas of application in diverse fields. In 1981, the physicists Gerd Binnig and Heinrich Rohrer at the IBM Zurich Research Laboratory, invented the Scanning Tunneling Microscope (STM), for which the duo were awarded the Nobel Prize in Physics, later in 1986 [1-3]. Few years later, in 1990, Don Eigler and his co-workers at the IBM Research Center, Almaden produced the IBM logo on a nickel surface by manipulating 35 individual xenon atoms [4]. The STM facilitated the manipulation of atoms and molecules to create structures by imaging surfaces at atomic

dimensions. This breakthrough led to the development of the atomic force microscope (AFM) and scanning probe microscopes (SPM) which serve as the key tools for nanotechnology in the present times [5,6]. Meanwhile, in 1985, Robert Curl, Richard Smalley, and Harold Kroto observed that carbon can exist as highly stable spheres, known as bucky balls or fullerenes, ushering in a new era of carbon chemistry [7]. A few years later, in 1991, Iijima and coworkers, discovered a novel member of the fullerene family which appeared as hollow graphitic tubes, termed as carbon nanotube. Due to their flexibility and strength, the carbon nanotubes emerged as promising agents for a wide range of nanotechnological applications [8]. Further in 2004, when purifying single-walled carbon nanotubes, Xu et al. accidentally discovered a new class of carbon nanomaterials bearing size less than 10 nm, called the carbon-dots (C-dots) [9]. Owing to their abundant, benign, and inexpensive nature, the C-dots gained immense popularity as a new class of nanocarbon member with intriguing features.

In the meantime, nanoscience progressed in other scientific domains as well. Bio-nanotechnology is described as one of the most fascinating areas of application of nanoscience in this respect. Numerous studies that emphasize the role of nanotechnologies in biology-related domains including drug delivery, diagnostics, and molecular imaging are being conducted. Since Nadrian Seeman established the conceptual foundation for DNA nanotechnology in the year 1982, nucleic acids represent one of the most significant areas of application of nanotechnology in molecular biology [10]. In the year 2006, Paul Rothermund developed the “scaffolded DNA origami” employing a ‘one-pot’ reaction by increasing the complexity and size of self-assembled DNA nanostructures [11]. Over time, DNA nanotechnology has therefore, evolved into an interdisciplinary area of research, with experts from chemistry, physics, computer science, material science, and medicine collaborating to address future challenges in nanotechnology [12, 13]. Remarkable progress have also been accomplished in the area of nanooncology, by enhancing the efficiency of conventional chemotherapeutic medications for a variety of aggressive human malignancies [14]. These developments are enabled by the application of various functional molecules such as antibodies, nanoparticles and cytotoxic substances to the tumor site. In this regard, many studies have demonstrated the use of nanomaterials either on their own or in conjunction with therapeutic compounds to control vital biological processes, such as, metabolism, autophagy, or oxidative stress, exhibiting anticancer properties [15]. Further, the application of nanotechnology have also been extended to the production of environmentally friendly and economically viable energy which includes, reduced pollution levels during the manufacturing of materials, solar cell production for generating electricity at a competitive cost, removal of organic pollutants from groundwater, and removal of volatile organic compounds (VOCs) from the air [16].

In only a few decades, nanoscience and nanotechnology has seen a prosperous growth in different directions. Technology has evolved in physics from observing objects at micro to nano-scale dimensions, or even smaller using different microscopes; in chemistry from micro-sized bulk matter to small-sized carbon dots; in computer science, from room-sized computers to portable, thin laptops; and in biological science, from gaining a deeper insight

into the behaviour of the cell nucleus to the study of single complicated biomolecules at the nanoscale. However, as nanotechnology continues to advance, it presents unprecedented opportunities to contribute to breakthroughs in drug delivery, materials science, environmental remediation, and beyond. This is simply why, nanotechnology is often referred to as “the future technology”. This article explores the future trajectory of nanotechnology, with a special focus on insights derived from a chemical perspective. It provides a comprehensive overview of potential future directions and challenges, emphasizing sustainable deployment and the role of chemistry in shaping the nanotechnology revolution.

Potential Applications and Future Prospects :

Precision Medicine :

Targeted Drug Delivery :

At present, 95% of all novel potential therapeutics possess inadequate pharmacokinetics and biopharmaceutical properties, necessitating the development of appropriate systems for drug delivery that selectively transfer the therapeutically active drug molecule to the site of action without affecting healthy tissues and organs [17]. Nanotechnology serves as a crucial part in future therapies as “nanomedicines” by allowing this situation by lowering the effective number of doses, while simultaneously boosting therapeutic indices and safety profiles of novel medications. Nanomedicines possess a size range that enables their administration, without clogging needles or capillaries and are therefore, excellent agents for targeted drug delivery and medical imaging in case of certain disorders such as inflammation and cancer [18]. Targeted drug delivery can be accomplished by modulating the nanocarriers with antibodies, ligands, or other targeting moieties that detect specific receptors or biomarkers on the surface of target cells. It reduces the exposure of healthy tissues to the therapeutic substances, lowering side effects and improving the safety profile of the treatment [19]. This subsequently, reduces drug toxicity due to preferential accumulation at target locations and decreased concentration in healthy tissues. In addition, many nanocarriers possess the ability to enhance solubility of hydrophobic compounds in aqueous media, and stabilize a wide range of therapeutics, rendering them suitable for parenteral administration [20]. Further, nanocarriers constituting of biocompatible materials are also being explored, as safe substitutes to existing formulation vehicles such as, Cremophor[®] EL (BASF, NJ, Mount Olive) which can produce hypersensitivity responses and peripheral neuropathy [21, 22]. However, despite their promises, the major challenges such as mass production, lack of long-term safety assessment studies and potential immune responses are few of the major challenges in the development of nanocarriers. Ongoing research continues to refine the design, functionality, and safety profiles of nanocarriers for various applications in medicine.

Diagnostic Nanoprobos :

Nanomaterials serve as versatile tools in chemical diagnostics, particularly when employed as nanoprobos for a range of sensing and imaging applications. They offer unique nanoscale features, making them ideal for imaging biological structures, and delivering

crucial diagnostics information. By enabling visualization and identification of specific biomarkers associated with a range of illnesses, including cancer, cardiovascular disorders, and certain infectious diseases, the nanoprobe exhibits the potential to be the next step-stone in medical imaging and diagnostics [23]. A list of ways in which nanomaterials function as chemical diagnostic nanoprobe are as follows: Fluorescent nanoparticles and quantum dots are designed to emit light at certain wavelengths. These nanoprobe when functionalized with targeted ligands can be employed in biological samples for imaging and detection. They are particularly useful in fluorescence imaging for diagnostics [24,25]. Magnetic nanoparticles can be employed in magnetic resonance imaging (MRI) as contrast agents for medical diagnostics. Functionalized magnetic nanoprobe can also be directed to specific targets, enhancing imaging sensitivity [26, 27]. Silica nanoparticles are versatile carriers that can be loaded with drugs, fluorescent dyes, or targeting ligands. Their porous structure enables controlled release and imaging, making them ideal for diagnostics [28]. Carbon nanotubes exhibit exceptional electrical conductivity as well as distinct optical characteristics. They may be functionalized for targeting specific biomolecules and employed in near-infrared fluorescence imaging [29]. Liposomes are lipid bilayer-based nanoscale vesicles that are frequently utilized as drugs and imaging agent carriers. For diagnostic applications, liposomes can be targeted to certain cells or tissues by surface modification using targeting ligands [30, 31]. The unique properties of nanomaterials contribute to the creation of portable, rapid, and sensitive diagnostic tools for use in various settings. One of the most significant benefits of nanoprobe is their ability to identify targets even at very low concentrations, resulting in extremely high sensitivity [32]. Nanoprobe holds the promise for non-invasive imaging reducing the need for invasive procedures. However, the development of diagnostic nanoprobe has its own challenges from lack of studies on biocompatibility to lack of understanding of clearance of nanoprobe from the body and the challenges faced due to regulatory approvals. Thus, continued research and development efforts are vital to optimize their performance, safety, and clinical applicability.

Catalysis and Synthesis:

Nanocatalysts:

Catalysis happens to be one of the pioneer utilizations of nanoparticles. Given the tremendous challenges faced by the world concerning the atmosphere, environment, and energy, catalysis technologies have all the earmarks of being increasingly important to synthesis process, energy, and environmental areas. The development of nanotechnology in recent years has led to a change in the study of improved catalysts and catalytic activity. Undoubtedly, the use of nanomaterials in catalysis particularly inorganic nanoparticles—has attracted several research initiatives worldwide to develop innovative and sustainable methods. Among the several potential applications, carbon capture and utilization [33, 34], fine chemicals and pharmaceutical manufacturing [35], are among the most significant prospective uses of nanocatalysts. Nanocatalysts aid in carbon capture and utilization efforts by helping in the conversion of captured carbon dioxide into valuable chemicals or fuels. This has the potential to play a critical role in reducing emissions of greenhouse gases and mitigating

climate change. Further, by facilitating the more efficient and selective synthesis of complex compounds, nanocatalysts have the potential to revolutionize the production of fine chemicals and medications, resulting in lower expenses, higher yields, and more general process efficiency [35]. Other uses include energy conversion and storage [36], catalytic conversion of biomass [37], catalytic nanomaterials for production of nanomedicine [38], water treatment [39], environmental remediation [40] etc. These nanoparticles can promote the reactant process in novel media, like water, by acting as a catalyst or mediator [41]. In addition, nanocatalysts have emerged as an intriguing option at the interface among homogeneous and heterogeneous catalysis, taking into account an increased reaction rate, due to their small size and increased surface area [42]. Furthermore, nanoparticles can provide additional functions to reactants due to their intriguing intrinsic characteristics, such as photocatalytic activity and nanomagnetism. Along these lines, in the quest for more economical and environmentally friendly catalysts, nanocatalysis is turning into a significant discipline of science, which finds widespread use in both academia and industry.

Green Synthesis :

Green chemistry has seen an explosive growth in terms of both preparing green nanocatalysts as well as green conditions during catalysis of industrially important reactions. The term “green nanocatalysis” describes the catalytic process carried out in environmentally friendly solvents, ideally with the aid of green nanocatalysts. According to a Scifinder Scholar search, the field of employing greener methods to catalyze processes with nanoparticles has grown rapidly from the year 2002 to present [43]. Green nanocatalysts are typically 23 nm in size since this is the ideal size range for many catalytic processes. This broad trend does, of course, contain numerous exceptions, and various greater sizes of green nanocatalyst have also been synthesized for these purposes. In recent years, a slew of industrially significant green processes have begun to use nanoparticles as catalysts. To mention a few, there is the Sonogashira cross-coupling reaction between aryl iodide and bromides with terminal alkynes using palladium nanoparticles as catalyst [44], the Suzuki cross-coupling reaction, which uses palladium nanoparticles adsorbed onto diatomite and is carried out in water as the solvent [45], the arene hydrogenation reaction in water, which uses rhodium nanoparticles adsorbed onto titanium dioxide [46] etc. As can be seen, the future of green chemistry is highly bright, with numerous paths of study accessible for investigation. The use of green chemistry in nanoparticle catalysis has grown rapidly, and discovering new green nanocatalysts or altering reaction conditions to be greener will likely continue this trend in the future.

Sensors and Detection :

Highly Sensitive Sensors :

Nanosensors are nanomaterial-based sensing devices that can convert chemical, physical or biological substances into detectable signals, allowing them to serve the purpose of detection and response to specific stimuli or analytes. Nanosensors are efficient and cost-effective methods capable of sensing changes in their environment, such as alterations in

temperature, concentration of biomolecules, pH, or the presence of specific chemicals, making them suitable for a wide range of scientific and technological applications [47]. In biomedical sensing, nanosensors can revolutionize medical diagnostics enabling the early detection of biomarkers associated with diseases [48]. Highly sensitive nanosensors can be employed to monitor pollution and other contaminants in the air, water, and soil in real time, promoting public health and environmental sustainability [49]. Additionally, foodborne illnesses can also be addressed by using nanosensors that can identify minute levels of pollutants in food items, improving food safety and quality control [50]. Owing to the high sensitivity to detect minute changes in chemical composition, temperature, and pressure, nanosensors can be extremely useful in monitoring and controlling industrial processes [47]. Further, highly sensitive nanosensors can be employed for the rapid detection of chemical and biological threats, enhancing security and defence capabilities [51]. Continuous advancements in nanotechnology are paving the way for the development of nanosensors with unprecedented sensitivity and versatility as they provide several benefits in sensitivity and response time over sensors based on traditional materials. Being a new technology, nanosensors have still not been adopted for general use due to many unanswered questions regarding nanotoxicology, separation procedures, calibration techniques, etc. which currently limit their applications in biological systems [47, 52]. In addition, research reveals that nanosensors can upset cell metabolism and homeostasis, affecting the cellular molecular profiles and making it challenging to distinguish between sensor-induced artefacts and fundamental biological processes [53]. Due to their small dimensions and sensitivity to various synthesis techniques, nanosensors require a high level of precision during manufacturing, which adds to the technological challenges. Despite the challenges, continuing research indicates ample application of nanosensors in detection of contaminants and pathogens, medicine and monitoring manufacturing processes and logistics systems, etc.

Point-of-Care Diagnostics :

Nanomaterials and nanotechnology offer intriguing prospects in point-of-care (POC) diagnostics and therapeutics due to their distinctive chemical and physical properties. POC systems aim to deliver rapid and portable therapeutic and diagnostic capabilities at the patient care site, while also providing cost-effective solutions [54]. Leveraging nanomaterials with specific electrical, optical, and magnetic characteristics has the potential to transform the POC sector, by significantly improving the effectiveness and efficiency of diagnostic and therapeutic devices [55]. Technological innovations that reduce invasiveness by enabling the study of many matrices, such as breath and saliva, complement and broaden the application spectrum of POCs. Nanomaterials, such as carbon nanotubes [56], gold nanoparticles [57], and carbon dots [58], can improve the specificity and sensitivity of the tests, enabling accurate detection of specific analytes or biomarkers. Using diverse techniques such as lateral flow assays, enzyme-linked immunosorbent assays (ELISAs), and other approaches, nanomaterial-based POC assays can be constructed for a variety of applications, which includes the detection of infectious illnesses, cancer, and other ailments. In general, POC devices have number of shortcomings that might compromise their accuracy, sensitivity, and

selectivity. One potential drawback is that POC devices may not be as sensitive or selective as laboratory-based diagnostic procedures, which might result in findings that are misleading. In addition, development of nanomaterial-based POC assays require the ensurety of the safety and biocompatibility of the nanomaterials utilized. However, as nanoscience and nanotechnology continue to evolve, there is significant potential for further improvements in POC platforms, enabling for precise and faster disease detection, personalized medications, and improved patient outcomes.

Energy Storage and Conversion :

Nanomaterials for Batteries :

Nanostructured materials have been extensively studied for energy-related applications such as solar cells, thermoelectrics, catalysts, supercapacitors, lithium ion batteries, and hydrogen storage systems, owing to their nanoscale dimensions, which offer huge surface to volume ratios with modified physical properties, favourable transport properties, and confinement effects. The nanostructured materials offer the advantages by, (1) offering a large surface area to increase the electrochemical reaction or molecular adsorption occurring at the solid–liquid or solid–gas interface, (2) generating optical effects to enhance optical absorption in solar cells, and (3) resulting in high crystallinity and/or porous structure to promote ion or electron transport as well as electrolyte diffusion, ensuring the high efficiency of the electrochemical process [59]. The following are a few instances that highlights the diverse applications of nanomaterials in various battery types, showcasing how nanotechnology contributes to enhancing energy storage technologies. The lithium-ion batteries use silicon nanoparticle [60], and titanium dioxide as anode materials [61], while Lithium Iron Phosphate (LiFePO_4) [62] and Lithium Nickel Cobalt Manganese Oxide (NCM) [63] as cathode materials. Nanostructured sulfur particles integrated into carbon matrices, such as carbon nanotubes or graphene, aid in the resolution of challenges related to sulfur's low electrical conductivity and volume expansion during cycling, and are therefore used in lithium-sulphur batteries. Nanostructured solid electrolytes, such as lithium-ion conductive ceramics are being investigated as ways to increase the safety, energy density, and cycle life of solid-state batteries. Further, graphene-based nanomaterials can be employed to improve the electrical conductivity and specific surface area of supercapacitor electrodes, resulting in higher energy and power density. To develop flexible batteries, nanomaterials such as nanowires or nanotubes can be incorporated into flexible substrates. This is particularly relevant for wearable devices and flexible electronics [59]. Quantum dots can be used in electrodes to enhance charge separation and efficiency in solar-charging technologies, such as quantum dot solar batteries [64, 65]. While, nanomaterials offer numerous advantages for improving the performance of batteries, their widespread use face several challenges due to synthesis and production costs, stability and durability, safety concerns, unintended side reaction, regulatory and environmental concerns, and so on. Therefore, addressing safety issues, optimizing synthesis techniques, and improving scalability are crucial steps toward realizing the full potential of nanomaterials in batteries. As research progresses, sustained innovation and multidisciplinary collaboration will contribute to overcoming these challenges

and advancing the integration of nanomaterials in battery technologies.

Nanocatalysts for Energy Conversion :

Energy conversion processes are greatly aided by nanocatalysts, which open up new avenues for sustainable and effective energy production and utilization. Nanocatalysts, often based on platinum nanoparticles supported on carbon, can be employed as catalysts in the anode and cathode of Proton Exchange Membrane Fuel Cells (PEMFCs) to facilitate the electrochemical processes involved in oxidation of hydrogen and reduction of hydrogen [66]. Nanocatalysts can also be used in water-splitting reactions for production of hydrogen via electrolysis. Transition metal-based nanoparticles, including iron, cobalt, and nickel, are employed in the process for both the oxygen evolution reaction (OER) and the hydrogen evolution reaction (HER) [67]. Photocatalytic methods employ nanocatalysts, specifically semiconductor-based nanoparticles such as titanium dioxide (TiO_2), to convert solar energy into chemical energy [68]. Nanocatalysts are explored for the electrochemical reduction of carbon dioxide (CO_2) to produce valuable fuels and chemicals. Nanocatalysts, such as copper-zinc oxide-based nanoparticles, can be employed in the synthesis of methanol through the hydrogenation of carbon dioxide [69]. These catalysts enhance the selectivity and efficiency of the methanol production process. Nanocatalysts are also investigated for ammonia synthesis, a crucial step in the production of fertilizers. Nanocatalysts, often based on solid acid catalysts or metal nanoparticles, are employed in transesterification reactions for the production of biodiesel. These catalysts enhance the efficiency of converting triglycerides into biodiesel and glycerol [70]. Nanocatalysts are utilized in photoelectrochemical cells for solar fuel production. These cells utilize sunlight to drive electrochemical reactions, and nanocatalysts facilitate the necessary redox reactions at the electrode interfaces [71]. As evident, nanocatalysts are anticipated to make a substantial contribution in addressing global energy challenges and transitioning to cleaner and more efficient energy systems. However, their practical implementation necessitates the resolution of several critical concerns, such as catalyst stability and durability, catalyst poisoning and deactivation, cost and scalability issues, reaction selectivity issues, and an emphasis on sustainable solutions.

Nanotechnology in Analytical Chemistry :

As research in nanotechnology has evolved, nanomaterials have gained substantial attention in analytical chemistry. Their distinct physical and chemical properties make them ideal candidates for augmenting sensitivity and selectivity, that can be exploited for diverse analytical applications, including single-molecule detection techniques [72]. Besides their utility as sensors, catalysts and probes explored in detail in the preceding discussion, nanoparticles, nanotubes and nanostructured surfaces find huge applications in sensitivity and selectivity methods like chromatography, spectroscopy, and mass spectrometry [73]. Nanoporous materials are also employed for adsorption and selective separation of analytes along with controlled release of encapsulated molecules, and enhancement of sample preparation methods [74]. Furthermore, Surface-Enhanced Raman Spectroscopy or SERS

uses metal nanoparticles for amplification of Raman signals in trace analysis, for highly sensitive molecular detection [75]. Nanochannels and nanowires are used in manipulation and analysis of minute sample volumes, as well as development of miniaturized analytical devices. Magnetic nanoparticles, molecularly imprinted polymers are employed for extraction, preconcentration, and clean-up of samples in order to increase sensitivity and selectivity in analytical methods [76]. Therefore, the use of nanomaterials in analytical chemistry has led to advancements in sensitivity, selectivity, and efficiency, opening the door to the development of more robust and miniaturized analytical devices.

Nanotechnology for Environmental Chemistry :

Water Treatment :

Given the importance of potable water in both the developed and developing nations, as well as concerns regarding the sustainability of current practices in meeting the growing demands of all water users, there is an obvious need for the development of novel technologies and materials, that can address the challenges associated with the provision of safe potable water [77]. Although new ways are constantly being investigated, they must be less expensive in the long run, durable, and more effective than current solutions for removing toxins from water, either in situ or in water purification systems.

Nanotechnology has been highlighted as a technology that has the potential to solve or improve many of the problems associated with water purification and quality [78]. Three categories comprise the potential impact areas of nanotechnology: treatment and remediation, sensing and detection, and pollution prevention [79]. Nanomaterial-based membranes with nanoscale pores can be used to remove pollutants from water such as bacteria, viruses, and nanoparticles, offering a greater level of filtration than standard approaches. Adsorbents based on nanoparticles with large surface area and functionalized surfaces can be utilized to remove heavy metals, organic pollutants, and other impurities. Semiconductor nanoparticles like titanium dioxide (TiO_2) may be utilized to break down organic contaminants through photocatalytic processes when exposed to light, providing a sustainable way to purify water [80]. Nanocatalysts, such as metal nanoparticles, can aid in the acceleration of chemical reactions for the breakdown of pollutants in water, including organic toxins. Further in water treatment processes, enhanced sedimentation and removal of suspended particles and colloids are made possible by nanoscale coagulants and flocculants. While significant emphasis has been placed on the development and possible benefits of nanomaterials in water treatment procedures, concerns have been raised about their potential toxicity to humans and the environment. Assessing the impact of nanomaterials on health and the environment is, however challenging since the tools and methodologies for doing so are not yet sufficiently developed. It is vital that the processes be developed and examined to guarantee that nanomaterials are as safe as feasible, while attaining their maximum potential. Regardless of these knowledge gaps, it is evident that emerging nanomaterials, notably in water and waste water treatment, will play critical roles in ensuring adequate and good quality water to fulfil the world's growing demand for potable water.

Air Purification by Nanotechnology :

With the world facing immense challenges in air quality as a result of ongoing economic growth and industrialization, recent advances in nanotechnology have resulted in excellent prospects for the use of nanomaterials for environmental remediation. Compared with bulk materials, nanomaterials possess larger surface areas and thereby, exhibit substantially different magnetic, electronic and catalytic properties for $PM < 100\text{ nm}$, making them ideal for filtration, adsorption, and photocatalysis [81]. Nanomaterials can be applied to catalytic degradation, gas adsorption, particle filtering, and bacterial disinfection processes. Nanomaterials can be applied to particle filtration, gas adsorption, bacterial disinfection, and catalytic degradation. Nanomaterials with a high surface area and specific functions can be employed in a variety of purification devices to minimize particulate matter (PM), gaseous precursors (i.e., NO_x, VOCs, etc.), and hazardous pollutants, like HCHO [82]. Early pollution identification is made possible by the real-time monitoring of air quality by nanosensors. In addition, the efficiency of catalytic converters in air purification systems can also be improved by nanocatalysts. Incorporating nanoparticles into HVAC (heating, ventilation, and air conditioning) can enhance filtration, air quality, and energy efficiency in structures and enclosed areas. Despite the promising achievements in air pollution control technologies, further research is desperately needed to address several issues for practical air pollution abatement applications considering nanotechnology. Efforts should be made to develop novel, cost-effective, and functional air cleaning nanomaterials that may be employed in ways that benefit our society. Given the ongoing challenges of air pollution and the predicted increase in the quantity of pollutants in the future, Air Purification by Nanotechnology or APN technology is anticipated to have a significant impact on mitigating air pollution globally.

Concluding Remarks :

The preceding discussion of the current status and future prospects of the science and technology of nanomaterials should suffice to highlight the remarkable vitality of the subject and the enormous opportunities it provides. The future of nanotechnology in applications for chemistry is poised for ground-breaking advancements, offering transformative solutions to address challenges and revolutionize numerous facets of life. The convergence of nanotechnology and chemistry has opened up new avenues for innovation, facilitating the development of novel materials, analytical techniques, and sustainable processes. From improving our knowledge of fundamental chemical processes to delivering practical solutions in the fields of medicine, energy, and the environment, nanotechnology is poised to shape the future of chemistry in profound and fascinating ways. Continued research, collaboration, and ethical considerations are crucial in navigating the evolving landscape of nanotechnology applications in chemistry.

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